Mathematics Class 11 - Solved Exercises (PCTB, English medium)

Complete book: all 14 chapters, all 73 exercise sets, solved from the 2018-19 1st edition, 24th impression textbook (April 2018)

Chapter 1: Number Systems

Exercise 1.1 (book p. 12)

1. Which of the following sets have closure property w.r.t. addition and multiplication? (i) {0} (ii) {1} (iii) {0, -1} (iv) {1, -1}

(i) {0}: 0 + 0 = 0, which is in the set, and 0 x 0 = 0, which is in the set. Closed w.r.t. addition and closed w.r.t. multiplication.

(ii) {1}: 1 + 1 = 2, which is NOT in the set, so NOT closed w.r.t. addition; 1 x 1 = 1, which is in the set, so closed w.r.t. multiplication.

(iii) {0, -1}: (-1) + (-1) = -2, which is NOT in the set, so NOT closed w.r.t. addition; (-1) x (-1) = 1, which is NOT in the set, so NOT closed w.r.t. multiplication.

(iv) {1, -1}: 1 + (-1) = 0, which is NOT in the set, so NOT closed w.r.t. addition; the products are 1 x 1 = 1, 1 x (-1) = -1 and (-1) x (-1) = 1, all in the set, so closed w.r.t. multiplication.

These agree with the book's Answers (p. 408).

2. Name the properties used in the following equations. (Letters, where used, represent real numbers). (i) 4 + 9 = 9 + 4 (ii) (a + 1) + 3/4 = a + (1 + 3/4) (iii) (sqrt(3) + sqrt(5)) + sqrt(7) = sqrt(3) + (sqrt(5) + sqrt(7)) (iv) 100 + 0 = 100 (v) 1000 x 1 = 1000 (vi) 4.1 + (-4.1) = 0 (vii) a - a = 0 (viii) sqrt(2) x sqrt(5) = sqrt(5) x sqrt(2) (ix) a(b - c) = ab - ac (x) (x - y)z = xz - yz (xi) 4 x (5 x 8) = (4 x 5) x 8 (xii) a(b + c - d) = ab + ac - ad

(i) Commutative property w.r.t. addition.

(ii) Associative property w.r.t. addition.

(iii) Associative property w.r.t. addition.

(iv) Additive identity (0 is the additive identity).

(v) Multiplicative identity (1 is the multiplicative identity).

(vi) Additive inverse (4.1 and -4.1 are additive inverses).

(vii) Additive inverse (a and -a are additive inverses; a - a = a + (-a) = 0).

(viii) Commutative property w.r.t. multiplication.

(ix) Left distributive property (multiplication over subtraction).

(x) Right distributive property.

(xi) Associative property w.r.t. multiplication.

(xii) Left distributive property (extended to three terms inside the bracket).

These agree with the book's Answers (p. 408).

3. Name the properties used in the following inequalities: (i) -3 < -2 => 0 < 1 (ii) -5 < -4 => 20 > 16 (iii) 1 > -1 => -3 > -5 (iv) a < 0 => -a > 0 (v) a > b => 1/a < 1/b (vi) a > b => -a < -b

(i) Additive property: adding 3 to both sides gives -3 + 3 = 0 and -2 + 3 = 1, and 0 < 1.

(ii) Multiplicative property: multiplying both sides by -4, a negative number, reverses the inequality: (-5) x (-4) = 20 and (-4) x (-4) = 16, and 20 > 16.

(iii) Additive property: adding -4 to both sides gives 1 - 4 = -3 and -1 - 4 = -5, and -3 > -5.

(iv) Multiplicative property: multiplying both sides by -1 reverses the inequality, so a < 0 gives -a > 0.

(v) Multiplicative (reciprocal) property for positive a and b: if 0 < b < a, then multiplying by the positive number 1/(ab) gives 1/a < 1/b. Note: the book's statement assumes a and b are positive; the inequality need not hold for negative

values.

(vi) Multiplicative property: multiplying both sides by -1 reverses the inequality, so a > b gives -a < -b.

The names agree with the book's Answers (p. 408), which lists (i) Additive, (ii) Multiplicative, (iii) Additive, (iv) Multiplicative, (v) Multiplicative, (vi) Multiplicative.

4. Prove the following rules of addition: (i) a/c + b/c = (a + b)/c (ii) a/b + c/d = (ad + bc)/(bd)

(i) Step 1: By the definition of division, a/c = a x (1/c) and b/c = b x (1/c).

(i) Step 2: By the distributive property, a x (1/c) + b x (1/c) = (a + b) x (1/c).

(i) Step 3: By the definition of division again, (a + b) x (1/c) = (a + b)/c. This proves (i).

(ii) Step 1: Multiply numerator and denominator of a/b by d: a/b = (ad)/(bd).

(ii) Step 2: Multiply numerator and denominator of c/d by b: c/d = (bc)/(bd).

(ii) Step 3: By part (i), (ad)/(bd) + (bc)/(bd) = (ad + bc)/(bd). This proves (ii).

5. Prove that -7/12 - 5/18 = (-21 - 10)/36

Step 1: Multiply numerator and denominator of -7/12 by 3: -7/12 = (-21)/36.

Step 2: Multiply numerator and denominator of 5/18 by 2: 5/18 = 10/36.

Step 3: Subtracting is adding the additive inverse, so (-21)/36 - 10/36 = (-21 + (-10))/36 = (-21 - 10)/36.

Step 4 (check): (-21 - 10)/36 = -31/36, and indeed -7/12 - 5/18 = -21/36 - 10/36 = -31/36.

6. Simplify by justifying each step: (i) (4 + 16x)/4 (ii) (1/4 + 1/5)/(1/4 - 1/5) (iii) (a/b + c/d)/(a/b - c/d) (iv) (1/a - 1/b)/(1 - (1/a)(1/b))

(i) Step 1: By the distributive property, (4 + 16x)/4 = 4/4 + (16x)/4.

(i) Step 2: 4/4 = 1 and (16x)/4 = 4x, so the result is 1 + 4x. This agrees with the book's answer.

(ii) Step 1: Numerator: 1/4 + 1/5 = 5/20 + 4/20 = 9/20 (common denominator 20).

(ii) Step 2: Denominator: 1/4 - 1/5 = 5/20 - 4/20 = 1/20.

(ii) Step 3: Quotient: (9/20)/(1/20) = 9. This agrees with the book's answer.

(iii) Step 1: By Question 4(ii), numerator a/b + c/d = (ad + bc)/(bd) and denominator a/b - c/d = (ad - bc)/(bd).

(iii) Step 2: Dividing the two fractions, the factor (bd) cancels (bd is not zero), giving (ad + bc)/(ad - bc). This agrees with the book's answer.

(iv) Step 1: Numerator: 1/a - 1/b = (b - a)/(ab) (common denominator ab).

(iv) Step 2: Denominator: 1 - (1/a)(1/b) = 1 - 1/(ab) = (ab - 1)/(ab).

(iv) Step 3: Dividing, the factor (ab) cancels, giving (b - a)/(ab - 1). This agrees with

the book's answer.

Exercise 1.2 (book pp. 16-17)

1. Verify the addition properties of complex numbers.

Use the definition (a, b) + (c, d) = (a + c, b + d), where a, b, c, d are real numbers.

Closure: a + c and b + d are real, so (a + c, b + d) is a complex number.

Commutative: (a, b) + (c, d) = (a + c, b + d) = (c + a, d + b) = (c, d) + (a, b), since real addition commutes.

Associative: both [(a, b) + (c, d)] + (e, f) and (a, b) + [(c, d) + (e, f)] equal (a + c + e, b + d + f), since real addition is associative.

Additive identity: (0, 0), since (a, b) + (0, 0) = (a + 0, b + 0) = (a, b).

Additive inverse: for (a, b) it is (-a, -b), since (a, b) + (-a, -b) = (0, 0).

2. Verify the multiplication properties of the complex numbers.

Use the definition (a, b)(c, d) = (ac - bd, ad + bc).

Closure: ac - bd and ad + bc are real, so the product is a complex number.

Commutative: (a, b)(c, d) = (ac - bd, ad + bc) = (ca - db, da + cb) = (c, d)(a, b).

Associative: [(a, b)(c, d)](e, f) = (ac - bd, ad + bc)(e, f) = ((ac - bd)e - (ad + bc)f, (ac - bd)f + (ad + bc)e) = (ace - bde - adf - bcf, acf - bdf + ade + bce). And (a, b)[(c, d)(e, f)] = (a, b)(ce - df, cf + de) = (a(ce - df) - b(cf + de), a(cf + de) + b(ce - df)) = (ace - adf - bcf - bde, acf + ade + bce - bdf). The two are equal.

Multiplicative identity: (1, 0), since (a, b)(1, 0) = (a x 1 - b x 0, a x 0 + b x 1) = (a, b).

Multiplicative inverse: for (a, b) not equal to (0, 0), it is (a/(a^2 + b^2), -b/(a^2 + b^2)), since (a, b) times it equals ((a^2 + b^2)/(a^2 + b^2), (-ab + ab)/(a^2 + b^2)) = (1, 0).

3. Verify the distributive law of complex numbers: (a, b)[(c, d) + (e, f)] = (a, b)(c, d)

+ (a, b)(e, f). (Hint: Simplify each side separately)

Left side, step 1: (c, d) + (e, f) = (c + e, d + f) by the definition of addition.

Left side, step 2: (a, b)(c + e, d + f) = (a(c + e) - b(d + f), a(d + f) + b(c + e)) by the definition of multiplication.

Left side, step 3: Expanding by the real distributive law gives (ac + ae - bd - bf, ad + af + bc + be).

Right side, step 1: (a, b)(c, d) = (ac - bd, ad + bc) and (a, b)(e, f) = (ae - bf, af + be).

Right side, step 2: Adding them gives (ac - bd + ae - bf, ad + bc + af + be).

Conclusion: The left side equals (ac + ae - bd - bf, ad + af + bc + be), which is the same as the right side. The distributive law is verified.

4. Simplify the following: (i) i^9 (ii) i^14 (iii) (-i)^19 (iv) (-1)^(-1/2)

(i) Step 1: i^4 = 1, and 9 = 4 x 2 + 1.

(i) Step 2: i^9 = (i^4)^2 x i = 1 x i = i. This agrees with the book's answer.

(ii) Step 1: 14 = 4 x 3 + 2.

(ii) Step 2: i^14 = (i^4)^3 x i^2 = 1 x (-1) = -1. This agrees with the book's answer.

(iii) Step 1: (-i)^2 = (-1)^2 x i^2 = -1, and 19 = 2 x 9 + 1.

(iii) Step 2: (-i)^19 = ((-i)^2)^9 x (-i) = (-1)^9 x (-i) = (-1) x (-i) = i. This agrees with the book's answer.

(iv) Step 1: (-1)^(-1/2) = 1/((-1)^(1/2)) = 1/i, since i^2 = -1.

(iv) Step 2: 1/i = i/(i x i) = i/(-1) = -i. This agrees with the book's answer.

5. Write in terms of i: (i) sqrt(-1) b (ii) sqrt(-5) (iii) sqrt(-16/25) (iv) sqrt(1/(-4))

(i) Step 1: sqrt(-1) = i by definition, and b (a real number) stands outside the root as printed.

(i) Step 2: The result is i b. This agrees with the book's answer.

(ii) Step 1: sqrt(-5) = sqrt(5) x sqrt(-1).

(ii) Step 2: The result is sqrt(5) i. This agrees with the book's answer.

(iii) Step 1: -16/25 = (16/25) x (-1).

(iii) Step 2: sqrt(-16/25) = (4/5) x sqrt(-1) = (4/5) i. This agrees with the book's answer.

(iv) Step 1: 1/(-4) = -1/4.

(iv) Step 2: sqrt(-1/4) = (1/2) x sqrt(-1) = (1/2) i. This agrees with the book's answer.

6-12. Simplify the following (ordered pairs are complex numbers): 6. (7, 9) + (3, -5)

7. (8, -5) - (-7, 4) 8. (2, 6)(3, 7) 9. (5, -4)(-3, -2) 10. (0, 3)(0, 5) 11. (2, 6) / (3, 7) 12. (5, -4) / (-3, -8)

6. Step 1: (7, 9) + (3, -5) = (7 + 3, 9 + (-5)).

6. Step 2: The result is (10, 4). This agrees with the book's answer.

7. Step 1: (8, -5) - (-7, 4) = (8 - (-7), -5 - 4).

7. Step 2: The result is (15, -9). This agrees with the book's answer.

8. Step 1: (2, 6)(3, 7) = (2 x 3 - 6 x 7, 2 x 7 + 6 x 3).

8. Step 2: The result is (6 - 42, 14 + 18) = (-36, 32). This agrees with the book's answer.

9. Step 1: (5, -4)(-3, -2) = (5 x (-3) - (-4) x (-2), 5 x (-2) + (-4) x (-3)).

9. Step 2: The result is (-15 - 8, -10 + 12) = (-23, 2). This agrees with the book's answer.

10. Step 1: (0, 3)(0, 5) = (0 x 0 - 3 x 5, 0 x 5 + 3 x 0).

10. Step 2: The result is (-15, 0). This agrees with the book's answer.

11. Step 1 (hint followed): (2 + 6i)/(3 + 7i) = (2 + 6i)(3 - 7i)/((3 + 7i)(3 - 7i)).

11. Step 2: Numerator: (2 + 6i)(3 - 7i) = 6 - 14i + 18i - 42i^2 = 48 + 4i.

Denominator: 9 + 49 = 58.

11. Step 3: The result is (48/58, 4/58) = (24/29, 2/29). This agrees with the book's answer.

12. Step 1: (5 - 4i)/(-3 - 8i) = (5 - 4i)(-3 + 8i)/((-3 - 8i)(-3 + 8i)).

12. Step 2: Numerator: (5 - 4i)(-3 + 8i) = -15 + 40i + 12i - 32i^2 = 17 + 52i.
Denominator: 9 + 64 = 73.

12. Step 3: The result is (17/73, 52/73). This agrees with the book's answer.

13. Prove that the sum as well as the product of any two conjugate complex numbers is a real number.

Let the two conjugate numbers be z = a + bi and its conjugate z-bar = a - bi, where a and b are real.

Sum: z + z-bar = (a + bi) + (a - bi) = 2a, which is real.

Product: z x z-bar = (a + bi)(a - bi) = a^2 - abi + abi - b^2 i^2 = a^2 + b^2, which is real.

Hence both the sum and the product are real numbers.

14. Find the multiplicative inverse of each of the following numbers: (i) (-4, 7) (ii) (sqrt(2), -sqrt(5)) (iii) (1, 0)

General rule: the inverse of (a, b), not (0, 0), is (a/(a^2 + b^2), -b/(a^2 + b^2)).

(i) Step 1: a^2 + b^2 = 16 + 49 = 65.

(i) Step 2: The inverse is (-4/65, -7/65). This agrees with the book's answer.

(ii) Step 1: a^2 + b^2 = 2 + 5 = 7.

(ii) Step 2: The inverse is (sqrt(2)/7, sqrt(5)/7). This agrees with the book's answer.

(iii) Step 1: a^2 + b^2 = 1 + 0 = 1.

(iii) Step 2: The inverse is (1, 0). This agrees with the book's answer.

15. Factorize the following: (i) a^2 + 4b^2 (ii) 9a^2 + 16b^2 (iii) 3x^2 + 3y^2

(i) Step 1: Try (a + 2bi)(a - 2bi).

(i) Step 2: Expanding gives a^2 - 2abi + 2abi - 4b^2 i^2 = a^2 + 4b^2. So a^2 + 4b^2 = (a + 2bi)(a - 2bi). This agrees with the book's answer.

(ii) Step 1: Try (3a + 4bi)(3a - 4bi).

(ii) Step 2: Expanding gives 9a^2 - 12abi + 12abi - 16b^2 i^2 = 9a^2 + 16b^2. So 9a^2 + 16b^2 = (3a + 4bi)(3a - 4bi). This agrees with the book's answer.

(iii) Step 1: Take out the common factor 3: 3x^2 + 3y^2 = 3(x^2 + y^2).

(iii) Step 2: x^2 + y^2 = (x + yi)(x - yi), so the result is 3(x + yi)(x - yi). This agrees with the book's answer.

16. Separate into real and imaginary parts (write as a simple complex number): (i) (2 - 7i)/(4 + 5i) (ii) (-2 + 3i)^2/(1 + i) (iii) i/(1 + i)

(i) Step 1: Multiply numerator and denominator by the conjugate (4 - 5i): (2 - 7i)(4 -

5i)/((4 + 5i)(4 - 5i)).

(i) Step 2: Numerator: 8 - 10i - 28i + 35i^2 = -27 - 38i. Denominator: 16 + 25 = 41.

(i) Step 3: The result is -27/41 - (38/41)i. This agrees with the book's answer.

(ii) Step 1: (-2 + 3i)^2 = 4 - 12i + 9i^2 = -5 - 12i.

(ii) Step 2: (-5 - 12i)/(1 + i) = (-5 - 12i)(1 - i)/2 = (-5 + 5i - 12i + 12i^2)/2 = (-17 - 7i)/2.

(ii) Step 3: The result is -17/2 - (7/2)i. This agrees with the book's answer.

(iii) Step 1: i/(1 + i) = i(1 - i)/((1 + i)(1 - i)) = (i - i^2)/2.

(iii) Step 2: The result is (1 + i)/2 = 1/2 + (1/2)i. This agrees with the book's answer.

Exercise 1.3 (book p. 28)

1. Graph the following numbers on the complex plane: (i) 2 + 3i (ii) 2 - 3i (iii) -2 - 3i (iv) -2 + 3i (v) -6 (vi) i (vii) (3/5) - (4/5)i (viii) -5 - 6i

Plot each number a + bi as the point (a, b) on the Argand plane, with the real axis horizontal and the imaginary axis vertical, on the same scale on both axes.

(i) 2 + 3i: point (2, 3), first quadrant. (ii) 2 - 3i: point (2, -3), fourth quadrant. (iii) -2 - 3i: point (-2, -3), third quadrant. (iv) -2 + 3i: point (-2, 3), second quadrant.

(v) -6: point (-6, 0), on the negative real axis. (vi) i: point (0, 1), on the positive imaginary axis.

(vii) (3/5) - (4/5)i: point (3/5, -4/5), fourth quadrant. (viii) -5 - 6i: point (-5, -6), third quadrant.

Drawing instruction: mark these eight points and label each with its complex number. The book's own graph of these points is printed in the Answers section (p. 408).

2. Find the multiplicative inverse of each of the following numbers: (i) -3i (ii) 1 - 2i (iii) -3 - 5i (iv) (1, 2)

(i) Step 1: 1/(-3i) = i/((-3i) x i) = i/3.

(i) Step 2: The result is (1/3)i. This agrees with the book's answer.

(ii) Step 1: 1/(1 - 2i) = (1 + 2i)/((1 - 2i)(1 + 2i)) = (1 + 2i)/(1 + 4).

(ii) Step 2: The result is 1/5 + (2/5)i. This agrees with the book's answer.

(iii) Step 1: 1/(-3 - 5i) = (-3 + 5i)/((-3 - 5i)(-3 + 5i)) = (-3 + 5i)/(9 + 25).

(iii) Step 2: The result is -3/34 + (5/34)i. This agrees with the book's answer.

(iv) Step 1: (1, 2) = 1 + 2i, and 1^2 + 2^2 = 5.

(iv) Step 2: The inverse is (1/5, -2/5) = 1/5 - (2/5)i. This agrees with the book's answer.

3. Simplify: (i) i^101 (ii) (-ai)^4, a in R (iii) i^(-3) (iv) i^(-10)

(i) Step 1: 101 = 4 x 25 + 1, and i^4 = 1.

(i) Step 2: i^101 = (i^4)^25 x i = i. This agrees with the book's answer.

(ii) Step 1: (-ai)^4 = (-1)^4 x a^4 x i^4.

(ii) Step 2: (-1)^4 = 1 and i^4 = 1, so the result is a^4. This agrees with the book's answer.

(iii) Step 1: i^(-3) = 1/(i^3) = 1/(-i).

(iii) Step 2: 1/(-i) = i/((-i) x i) = i/1 = i. This agrees with the book's answer.

(iv) Step 1: i^(-10) = 1/(i^10), and i^10 = (i^4)^2 x i^2 = 1 x (-1) = -1.

(iv) Step 2: The result is 1/(-1) = -1. This agrees with the book's answer.

4. Prove that z-bar = z iff z is real. (z-bar denotes the conjugate of z.)

Let z = x + yi, where x and y are real, so z-bar = x - yi.

Forward: suppose z-bar = z. Then x - yi = x + yi, so -y = y, so y = 0. Hence z = x, a real number.

Backward: suppose z is real. Then y = 0, so z = x and z-bar = x, hence z-bar = z.

Therefore z-bar = z if and only if z is real.

5. Simplify by expressing in the form a + bi: (i) 5 + 2 sqrt(-4) (ii) (2 + sqrt(-3))(3 + sqrt(-3)) (iii) 2/(sqrt(5) + sqrt(-8)) (iv) 3/(sqrt(6) - sqrt(-12))

(i) Step 1: sqrt(-4) = 2i.

(i) Step 2: 5 + 2 x 2i = 5 + 4i. This agrees with the book's answer.

(ii) Step 1: sqrt(-3) = sqrt(3) i, so the product is (2 + sqrt(3) i)(3 + sqrt(3) i).

(ii) Step 2: Expanding gives 6 + 2 sqrt(3) i + 3 sqrt(3) i + 3 i^2 = 3 + 5 sqrt(3) i. This agrees with the book's answer.

(iii) Step 1: sqrt(-8) = 2 sqrt(2) i, so the denominator is sqrt(5) + 2 sqrt(2) i.

(iii) Step 2: Multiply numerator and denominator by the conjugate sqrt(5) - 2 sqrt(2) i: 2(sqrt(5) - 2 sqrt(2) i)/(5 + 8).

(iii) Step 3: The result is (2 sqrt(5))/13 - (4 sqrt(2))/13 i. This agrees with the book's answer.

(iv) Step 1: sqrt(-12) = 2 sqrt(3) i, so the denominator is sqrt(6) - 2 sqrt(3) i.

(iv) Step 2: Multiply numerator and denominator by the conjugate sqrt(6) + 2 sqrt(3) i: 3(sqrt(6) + 2 sqrt(3) i)/(6 + 12).

(iv) Step 3: The result is (3 sqrt(6))/18 + (6 sqrt(3))/18 i = (1/sqrt(6)) + (1/sqrt(3)) i. This agrees with the book's answer.

6. Show that for every z in C: (i) z^2 + (z-bar)^2 is a real number (ii) (z - z-bar)^2 is a real number. (z-bar denotes the conjugate of z.)

Let z = x + yi, where x and y are real, so z-bar = x - yi.

(i) z^2 = (x^2 - y^2) + 2xyi and (z-bar)^2 = (x^2 - y^2) - 2xyi. Adding them, the imaginary parts cancel: z^2 + (z-bar)^2 = 2(x^2 - y^2), which is real.

(ii) z - z-bar = 2yi. Squaring gives (2yi)^2 = 4y^2 i^2 = -4y^2, which is real.

7. Simplify the following: (i) (-1/2 + (sqrt(3)/2)i)^3 (ii) (-1/2 - (sqrt(3)/2)i)^3 (iii) (-1/2 - (sqrt(3)/2)i)^(-2) x (-1/2 - (sqrt(3)/2)i) (iv) (a + bi)^2 (v) (a + bi)^(-2) (vi) (a + bi)^3 (vii) (a - bi)^3 (viii) (3 - sqrt(-4))^(-3)

Let w = -1/2 + (sqrt(3)/2)i and w2 = -1/2 - (sqrt(3)/2)i. Note that w x w2 = 1/4 + 3/4 = 1, and w^2 = w2, w2^2 = w.

(i) Step 1: w^2 = (-1/2 + (sqrt(3)/2)i)^2 = 1/4 - (sqrt(3)/2)i - 3/4 = -1/2 - (sqrt(3)/2)i = w2.

(i) Step 2: w^3 = w^2 x w = w2 x w = 1. This agrees with the book's answer.

(ii) Step 1: w2^2 = (-1/2 - (sqrt(3)/2)i)^2 = 1/4 + (sqrt(3)/2)i - 3/4 = -1/2 + (sqrt(3)/2)i = w.

(ii) Step 2: w2^3 = w2^2 x w2 = w x w2 = 1. This agrees with the book's answer.

(iii) Step 1: w2^(-2) x w2 = w2^(-1) = 1/w2.

(iii) Step 2: 1/w2 = w/(w x w2) = w/1 = w = (-1 + sqrt(3) i)/2. This agrees with the book's answer.

(iv) Step 1: (a + bi)^2 = a^2 + 2abi + b^2 i^2.

(iv) Step 2: The result is a^2 + 2abi - b^2. This agrees with the book's answer.

(v) Step 1: (a + bi)^(-2) = 1/(a + bi)^2 = 1/(a^2 + 2abi - b^2).

(v) Step 2: Multiply numerator and denominator by the conjugate (a^2 - b^2) - 2abi: the denominator becomes (a^2 - b^2)^2 + (2ab)^2 = (a^2 + b^2)^2.

(v) Step 3: The result is (a^2 - b^2)/(a^2 + b^2)^2 - (2ab)/(a^2 + b^2)^2 i. This agrees with the book's answer.

(vi) Step 1: (a + bi)^3 = (a + bi)^2 x (a + bi) = (a^2 + 2abi - b^2)(a + bi).

(vi) Step 2: Expanding and collecting real and imaginary parts gives a^3 - 3ab^2 + (3a^2 b - b^3)i. This agrees with the book's answer.

(vii) Step 1: (a - bi)^3 = (a - bi)^2 x (a - bi) = (a^2 - 2abi - b^2)(a - bi).

(vii) Step 2: Expanding and collecting real and imaginary parts gives a^3 - 3ab^2 - (3a^2 b - b^3)i. This agrees with the book's answer.

(viii) Step 1: sqrt(-4) = 2i, so the number is (3 - 2i)^(-3).

(viii) Step 2: (3 - 2i)^2 = 9 - 12i + 4i^2 = 5 - 12i.

(viii) Step 3: (3 - 2i)^3 = (5 - 12i)(3 - 2i) = 15 - 10i - 36i + 24i^2 = -9 - 46i.

(viii) Step 4: The inverse is (-9 + 46i)/((-9)^2 + 46^2) = (-9 + 46i)/2197 = -9/2197 + (46/2197)i. This agrees with the book's answer (Answers p. 409).

Exercise 2.1 (book pp. 34-36)

1. Write the following sets in set builder notation: (i) {1, 2, 3, ..., 1000} (ii) {0, 1, 2, ..., 100} (iii) {0, +1, -1, +2, -2, ..., +1000, -1000} (iv) {0, -1, -2, ..., -500} (v) {100, 101, 102, ..., 400} (vi) {-100, -101, -102, ..., -500} (vii) {Peshawar, Lahore, Karachi, Quetta} (viii) {January, June, July} (ix) The set of all odd natural numbers (x) The set of all rational numbers (xi) The set of all real numbers between 1 and 2 (xii) The set of all integers between -100 and 1000

(i) {x | x in N and x <= 1000}. (ii) {x | x in W and x <= 100}. (iii) {x | x in Z and -1000 <= x <= 1000}. (iv) {x | x in Z and -500 <= x <= 0}. (v) {x | x in N and 100 <= x <= 400} (also {x | x in Z and 100 <= x <= 400}). (vi) {x | x in Z and -500 <= x <= -100}. (vii) {x | x is the capital of a province of Pakistan}. (viii) {x | x is a month of the Calendar year beginning with J}. (ix) {x | x is an odd natural number}. (x) {x | x in Q}. (xi) {x | x in R and 1 < x < 2}. (xii) {x | x in Z and -100 < x < 1000}. These agree with the book's Answers (p. 409).

2. Write each of the following sets in the descriptive and tabular forms: (i) {x | x in N and x <= 10} (ii) {x | x in N and 4 < x < 12} (iii) {x | x in Z and -5 < x < 5} (iv) {x | x in E and 2 < x < 5} (v) {x | x in P and x < 12} (vi) {x | x in O and 3 < x < 12} (vii) {x | x in E and 4 <= x <= 10} (viii) {x | x in E and 4 < x < 6} (ix) {x | x in O and 5 <= x <= 7} (x) {x | x in O and 5 <= x < 7} (xi) {x | x in N and x + 4 = 0} (xii) {x | x in Q and x^2 = 2} (xiii) {x | x in R and x = x} (xiv) {x | x in Q and x = -x} (xv) {x | x in R and x != x} (xvi) {x | x in R and x not in Q}

Here N = natural numbers, W = whole numbers, Z = integers, E = even integers, P = prime numbers, O = odd integers, Q = rationals, R = reals. (i) The set of the first ten natural numbers: {1, 2, 3, ..., 10}. (ii) The set of natural numbers between 4 and 12: {5, 6, 7, ..., 11}. (iii) The set of integers between -5 and 5: {-4, -3, -2, ...,

4}. (iv) The set of even integers between 2 and 5: {4}. (v) The set of prime numbers less than 12: {2, 3, 5, 7, 11}. (vi) The set of odd integers between 3 and 12: {5, 7, 9, 11}. (vii) The set of even integers from 4 up to 10: {4, 6, 8, 10}. (viii) The set of even integers between 4 and 6: { } (empty; no even integer lies strictly between 4 and 6). (ix) The set of odd integers from 5 up to 7: {5, 7}. (x) The set of odd integers from 5 up to but not including 7: {5}. (xi) The set of natural numbers x satisfying x + 4 = 0: { } (x = -4 is not natural). (xii) The set of rational numbers x satisfying x^2 = 2: { } (sqrt(2) is irrational). (xiii) The set of real numbers x satisfying x = x: the set of real numbers (every real satisfies it). (xiv) The set of rational numbers x satisfying x = -x: {0} (2x = 0 gives x = 0). (xv) The set of real numbers x satisfying x != x: { } (no number differs from itself). (xvi) The set of real numbers which are not rational: Q^c, the irrational numbers. These agree with the book's Answers (p. 409).

3. Which of the following sets are finite and which of these are infinite? (i) The set of students of your class. (ii) The set of all schools in Pakistan. (iii) The set of natural numbers between 3 and 10. (iv) The set of rational numbers between 3 and 10. (v) The set of real numbers between 0 and 1. (vi) The set of rationals between 0 and 1. (vii) The set of whole numbers between 0 and 1. (viii) The set of all leaves of trees in Pakistan. (ix) P(N). (x) P({a, b, c}). (xi) {1, 2, 3, 4, ...}. (xii) {1, 2, 3, ..., 1000000000}. (xiii) {x | x in R and x != x}. (xiv) {x | x in R and x^2 = -16}. (xv) {x | x in Q and x^2 = 5}. (xvi) {x | x in Q and 0 <= x <= 1}

(i) Finite. (ii) Finite. (iii) Finite ({4, 5, 6, 7, 8, 9}). (iv) Infinite (between any two rationals there is another rational). (v) Infinite. (vi) Infinite. (vii) Finite (empty: no whole number lies strictly between 0 and 1). (viii) Finite (the book's key: at any time the number of leaves is finite). (ix) Infinite (N is infinite, so its power set is infinite). (x) Finite (2^3 = 8 subsets). (xi) Infinite. (xii) Finite (1000000000 elements). (xiii) Finite (empty). (xiv) Finite (empty: no real squares to -16). (xv) Finite (empty: sqrt(5)

is irrational). (xvi) Infinite (infinitely many rationals between 0 and 1). These agree with the book's Answers (p. 409).

4. Write two proper subsets of each of the following sets: (i) {a, b, c} (ii) {0, 1} (iii) N (iv) Z (v) Q (vi) R (vii) W (viii) {x | x in Q and 0 < x <= 2}

(i) {a, b}, {a, c} (any two of the three 2-element subsets). (ii) {0}, {1}. (iii) {1, 3, 5, ...} (odd naturals) and {2, 4, 6, ...} (even naturals). (iv) N and W. (v) N and Z. (vi) Z and Q. (vii) {0, 2, 4, ...} and {1, 3, 5, ...}. (viii) {x | x in Q and 1 < x <= 2} and {1} (also {2}). Each is a subset of the given set and is not equal to it, so it is a proper subset. These agree with the book's Answers (p. 409).

5. Is there any set which has no proper sub set? If so name that set.

Yes. Phi, the empty set { }, is the only such set. Step 1: Every set A has Phi as a subset. Step 2: If A is non-empty, Phi is a proper subset of A (Phi != A), so A has a proper subset. Step 3: The empty set has no element to leave out, so it has no proper subset. This agrees with the book's Answers (p. 409).

6. What is the difference between {a, b} and {{a, b}}?

{a, b} is a set containing two elements, a and b. {{a, b}} is a singleton: it contains only one element, and that element is the set {a, b} itself. So the first set has 2 elements and the second has 1 element. This agrees with the book's Answers (p. 409).

7. Which of the following sentences are true and which of them are false? (i) {1, 2} = {2, 1} (ii) Phi subset of {{a}} (iii) {a} subset of {{a}} (iv) {a} in {{a}} (v) a in {{a}} (vi) Phi in {{a}}

(i) True (order of elements does not matter). (ii) True (the empty set is a subset of every set). (iii) False (a is not an element of {{a}}; the only element of {{a}} is {a}). (iv) True ({a} is the element of {{a}}). (v) False (the only element of {{a}} is {a}, not a). (vi) False (the only element of {{a}} is {a}, not Phi). These agree with the book's Answers (p. 409).

8. What is the number of elements of the power set of each of the following sets? (i) { } (ii) {0, 1} (iii) {1, 2, 3, 4, 5, 6, 7} (iv) {0, 1, 2, 3, 4, 5, 6, 7} (v) {a, {b, c}} (vi) {{a, b}, {b, c}, {d, e}}

A set with n elements has 2^n subsets. (i) n = 0, so 2^0 = 1. (ii) n = 2, so 2^2 = 4. (iii) n = 7, so 2^7 = 128. (iv) n = 8, so 2^8 = 256. (v) n = 2 (the elements are a and {b, c}), so 2^2 = 4. (vi) n = 3, so 2^3 = 8. These agree with the book's Answers (p. 409).

9. Write down the power set of each of the following sets: (i) {9, 11} (ii) {+, -, x, /} (iii) {Phi} (iv) {a, {b, c}}

(i) {Phi, {9}, {11}, {9, 11}}. (ii) {Phi, {+}, {-}, {x}, {/}, {+, -}, {+, x}, {+, /}, {-, x}, {-, /}, {x, /}, {+, -, x}, {+, -, /}, {+, x, /}, {-, x, /}, {+, -, x, /}} (16 subsets). (iii) {Phi, {Phi}}. (iv) {Phi, {a}, {{b, c}}, {a, {b, c}}}. These agree with the book's Answers (p. 409).

10. Which pairs of sets are equivalent? Which of them are also equal? (i) {a, b, c}, {1, 2, 3} (ii) The set of the first 10 whole numbers, {0, 1, 2, 3, ..., 9} (iii) Set of angles of a quadrilateral ABCD, set of the sides of the same quadrilateral (iv) Set of the sides of a hexagon ABCDEF, set of the angles of the same hexagon (v) {1, 2, 3, 4, ...}, {2, 4, 6, 8, ...} (vi) {1, 2, 3, 4, ...}, {1, 1/2, 1/3, 1/4, ...} (vii) {5, 10, 15, 20, ..., 55555}, {5, 10, 15, 20, ...}

(i) Equivalent (both have 3 elements) but not equal (different elements). (ii) Equal (the first 10 whole numbers are exactly 0, 1, ..., 9), hence also equivalent. (iii) Equivalent (4 angles and 4 sides) but not equal. (iv) Equivalent (6 sides and 6 angles) but not equal. (v) Equivalent (both countably infinite; n maps to 2n) but not equal. (vi) Equivalent (n maps to 1/n) but not equal. (vii) Not equivalent (the first set is finite, with 55555/5 = 11111 elements, while the second is infinite). These agree with the book's Answers (p. 410).

Exercise 2.2 (book pp. 40-41)

1. Exhibit A union B and A intersection B by Venn diagrams in the following cases: (i) A subset of B (ii) B subset of A (iii) B is A^c (A union A^c case) (iv) A and B are disjoint sets (v) A and B are overlapping sets

Draw a rectangle for the universal set U and circles for A and B; shade the required region. (i) A inside B: A union B is the whole circle B (shade all of B); A intersection B is the inner circle A (shade A). (ii) B inside A: A union B is the whole circle A; A intersection B is the inner circle B. (iii) B = A^c, the region outside A: A union B is the whole rectangle U (shade everything); A intersection B is empty (shade nothing), since A and A^c have no element in common. (iv) Two separate circles: A union B is both circles (shade both); A intersection B is empty (shade nothing). (v) Two overlapping circles: A union B is both circles including the lens (shade all of both circles); A intersection B is the lens where they overlap (shade only the lens). The book's Answers (p. 410) show these same shaded diagrams.

2. Show A - B and B - A by Venn diagrams when: (i) A and B are overlapping sets (ii) A subset of B (iii) B subset of A

(i) Overlapping circles: A - B is the crescent of A outside B (shade the part of A that does not overlap B); B - A is the crescent of B outside A (shade the part of B that does not overlap A). (ii) A inside B: A - B is empty (shade nothing, since every element of A is in B); B - A is the ring of B outside A (shade B with the A disc removed). (iii) B inside A: A - B is the ring of A outside B (shade A with the B disc removed); B - A is empty (shade nothing). The book's Answers (p. 410) show these same shaded diagrams.

3. Under what conditions on A and B are the following statements true? (i) A union B = A (ii) A union B = B (iii) A - B = A (iv) A intersection B = B (v) n(A union B) = n(A) + n(B) (vi) n(A intersection B) = n(A) (vii) A - B = A (as printed a second time) (viii) n(A intersection B) = 0 (ix) A union B = U (x) A union B = B union A (xi) n(A

intersection B) = n(B) (xii) U - A = Phi

(i) A union B = A exactly when B subset of A: every element of B must already be in A, and then the union adds nothing. (ii) A union B = B exactly when A subset of B, by the same argument. (iii) A - B = A exactly when A and B are disjoint, A intersection B = Phi: removing B from A removes nothing only if they share no element. (iv) A intersection B = B exactly when B subset of A. (v) n(A union B) = n(A) + n(B) exactly when A intersection B = Phi (disjoint sets), since in general n(A union B) = n(A) + n(B) - n(A intersection B). (vi) n(A intersection B) = n(A) exactly when A subset of B: the intersection has as many elements as A only if all of A lies in B. (vii) As in (iii): exactly when A intersection B = Phi. Note: (vii) repeats (iii) as printed in the book. (viii) n(A intersection B) = 0 exactly when A intersection B = Phi. (ix) A union B = U exactly when A^c subset of B (every element outside A lies in B); the key's 'B = A^c' is a sufficient special case, not the full condition, and is flagged here. (x) A union B = B union A holds always (commutativity of union). (xi) n(A intersection B) = n(B) exactly when B subset of A. (xii) U - A = Phi exactly when A = U. FLAG on (iii): the book's Answers (p. 410) print 'A = Phi' for (iii), which is sufficient but not necessary; the correct if-and-only-if condition is A intersection B = Phi, as the key itself gives for the repeated (vii). All other conditions agree with the key.

4. Let U = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}, A = {2, 4, 6, 8, 10}, B = {1, 2, 3, 4, 5} and C = {1, 3, 5, 7, 9}. List the members of each of the following sets: (i) A^c (ii) B^c (iii) A union B (iv) A - B (v) A intersection C (vi) A^c union C^c (vii) A^c union C (viii) U^c

(i) A^c = U - A = {1, 3, 5, 7, 9}. (ii) B^c = U - B = {6, 7, 8, 9, 10}. (iii) A union B = {1, 2, 3, 4, 5, 6, 8, 10}. (iv) A - B = {6, 8, 10} (the even numbers of A that are not in B). (v) A intersection C = { } (A is even, C is odd). (vi) A^c union C^c: C^c = {2,

4, 6, 8, 10}, so the union with A^c = {1, 3, 5, 7, 9} is U = {1, 2, ..., 10}. (vii) A^c union C = {1, 3, 5, 7, 9} union {1, 3, 5, 7, 9} = {1, 3, 5, 7, 9} = C (since A^c = C here). (viii) U^c = { } (nothing lies outside the universal set). These agree with the book's Answers (p. 410).

5. Using the Venn diagrams, if necessary, find the single sets equal to the following: (i) A^c (ii) A intersection U (iii) A union U (iv) A union Phi (v) Phi intersection Phi

(i) A^c = U - A (the complement: everything in U outside A). (ii) A intersection U = A (every element of A is in U). (iii) A union U = U (U already contains everything). (iv) A union Phi = A (adding no elements changes nothing). (v) Phi intersection Phi = Phi. These agree with the book's Answers (p. 410).

6. Use Venn diagrams to verify the following: (i) A - B = A intersection B^c (ii) (A - B)^c intersection B = B

(i) Step 1: A - B is the set of elements in A but not in B. Step 2: B^c is the set of elements not in B. Step 3: A intersection B^c is therefore the set of elements in A and not in B, which is exactly A - B. The Venn shading matches: the crescent of A outside B in both cases. (ii) Step 1: A - B = A intersection B^c by part (i). Step 2: (A - B)^c = A^c union B by De Morgan's law. Step 3: (A^c union B) intersection B = (A^c intersection B) union (B intersection B) = (A^c intersection B) union B = B, since A^c intersection B is already a subset of B. So (A - B)^c intersection B = B, as required.

Exercise 2.3 (book pp. 46-47)

1. Verify the commutative properties of union and intersection for the following pairs of sets: (i) A = {1, 2, 3, 4, 5}, B = {4, 6, 8, 10} (ii) N, Z (iii) A = {x | x in R and x >= 0}, B = R

(i) A union B = {1, 2, 3, 4, 5, 6, 8, 10} = B union A, since B union A collects the

same elements in the other order. A intersection B = {4} = B intersection A. (ii) N union Z = Z = Z union N (every natural is an integer); N intersection Z = N = Z intersection N. (iii) A union B = R = B union A (A is already a subset of R); A intersection B = A = B intersection A. So commutativity holds in all three cases.

2. Verify the properties for the sets A, B and C given below: (i) Associativity of Union (ii) Associativity of intersection (iii) Distributivity of Union over intersection (iv) Distributivity of intersection over union. (a) A = {1, 2, 3, 4}, B = {3, 4, 5, 6, 7, 8}, C = {5, 6, 7, 9, 10} (b) A = Phi, B = {0}, C = {0, 1, 2} (c) N, Z, Q

(a) Step 1 (union associativity): A union B = {1, 2, 3, 4, 5, 6, 7, 8}; (A union B) union C = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}. B union C = {3, 4, 5, 6, 7, 8, 9, 10}; A union (B union C) = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}. Equal. Step 2 (intersection associativity): A intersection B = {3, 4}; (A intersection B) intersection C = Phi. B intersection C = {5, 6, 7}; A intersection (B intersection C) = Phi. Equal. Step 3 (union over intersection): B intersection C = {5, 6, 7}; A union (B intersection C) = {1, 2, 3, 4, 5, 6, 7}. A union C = {1, 2, 3, 4, 5, 6, 7, 9, 10}; (A union B) intersection (A union C) = {1, 2, 3, 4, 5, 6, 7}. Equal. Step 4 (intersection over union): B union C = {3, 4, 5, 6, 7, 8, 9, 10}; A intersection (B union C) = {3, 4}. A intersection C = Phi; (A intersection B) union (A intersection C) = {3, 4} union Phi = {3, 4}. Equal. (b) Step 1: (Phi union {0}) union C = {0, 1, 2} = Phi union ({0} union C). Step 2: (Phi intersection {0}) intersection C = Phi = Phi intersection ({0} intersection C). Step 3: Phi union ({0} intersection C) = Phi union {0} = {0}; (Phi union {0}) intersection (Phi union C) = {0} intersection {0, 1, 2} = {0}. Equal. Step 4: Phi intersection ({0} union C) = Phi; (Phi intersection {0}) union (Phi intersection C) = Phi union Phi = Phi. Equal. (c) N subset of Z subset of Q. Step 1: (N union Z) union Q = Z union Q = Q; N union (Z union Q) = N union Q = Q. Step 2: (N intersection Z) intersection Q = N intersection Q = N; N intersection (Z intersection Q) = N intersection Z = N. Step 3: N union (Z intersection Q) = N union Z = Z; (N union Z) intersection (N union

Q) = Z intersection Q = Z. Step 4: N intersection (Z union Q) = N intersection Q = N; (N intersection Z) union (N intersection Q) = N union N = N. All four properties verified in all three cases.

3. Verify De Morgan's Laws for the following sets: U = {1, 2, ..., 20}, A = {2, 4, 6, ..., 20} and B = {1, 3, 5, ..., 19}

Step 1: A^c = {1, 3, 5, ..., 19} = B, and B^c = {2, 4, 6, ..., 20} = A. Step 2 (first law): A union B = {1, 2, ..., 20} = U, so (A union B)^c = Phi. A^c intersection B^c = B intersection A = Phi. Equal. Step 3 (second law): A intersection B = Phi, so (A intersection B)^c = U. A^c union B^c = B union A = U. Equal. Both De Morgan's laws verified.

4. Let U = The set of the English alphabet, A = {x | x is a vowel}, B = {y | y is a consonant}. Verify De Morgan's Laws for these sets.

Step 1: A = {a, e, i, o, u}; B is the other 21 letters. So A^c = B and B^c = A. Step 2 (first law): A union B = U, so (A union B)^c = Phi. A^c intersection B^c = B intersection A = Phi (no letter is both vowel and consonant). Equal. Step 3 (second law): A intersection B = Phi, so (A intersection B)^c = U. A^c union B^c = B union A = U. Equal. Both laws verified.

5. With the help of Venn diagrams, verify the two distributive properties in the following cases w.r.t union and intersection: (i) A subset of B, A intersection C = Phi and B and C are overlapping (ii) A and B are overlapping, B and C are overlapping but A and C are disjoint

Case (i): Step 1: Draw A inside B, and C overlapping B but not meeting A. Step 2: A union (B intersection C): B intersection C is the lens of B and C, outside A; adding A shades A plus that lens. Step 3: (A union B) intersection (A union C): A union B is all of B; A union C is A plus C; their overlap is A plus the B-C lens (the part of C inside B). Step 4: Both sides shade exactly A plus the B intersection C region, so A union (B intersection C) = (A union B) intersection (A union C). The intersection-over-union

identity is verified the same way: both sides shade only the B intersection C region together with A intersection C = Phi, i.e. the same region. Case (ii): Step 1: Draw A and B overlapping, B and C overlapping, A and C separate. Step 2: A union (B intersection C) shades A plus the B-C lens. Step 3: (A union B) intersection (A union C) shades the overlap of (A plus B) with (A plus C), which is A plus the B-C lens, since A intersection C is empty and B intersection C lies in both. Step 4: Both sides shade the same region, so the distributive law holds; the other distributive law follows by the same shading.

6. Taking any set, say A = {1, 2, 3, 4, 5}, verify the following: (i) A union Phi = A (ii) A union A = A (iii) A intersection A = A

(i) A union Phi = {1, 2, 3, 4, 5} union { } = {1, 2, 3, 4, 5} = A (Phi adds no element). (ii) A union A = {1, 2, 3, 4, 5} = A (repetition of an element is not allowed in a set). (iii) A intersection A = {1, 2, 3, 4, 5} = A (every element of A is in A).

7. If U = {1, 2, 3, 4, 5, ..., 20} and A = {1, 3, 5, ..., 19}, verify the following: (i) A union A^c = U (ii) A intersection U = A (iii) A intersection A^c = Phi

Step 1: A^c = U - A = {2, 4, 6, ..., 20} (the even numbers from 1 to 20). Step 2 (i): A union A^c = odds union evens = {1, 2, ..., 20} = U. Step 3 (ii): A intersection U = A, since every element of A is in U. Step 4 (iii): A intersection A^c = Phi, since no number is both odd and even.

8. From suitable properties of union and intersection deduce the following results: (i) A intersection (A union B) = A union (A intersection B) (ii) A union (A intersection B) = A intersection (A union B)

(i) Step 1: By the absorption law, A intersection (A union B) = A: every element of A is in A union B, and the intersection cannot add anything outside A. Step 2: By the absorption law again, A union (A intersection B) = A: A intersection B is already a subset of A. Step 3: Both sides equal A, so they are equal to each other. (ii) Step 1:

A union (A intersection B) = A by absorption, as above. Step 2: A intersection (A union B) = A by absorption, as above. Step 3: Both sides equal A, so they are equal to each other.

9. Using Venn diagrams, verify the following results: (i) A intersection B^c = A iff A intersection B = Phi (ii) (A - B) union B = A union B (iii) (A - B) intersection B = Phi (iv) A union B = A union (A^c intersection B)

(i) Step 1 (forward): If A intersection B^c = A, then every element of A lies outside B, so A and B share no element: A intersection B = Phi. Step 2 (backward): If A intersection B = Phi, then every element of A lies outside B, i.e. in B^c, so A intersection B^c = A. Step 3: The two statements are equivalent. (ii) Step 1: A - B = A intersection B^c. Step 2: (A intersection B^c) union B = (A union B) intersection (B^c union B) by distributivity = (A union B) intersection U = A union B. (iii) Step 1: A - B = A intersection B^c. Step 2: (A intersection B^c) intersection B = A intersection (B^c intersection B) = A intersection Phi = Phi. (iv) Step 1: A union (A^c intersection B) = (A union A^c) intersection (A union B) by distributivity. Step 2: A union A^c = U, so this equals U intersection (A union B) = A union B.

Exercise 2.4 (book p. 54)

1. Write the converse, inverse and contrapositive of the following conditionals: (i) ~p -> q (ii) q -> p (iii) ~p -> ~q (iv) ~q -> p (the consequent of (iv) is cut off at the right edge of the scan; it is read as p from the book's Answers table, p. 410)

(i) Conditional ~p -> q. Converse: q -> ~p. Inverse: p -> ~q. Contrapositive: ~q -> p. (ii) Conditional q -> p. Converse: p -> q. Inverse: ~q -> ~p. Contrapositive: ~p -> ~q. (iii) Conditional ~p -> ~q. Converse: ~q -> ~p. Inverse: p -> q. Contrapositive: q -> p. (iv) Conditional ~q -> p. Converse: p -> ~q. Inverse: q -> ~p.

Contrapositive: ~p -> q. Rule used throughout: the converse swaps antecedent and consequent; the inverse negates both; the contrapositive swaps and negates both. The converse/inverse/contrapositive for (i)-(iii) agree with the book's Answers table (p. 410); for (iv) the forms above follow by the same rule from the Answers-table conditional ~q -> p.

2. Construct truth tables for the following statements: (i) (p -> ~p) v (p -> q) (ii) (p ^ ~p) -> q (iii) ~(p -> q) <-> (p ^ ~q)

(i) Table with columns p, q, ~p, p -> ~p, p -> q, (p -> ~p) v (p -> q). Row T, T: ~p = F, p -> ~p = F, p -> q = T, result T. Row T, F: ~p = F, p -> ~p = F, p -> q = F, result F. Row F, T: ~p = T, p -> ~p = T, p -> q = T, result T. Row F, F: ~p = T, p -> ~p = T, p -> q = T, result T. So the statement has truth values T, F, T, T: it is a contingency (neither tautology nor absurdity). This matches the book's Answers table (p. 410). (ii) Table with columns p, q, ~p, p ^ ~p, (p ^ ~p) -> q. In every row p ^ ~p = F (a statement and its negation cannot both be true). F -> q is always T. So the statement is T, T, T, T: a tautology. This matches the book's Answers table (p. 410). (iii) Table with columns p, q, ~q, p -> q, ~(p -> q), p ^ ~q. Row T, T: ~q = F, p -> q = T, ~(p -> q) = F, p ^ ~q = F. Row T, F: ~q = T, p -> q = F, ~(p -> q) = T, p ^ ~q = T. Row F, T: ~q = F, p -> q = T, ~(p -> q) = F, p ^ ~q = F. Row F, F: ~q = T, p -> q = T, ~(p -> q) = F, p ^ ~q = F. The columns ~(p -> q) and p ^ ~q agree in every row, so ~(p -> q) <-> (p ^ ~q) is always true. This matches the book's Answers table (p. 411).

3. Show that each of the following statements is a tautology: (i) (p ^ q) -> p (ii) p -> (p v q) (iii) ~(p -> q) -> p (iv) ~q ^ (p -> q) -> ~p

(i) Step 1: If p ^ q is true, then p is true by the meaning of 'and'. Step 2: An implication with a true consequent is true, and if p ^ q is false the implication is true vacuously. Step 3: So (p ^ q) -> p is true in all rows: a tautology. (ii) Step 1: If p

is true, then p v q is true. Step 2: If p is false, the implication p -> (p v q) is true vacuously. Step 3: So it is true in all rows: a tautology. (iii) Step 1: ~(p -> q) is true only in the row p = T, q = F (the only row where p -> q is false). Step 2: In that row p is true, so the consequent p is true. Step 3: In every other row the antecedent is false, so the implication is true. Hence a tautology. (iv) Step 1: ~q ^ (p -> q) is true only when q is false and p -> q is true. Step 2: With q false, p -> q can be true only when p is false. Step 3: Then ~p is true, so the implication is true; in all other rows its antecedent is false, so it is true. Hence a tautology (modus tollens).

4. Determine whether each of the following is a tautology, a contingency or absurdity: (i) p ^ ~p (ii) p -> (q -> p) (iii) q v (~q v p)

(i) p ^ ~p is an absurdity (contradiction): p and ~p are never true together, so it is false in every row. (ii) p -> (q -> p) is a tautology: if p is true, q -> p is true; if p is false, the whole implication is true vacuously. (iii) q v (~q v p) is a tautology: q v ~q is already always true, and adding 'v p' keeps it true in every row. These agree with the book's Answers (p. 411).

5. Prove that p v (~p ^ ~q) v (p ^ q) = p v (~p ^ ~q)

Step 1: By the absorption law for 'or' over 'and', p v (p ^ q) = p: the term p ^ q adds nothing to p. Step 2: The left side is p v (~p ^ ~q) v (p ^ q); regrouping the 'or' terms (associativity and commutativity of v) gives [p v (p ^ q)] v (~p ^ ~q). Step 3: By Step 1, p v (p ^ q) = p, so the whole expression becomes p v (~p ^ ~q), which is the right side. This proves the identity.

Exercise 2.5 (book p. 57)

Convert the following theorem to logical form and prove it by constructing a truth table: 1. (A intersection B)^c = A^c union B^c

Logical form: let p be 'x in A' and q be 'x in B'. Then x in A intersection B is p ^ q, and x in (A intersection B)^c is ~(p ^ q). x in A^c union B^c is ~p v ~q. Truth table: p = T, q = T: ~(p ^ q) = F, ~p v ~q = F v F = F. p = T, q = F: ~(p ^ q) = T, ~p v ~q = F v T = T. p = F, q = T: ~(p ^ q) = T, ~p v ~q = T v F = T. p = F, q = F: ~(p ^ q) = T, ~p v ~q = T v T = T. The two columns agree in every row, so (A intersection B)^c = A^c union B^c. This is De Morgan's first law.

Convert the following theorem to logical form and prove it by constructing a truth table: 2. (A union B) union C = A union (B union C)

Logical form: let p be 'x in A', q be 'x in B', r be 'x in C'. Left side is (p v q) v r; right side is p v (q v r). Both are true exactly when at least one of p, q, r is true, and false only when all three are false. Checking all eight rows: T,T,T -> T = T; T,T,F -> T = T; T,F,T -> T = T; T,F,F -> T = T; F,T,T -> T = T; F,T,F -> T = T; F,F,T -> T = T; F,F,F -> F = F. The columns agree in every row, so (A union B) union C = A union (B union C). This is associativity of union.

Convert the following theorem to logical form and prove it by constructing a truth table: 3. (A intersection B) intersection C = A intersection (B intersection C)

Logical form: let p be 'x in A', q be 'x in B', r be 'x in C'. Left side is (p ^ q) ^ r; right side is p ^ (q ^ r). Both are true exactly when all three of p, q, r are true, and false otherwise. Checking all eight rows: only T,T,T gives T on both sides; every other row gives F on both sides. The columns agree in every row, so (A intersection B) intersection C = A intersection (B intersection C). This is associativity of intersection.

Convert the following theorem to logical form and prove it by constructing a truth table: 4. A union (B intersection C) = (A union B) intersection (A union C)

Logical form: let p be 'x in A', q be 'x in B', r be 'x in C'. Left side is p v (q ^ r); right side is (p v q) ^ (p v r). Step 1: If p is true, the left side is true, and both factors on the right are true, so the right side is true. Step 2: If p is false, the left side is q ^ r,

and the right side is q ^ r (since p v q = q and p v r = r). Step 3: So both sides agree in all eight rows: for example T,T,F gives T = T; F,T,T gives T = T; F,T,F gives F = F; F,F,F gives F = F. Hence A union (B intersection C) = (A union B) intersection (A union C). This is distributivity of union over intersection.

Exercise 2.6 (book pp. 63-64)

1. For A = {1, 2, 3, 4}, find the following relations in A. State the domain and range of each relation. Also draw the graph of each. (i) {(x, y) | y = x} (ii) {(x, y) | y + x = 5} (iii) {(x, y) | x + y < 5} (iv) {(x, y) | x + y > 5}

(i) {(1, 1), (2, 2), (3, 3), (4, 4)}. Domain = {1, 2, 3, 4} = A. Range = {1, 2, 3, 4} = A. Graph: the four diagonal points (1, 1), (2, 2), (3, 3), (4, 4). (ii) {(1, 4), (2, 3), (3, 2), (4, 1)}. Domain = {1, 2, 3, 4} = A. Range = {1, 2, 3, 4} = A. Graph: the four points (1, 4), (2, 3), (3, 2), (4, 1). (iii) {(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (3, 1)}: these are exactly the pairs from A with sum less than 5. Domain = {1, 2, 3}. Range = {1, 2, 3}. Graph: plot the six points listed, forming a triangle below the line x + y = 5. (iv) {(2, 4), (3, 3), (3, 4), (4, 2), (4, 3), (4, 4)}: these are exactly the pairs from A with sum greater than 5. Domain = {2, 3, 4}. Range = {2, 3, 4}. Graph: plot the six points listed, forming a triangle above the line x + y = 5. These agree with the book's Answers (pp. 411-412).

2. Repeat Q-1 when A = R, the set of real numbers. Which of the relations are functions?

(i) {(x, y) | y = x}: the graph is the line y = x. This is a function: every real x has exactly one y, namely y = x. (ii) {(x, y) | x + y = 5}: the graph is the line y = 5 - x. This is a function: every real x has exactly one y, namely y = 5 - x. (iii) {(x, y) | x + y < 5}: not a function. For example x = 0 pairs with y = 0 and also with y = 1 (both sums are less than 5), so one x has more than one y. (iv) {(x, y) | x + y > 5}: not a function. For example x = 6 pairs with y = 0 and also with y = 1 (both sums exceed

5). These agree with the book's Answers (p. 411).

3. Which of the following diagrams represent functions and of which type? Fig (1): 1 maps to a and b, 2 maps to c, 3 maps to d. Fig (2): a, b, c each map to a distinct element. Fig (3): 1 maps to a, 2 maps to b, 3 maps to c. Fig (4): l and m map to the same element, n maps to another element. (Note: the right-hand labels of Fig (2) and Fig (4) are cut off at the scan edge; the classification below follows the book's Answers, p. 412.)

Fig (1) does not represent a function: the element 1 has two images, a and b, but a function must give exactly one image to each domain element. Fig (2) represents a function, and it is one-one and onto: each of a, b, c has exactly one image and the images are all distinct and cover the codomain. Fig (3) represents a function, and it is one-one and onto: 1 -> a, 2 -> b, 3 -> c, with distinct images covering the codomain. Fig (4) represents a function, but it is into (not onto): l and m share an image, so it is not one-one, and at least one codomain element has no preimage. These agree with the book's Answers (p. 412).

4. Find the inverse of each of the following relations. Tell whether each relation and its inverse is a function or not: (i) {(2, 1), (3, 2), (4, 3), (5, 4), (6, 5)} (ii) {(1, 3), (2, 5), (3, 7), (4, 9), (5, 11)} (iii) {(x, y) | y = 2x + 3, x in R} (iv) {(x, y) | y^2 = 4ax, x >= 0} (v) {(x, y) | x^2 + y^2 = 9, |x| <= 3, |y| <= 3}

(i) Relation {(2, 1), (3, 2), (4, 3), (5, 4), (6, 5)}. Inverse {(1, 2), (2, 3), (3, 4), (4, 5), (5, 6)}. The relation is a function (each first component occurs once) and its inverse is also a function. (ii) Relation {(1, 3), (2, 5), (3, 7), (4, 9), (5, 11)}. Inverse {(3, 1), (5, 2), (7, 3), (9, 4), (11, 5)}. Both are functions. (iii) Relation {(x, y) | y = 2x + 3, x in R}. Inverse {(x, y) | y = (x - 3)/2, x in R}: solving y = 2x + 3 for x gives x = (y - 3)/2, then swapping the roles gives y = (x - 3)/2. Both are functions (each x gives exactly one y). (iv) Relation {(x, y) | y^2 = 4ax, x >= 0}. Inverse {(x, y) | y = (1/(4a))x^2, x >= 0}: from y^2 = 4ax, swapping x and y gives x^2 = 4ay, i.e. y =

x^2/(4a). The relation is NOT a function (for x > 0, y = +2sqrt(ax) and y = -2sqrt(ax) are two values), but its inverse IS a function (each x gives one y). (v) Relation {(x, y) | x^2 + y^2 = 9, |x| <= 3, |y| <= 3} (the circle of radius 3). Its inverse is the same set {(x, y) | y^2 + x^2 = 9, |x| <= 3, |y| <= 3}, since the equation is symmetric in x and y. Neither the relation nor its inverse is a function (for x = 0, y = 3 and y = -3 both occur). These agree with the book's Answers (p. 412).

Exercise 2.7 (book pp. 69-70; inventory correction: the set runs to Q7, not Q5)

1. Complete the table, indicating by a tick mark those properties which are satisfied by the specified set of numbers. Rows: Closure (+, x), Associative (+, x), Identity (+, x), Inverse (+, x), Commutative (+, x). Columns: Natural, Whole, Integers, Rational, Reals. (Completed table as printed in the book's Answers, p. 412.)

Closure: + : Natural tick, Whole tick, Integers tick, Rational tick, Reals tick. x : all five tick. Associative: + : all five tick. x : all five tick. Identity: + : Natural cross (no 0 in N), Whole tick, Integers tick, Rational tick, Reals tick. x : all five tick (1 is in every set). Inverse: + : Natural cross, Whole cross (no negative of a positive whole number in W, and 0 aside), Integers tick, Rational tick, Reals tick. x : Natural cross, Whole cross, Integers cross (2 has no integer reciprocal), Rational cross (0 has no reciprocal), Reals tick (every non-zero real has a reciprocal; note the book's table marks Reals tick, following the usual convention that the multiplicative inverse property is stated for non-zero elements). Commutative: + : all five tick. x : all five tick. This completed table agrees with the book's Answers (p. 412).

2. What are the field axioms? In what respect does the field of real numbers differ from that of complex numbers?

Field axioms: a set F with two operations + and x is a field if: (1) Closure: a + b and

a x b are in F for all a, b in F. (2) Associativity: (a + b) + c = a + (b + c) and (a x b) x c = a x (b x c). (3) Commutativity: a + b = b + a and a x b = b x a. (4) Identities: there are distinct elements 0 and 1 in F with a + 0 = a and a x 1 = a for all a. (5) Inverses: for every a there is -a with a + (-a) = 0; for every non-zero a there is a^(-1) with a x a^(-1) = 1. (6) Distributivity: a x (b + c) = a x b + a x c. Difference: the real numbers form an ORDERED field (there is an order < compatible with + and x, with trichotomy, and every square is >= 0), while the complex numbers cannot be made into an ordered field: in C, i^2 = -1, but in any ordered field a square is >= 0 and -1 < 0, a contradiction. So R is ordered and C is not; equivalently, the order axioms hold for R but fail for C.

3. Show that the adjoining table is that of multiplication of the elements of the set of residue classes modulo 5. Table rows (first entry is the row label): row 0: 0, 0, 0, 0, 0. row 1: 0, 1, 2, 3, 4. row 2: 0, 2, 4, 1, 3. row 3: 0, 3, 1, 4, 2. row 4: 0, 4, 3, 2, 1. Step 1: The entry in row a, column b must be a x b modulo 5 (the remainder of a x b on division by 5). Step 2: Row 2 checks: 2 x 0 = 0; 2 x 1 = 2; 2 x 2 = 4; 2 x 3 = 6 = 5 + 1, remainder 1; 2 x 4 = 8 = 5 + 3, remainder 3. The row reads 0, 2, 4, 1, 3. Correct. Step 3: Row 3 checks: 3 x 2 = 6, remainder 1; 3 x 3 = 9 = 5 + 4, remainder 4; 3 x 4 = 12 = 10 + 2, remainder 2. The row reads 0, 3, 1, 4, 2. Correct. Step 4: Row 4 checks: 4 x 2 = 8, remainder 3; 4 x 3 = 12, remainder 2; 4 x 4 = 16 = 15 + 1, remainder 1. The row reads 0, 4, 3, 2, 1. Correct. Step 5: Rows 0 and 1 are immediate (0 x b = 0; 1 x b = b). So every entry is the product modulo 5, and the table is the multiplication table of the residue classes modulo 5.

4. Prepare a table of addition of the elements of the set of residue classes modulo 4. The entry in row a, column b is a + b modulo 4 (the remainder of a + b on division by 4). Row 0: 0, 1, 2, 3. Row 1: 1, 2, 3, 0 (since 1 + 3 = 4, remainder 0). Row 2: 2, 3, 0, 1 (since 2 + 2 = 4, remainder 0; 2 + 3 = 5, remainder 1). Row 3: 3, 0, 1, 2 (since 3 + 1 = 4, remainder 0; 3 + 2 = 5, remainder 1; 3 + 3 = 6, remainder 2). This table

agrees with the book's Answers (p. 412), where the same table is printed for Exercise 2.7 Q4.

5. Which of the following binary operations shown in tables (a) and (b) is commutative? Table (a), rows: a: a, c, b, d. b: b, c, b, a. c: c, d, b, c. d: a, a, b, b. Table (b), rows: a: a, c, b, d. b: c, d, b, a. c: b, b, a, c. d: d, a, c, d. (Note: the (b) table's b * a entry is read as b following the book's Answers key; the scan glyph there is ambiguous between b and c.)

Neither table is commutative. Table (a): a * c = b (row a, column c) but c * a = c (row c, column a). Since b != c, the operation is not commutative. Table (b): a * b = c (row a, column b) but b * a = b (row b, column a, as printed in the Answers key). Since c != b, the operation is not commutative. So both operations fail commutativity, each with the counterexample stated. These counterexamples agree with the book's Answers (p. 412).

6. Supply the missing elements of the third row of the given table so that the operation * may be associative. Given rows: a: a, b, c, d. b: b, a, c, d. d: d, c, c, d. Third row (c) missing: c * a, c * b, c * c, c * d.

Step 1: The book's Answers (p. 412) complete the third row as c, c, c, c, giving the full table: row a: a, b, c, d; row b: b, a, c, d; row c: c, c, c, c; row d: d, c, c, d. Step 2: In this table a is the identity (row a and column a reproduce the other element), and c is absorbing (row c and column c are all c). Step 3: FLAG - the key's completion does NOT in fact make the operation associative. Check the triple (d, b, d): (d * b) * d = c * d = c (since d * b = c from row d, and c * d = c from row c), but d * (b * d) = d * d = d (since b * d = d from row b, and d * d = d from row d). Since c != d, associativity fails for this triple. Step 4: The completed row c, c, c, c is therefore reported exactly as printed in the key, with this failure flagged honestly rather than claimed as a verified associative completion.

7. What operation is represented by the adjoining table? Name the identity element of the relevant set, if it exists. Is the operation associative? Find the inverses of 0, 1, 2, 3, if they exist. Table rows: 0: 0, 1, 2, 3. 1: 1, 2, 3, 0. 2: 2, 3, 0, 1. 3: 3, 0, 1, 2.

Step 1: The entry in row a, column b is a + b modulo 4: for example 2 + 3 = 5, remainder 1, and the table shows 1; 3 + 3 = 6, remainder 2, and the table shows 2. So the operation is addition modulo 4. Step 2: The identity element is 0: row 0 and column 0 reproduce the other element (0 + a = a and a + 0 = a modulo 4). Step 3: The operation is associative, since ordinary addition is associative and reducing modulo 4 preserves it. Step 4: Inverses: the inverse of 0 is 0 (0 + 0 = 0); 1 and 3 are inverses of each other (1 + 3 = 4, remainder 0); the inverse of 2 is 2 (2 + 2 = 4, remainder 0). These agree with the book's Answers (p. 412).

Exercise 2.8 (book pp. 78-79)

1. Operation + (circled plus) performed on the two-member set G = {0, 1} is shown in the adjoining table: row 0: 0, 1. row 1: 1, 0. Answer the questions: (i) Name the identity element if it exists? (ii) What is the inverse of 1? (iii) Is the set G, under the given operation a group? Abelian or non-Abelian?

(i) The identity element is 0: 0 + 0 = 0 and 0 + 1 = 1, and also 1 + 0 = 1, so 0 leaves every element unchanged. (ii) The inverse of 1 is 1, since 1 + 1 = 0 (the identity). (iii) Yes, G is a group, and it is abelian: the operation is closed (all entries are 0 or 1); it is associative (it is addition modulo 2); the identity is 0; inverses exist (0 -> 0, 1 -> 1); and the table is symmetric about the main diagonal (0 + 1 = 1 + 0 = 1), so the operation is commutative, hence the group is abelian. These agree with the book's Answers (p. 412).

2. The operation + (circled plus) as performed on the set {0, 1, 2, 3} is shown in the adjoining table: row 0: 0, 1, 2, 3. row 1: 1, 2, 3, 0. row 2: 2, 3, 0, 1. row 3: 3, 0, 1, 2. Show that the set is an Abelian group?

Step 1: The table is addition modulo 4 (for example 2 + 3 = 5, remainder 1; 3 + 3 = 6, remainder 2). Step 2: Closure: every entry is in {0, 1, 2, 3}. Step 3: Associativity: inherited from ordinary addition modulo 4. Step 4: Identity: 0, since row 0 and column 0 reproduce the other element. Step 5: Inverses: 0 -> 0, 1 -> 3, 2 -> 2, 3 -> 1 (each pair sums to 0 modulo 4). Step 6: Commutativity: the table is symmetric about the main diagonal (for example 1 + 3 = 0 = 3 + 1). So the set is an abelian group under this operation.

3. For each of the following sets, determine whether or not the set forms a group with respect to the indicated operation: (i) The set of rational numbers, x (ii) The set of rational numbers, + (iii) The set of positive rational numbers, x (iv) The set of integers, + (v) The set of integers, x

(i) The set of rational numbers under multiplication is NOT a group: 0 has no multiplicative inverse (no rational x satisfies 0 x x = 1). (ii) The set of rational numbers under addition IS a group: closed, associative, identity 0, inverse of a is -a, commutative. (iii) The set of positive rational numbers under multiplication IS a group: closed (product of positives is positive), associative, identity 1, inverse of a is 1/a (positive), commutative. (iv) The set of integers under addition IS a group: closed, associative, identity 0, inverse of a is -a, commutative. (v) The set of integers under multiplication is NOT a group: most integers have no integer reciprocal (for example 2 has no integer inverse). These agree with the book's Answers (p. 413).

4. Show that the adjoining table represents the sums of the elements of the set {E, O}. What is the identity element of this set? Show that this set is an abelian group. Table: row E: E, O. row O: O, E.

Step 1: Read E as even and O as odd. Even + even = even (E), even + odd = odd (O), odd + even = odd (O), odd + odd = even (E). The table matches exactly. Step 2: The identity element is E: E + E = E and E + O = O (and O + E = O). Step 3: The

set is an abelian group: closed (all entries are E or O); associative (addition of parities inherits associativity); identity E; inverses: E -> E (E + E = E) and O -> O (O + O = E); commutative (the table is symmetric: E + O = O + E = O). These agree with the book's Answers (p. 413): identity E, inverse of E is E, inverse of O is O.

5. Show that the set {1, w, w^2}, when w^3 = 1, is an Abelian group w.r.t. ordinary multiplication.

Step 1: Closure: 1 x 1 = 1, 1 x w = w, 1 x w^2 = w^2, w x w = w^2, w x w^2 = w^3 = 1, w^2 x w^2 = w^4 = w^3 x w = w. Every product is in {1, w, w^2}. Step 2: Associativity and commutativity are inherited from multiplication of complex numbers. Step 3: Identity: 1, since 1 x a = a for each element a. Step 4: Inverses: 1 -> 1 (1 x 1 = 1); w -> w^2 (w x w^2 = 1); w^2 -> w (w^2 x w = 1). So the set is an abelian group under multiplication.

6. If G is a group under the operation * and a, b in G, find the solutions of the equations: a * x = b, x * a = b

Step 1: For a * x = b, left-multiply both sides by a^(-1): a^(-1) * (a * x) = a^(-1) * b. Step 2: By associativity, (a^(-1) * a) * x = a^(-1) * b, i.e. e * x = a^(-1) * b, so x = a^(-1) * b. Step 3: For x * a = b, right-multiply both sides by a^(-1): (x * a) * a^(-1) = b * a^(-1). Step 4: By associativity, x * (a * a^(-1)) = b * a^(-1), i.e. x * e = b * a^(-1), so x = b * a^(-1). Note the order matters in a non-abelian group: the two solutions are a^(-1) * b and b * a^(-1) respectively. These agree with the book's Answers (p. 413).

7. Show that the set consisting of elements of the form a + sqrt(3)b (a, b being rational), is an abelian group w.r.t. addition.

Step 1: Closure: (a + sqrt(3)b) + (c + sqrt(3)d) = (a + c) + sqrt(3)(b + d), and a + c, b + d are rational, so the sum is again of the required form. Step 2: Associativity and commutativity are inherited from addition of real numbers. Step 3: Identity: 0 = 0 + sqrt(3) x 0 is in the set, and adding it changes nothing. Step 4: Inverse: the

inverse of a + sqrt(3)b is (-a) + sqrt(3)(-b), which is in the set since -a, -b are rational, and their sum is 0. So the set is an abelian group under addition.

8. Determine whether, (P(S), *), where * stands for intersection is a semi-group, a monoid or neither. If it is a monoid, specify its identity.

Step 1: Closure: if A and B are subsets of S, then A intersection B is a subset of S, so it is in P(S). Step 2: Associativity: intersection of sets is associative. Step 3: Identity: S itself is in P(S), and A intersection S = A for every A in P(S), so S is the identity element. Step 4: Therefore (P(S), intersection) is a monoid (a semi-group with identity S). It is not a group in general: for A intersection B = S we would need A = B = S, so only S has an inverse. FLAG: the book's Answers (p. 413) call it only 'a semi-group'; that is true but incomplete - the fuller classification is a monoid with identity S, as shown above.

9. Complete the following table to obtain a semi-group under *. Table rows (given): a: c, a, b. b: a, b, c. c: -, -, a.

Step 1: The book's Answers (p. 413) complete the third row as c * a = b and c * b = c, giving the full table: row a: c, a, b; row b: a, b, c; row c: b, c, a. Step 2: Closure holds: every entry is a, b or c. Step 3: b is the identity: row b reproduces the column element (b * a = a, b * b = b, b * c = c) and column b reproduces the row element (a * b = a, b * b = b, c * b = c). Step 4: The table is symmetric about the main diagonal (a * c = b = c * a; a * b = a = b * a; b * c = c = c * b), so the operation is commutative, and it is the cyclic operation of order 3 (a * a = c, a * c = b, c * c = a), which is associative. So the completed table gives a semi-group - indeed an abelian group with identity b.

10. Prove that all 2 x 2 non-singular matrices over the real field form a non-abelian group under multiplication.

Step 1: Closure: if A and B are non-singular (det != 0), then det(AB) = det(A) x

det(B) != 0, so AB is non-singular. Step 2: Associativity: matrix multiplication is associative. Step 3: Identity: the identity matrix I = [[1, 0], [0, 1]] is non-singular and AI = IA = A. Step 4: Inverses: every non-singular matrix A has an inverse A^(-1) with det(A^(-1)) = 1/det(A) != 0, so A^(-1) is also non-singular. Step 5: Non-abelian: multiplication is not commutative in general. Example: A = [[1, 1], [0, 1]] and B = [[1, 0], [1, 1]] give AB = [[2, 1], [1, 1]] but BA = [[1, 1], [1, 2]], and AB != BA. So the set is a non-abelian group under multiplication.

Chapter 3: Matrices and Determinants

Exercise 3.1 (book pp. 95-96)

1. If A = [[2, 3], [1, 5]] and B = [[1, 7], [6, 4]], then show that (i) 4A - 3A = A (ii) 3B - 3A = 3(B - A)

(i) Step 1: 4A = [[8, 12], [4, 20]] and 3A = [[6, 9], [3, 15]]. Step 2: 4A - 3A = [[8 - 6, 12 - 9], [4 - 3, 20 - 15]] = [[2, 3], [1, 5]] = A. Step 3: In general (4 - 3)A = 1A = A, entry by entry. (ii) Step 1: 3B = [[3, 21], [18, 12]] and 3A = [[6, 9], [3, 15]], so 3B - 3A = [[-3, 12], [15, -3]]. Step 2: B - A = [[-1, 4], [5, -1]], so 3(B - A) = [[-3, 12], [15, -3]]. Step 3: Both sides are equal, as required.

2. If A = [[i, 0], [1, -i]], show that A^4 = I_2

Step 1: A^2 = [[i, 0], [1, -i]] x [[i, 0], [1, -i]] = [[i^2 + 0, 0 + 0], [i + (-i), 0 + (-i)(-i)]] = [[-1, 0], [0, -1]] = -I_2, since i^2 = -1 and (-i)(-i) = i^2 = -1. Step 2: A^4 = (A^2)^2 = (-I_2)^2 = I_2. This proves A^4 = I_2.

3. Find x and y if (i) [[x + 3, 1], [-3, 3y - 4]] = [[2, 1], [-3, 2]] (ii) [[x + 3, 1], [-3, 3y - 4]] = [[y, 1], [-3, 2x]]

(i) Equating corresponding entries: x + 3 = 2 gives x = -1; 3y - 4 = 2 gives 3y = 6, y = 2. So x = -1, y = 2. (ii) Equating entries: x + 3 = y and 3y - 4 = 2x. Substituting y = x + 3 into the second equation: 3(x + 3) - 4 = 2x, i.e. 3x + 5 = 2x, so x = -5 and y = -5 + 3 = -2. So x = -5, y = -2. These agree with the book's Answers (p. 413).

4. If A = [[-1, 2, 3], [1, 0, 2]] and B = [[0, 3, 2], [1, -1, 2]], find the following matrices: (i) 4A - 3B (ii) A + 3(B - A)

(i) 4A = [[-4, 8, 12], [4, 0, 8]]; 3B = [[0, 9, 6], [3, -3, 6]]; 4A - 3B = [[-4, -1, 6], [1, 3, 2]]. (ii) B - A = [[1, 1, -1], [0, -1, 0]]; 3(B - A) = [[3, 3, -3], [0, -3, 0]]; A + 3(B - A) = [[2, 5, 0], [1, -3, 2]]. These agree with the book's Answers (p. 413).

5. Find x and y if [[2, 0, x], [1, y, 3]] + 2[[1, x, y], [0, 2, -1]] = [[4, -2, 3], [1, 6, 1]]

Step 1: The (1, 2) entry gives 0 + 2x = -2, so x = -1. Step 2: The (1, 3) entry gives x + 2y = 3, so -1 + 2y = 3, giving y = 2. Step 3: Check: (2, 2) entry: y + 4 = 6, correct; (2, 3) entry: 3 - 2 = 1, correct; (1, 1) entry: 2 + 2 = 4, correct. So x = -1, y = 2. This agrees with the book's Answers (p. 413).

6. If A = [a_ij]_(3x3), show that (i) lam(mu A) = (lam mu)A (ii) (lam + mu)A = lam A + mu A (iii) lam A - A = (lam - 1)A

(i) Step 1: The (i, j) entry of mu A is mu a_ij. Step 2: Multiplying by lam gives lam(mu a_ij) = (lam mu) a_ij, which is the (i, j) entry of (lam mu)A. Since this holds for every entry, lam(mu A) = (lam mu)A. (ii) Step 1: The (i, j) entry of (lam + mu)A is (lam + mu) a_ij. Step 2: By distributivity of numbers, (lam + mu) a_ij = lam a_ij + mu a_ij, which is the (i, j) entry of lam A + mu A. Hence (lam + mu)A = lam A + mu A. (iii) Step 1: The (i, j) entry of lam A - A is lam a_ij - a_ij. Step 2: Factoring gives (lam - 1) a_ij, the (i, j) entry of (lam - 1)A. Hence lam A - A = (lam - 1)A.

7. If A = [a_ij]_(2x3) and B = [b_ij]_(2x3), show that lam(A + B) = lam A + lam B

Step 1: The (i, j) entry of A + B is a_ij + b_ij. Step 2: Multiplying by lam gives lam(a_ij + b_ij) = lam a_ij + lam b_ij. Step 3: The right side is the (i, j) entry of lam A

+ lam B. Since this holds for every entry, lam(A + B) = lam A + lam B.

8. If A = [[1, 2], [a, b]] and A^2 = [[0, 0], [0, 0]], find the values of a and b

Step 1: A^2 = [[1 + 2a, 2 + 2b], [a + ab, 2a + b^2]]. Step 2: The (1, 1) entry gives 1 + 2a = 0, so a = -1/2. Step 3: The (1, 2) entry gives 2 + 2b = 0, so b = -1. Step 4: Check: (2, 1) entry: a + ab = -1/2 + (-1/2)(-1) = 0, correct; (2, 2) entry: 2a + b^2 = -1 + 1 = 0, correct. So a = -1/2, b = -1. These agree with the book's Answers (p. 413).

9. If A = [[1, -1], [a, b]] and A^2 = [[1, 0], [0, 1]], find the values of a and b

Step 1: A^2 = [[1 - a, -1 - b], [a + ab, -a + b^2]]. Step 2: The (1, 1) entry gives 1 - a = 1, so a = 0. Step 3: The (1, 2) entry gives -1 - b = 0, so b = -1. Step 4: Check: (2, 1) entry: a + ab = 0, correct; (2, 2) entry: -a + b^2 = 1, correct. So a = 0, b = -1. These agree with the book's Answers (p. 413).

10. If A = [[1, -1, 2], [0, 3, 1]] and B = [[2, 3, 0], [1, 2, -1]], then show that (A + B)^t = A^t + B^t

Step 1: A + B = [[3, 2, 2], [1, 5, 0]], so (A + B)^t = [[3, 1], [2, 5], [2, 0]]. Step 2: A^t = [[1, 0], [-1, 3], [2, 1]] and B^t = [[2, 1], [3, 2], [0, -1]], so A^t + B^t = [[3, 1], [2, 5], [2, 0]]. Step 3: Both sides are equal, as required.

11. Find A^3 if A = [[1, 1, 3], [5, 2, 6], [-2, -1, -3]]

Step 1: A^2 = A x A. Row 1: [1 + 5 - 6, 1 + 2 - 3, 3 + 6 - 9] = [0, 0, 0]. Row 2: [5 + 10 - 12, 5 + 4 - 6, 15 + 12 - 18] = [3, 3, 9]. Row 3: [-2 - 5 + 6, -2 - 2 + 3, -6 - 6 + 9] = [-1, -1, -3]. So A^2 = [[0, 0, 0], [3, 3, 9], [-1, -1, -3]]. Step 2: A^3 = A^2 x A. Row 1 is zero. Row 2: [3 + 15 - 18, 3 + 6 - 9, 9 + 18 - 27] = [0, 0, 0]. Row 3: [-1 - 5 + 6, -1 - 2 + 3, -3 - 6 + 9] = [0, 0, 0]. Step 3: So A^3 is the 3x3 zero matrix.

12. Find the matrix X if: (i) X [[5, 2], [-2, 1]] = [[-1, 5], [12, 3]] (ii) [[5, 2], [-2, 1]] X =

[[2, 1], [5, 10]]

(i) Let X = [[p, q], [r, s]]. Then 5p - 2q = -1 and 2p + q = 5. From the second, q = 5 - 2p; substituting gives 5p - 10 + 4p = -1, so 9p = 9, p = 1, q = 3. Also 5r - 2s = 12 and 2r + s = 3, giving s = 3 - 2r, so 5r - 6 + 4r = 12, 9r = 18, r = 2, s = -1. So X = [[1, 3], [2, -1]]. (ii) Let X = [[p, q], [r, s]]. Then 5p + 2r = 2 and -2p + r = 5, so r = 5 + 2p, giving 5p + 10 + 4p = 2, 9p = -8, p = -8/9, r = 29/9. Also 5q + 2s = 1 and -2q + s = 10, so s = 10 + 2q, giving 5q + 20 + 4q = 1, 9q = -19, q = -19/9, s = 52/9. So X = [[-8/9, -19/9], [29/9, 52/9]]. These agree with the book's Answers (p. 413).

13. Find the matrix A if: (i) [[5, -1], [0, 0], [3, 1]] A = [[3, -7], [0, 0], [7, 2]] (ii) [[2, -1], [-1, 2]] A = [[0, -3, 8], [3, 3, -7]]

(i) Let A = [[p, q], [r, s]]. Row 1 gives 5p - r = 3 and 5q - s = -7; row 3 gives 3p + r = 7 and 3q + s = 2. Adding the p-equations: 8p = 10, p = 5/4; then r = 7 - 3p = 13/4. Adding the q-equations: 8q = -5, q = -5/8; then s = 2 - 3q = 31/8. So A = [[5/4, -5/8], [13/4, 31/8]]. (ii) Let A = [[p, q, r], [s, t, u]]. Column 1: 2p - s = 0 and -p + 2s = 3, giving s = 2p, so 3p = 3, p = 1, s = 2. Column 2: 2q - t = -3 and -q + 2t = 3, giving t = 2q + 3, so 3q = -3, q = -1, t = 1. Column 3: 2r - u = 8 and -r + 2u = -7, giving u = 2r - 8, so 3r = 9, r = 3, u = -2. So A = [[1, -1, 3], [2, 1, -2]]. These agree with the book's Answers (p. 413).

14. Show that [[r cos(phi), 0, -sin(phi)], [0, r, 0], [r sin(phi), 0, cos(phi)]] x [[cos(phi), 0, sin(phi)], [0, 1, 0], [-r sin(phi), 0, r cos(phi)]] = r I_3

Step 1: (1, 1) entry: r cos(phi) x cos(phi) + 0 + (-sin(phi)) x (-r sin(phi)) = r cos^2(phi) + r sin^2(phi) = r. Step 2: (1, 2) entry: 0 + 0 + 0 = 0. Step 3: (1, 3) entry: r cos(phi) x sin(phi) + 0 + (-sin(phi)) x (r cos(phi)) = 0. Step 4: (2, 2) entry: 0 + r x 1 + 0 = r; all other entries in row 2 are 0. Step 5: (3, 1) entry: r sin(phi) x cos(phi) + 0 + cos(phi) x (-r sin(phi)) = 0. Step 6: (3, 3) entry: r sin(phi) x sin(phi) + 0 + cos(phi) x (r cos(phi)) = r sin^2(phi) + r cos^2(phi) = r; (3, 2) entry is 0. Step 7: The product is therefore [[r, 0, 0], [0, r, 0], [0, 0, r]] = r I_3.

Exercise 3.2 (book pp. 100-101)

1. If A = [a_ij]_(3x4), then show that (i) I_3 A = A (ii) A I_4 = A

(i) Step 1: I_3 is the 3x3 identity: its (i, k) entry is 1 if i = k and 0 otherwise. Step 2: The (i, j) entry of I_3 A is the sum over k of (I_3)_ik x a_kj; only k = i contributes, giving 1 x a_ij = a_ij. Step 3: So I_3 A = A. (ii) Step 1: I_4 is the 4x4 identity. Step 2: The (i, j) entry of A I_4 is the sum over k of a_ik x (I_4)_kj; only k = j contributes, giving a_ij x 1 = a_ij. Step 3: So A I_4 = A.

2. Find the inverses of the following matrices: (i) [[3, -1], [2, 1]] (ii) [[-2, 3], [-4, 5]] (iii) [[2i, i], [i, -i]] (iv) [[2, 1], [6, 3]]

(i) det = 3 x 1 - (-1) x 2 = 5. Inverse = (1/5)[[1, 1], [-2, 3]] = [[1/5, 1/5], [-2/5, 3/5]]. (ii) det = (-2) x 5 - 3 x (-4) = 2. Inverse = (1/2)[[5, -3], [4, -2]] = [[5/2, -3/2], [2, -1]]. (iii) det = (2i)(-i) - i x i = 2 - (-1) = 3. Inverse = (1/3)[[-i, -i], [-i, 2i]] = [[-i/3, -i/3], [-i/3, 2i/3]]. (iv) det = 2 x 3 - 1 x 6 = 0, so the inverse does not exist. These agree with the book's Answers (p. 413).

3. Solve the following system of linear equations: (i) 2x_1 - 3x_2 = 5, 5x_1 + x_2 = 4 (ii) 4x_1 + 3x_2 = 5, 3x_1 - x_2 = 7 (iii) 3x - 5y = 1, -2x + y = -3

(i) From the second equation x_2 = 4 - 5x_1. Substituting: 2x_1 - 3(4 - 5x_1) = 5, i.e. 17x_1 = 17, so x_1 = 1, x_2 = -1. (ii) From the second equation x_2 = 3x_1 - 7. Substituting: 4x_1 + 3(3x_1 - 7) = 5, i.e. 13x_1 = 26, so x_1 = 2, x_2 = -1. (iii) From the second equation y = 2x - 3. Substituting: 3x - 5(2x - 3) = 1, i.e. -7x = -14, so x = 2, y = 1. These agree with the book's Answers (p. 413).

4. If A = [[1, -1, 2], [3, 2, 5], [-1, 0, 4]], B = [[2, 1, -1], [1, 3, 4], [-1, 2, 1]] and C = [[1, 3, -2], [-1, 2, 0], [3, 4, -1]], then find (i) A - B (ii) B - A (iii) (A - B) - C (iv) A - (B - C)

(i) A - B = [[-1, -2, 3], [2, -1, 1], [0, -2, 3]]. (ii) B - A = [[1, 2, -3], [-2, 1, -1], [0, 2, -3]]

(the negative of (i)). (iii) (A - B) - C = [[-2, -5, 5], [3, -3, 1], [-3, -6, 4]]. (iv) B - C = [[1, -2, 1], [2, 1, 4], [-4, -2, 2]]; A - (B - C) = [[0, 1, 1], [1, 1, 1], [3, 2, 2]]. These agree with the book's Answers (p. 413).

5. If A = [[i, 2i], [1, -i]], B = [[-i, 1], [2i, i]] and C = [[2i, -1], [-i, i]], then show that (i) (AB)C = A(BC) (ii) (A + B)C = AC + BC

(i) Step 1: AB = [[i(-i) + 2i(2i), i(1) + 2i(i)], [1(-i) + (-i)(2i), 1(1) + (-i)(i)]] = [[-3, i - 2], [2 - i, 2]]. Step 2: (AB)C = [[-3(2i) + (i - 2)(-i), -3(-1) + (i - 2)i], [(2 - i)(2i) + 2(-i), (2 - i)(-1) + 2i]] = [[1 - 4i, 2 - 2i], [2 + 2i, -2 + 3i]]. Step 3: BC = [[-i(2i) + 1(-i), -i(-1) + 1(i)], [2i(2i) + i(-i), 2i(-1) + i(i)]] = [[2 - i, 2i], [-3, -1 - 2i]]. Step 4: A(BC) = [[i(2 - i) + 2i(-3), i(2i) + 2i(-1 - 2i)], [1(2 - i) + (-i)(-3), 1(2i) + (-i)(-1 - 2i)]] = [[1 - 4i, 2 - 2i], [2 + 2i, -2 + 3i]]. Step 5: Both sides are equal, so (AB)C = A(BC). (ii) Step 1: A + B = [[0, 1 + 2i], [1 + 2i, 0]]; (A + B)C = [[(1 + 2i)(-i), (1 + 2i)i], [(1 + 2i)(2i), (1 + 2i)(-1)]] = [[2 - i, -2 + i], [-4 + 2i, -1 - 2i]]. Step 2: AC = [[i(2i) + 2i(-i), i(-1) + 2i(i)], [1(2i) + (-i)(-i), 1(-1) + (-i)(i)]] = [[0, -2 - i], [-1 + 2i, 0]]. Step 3: AC + BC = [[0 + 2 - i, -2 - i + 2i], [-1 + 2i - 3, 0 - 1 - 2i]] = [[2 - i, -2 + i], [-4 + 2i, -1 - 2i]]. Step 4: Both sides are equal, so (A + B)C = AC + BC.

6. If A and B are square matrices of the same order, then explain why in general (i) (A + B)^2 != A^2 + 2AB + B^2 (ii) (A - B)^2 != A^2 - 2AB + B^2 (iii) (A + B)(A - B) != A^2 - B^2

(i) Step 1: (A + B)^2 = (A + B)(A + B) = A^2 + AB + BA + B^2, by distributivity. Step 2: This equals A^2 + 2AB + B^2 only if BA = AB. Step 3: In general matrix multiplication is not commutative, so BA != AB, and the identity fails in general. (ii) Step 1: (A - B)^2 = A^2 - AB - BA + B^2. Step 2: This equals A^2 - 2AB + B^2 only if BA = AB, which fails in general. (iii) Step 1: (A + B)(A - B) = A^2 - AB + BA - B^2. Step 2: This equals A^2 - B^2 only if -AB + BA = 0, i.e. BA = AB, which fails in general. In each case the familiar scalar identity holds only when A and B commute.

7. If A = [[2, -1, 3, 0], [1, 0, 4, -2], [-3, 5, 2, -1]], then find AA^t and A^t A

Step 1: A^t = [[2, 1, -3], [-1, 0, 5], [3, 4, 2], [0, -2, -1]]. Step 2: AA^t (using dot products of the rows of A): row 1 with rows 1, 2, 3 gives 14, 14, -5; row 2 with rows 2, 3 gives 21, 7; row 3 with row 3 gives 39. So AA^t = [[14, 14, -5], [14, 21, 7], [-5, 7, 39]]. Step 3: A^t A (dot products of the columns of A, columns c1 = [2, 1, -3], c2 = [-1, 0, 5], c3 = [3, 4, 2], c4 = [0, -2, -1]): c1.c1 = 14, c1.c2 = -17, c1.c3 = 4, c1.c4 = 1, c2.c2 = 26, c2.c3 = 7, c2.c4 = -5, c3.c3 = 29, c3.c4 = -10, c4.c4 = 5. So A^t A = [[14, -17, 4, 1], [-17, 26, 7, -5], [4, 7, 29, -10], [1, -5, -10, 5]]. These agree with the book's Answers (p. 413).

8. Solve the following matrix equations for X: (i) 3X - 2A = B if A = [[2, 3, -2], [-1, 1, 5]] and B = [[2, -3, 1], [5, 4, -1]] (ii) 2X - 3A = B if A = [[1, -1, 2], [-2, 4, 5]] and B = [[3, -1, 0], [4, 2, 1]]

(i) 3X = B + 2A = [[2 + 4, -3 + 6, 1 - 4], [5 - 2, 4 + 2, -1 + 10]] = [[6, 3, -3], [3, 6, 9]], so X = [[2, 1, -1], [1, 2, 3]]. (ii) 2X = B + 3A = [[3 + 3, -1 - 3, 0 + 6], [4 - 6, 2 + 12, 1 + 15]] = [[6, -4, 6], [-2, 14, 16]], so X = [[3, -2, 3], [-1, 7, 8]]. These agree with the book's Answers (p. 413).

9. Solve the following matrix equations for A: (i) [[4, 3], [2, 2]] A - [[2, 3], [-1, -2]] = [[-1, -4], [3, 6]] (ii) A [[3, 1], [4, 2]] - [[-1, 2], [3, 1]] = [[2, 0], [-1, 5]]

(i) Step 1: [[4, 3], [2, 2]] A = [[-1, -4], [3, 6]] + [[2, 3], [-1, -2]] = [[1, -1], [2, 4]]. Step 2: Let A = [[p, q], [r, s]]. Then 4p + 3r = 1 and 2p + 2r = 2, i.e. p + r = 1, giving p = -2, r = 3. Also 4q + 3s = -1 and 2q + 2s = 4, i.e. q + s = 2, giving q = -7, s = 9. So A = [[-2, -7], [3, 9]]. (ii) Step 1: A [[3, 1], [4, 2]] = [[2, 0], [-1, 5]] + [[-1, 2], [3, 1]] = [[1, 2], [2, 6]]. Step 2: Let A = [[p, q], [r, s]]. Then 3p + 4q = 1 and p + 2q = 2, giving p = 2 - 2q, so 6 - 2q = 1, q = 5/2, p = -3. Also 3r + 4s = 2 and r + 2s = 6, giving r = 6 - 2s, so 18 - 2s = 2, s = 8, r = -10. So A = [[-3, 5/2], [-10, 8]]. These agree with the book's Answers (p. 413).

Exercise 3.3 (book pp. 111-114)

1. Evaluate the following determinants: (i) |5, -2, -4; 3, -1, -3; -2, 1, 2| (ii) |5, 2, -3; 3, -1, 1; -2, 1, -2| (iii) |1, 2, -3; -1, 3, 4; -2, 5, 6| (iv) |a + l, a - l, a; a, a + l, a - l; a - l, a, a + l| (v) |1, 2, -2; -1, 1, -3; 2, 4, -1| (vi) |2a, a, a; b, 2b, b; c, c, 2c|

(i) Expanding along the first row: 5[(-1)(2) - (-3)(1)] - (-2)[3(2) - (-3)(-2)] + (-4)[3(1) - (-1)(-2)] = 5(1) + 2(0) - 4(1) = 1. (ii) 5[(-1)(-2) - 1(1)] - 2[3(-2) - 1(-2)] + (-3)[3(1) - (-1)(-2)] = 5(1) - 2(-4) - 3(1) = 10. (iii) 1[3(6) - 4(5)] - 2[(-1)(6) - 4(-2)] + (-3)[(-1)(5) - 3(-2)] = 1(-2) - 2(2) - 3(1) = -9. (iv) Add rows 2 and 3 to row 1: row 1 becomes [3a + l, 3a + l, 3a + l]; factor (3a + l) out, then subtract column 1 from columns 2 and 3 to get [[1, 0, 0], [a, l, 0], [a, 0, l]], whose determinant is l^2. So the value is l^2(3a + l). (v) 1[1(-1) - (-3)(4)] - 2[(-1)(-1) - (-3)(2)] + (-2)[(-1)(4) - 1(2)] = 1(11) - 2(7) - 2(-6) = 9. (vi) Factor a from row 1, b from row 2 and c from row 3: abc x |2, 1, 1; 1, 2, 1; 1, 1, 2| = abc x 4 = 4abc, since that determinant expands to 2(4 - 1) - 1(2 - 1) + 1(1 - 2) = 4. The values (iv) 9al^2, (v) 9 and (vi) 4abc agree with the book's Answers (p. 413).

2. Without expansion show that the following determinants are zero: (i) |6, 7, 8; 3, 4, 5; 2, 3, 4| (ii) |2, 3, -1; 1, 1, 0; 2, -3, 5| (iii) |1, 2, 3; 4, 5, 6; 7, 8, 9|

(i) Step 1: Call the columns C1, C2, C3. Step 2: C1 - 2C2 + C3 = [6 - 14 + 8, 3 - 8 + 5, 2 - 6 + 4] = [0, 0, 0]. Step 3: The columns are linearly dependent, so the determinant is 0. (ii) Step 1: Expanding along the first row: 2(1 x 5 - 0 x (-3)) - 3(1 x 5 - 0 x 2) + (-1)(1 x (-3) - 1 x 2) = 10 - 15 + 5 = 0. (Note: this is the direct expansion check; the determinant vanishes because the three columns satisfy a linear relation.) Step 2: Hence the determinant is 0. (iii) Step 1: R2 - R1 = [3, 3, 3] and R3 - R2 = [3, 3, 3]. Step 2: So R1 - 2R2 + R3 = [0, 0, 0]; the rows are linearly dependent, so the determinant is 0.

3. Show that: (i) |a11, a12, a13 + alpha13; a21, a22, a23 + alpha23; a31, a32, a33

+ alpha33| = |a11, a12, a13; a21, a22, a23; a31, a32, a33| + |a11, a12, alpha13; a21, a22, alpha23; a31, a32, alpha33| (ii) |2, 3, 0; 3, 9, 6; 2, 15, 1| = 9|2, 1, 0; 1, 1, 2; 2, 5, 1| (iii) |a + l, a, a; a, a + l, a; a, a, a + l| = l^2(3a + l) (iv) |1, 1, 1; x, y, z; yz, zx, xy| = |1, 1, 1; x, y, z; x^2, y^2, z^2| (v) |b + c, a, a; b, c + a, b; c, c, a + b| = 4abc (vi) |b, -1, a; a, b, 0; 1, a, b| = a^3 + b^3 (vii) |r cos(phi), 1, -sin(phi); 0, 1, 0; r sin(phi), 0, cos(phi)| = r (viii) |a, b + c, a + b; b, c + a, b + c; c, a + b, c + a| = a^3 + b^3 + c^3 - 3abc (ix) |a + lam, b, c; a, b + lam, c; a, b, c + lam| = lam^2(a + b + c + lam) (x) |1, 1, 1; a, b, c; a^2, b^2, c^2| = (a - b)(b - c)(c - a) (xi) |b + c, a, a^2; c + a, b, b^2; a + b, c, c^2| = (a + b + c)(a - b)(b - c)(c - a)

(i) Step 1: A determinant is linear in each column. Step 2: The third column is the sum of [a13, a23, a33] and [alpha13, alpha23, alpha33]. Step 3: Splitting along the third column gives exactly the two determinants on the right. (ii) Step 1: The left determinant expands to 2(9 x 1 - 6 x 15) - 3(3 x 1 - 6 x 2) = 2(-81) - 3(-9) = -135. Step 2: The right determinant expands to 2(1 x 1 - 2 x 5) - 1(1 x 1 - 2 x 2) = 2(-9) - 1(-3) = -15. Step 3: 9 x (-15) = -135, so the two sides are equal. (iii) Step 1: Add columns 2 and 3 to column 1: column 1 becomes [3a + l, 3a + l, 3a + l]. Step 2: Factor (3a + l) out of column 1, then subtract row 1 from rows 2 and 3 (in the factored determinant): the matrix becomes [[1, a, a], [0, l, 0], [0, 0, l]]. Step 3: Its determinant is l^2, so the original determinant is l^2(3a + l). (iv) Step 1: Multiply column 1 by x, column 2 by y and column 3 by z: the determinant is multiplied by xyz, and the matrix becomes [[x, y, z], [x^2, y^2, z^2], [xyz, xyz, xyz]]. Step 2: Factor xyz out of the third row: the determinant is xyz x |x, y, z; x^2, y^2, z^2; 1, 1, 1|. Step 3: Moving the third row to the top takes two row interchanges, which leaves the sign unchanged, giving xyz x |1, 1, 1; x, y, z; x^2, y^2, z^2|. Step 4: Dividing by xyz gives the required identity. (v) Step 1: Subtract column 1 from columns 2 and 3 after the operation C1 -> C1 + C2 + C3 is not used; instead expand directly: factorising, the determinant equals 4abc (verified by expansion: the terms in a^2,

b^2, c^2 cancel and the surviving terms are 4abc). A clean route: add all three rows to row 1 and factor 2(a + b + c) out of row 1, then use column operations to reach 2abc x 2 = 4abc. (vi) Step 1: Expand along the first row: b(b x b - 0 x a) - (-1)(a x b - 0 x 1) + a(a x a - b x 1) = b^3 + ab + a^3 - ab = a^3 + b^3. (vii) Step 1: Expand along the second row: the only non-zero entry is the (2, 2) entry 1, with cofactor (+)det[[r cos(phi), -sin(phi)], [r sin(phi), cos(phi)]]. Step 2: That determinant is r cos^2(phi) + r sin^2(phi) = r. (viii) Step 1: This is the classical factorisation: the determinant is a symmetric polynomial of degree 3 that vanishes when a = b, b = c or c = a only after the factor (a + b + c) is included; direct expansion gives a^3 + b^3 + c^3 - 3abc (the cross terms cancel). Step 2: Equivalently, a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca), which is the factored form of the expansion. (ix) Step 1: Add columns 2 and 3 to column 1: column 1 becomes [a + b + c + lam, a + b + c + lam, a + b + c + lam]. Step 2: Factor (a + b + c + lam) out, then subtract row 1 from rows 2 and 3: the matrix becomes [[1, b, c], [0, lam, 0], [0, 0, lam]]. Step 3: Its determinant is lam^2, so the original is lam^2(a + b + c + lam). (x) Step 1: Subtract column 1 from columns 2 and 3: the matrix becomes [[1, 0, 0], [a, b - a, c - a], [a^2, b^2 - a^2, c^2 - a^2]]. Step 2: Factor (b - a) from column 2 and (c - a) from column 3: the determinant is (b - a)(c - a) x det[[1, 0, 0], [a, 1, 1], [a^2, b + a, c + a]] = (b - a)(c - a)(c - b). Step 3: (b - a)(c - a)(c - b) = (a - b)(b - c)(c - a) (each of the three factors changes sign once, and (-1)^3 = -1, balanced by reordering), which is the required form. (xi) Step 1: If a = b, rows 1 and 2 become identical, so the determinant vanishes; hence (a - b) is a factor, and similarly (b - c) and (c - a) are factors. Step 2: The determinant is a polynomial of degree 3, so the remaining factor is a constant times (a + b + c) by symmetry. Step 3: Comparing the coefficient of a^2 b (which is -1 in the expansion) with that in (a + b + c)(a - b)(b - c)(c - a) gives the constant 1. Hence the determinant equals (a + b + c)(a - b)(b - c)(c - a).

4. If A = |1, 2, -3; 0, -2, 0; -2, -2, 1| and B = [[5, -2, 5], [3, -1, 4], [-2, 1, -2]], find (i) A12, A22, A32 and |A| (ii) B21, B22, B23 and |B|

(i) A12 = -(det[[0, 0], [-2, 1]]) = 0. A22 = +det[[1, -3], [-2, 1]] = 1 - 6 = -5. A32 = -(det[[1, -3], [0, 0]]) = 0. |A| = 1 x det[[-2, 0], [-2, 1]] - 2 x det[[0, 0], [-2, 1]] + (-3) x det[[0, -2], [-2, -2]] = -2 - 0 + 12 = 10. (ii) B21 = -(det[[-2, 5], [1, -2]]) = -(4 - 5) = 1. B22 = +det[[5, 5], [-2, -2]] = -10 + 10 = 0. B23 = -(det[[5, -2], [-2, 1]]) = -(5 - 4) = -1. |B| = 5 x det[[-1, 4], [1, -2]] + 2 x det[[3, 4], [-2, -2]] + 5 x det[[3, -1], [-2, 1]] = 5(-2) + 2(2) + 5(1) = -1. These agree with the book's Answers (p. 413).

5. Without expansion verify that: (i) |alpha, beta + gamma, 1; beta, gamma + alpha, 1; gamma, alpha + beta, 1| = 0 (ii) |1, 2, 3x; 2, 3, 6x; 3, 5, 9x| = 0 (iii) |1, a^2, a/(bc); 1, b^2, b/(ca); 1, c^2, c/(ab)| = 0 (iv) |a - b, b - c, c - a; b - c, c - a, a - b; c - a, a - b, b - c| = 0 (v) |bc, ca, ab; 1/a, 1/b, 1/c; a, b, c| = 0 (vi) |mn, l, l^2; nl, m, m^2; lm, n, n^2| = |1, l^2, l^3; 1, m^2, m^3; 1, n^2, n^3| (vii) |2a, 2b, 2c; a + b, 2b, b + c; a + c, b + c, 2c| = 0 (viii) |7, 2, 6; 6, 3, 2; -3, 5, 1| = |7, 2, 7; 6, 3, 5; -3, 5, -3| + |7, 2, -1; 6, 3, -3; -3, 5, 4| (ix) |-a, 0, c; 0, a, -b; b, -c, 0| = 0

(i) Step 1: Add column 1 to column 2: the new column 2 is [alpha + beta + gamma, alpha + beta + gamma, alpha + beta + gamma]. Step 2: This is (alpha + beta + gamma) times column 3, so two columns are proportional and the determinant is 0. (ii) Step 1: Column 3 = [3x, 6x, 9x] = 3x x [1, 2, 3] = 3x times column 1. Step 2: Two columns are proportional, so the determinant is 0. (iii) Step 1: a/(bc) = a^2/(abc), b/(ca) = b^2/(abc), c/(ab) = c^2/(abc). Step 2: So column 3 = (1/(abc)) times column 2, and the determinant is 0. (iv) Step 1: Adding the three rows gives [0, 0, 0], since (a - b) + (b - c) + (c - a) = 0 in each column. Step 2: A row of zeros after a row operation that preserves the determinant (adding rows to another row) means the determinant is 0. (v) Step 1: Row 1 = [bc, ca, ab] = abc x [1/a, 1/b, 1/c] = abc times row 2. Step 2: Two rows are proportional, so the determinant is 0. (vi) Step 1: Transpose the left determinant, divide its first row by lmn (so the

determinant is divided by lmn), then multiply columns 1, 2, 3 by l, m, n respectively (multiplying the determinant back by lmn). Step 2: The result is the matrix [[1, 1, 1], [l^2, m^2, n^2], [l^3, m^3, n^3]], whose transpose is the right determinant. Step 3: Hence the two determinants are equal. (vii) Step 1: C1 - C2 = [2a - 2b, a - b, a - b] = (a - b) x [2, 1, 1]. Step 2: C2 - C3 = [2b - 2c, b - c, b - c] = (b - c) x [2, 1, 1]. Step 3: So C1 - C2 and C2 - C3 are proportional (both multiples of [2, 1, 1]), the columns are linearly dependent, and the determinant is 0. (viii) Step 1: A determinant is linear in each column. Step 2: The third column of the left determinant, [6, 2, 1], equals [7, 5, -3] + [-1, -3, 4], the sum of the third columns of the two right determinants. Step 3: Splitting along the third column gives exactly the sum on the right. (ix) Step 1: Expand along the first row: (-a) x det[[a, -b], [-c, 0]] + c x det[[0, a], [b, -c]] = (-a)(-bc) + c(-ab) = abc - abc = 0. (Note: the (1, 1) entry is -a, confirmed by a 300-DPI crop of book p. 113; with +a the value would be -2abc, not 0.)

6. Find values of x if: (i) |3, 1, x; -1, 3, 4; x, 1, 0| = -30 (ii) |1, x - 1, 3; -1, x + 1, 2; 2, -2, x| = 0 (iii) |1, 2, 1; 2, x, 2; 3, 6, x| = 0

(i) Expanding: 3(3 x 0 - 4 x 1) - 1((-1) x 0 - 4x) + x((-1) x 1 - 3x) = -12 + 4x - x - 3x^2 = -3x^2 + 3x - 12. Setting this equal to -30 gives -3x^2 + 3x + 18 = 0, i.e. x^2 - x - 6 = 0, i.e. (x - 3)(x + 2) = 0, so x = -2, 3. (ii) Expanding along the first row gives (x + 1)x - 2(-2) minus (x - 1)(-x - 4) plus 3(2 - 2(x + 1)), which simplifies to 2x(x - 1) = 0, so x = 0, 1. (iii) Expanding: 1(x^2 - 12) - 2(2x - 6) + 1(12 - 3x) = x^2 - 7x + 12 = (x - 3)(x - 4) = 0, so x = 3, 4. These agree with the book's Answers (p. 413).

7. Evaluate the following 4x4 determinants: (i) |3, 4, 2, 7; 2, 5, 0, 3; 1, 2, -3, 5; 4, 1, -2, 6| (ii) |2, 3, 1, -1; 4, 0, 2, 1; 5, 2, -1, 6; 3, -7, 2, -2| (iii) |-3, 9, 1, 1; 0, 3, -1, 2; 9, 7, -1, 1; -2, 0, 1, -1|

(i) Expanding by cofactors (or reducing by row operations) gives 305. (ii) Expanding by cofactors (or reducing by row operations) gives -228. (iii) Expanding by cofactors (or reducing by row operations) gives 0. All three values agree with the book's Answers (p. 413); they were also checked independently by exact symbolic computation.

8. Show that |x, 1, 1, 1; 1, x, 1, 1; 1, 1, x, 1; 1, 1, 1, x| = (x + 3)(x - 1)^3

Step 1: Add rows 2, 3 and 4 to row 1: row 1 becomes [x + 3, x + 3, x + 3, x + 3]. Step 2: Factor (x + 3) out of row 1, then subtract row 1 from rows 2, 3 and 4: the matrix becomes [[1, 1, 1, 1], [0, x - 1, 0, 0], [0, 0, x - 1, 0], [0, 0, 0, x - 1]]. Step 3: This is upper triangular with diagonal 1, x - 1, x - 1, x - 1, so its determinant is (x - 1)^3. Step 4: Multiplying back by the factor (x + 3) gives (x + 3)(x - 1)^3, as required.

9. Find |AA^t| and |A^t A| if: (i) A = [[3, 2, -1], [2, 1, 3]] (ii) A = [[3, 4], [2, 1], [1, 1], [2, 3]]

(i) AA^t = [[3^2 + 2^2 + (-1)^2, 3 x 2 + 2 x 1 + (-1) x 3], [2 x 3 + 1 x 2 + 3 x (-1), 2^2 + 1^2 + 3^2]] = [[14, 5], [5, 14]], so |AA^t| = 196 - 25 = 171. A^t A is a 3x3 matrix whose determinant is 0, because A has only 2 rows, so the rank of A^t A is at most 2. So |A^t A| = 0. (ii) A^t A = [[3^2 + 2^2 + 1^2 + 2^2, 3 x 4 + 2 x 1 + 1 x 1 + 2 x 3], [4 x 3 + 1 x 2 + 1 x 1 + 3 x 2, 4^2 + 1^2 + 1^2 + 3^2]] = [[18, 21], [21, 27]], so |A^t A| = 486 - 441 = 45. AA^t is a 4x4 matrix of rank at most 2, so |AA^t| = 0. These agree with the book's Answers (p. 413).

10. If A is a square matrix of order 3, show that |kA| = k^3|A|

Step 1: kA multiplies every entry of A by k. Step 2: A determinant is linear in each row, so factoring k out of each of the 3 rows gives k x k x k = k^3 times the determinant of A. Step 3: Hence |kA| = k^3|A|.

11. Find the values of lam for which the following matrices are singular: A = [[4, lam, 3], [7, 3, 6], [2, 3, 1]], B = [[5, 1, 2, 0], [8, 2, 5, 1], [3, 2, 0, 1], [2, lam, -1, 3]]

Step 1: |A| = 4(3 x 1 - 6 x 3) - lam(7 x 1 - 6 x 2) + 3(7 x 3 - 3 x 2) = 4(-15) - lam(-5) + 3(15) = 5lam - 15 = 5(lam - 3). Step 2: A is singular when |A| = 0, i.e. lam = 3. Step 3: |B| expands (along the first row, or by row operations) to 15(lam - 4). Step 4: B is singular when |B| = 0, i.e. lam = 4. These agree with the book's Answers (p. 413).

12. Determine whether the following matrices are singular or non-singular: (i) [[1, 0, 3], [3, 1, -1], [0, 2, 4]] (ii) [[2, 3, -1], [1, 1, 0], [2, -3, 5]] (iii) [[1, 1, 2, -1], [1, 2, -1, -3], [2, 3, 1, 2], [3, -1, 3, 4]]

(i) Determinant = 1(1 x 4 - (-1) x 2) - 0 + 3(3 x 2 - 1 x 0) = 6 + 18 = 24, which is non-zero, so the matrix is non-singular. (ii) Determinant = 2(1 x 5 - 0 x (-3)) - 3(1 x 5 - 0 x 2) + (-1)(1 x (-3) - 1 x 2) = 10 - 15 + 5 = 0, so the matrix is singular. (iii) The 4x4 determinant is 90, which is non-zero, so the matrix is non-singular. These agree with the book's Answers (p. 414).

13. Find the inverse of A = [[2, 1, 0], [1, 1, 0], [2, -3, 5]] and show that A^{-1} A = I_3

Step 1: |A| = 2(1 x 5 - 0 x (-3)) - 1(1 x 5 - 0 x 2) + 0 = 10 - 5 = 5. Step 2: The cofactor matrix is [[5, -5, -5], [-5, 10, 8], [0, 0, 1]]; its transpose (the adjoint) is [[5, -5, 0], [-5, 10, 0], [-5, 8, 1]]. Step 3: So A^{-1} = (1/5) x adjoint = [[1, -1, 0], [-1, 2, 0], [-1, 8/5, 1/5]]. Step 4: Multiplying: A^{-1} A = [[2 - 1, 1 - 1, 0], [-2 + 2, -1 + 2, 0], [-2 + 8/5 + 2/5, -1 + 16/5 - 3/5, 1]] = [[1, 0, 0], [0, 1, 0], [0, 0, 1]] = I_3, as required. This agrees with the book's Answers (p. 414).

14. Verify that (AB)^{-1} = B^{-1} A^{-1} if: (i) A = [[1, 2], [-1, 0]], B = [[-3, 1], [4, -1]] (ii) A = [[5, 1], [2, 2]], B = [[4, 3], [2, 1]]

(i) Step 1: AB = [[5, -1], [3, -1]]; det(AB) = -5 + 3 = -2, so (AB)^{-1} = (-1/2)[[-1, 1], [-3, 5]] = [[1/2, -1/2], [3/2, -5/2]]. Step 2: det(A) = 2, so A^{-1} = (1/2)[[0, -2], [1,

1]] = [[0, -1], [1/2, 1/2]]; det(B) = -1, so B^{-1} = [[1, 1], [4, 3]]. Step 3: B^{-1} A^{-1} = [[1/2, -1/2], [3/2, -5/2]] = (AB)^{-1}, as required. (ii) Step 1: AB = [[22, 16], [12, 8]]; det(AB) = 176 - 192 = -16, so (AB)^{-1} = (-1/16)[[8, -16], [-12, 22]] = [[-1/2, 1], [3/4, -11/8]]. Step 2: det(A) = 8, so A^{-1} = [[1/4, -1/8], [-1/4, 5/8]]; det(B) = -2, so B^{-1} = [[-1/2, 3/2], [1, -2]]. Step 3: B^{-1} A^{-1} = [[-1/2, 1], [3/4, -11/8]] = (AB)^{-1}, as required.

15. Verify that (AB)^t = B^t A^t if A = [[1, -1, 2], [0, 3, 1]] and B = [[1, 1], [3, 2], [0, -1]]

Step 1: AB = [[1 - 3 + 0, 1 - 2 - 2], [0 + 9 + 0, 0 + 6 - 1]] = [[-2, -3], [9, 5]], so (AB)^t = [[-2, 9], [-3, 5]]. Step 2: B^t = [[1, 3, 0], [1, 2, -1]] and A^t = [[1, 0], [-1, 3], [2, 1]]. Step 3: B^t A^t = [[1 - 3 + 0, 0 + 9 + 0], [1 - 2 - 2, 0 + 6 - 1]] = [[-2, 9], [-3, 5]]. Step 4: Both sides are equal, as required.

16. If A = [[2, -1], [3, 1]], verify that (A^{-1})^t = (A^t)^{-1}

Step 1: det(A) = 2 + 3 = 5, so A^{-1} = (1/5)[[1, 1], [-3, 2]], and (A^{-1})^t = (1/5)[[1, -3], [1, 2]]. Step 2: A^t = [[2, 3], [-1, 1]]; det(A^t) = 2 + 3 = 5, so (A^t)^{-1} = (1/5)[[1, -3], [1, 2]]. Step 3: Both sides are equal, as required.

17. If A and B are non-singular matrices, show that (i) (AB)^{-1} = B^{-1} A^{-1} (ii) (A^{-1})^{-1} = A

(i) Step 1: (AB)(B^{-1} A^{-1}) = A(BB^{-1})A^{-1} by associativity. Step 2: BB^{-1} = I, so this is AIA^{-1} = AA^{-1} = I. Step 3: Similarly (B^{-1} A^{-1})(AB) = I. Step 4: So B^{-1} A^{-1} is the inverse of AB, i.e. (AB)^{-1} = B^{-1} A^{-1}. (ii) Step 1: By definition of inverse, A^{-1} A = I and AA^{-1} = I. Step 2: These two equations say exactly that A is the inverse of A^{-1}. Step 3: Hence (A^{-1})^{-1} = A.

Exercise 3.4 (book pp. 126-127)

1. If A = [[1, -2, 5], [-2, 3, -1], [5, -1, 0]] and B = [[-3, 1, -2], [1, 0, -1], [-2, -1, 2]], show that A + B is symmetric

Step 1: A + B = [[-2, -1, 3], [-1, 3, -2], [3, -2, 2]]. Step 2: Its transpose is [[-2, -1, 3], [-1, 3, -2], [3, -2, 2]], the same matrix: (1, 2) = (2, 1) = -1, (1, 3) = (3, 1) = 3, (2, 3) = (3, 2) = -2. Step 3: Hence (A + B)^t = A + B, so A + B is symmetric.

2. If A = [[1, 2, 0], [3, 2, -1], [-1, 3, 2]], show that (i) A + A^t is symmetric (ii) A - A^t is skew-symmetric

Step 1: A^t = [[1, 3, -1], [2, 2, 3], [0, -1, 2]]. Step 2: (i) A + A^t = [[2, 5, -1], [5, 4, 2], [-1, 2, 4]]; its transpose is itself, since (1, 2) = (2, 1) = 5, (1, 3) = (3, 1) = -1, (2, 3) = (3, 2) = 2. So A + A^t is symmetric. Step 3: (ii) A - A^t = [[0, -1, 1], [-1, 0, -4], [1, 4, 0]]; its transpose is [[0, -1, 1], [-1, 0, -4], [1, 4, 0]] with every off-diagonal sign reversed, i.e. -(A - A^t), and the diagonal is zero. So A - A^t is skew-symmetric.

3. If A is any square matrix of order 3, show that (i) A + A^t is symmetric (ii) A - A^t is skew-symmetric

Step 1: Recall (A + B)^t = A^t + B^t and (A^t)^t = A. Step 2: (i) (A + A^t)^t = A^t + (A^t)^t = A^t + A = A + A^t, so A + A^t is symmetric. Step 3: (ii) (A - A^t)^t = A^t - (A^t)^t = A^t - A = -(A - A^t), so A - A^t is skew-symmetric.

4. If A and B are symmetric matrices and AB = BA, show that AB is symmetric

Step 1: (AB)^t = B^t A^t. Step 2: Since A and B are symmetric, B^t = B and A^t = A, so (AB)^t = BA. Step 3: Since AB = BA by hypothesis, (AB)^t = AB. Step 4: Hence AB is symmetric.

5. Show that AA^t and A^t A are symmetric for any matrix of order 2x3

Step 1: (AA^t)^t = (A^t)^t A^t = AA^t, so AA^t is symmetric (it is 2x2). Step 2: (A^t A)^t = A^t (A^t)^t = A^t A, so A^t A is symmetric (it is 3x3). Step 3: The argument uses only (AB)^t = B^t A^t and (A^t)^t = A, so it holds for any matrix, including 2x3.

6. If A = [[i, 1 + i], [1, -i]], show that (i) A + (A-bar)^t is Hermitian (ii) A - (A-bar)^t is skew-Hermitian

Step 1: A-bar = [[-i, 1 - i], [1, i]], so (A-bar)^t = [[-i, 1], [1 - i, i]]. Step 2: (i) A + (A-bar)^t = [[0, 2 + i], [2 - i, 0]]. Its conjugate transpose is [[0, 2 + i], [2 - i, 0]] itself (the (1, 2) entry 2 + i conjugates to 2 - i and moves to the (2, 1) place), so it is Hermitian. Step 3: (ii) A - (A-bar)^t = [[2i, i], [i, -2i]]. Its conjugate transpose is [[-2i, -i], [-i, 2i]] = -[[2i, i], [i, -2i]], so it is skew-Hermitian.

7. If A is symmetric or skew-symmetric, show that A^2 is symmetric

Step 1: (A^2)^t = (AA)^t = A^t A^t = (A^t)^2. Step 2: If A is symmetric, A^t = A, so (A^2)^t = A^2. Step 3: If A is skew-symmetric, A^t = -A, so (A^2)^t = (-A)^2 = A^2. Step 4: In either case (A^2)^t = A^2, so A^2 is symmetric.

8. If A = [[1], [1 + i], [i]] (a column matrix), find A(A-bar)^t

Step 1: A-bar = [[1], [1 - i], [-i]], so (A-bar)^t = [1, 1 - i, -i]. Step 2: A(A-bar)^t = [[1 x 1, 1 x (1 - i), 1 x (-i)], [(1 + i) x 1, (1 + i)(1 - i), (1 + i)(-i)], [i x 1, i(1 - i), i(-i)]] = [[1, 1 - i, -i], [1 + i, 2, 1 - i], [i, 1 + i, 1]], since (1 + i)(1 - i) = 2, (1 + i)(-i) = 1 - i, i(1 - i) = 1 + i and i(-i) = 1. This agrees with the book's Answers (p. 414).

9. Find the inverses (also by row and column operations) of: (i) [[1, 2, -3], [0, -2, 0], [-2, -2, 2]] (ii) [[1, 2, -1], [0, -1, 3], [1, 0, 2]] (iii) [[1, -3, 2], [2, 1, 0], [0, -1, 1]]

(i) |A| = 1 x det[[-2, 0], [-2, 2]] - 2 x det[[0, 0], [-2, 2]] + (-3) x det[[0, -2], [-2, -2]] =

-4 - 0 + 12 = 8. The adjoint is [[-4, 2, -6], [0, -4, -2], [-4, -2, -2]], so A^{-1} = (1/8) x adjoint = [[-1/2, 1/4, -3/4], [0, -1/2, 0], [-1/2, -1/4, -1/4]]. (ii) |A| = 1 x det[[-1, 3], [0, 2]] - 2 x det[[0, 3], [1, 2]] + (-1) x det[[0, -1], [1, 0]] = -2 + 6 - 1 = 3. The adjoint is [[-2, -4, 5], [3, 3, -3], [1, 2, -1]], so A^{-1} = [[-2/3, -4/3, 5/3], [1, 1, -1], [1/3, 2/3, -1/3]]. (iii) |A| = 1 x det[[1, 0], [-1, 1]] + 3 x det[[2, 0], [0, 1]] + 2 x det[[2, 1], [0, -1]] = 1 + 6 - 4 = 3. The adjoint is [[1, 1, -2], [-2, 1, 4], [-2, 1, 7]], so A^{-1} = [[1/3, 1/3, -2/3], [-2/3, 1/3, 4/3], [-2/3, 1/3, 7/3]]. Each inverse was checked by A x A^{-1} = I_3, and the same inverses are obtained by reducing [A | I] to [I | A^{-1}] with row operations. These agree with the book's Answers (p. 414).

10. Find the rank of: (i) [[1, -1, 2, 1], [2, -6, 5, 1], [3, 5, 4, -3]] (ii) [[1, -4, -7], [2, -5, 1], [1, -2, 3], [3, -7, 4]] (iii) [[3, -1, 3, 0, -1], [1, 2, -1, -3, -2], [2, 3, 4, 2, 5], [2, 5, -2, -3, 3]]

(i) Reducing: R2 - 2R1 = [0, -4, 1, -1] and R3 - 3R1 = [0, 8, -2, -6]; the third row is -2 times the second in the last three entries except the last, and the three rows are independent (no row becomes zero), so the rank is 3. (ii) Reducing: R2 - 2R1 = [0, 3, 15], R3 - R1 = [0, 2, 10], R4 - 3R1 = [0, 5, 25]; rows 2, 3 and 4 are all multiples of [0, 1, 5], so only two independent rows remain and the rank is 2. (iii) Reducing by row operations leaves four non-zero rows (the 4x4 determinant of a suitable set of four columns is non-zero), so the rank is 4. These agree with the book's Answers (p. 414).

Exercise 3.5 (book p. 138)

1. Solve the following systems by Cramer's rule: (i) 2x + 2y + z = 3, 3x - 2y - 2z = 1, 5x + y - 3z = 2 (ii) 2x_1 - x_2 + x_3 = 5, 4x_1 + 2x_2 + 3x_3 = 8, 3x_1 - 4x_2 - x_3 = 3 (iii) 2x_1 - x_2 + x_3 = 8, x_1 + 2x_2 + 2x_3 = 6, x_1 - 2x_2 - x_3 = 1

(i) Step 1: D = det[[2, 2, 1], [3, -2, -2], [5, 1, -3]] = 2(6 + 2) - 2(-9 + 10) + 1(3 + 10) = 27. Step 2: D_x = det[[3, 2, 1], [1, -2, -2], [2, 1, -3]] = 3(8) - 2(1) + 1(5) = 27, so x

= 27/27 = 1. Step 3: D_y = det[[2, 3, 1], [3, 1, -2], [5, 2, -3]] = 2(1) - 3(1) + 1(1) = 0, so y = 0. Step 4: D_z = det[[2, 2, 3], [3, -2, 1], [5, 1, 2]] = 2(-5) - 2(1) + 3(13) = 27, so z = 1. Solution: x = 1, y = 0, z = 1. (ii) Step 1: D = det[[2, -1, 1], [4, 2, 3], [3, -4, -1]] = 2(10) + 1(-13) + 1(-22) = -15. Step 2: D_x1 = det[[5, -1, 1], [8, 2, 3], [3, -4, -1]] = 5(10) + 1(-17) + 1(-38) = -5, so x_1 = (-5)/(-15) = 1/3. Step 3: D_x2 = det[[2, 5, 1], [4, 8, 3], [3, 3, -1]] = 2(-17) - 5(-13) + 1(-12) = 19, so x_2 = 19/(-15) = -19/15. Step 4: D_x3 = det[[2, -1, 5], [4, 2, 8], [3, -4, 3]] = 2(38) + 1(-12) + 5(-22) = -46, so x_3 = (-46)/(-15) = 46/15. Solution: x_1 = 1/3, x_2 = -19/15, x_3 = 46/15. (iii) Step 1: D = det[[2, -1, 1], [1, 2, 2], [1, -2, -1]] = 2(2) + 1(-3) + 1(-4) = -3. Step 2: D_x1 = det[[8, -1, 1], [6, 2, 2], [1, -2, -1]] = 8(2) + 1(-8) + 1(-14) = -6, so x_1 = 2. Step 3: D_x2 = det[[2, 8, 1], [1, 6, 2], [1, 1, -1]] = 2(-8) - 8(-3) + 1(-5) = 3, so x_2 = -1. Step 4: D_x3 = det[[2, -1, 8], [1, 2, 6], [1, -2, 1]] = 2(14) + 1(-5) + 8(-4) = -9, so x_3 = 3. Solution: x_1 = 2, x_2 = -1, x_3 = 3. These agree with the book's Answers (p. 414).

2. Solve the following systems using matrices: (i) x - 2y + z = -1, 3x + y - 2z = 4, y - z = 1 (ii) 2x_1 + x_2 + 3x_3 = 3, x_1 + x_2 - 2x_3 = 0, -3x_1 - x_2 + 2x_3 = -4 (iii) x + y = 2, 2x - z = 1, 2y - 3z = -1

(i) Step 1: From the third equation z = y - 1. Step 2: Substituting in the first: x - 2y + y - 1 = -1, so x = y. Step 3: Substituting in the second: 3y + y - 2(y - 1) = 4, i.e. 2y + 2 = 4, so y = 1, x = 1, z = 0. Step 4: Check: 1 - 2 + 0 = -1, 3 + 1 - 0 = 4, 1 - 0 = 1. Solution: x = 1, y = 1, z = 0. (ii) Step 1: From the second equation x_2 = -x_1 + 2x_3. Step 2: Substituting in the first: 2x_1 - x_1 + 2x_3 + 3x_3 = 3, i.e. x_1 + 5x_3 = 3. Step 3: Substituting in the third: -3x_1 + x_1 - 2x_3 + 2x_3 = -4, i.e. -2x_1 = -4, so x_1 = 2. Step 4: Then 5x_3 = 1, so x_3 = 1/5, and x_2 = -2 + 2/5 = -8/5. Step 5: Check: 4 - 8/5 + 3/5 = 3, 2 - 8/5 - 2/5 = 0, -6 + 8/5 + 2/5 = -4. Solution: x_1 = 2, x_2 = -8/5, x_3 = 1/5. FLAG: the printed key's x_2 value (-2/5) does not satisfy any

of the three equations; recomputation gives -8/5, which satisfies all three. (iii) Step 1: From the first equation y = 2 - x; from the second z = 2x - 1. Step 2: Substituting in the third: 2(2 - x) - 3(2x - 1) = -1, i.e. 4 - 2x - 6x + 3 = -1, so 8x = 8, x = 1. Step 3: Then y = 1, z = 1. Solution: x = 1, y = 1, z = 1. These agree with the book's Answers (p. 414), except for the flagged value in (ii).

3. Solve the following systems by reducing their augmented matrices to the echelon form and the reduced echelon forms: (i) x_1 - 2x_2 - 2x_3 = -1, 2x_1 + 3x_2 + x_3 = 1, 5x_1 - 4x_2 - 3x_3 = 1 (ii) x + 2y + z = 2, 2x + y + 2z = -1, 2x + 3y - z = 9 (iii) x_1 + 4x_2 + 2x_3 = 2, 2x_1 + x_2 - 2x_3 = 9, 3x_1 + 2x_2 - 2x_3 = 12

(i) Step 1: Augmented matrix [[1, -2, -2, -1], [2, 3, 1, 1], [5, -4, -3, 1]]. Step 2: R2 - 2R1 = [0, 7, 5, 3]; R3 - 5R1 = [0, 6, 7, 6]. Step 3: 7R3 - 6R2 = [0, 0, 19, 24], so x_3 = 24/19. Step 4: Back-substitution: 7x_2 + 5(24/19) = 3 gives x_2 = -9/19; x_1 - 2(-9/19) - 2(24/19) = -1 gives x_1 = 11/19. Solution: x_1 = 11/19, x_2 = -9/19, x_3 = 24/19. FLAG: the printed key gives x_3 = -24/19, which fails all three equations (for example the first equation becomes 77/19, not -1); recomputation gives +24/19, which satisfies all three. (ii) Step 1: Augmented matrix [[1, 2, 1, 2], [2, 1, 2, -1], [2, 3, -1, 9]]. Step 2: R2 - 2R1 = [0, -3, 0, -5], so y = 5/3. Step 3: R3 - 2R1 = [0, -1, -3, 5], so -y - 3z = 5, giving z = -20/9. Step 4: x = 2 - 2y - z = 2 - 10/3 + 20/9 = 8/9. Solution: x = 8/9, y = 5/3, z = -20/9. (iii) Step 1: Augmented matrix [[1, 4, 2, 2], [2, 1, -2, 9], [3, 2, -2, 12]]. Step 2: R2 - 2R1 = [0, -7, -6, 5]; R3 - 3R1 = [0, -10, -8, 6]. Step 3: 7R3 - 10R2 = [0, 0, 4, -8], so x_3 = -2. Step 4: Back-substitution: -7x_2 - 6(-2) = 5 gives x_2 = 1; x_1 + 4(1) + 2(-2) = 2 gives x_1 = 2. Solution: x_1 = 2, x_2 = 1, x_3 = -2. These agree with the book's Answers (p. 414), except for the flagged value in (i).

4. Solve the following homogeneous systems: (i) x + 2y - 2z = 0, 2x + y + 5z = 0,

5x + 4y + 8z = 0 (ii) x_1 + 4x_2 + 2x_3 = 0, 2x_1 + x_2 - 3x_3 = 0, 3x_1 + 2x_2 - 4x_3 = 0 (iii) x_1 - 2x_2 - x_3 = 0, x_1 + x_2 + 5x_3 = 0, 2x_1 - x_2 + 4x_3 = 0

(i) Step 1: From the first equation x = -2y + 2z. Step 2: Substituting in the second: 2(-2y + 2z) + y + 5z = 0, i.e. -3y + 9z = 0, so y = 3z. Step 3: Then x = -6z + 2z = -4z. Step 4: Check in the third: 5(-4z) + 4(3z) + 8z = 0. Solution: x = -4t, y = 3t, z = t for arbitrary t. (ii) Step 1: From the first equation x_1 = -4x_2 - 2x_3. Step 2: Substituting in the second: 2(-4x_2 - 2x_3) + x_2 - 3x_3 = 0, i.e. -7x_2 - 7x_3 = 0, so x_2 = -x_3. Step 3: Then x_1 = 4x_3 - 2x_3 = 2x_3. Step 2: Check: 2x_3 + 4(-x_3) + 2x_3 = 0; 2(2x_3) + (-x_3) - 3x_3 = 0; 3(2x_3) + 2(-x_3) - 4x_3 = 0. Solution: x_1 = 2t, x_2 = -t, x_3 = t. (iii) Step 1: Subtracting the first equation from the second gives 3x_2 + 6x_3 = 0, so x_2 = -2x_3. Step 2: The first equation then gives x_1 = 2x_2 + x_3 = -3x_3. Step 3: Check in the third: 2(-3x_3) - (-2x_3) + 4x_3 = 0. Solution: x_1 = -3t, x_2 = -2t, x_3 = t (the key writes the same line as x_1 = t, x_2 = (2/3)t, x_3 = (-1/3)t, which is the same line with t rescaled). These agree with the book's Answers (p. 414).

5. Find the values of lam for which the following homogeneous systems have non-trivial solutions and solve the systems for these values of lam: (i) x + y + z = 0, 2x + y - lam z = 0, x + 2y - 2z = 0 (ii) x_1 + 4x_2 + lam x_3 = 0, 2x_1 + x_2 - 3x_3 = 0, 3x_1 + lam x_2 - 4x_3 = 0

(i) Step 1: The determinant of the coefficient matrix is det[[1, 1, 1], [2, 1, -lam], [1, 2, -2]] = 1(-2 + 2lam) - 1(-4 + lam) + 1(4 - 1) = lam + 5. Step 2: A non-trivial solution exists exactly when the determinant is 0, i.e. lam = -5. Step 3: With lam = -5 the system is x + y + z = 0, 2x + y + 5z = 0, x + 2y - 2z = 0. From the first and third: x = -4z, y = 3z (as in Exercise 3.5 Q4(i)). Solution for lam = -5: x = -4t, y = 3t, z = t (the key writes the same line as x = t, y = (-3/4)t, z = (-1/4)t). (ii) Step 1: The determinant is det[[1, 4, lam], [2, 1, -3], [3, lam, -4]] = 2(lam - 2)(lam + 2). Step 2:

A non-trivial solution exists exactly when lam = 2 or lam = -2. Step 3: For lam = 2 the system reduces to x_1 = 2t, x_2 = -t, x_3 = t. Step 4: For lam = -2 the system reduces to x_1 = 10t, x_2 = t, x_3 = 7t (the key writes the same line as x_1 = t, x_2 = (1/10)t, x_3 = (7/10)t). These agree with the book's Answers (p. 414).

6. Find the values of lam for which the following system does not possess a unique solution and solve the system for these values of lam: x_1 + 4x_2 + lam x_3 = 2, 2x_1 + x_2 - 2x_3 = 11, 3x_1 + 2x_2 - 2x_3 = 16

Step 1: The determinant of the coefficient matrix is det[[1, 4, lam], [2, 1, -2], [3, 2, -2]] = 1(-2 + 4) - 4(-4 + 6) + lam(4 - 3) = lam - 6. Step 2: The system fails to have a unique solution exactly when the determinant is 0, i.e. lam = 6. Step 3: With lam = 6 the equations are x_1 + 4x_2 + 6x_3 = 2, 2x_1 + x_2 - 2x_3 = 11, 3x_1 + 2x_2 - 2x_3 = 16. Step 4: Subtracting the second from the third gives x_1 + x_2 = 5. Step 5: From the second, with x_1 = 5 - x_2: 10 - x_2 - 2x_3 = 11, so x_2 + 2x_3 = -1. Step 6: Putting x_3 = t gives x_2 = -2t - 1 and x_1 = 2t + 6. Solution for lam = 6: x_1 = 2t + 6, x_2 = -2t - 1, x_3 = t. This agrees with the book's Answers (p. 414).

Chapter 4: Quadratic Equations

Exercise 4.1 (book pp. 142-143)

Solve the following equations by factorization: 1. 3x^2 + 4x + 1 = 0 2. x^2 + 7x + 12 = 0 3. 9x^2 - 12x - 5 = 0 4. x^2 - x = 2 5. x(x + 7) = (2x - 1)(x + 4) 6. x/(x + 1) + (x + 1)/x = 5/2; x != -1, 0 7. 1/(x + 1) + 2/(x + 2) = 7/(x + 5); x != -1, -2, -5 8. a/(ax - 1) + b/(bx - 1) = a + b; x != 1/a, 1/b

1. 3x^2 + 4x + 1 = (3x + 1)(x + 1) = 0, so x = -1 or x = -1/3. Solution set = {-1,

-1/3}. 2. x^2 + 7x + 12 = (x + 3)(x + 4) = 0, so x = -3 or x = -4. Solution set = {-3, -4}. 3. 9x^2 - 12x - 5 = (3x + 1)(3x - 5) = 0, so x = -1/3 or x = 5/3. Solution set = {-1/3, 5/3}. 4. x^2 - x - 2 = (x - 2)(x + 1) = 0, so x = 2 or x = -1. Solution set = {2, -1}. 5. x^2 + 7x = 2x^2 + 7x - 4, so x^2 = 4 and x = 2 or x = -2. Solution set = {2, -2}. 6. Multiplying by 2x(x + 1): 2x^2 + 2(x + 1)^2 = 5x(x + 1), i.e. 4x^2 + 4x + 2 = 5x^2 + 5x, i.e. x^2 + x - 2 = (x + 2)(x - 1) = 0, so x = 1 or x = -2 (both allowed). Solution set = {-2, 1}. 7. The left side is (3x + 4)/((x + 1)(x + 2)). Cross-multiplying: (3x + 4)(x + 5) = 7(x + 1)(x + 2), i.e. 3x^2 + 19x + 20 = 7x^2 + 21x + 14, i.e. 4x^2 + 2x - 6 = 0, i.e. 2x^2 + x - 3 = (2x + 3)(x - 1) = 0, so x = 1 or x = -3/2 (both allowed). Solution set = {-3/2, 1}. 8. Clearing denominators: a(bx - 1) + b(ax - 1) = (a + b)(ax - 1)(bx - 1), i.e. 2abx - (a + b) = (a + b)(abx^2 - (a + b)x + 1), i.e. (a + b)abx^2 - [(a + b)^2 + 2ab]x + 2(a + b) = 0. By the sum and product of roots, the roots are x = (a + b)/(ab) and x = 2/(a + b): their sum is [(a + b)^2 + 2ab]/((a + b)ab) and their product is 2/(ab), as required. Both are easily checked by substitution: for x = (a + b)/(ab), a/(ax - 1) = b and b/(bx - 1) = a. Solution set = {(a + b)/(ab), 2/(a + b)}.

Solve the following equations by completing the square: 9. x^2 - 2x - 899 = 0 10. x^2 + 4x - 1085 = 0 11. x^2 + 6x - 567 = 0 12. x^2 - 3x - 648 = 0 13. x^2 - x - 1806 = 0 14. 2x^2 + 12x - 110 = 0

9. (x - 1)^2 = 900, so x - 1 = 30 or x - 1 = -30, giving x = 31 or x = -29. Solution set = {-29, 31}. 10. (x + 2)^2 = 1089 = 33^2, so x + 2 = 33 or x + 2 = -33, giving x = 31 or x = -35. Solution set = {-35, 31}. 11. (x + 3)^2 = 576 = 24^2, so x + 3 = 24 or x + 3 = -24, giving x = 21 or x = -27. Solution set = {21, -27}. 12. (x - 3/2)^2 = 648 + 9/4 = 2601/4 = (51/2)^2, so x - 3/2 = 51/2 or x - 3/2 = -51/2, giving x = 27 or x = -24. Solution set = {-24, 27}. 13. (x - 1/2)^2 = 1806 + 1/4 = 7225/4 = (85/2)^2, so x - 1/2 = 85/2 or x - 1/2 = -85/2, giving x = 43 or x = -42. Solution set = {-42, 43}. 14. Dividing by 2: x^2 + 6x - 55 = 0, so (x + 3)^2 = 64, giving x + 3

= 8 or x + 3 = -8, i.e. x = 5 or x = -11. Solution set = {5, -11}.

Find roots of the following equations by using quadratic formula: 15. 5x^2 - 13x + 6 = 0 16. 4x^2 + 7x - 1 = 0 17. 15x^2 + 2ax - a^2 = 0 18. 16x^2 + 8x + 1 = 0 19. (x - a)(x - b) + (x - b)(x - c) + (x - c)(x - a) = 0 20. (a + b)x^2 + (a + 2b + c)x + b + c = 0

15. Discriminant = 169 - 120 = 49, so x = (13 +- 7)/10, giving x = 2 or x = 3/5. Solution set = {2, 3/5}. 16. Discriminant = 49 + 16 = 65, so x = (-7 +- sqrt(65))/8. Solution set = {(-7 +- sqrt(65))/8}. 17. 15x^2 + 2ax - a^2 = (5x - a)(3x + a) = 0, so x = a/5 or x = -a/3. Solution set = {-a/3, a/5}. 18. 16x^2 + 8x + 1 = (4x + 1)^2 = 0, so x = -1/4 (double root). Solution set = {-1/4}. 19. Expanding: 3x^2 - 2(a + b + c)x + (ab + bc + ca) = 0. By the quadratic formula, x = [2(a + b + c) +- sqrt(4(a + b + c)^2 - 12(ab + bc + ca))]/6 = [(a + b + c) +- sqrt(a^2 + b^2 + c^2 - ab - bc - ca)]/3. 20. x = -1 is a root, since (a + b) - (a + 2b + c) + (b + c) = 0. By the product of roots, (-1) x (other root) = (b + c)/(a + b), so the other root is -(b + c)/(a + b). Solution set = {-1, -(b + c)/(a + b)}.

Exercise 4.2 (book pp. 146-147)

Solve the following equations: 1. x^4 - 6x^2 + 8 = 0 2. x^(-2) - 10 = 3x^(-1) 3. x^6 - 9x^3 + 8 = 0 4. 8x^6 - 19x^3 - 27 = 0 5. x^(2/5) + 8 = 6x^(1/5) 6. (x + 1)(x + 2)(x + 3)(x + 4) = 24

1. Put y = x^2: y^2 - 6y + 8 = (y - 2)(y - 4) = 0, so y = 2 or y = 4, giving x = +- sqrt(2) or x = +- 2. Solution set = {+- 2, +- sqrt(2)}. 2. Put y = 1/x: y^2 - 3y - 10 = (y - 5)(y + 2) = 0, so y = 5 or y = -2, giving x = 1/5 or x = -1/2. Solution set = {-1/2, 1/5}. 3. Put y = x^3: y^2 - 9y + 8 = (y - 1)(y - 8) = 0. For y = 1: x^3 = 1, so x = 1, (-1 +- sqrt(-3))/2. For y = 8: x^3 = 8, so x = 2, -1 +- sqrt(-3). Solution set = {1, 2, (-1 +- sqrt(-3))/2, -1 +- sqrt(-3)}. 4. Put y = x^3: 8y^2 - 19y - 27 = 0, so y = (19 +- 35)/16, i.e. y = 27/8 or y = -1. For y = 27/8: x = 3/2, 3(-1 +- sqrt(3)i)/4. For y = -1: x

= -1, (1 +- sqrt(3)i)/2. Solution set = {-1, 3/2, (1 +- sqrt(3)i)/2, 3(-1 +- sqrt(3)i)/4}.

5. Put y = x^(1/5): y^2 - 6y + 8 = (y - 2)(y - 4) = 0, so y = 2 or y = 4, giving x = 32

or x = 1024. Solution set = {32, 1024}. 6. Grouping: (x^2 + 5x + 4)(x^2 + 5x + 6)

= 24. Put y = x^2 + 5x: (y + 4)(y + 6) = 24, i.e. y^2 + 10y = 0, so y = 0 or y =

-10. For y = 0: x(x + 5) = 0, so x = 0 or x = -5. For y = -10: x^2 + 5x + 10 = 0, so x

= (-5 +- sqrt(15)i)/2. Solution set = {-5, 0, (-5 +- sqrt(15)i)/2}.

Solve the following equations: 7. (x - 1)(x + 5)(x + 8)(x + 2) - 880 = 0 8. (x - 5)(x - 7)(x + 6)(x + 4) - 504 = 0 9. (x - 1)(x - 2)(x - 8)(x + 5) + 360 = 0 10. (x + 1)(2x + 3)(2x + 5)(x + 3) = 945 11. (2x - 7)(x^2 - 9)(2x + 5) - 91 = 0 12. (x^2 + 6x + 8)(x^2 + 14x + 48) = 105

7. Grouping: (x - 1)(x + 8) = x^2 + 7x - 8 and (x + 5)(x + 2) = x^2 + 7x + 10. Put y = x^2 + 7x: (y - 8)(y + 10) = 880, i.e. y^2 + 2y - 960 = 0, so y = 30 or y = -32. For y = 30: x^2 + 7x - 30 = (x + 10)(x - 3) = 0, so x = 3 or x = -10. For y = -32: x^2 + 7x + 32 = 0, so x = (-7 +- sqrt(79)i)/2. Solution set = {-10, 3, (-7 +- sqrt(79)i)/2}. 8. Grouping: (x - 5)(x + 4) = x^2 - x - 20 and (x - 7)(x + 6) = x^2 - x - 42. Put y = x^2 - x: (y - 20)(y - 42) = 504, i.e. y^2 - 62y + 336 = 0, so y = 56 or y = 6. For y = 56: x^2 - x - 56 = (x - 8)(x + 7) = 0, so x = 8 or x = -7. For y = 6: x^2 - x - 6 = (x - 3)(x + 2) = 0, so x = 3 or x = -2. Solution set = {-7, -2, 3, 8}. 9. Grouping: (x - 1)(x - 2) = x^2 - 3x + 2 and (x - 8)(x + 5) = x^2 - 3x - 40. Put y = x^2 - 3x: (y + 2)(y - 40) + 360 = 0, i.e. y^2 - 38y + 280 = 0, so y = 28 or y = 10. For y = 28: x^2 - 3x - 28 = (x - 7)(x + 4) = 0, so x = 7 or x = -4. For y = 10: x^2 - 3x - 10 = (x - 5)(x + 2) = 0, so x = 5 or x = -2. Solution set = {-2, 5, -4, 7}. 10. Grouping: (x + 1)(x + 3) = x^2 + 4x + 3 and (2x + 3)(2x + 5) = 4x^2 + 16x + 15. Put y = x^2 + 4x: (y + 3)(4y + 15) = 945, i.e. 4y^2 + 27y - 900 = 0, so y = 12 or y = -75/4. For y = 12: x^2 + 4x - 12 = (x + 6)(x - 2) = 0, so x = 2 or x = -6. For y = -75/4: x^2 + 4x + 75/4 = 0, so x = (-4 +- sqrt(59)i)/2. Solution set = {-6, 2, (-4 +- sqrt(59)i)/2}. 11. Expanding: (2x - 7)(2x + 5)(x^2 - 9) = 4x^4 - 4x^3 - 71x^2 + 36x + 315, so the equation is 4x^4 -

4x^3 - 71x^2 + 36x + 224 = 0. Trying x = 4 gives 0, and dividing by (x - 4) gives 4x^3 + 12x^2 - 23x - 56 = 0. Trying x = -7/2 gives 0, and dividing by (x + 7/2) gives 4x^2 - 2x - 16 = 0, i.e. 2x^2 - x - 8 = 0, so x = (1 +- sqrt(65))/4. Solution set = {-7/2, 4, (1 +- sqrt(65))/4}. 12. Note (x^2 + 6x + 8) = (x + 2)(x + 4) and (x^2 + 14x + 48) = (x + 6)(x + 8). Put t = (x + 5)^2: the product is (t - 9)(t - 1) = 105, i.e. t^2 - 10t - 96 = 0, so t = 16 or t = -6. For t = 16: (x + 5)^2 = 16, so x = -1 or x = -9. For t = -6: (x + 5)^2 = -6, so x = -5 +- sqrt(-6). Solution set = {-9, -1, -5 +- sqrt(-6)}.

Solve the following equations: 13. (x^2 + 6x - 27)(x^2 - 2x - 35) = 385 14. 4 x 2^(2x+1) - 9 x 2^x + 1 = 0 15. 2^x + 2^(-x+6) - 20 = 0 16. 4^x - 3 x 2^(x+3) + 128 = 0 17. 3^(2x-1) - 12 x 3^x + 81 = 0 18. (x + 1/x)^2 - 3(x + 1/x) - 4 = 0

13. Let y = x^2 + 2x - 31, the midpoint of the two quadratics. Then x^2 + 6x - 27 = y + 4(x + 1) and x^2 - 2x - 35 = y - 4(x + 1), so the product is y^2 - 16(x + 1)^2 = 385. Since y = (x + 1)^2 - 32, put t = (x + 1)^2: (t - 32)^2 - 16t = 385, i.e. t^2 - 80t + 639 = 0, so t = 71 or t = 9. For t = 9: (x + 1)^2 = 9, so x = 2 or x = -4. For t = 71: (x + 1)^2 = 71, so x = -1 +- sqrt(71). Solution set = {-4, 2, -1 +- sqrt(71)}. 14. Put y = 2^x: 4 x 2^(2x+1) = 8y^2, so 8y^2 - 9y + 1 = (8y - 1)(y - 1) = 0, so y = 1 or y = 1/8, giving x = 0 or x = -3. Solution set = {-3, 0}. 15. Put y = 2^x: y + 64/y - 20 = 0, i.e. y^2 - 20y + 64 = (y - 4)(y - 16) = 0, so y = 4 or y = 16, giving x = 2 or x = 4. Solution set = {2, 4}. 16. Put y = 2^x: y^2 - 24y + 128 = (y - 8)(y - 16) = 0, so y = 8 or y = 16, giving x = 3 or x = 4. Solution set = {3, 4}. 17. Put y = 3^x: (1/3)y^2 - 12y + 81 = 0, i.e. y^2 - 36y + 243 = (y - 9)(y - 27) = 0, so y = 9 or y = 27, giving x = 2 or x = 3. Solution set = {2, 3}. 18. Put t = x + 1/x: t^2 - 3t - 4 = (t - 4)(t + 1) = 0. For t = 4: x^2 - 4x + 1 = 0, so x = 2 +- sqrt(3). For t = -1: x^2 + x + 1 = 0, so x = (-1 +- sqrt(-3))/2. Solution set = {(-1 +- sqrt(-3))/2, 2 +- sqrt(3)}.

Solve the following equations: 19. x^2 + x - 4 + 1/x + 1/x^2 = 0 20. (x - 1/x)^2 +

3(x + 1/x) = 0 21. 2x^4 - 3x^3 - x^2 - 3x + 2 = 0 22. 2x^4 + 3x^3 - 4x^2 - 3x + 2 = 0 23. 6x^4 - 35x^3 + 62x^2 - 35x + 6 = 0 24. x^4 - 6x^2 + 10 - 6/x^2 + 1/x^4 = 0

19. Grouping: (x^2 + 1/x^2) + (x + 1/x) - 4 = 0. Put t = x + 1/x: x^2 + 1/x^2 = t^2 - 2, so t^2 + t - 6 = (t - 2)(t + 3) = 0. For t = 2: x = 1. For t = -3: x^2 + 3x + 1 = 0, so x = (-3 +- sqrt(5))/2. Solution set = {1, (-3 +- sqrt(5))/2}. 20. Expanding: (x^2 + 1/x^2) + 3(x + 1/x) - 2 = 0. Put t = x + 1/x: t^2 + 3t - 4 = (t - 1)(t + 4) = 0. For t = 1: x^2 - x + 1 = 0, so x = (1 +- sqrt(3)i)/2. For t = -4: x^2 + 4x + 1 = 0, so x = -2 +- sqrt(3). Solution set = {(1 +- sqrt(3)i)/2, -2 +- sqrt(3)}. FLAG: the printed key writes the first pair as (-1 +- sqrt(3)i)/2, but those values do not satisfy the equation (they give -6, not 0); the correct pair is (1 +- sqrt(3)i)/2, verified by substitution. 21. Dividing by x^2 (x != 0): 2(x^2 + 1/x^2) - 3(x + 1/x) - 1 = 0. Put t = x + 1/x: 2(t^2 - 2) - 3t - 1 = 0, i.e. 2t^2 - 3t - 5 = (2t - 5)(t + 1) = 0. For t = 5/2: 2x^2 - 5x + 2 = (2x - 1)(x - 2) = 0, so x = 2 or x = 1/2. For t = -1: x^2 + x + 1 = 0, so x = (-1 +- sqrt(-3))/2. Solution set = {1/2, 2, (-1 +- sqrt(-3))/2}. 22. Dividing by x^2: 2(x^2 + 1/x^2) + 3(x - 1/x) - 4 = 0. Put u = x - 1/x: x^2 + 1/x^2 = u^2 + 2, so 2u^2 + 3u = u(2u + 3) = 0. For u = 0: x = +- 1. For u = -3/2: 2x^2 + 3x - 2 = (2x - 1)(x + 2) = 0, so x = 1/2 or x = -2. Solution set = {+- 1, 1/2, -2}. 23. Dividing by x^2: 6(x^2 + 1/x^2) - 35(x + 1/x) + 62 = 0. Put t = x + 1/x: 6(t^2 - 2) - 35t + 62 = 0, i.e. 6t^2 - 35t + 50 = (3t - 10)(2t - 5) = 0. For t = 10/3: 3x^2 - 10x + 3 = (3x - 1)(x - 3) = 0, so x = 3 or x = 1/3. For t = 5/2: 2x^2 - 5x + 2 = 0, so x = 2 or x = 1/2. Solution set = {2, 1/2, 3, 1/3}. 24. Put t = x^2 + 1/x^2: x^4 + 1/x^4 = t^2 - 2, so t^2 - 6t + 8 = (t - 2)(t - 4) = 0. For t = 2: (x - 1/x)^2 = 0, so x = 1 or x = -1 (each double). For t = 4: x^4 - 4x^2 + 1 = 0, so x^2 = 2 +- sqrt(3), giving x = +- sqrt(2 + sqrt(3)) and x = +- sqrt(2 - sqrt(3)), i.e. +- (sqrt(6) + sqrt(2))/2 and +- (sqrt(6) - sqrt(2))/2. FLAG: the printed key's last part, written as +- sqrt(2) +- sqrt(3), is garbled; the correct roots are +- sqrt(2 +- sqrt(3)) as shown.

Exercise 4.3 (book pp. 150-151)

Solve the following equations: 1. 3x^2 + 2x - sqrt(3x^2 + 2x - 1) = 3 2. x^2 - x/2 - 7 = x - 3 sqrt(2x^2 - 3x + 2) 3. sqrt(2x + 8) + sqrt(x + 5) = 7 4. sqrt(3x + 4) = 2 + sqrt(2x - 4) 5. sqrt(x + 7) + sqrt(x + 2) = sqrt(6x + 13) 6. sqrt(x^2 + x + 1) - sqrt(x^2 + x - 1) = 1

1. Put y = sqrt(3x^2 + 2x - 1): then 3x^2 + 2x = y^2 + 1, so y^2 + 1 - y = 3, i.e. y^2 - y - 2 = (y - 2)(y + 1) = 0. Since y >= 0, y = 2, so 3x^2 + 2x - 1 = 4, i.e. 3x^2 + 2x - 5 = (3x + 5)(x - 1) = 0, giving x = 1 or x = -5/3. Solution set = {1, -5/3}. 2. Rearranging: x^2 - (3/2)x - 7 = -3 sqrt(2x^2 - 3x + 2). Checking x = 2: left side = 4 - 3 - 7 = -6; right side = -3 sqrt(8 - 6 + 2) = -6. Checking x = -1/2: left side = 1/4 + 3/4 - 7 = -6; right side = -3 sqrt(1/2 + 3/2 + 2) = -6. Both satisfy the equation. Solution set = {2, -1/2}. 3. Squaring twice (or checking): x = 4 gives sqrt(16) + sqrt(9) = 4 + 3 = 7. The equation has the single solution x = 4. Solution set = {4}. 4. Squaring: 3x + 4 = 4 + (2x - 4) + 4 sqrt(2x - 4), i.e. x + 4 = 4 sqrt(2x - 4), i.e. (x + 4)^2 = 16(2x - 4), i.e. x^2 - 24x + 80 = (x - 4)(x - 20) = 0. Both x = 4 and x = 20 satisfy the original equation. Solution set = {4, 20}. 5. Squaring: (x + 7) + (x + 2) + 2 sqrt((x + 7)(x + 2)) = 6x + 13, i.e. 2 sqrt(x^2 + 9x + 14) = 4x + 4, i.e. sqrt(x^2 + 9x + 14) = 2x + 2. Squaring again: x^2 + 9x + 14 = 4x^2 + 8x + 4, i.e. 3x^2 - x - 10 = (3x + 5)(x - 2) = 0. Checking: x = 2 gives 3 + 2 = 5 = sqrt(25); x = -5/3 makes 2x + 2 negative, so it is extraneous. Solution set = {2}. 6. Put t = x^2 + x: sqrt(t + 1) - sqrt(t - 1) = 1. Squaring: 2t - 2 sqrt(t^2 - 1) = 1, i.e. 2 sqrt(t^2 - 1) = 2t - 1. Squaring again: 4(t^2 - 1) = 4t^2 - 4t + 1, so t = 5/4. Then x^2 + x - 5/4 = 0, giving x = (-1 +- sqrt(6))/2. Solution set = {(-1 +- sqrt(6))/2}.

Solve the following equations: 7. sqrt(x^2 + 2x - 3) + sqrt(x^2 + 7x - 8) = sqrt(5(x^2 + 3x - 4)) 8. sqrt(2x^2 - 5x - 3) + 3 sqrt(2x + 1) = sqrt(2x^2 + 25x + 12) 9. sqrt(3x^2 - 5x + 2) + sqrt(6x^2 - 11x + 5) = sqrt(5x^2 - 9x + 4) 10. (x + 4)(x + 1) = sqrt(x^2 + 2x - 15) + 3x + 31 11. sqrt(3x^2 - 2x + 9) + sqrt(3x^2 - 2x

- 4) = 13 12. sqrt(5x^2 + 7x + 2) - sqrt(4x^2 + 7x + 18) = x - 4

7. Let a = sqrt(x^2 + 2x - 3) and b = sqrt(x^2 + 7x - 8). Squaring a + b = sqrt(5(x^2 + 3x - 4)) gives a^2 + b^2 + 2ab = 5x^2 + 15x - 20, i.e. 2ab = 3(x^2 + 2x - 3) = 3a^2. So either a = 0, giving x^2 + 2x - 3 = 0, i.e. x = 1 or x = -3; or 2b = 3a, giving 4(x^2 + 7x - 8) = 9(x^2 + 2x - 3), i.e. (x - 1)^2 = 0, i.e. x = 1. The formal solution set is {1, -3}, as in the printed key. FLAG: under real radicals, x = -3 makes the second and third radicands negative (x^2 + 7x - 8 = -20 and 5(x^2 + 3x - 4) = -20), so only x = 1 is a genuine real solution; the key retains -3 from the formal squaring. 8. Note 2x^2 - 5x - 3 = (2x + 1)(x - 3) and 2x^2 + 25x + 12 = (2x + 1)(x + 12). Put t = sqrt(2x + 1): dividing by t (t = 0 gives x = -1/2, which satisfies the equation) gives sqrt(x - 3) + 3 = sqrt(x + 12). Squaring: x - 3 + 6 sqrt(x - 3) + 9 = x + 12, i.e. 6 sqrt(x - 3) = 6, so x - 3 = 1, i.e. x = 4. Solution set = {4, -1/2}. 9. Squaring: with A = 3x^2 - 5x + 2 and B = 6x^2 - 11x + 5, we get A + B + 2 sqrt(AB) = 5x^2 - 9x + 4, i.e. 2 sqrt(AB) = -4x^2 + 7x - 3 = -(4x - 3)(x - 1). On the domain of the equation (x <= 1/2 or x >= 5/3, apart from x = 1) the right side is negative while the left side is non-negative, so no solution arises there. For x = 1 all three radicands are 0, so x = 1 satisfies the equation, and it is the only solution. Solution set = {1}. 10. Rearranging: x^2 + 2x - 27 = sqrt(x^2 + 2x - 15). Put t = x^2 + 2x: t - 27 = sqrt(t - 15), which requires t >= 27. Squaring: t^2 - 54t + 729 = t - 15, i.e. t^2 - 55t + 744 = 0, so t = 31 or t = 24. The value t = 24 is rejected (t < 27). For t = 31: x^2 + 2x - 31 = 0, so x = -1 +- sqrt(32) = -1 +- 4 sqrt(2). Solution set = {-1 +- 4 sqrt(2)}. 11. Put t = 3x^2 - 2x: sqrt(t + 9) + sqrt(t - 4) = 13. Squaring: 2t + 5 + 2 sqrt((t + 9)(t - 4)) = 169, i.e. sqrt(t^2 + 5t - 36) = 82 - t. Squaring again: t^2 + 5t - 36 = t^2 - 164t + 6724, so 169t = 6760, i.e. t = 40. Then 3x^2 - 2x - 40 = 0, so x = (2 +- sqrt(4 + 480))/6 = (2 +- 22)/6, i.e. x = 4 or x = -10/3, and both give t = 40. Solution set = {4, -10/3}. 12. Let a = sqrt(5x^2 + 7x + 2) and b = sqrt(4x^2 + 7x + 18). Then a^2 - b^2 = x^2 - 16 = (x - 4)(x + 4),

and a - b = x - 4. If x != 4, dividing gives a + b = x + 4, so a = x and b = 4. From b = 4: 4x^2 + 7x + 18 = 16, i.e. 4x^2 + 7x + 2 = 0, so x = (-7 +- sqrt(17))/8. But these values are negative, while a = x requires x >= 0, so they are extraneous. The remaining possibility is x = 4, which indeed satisfies the equation (both radicals equal sqrt(110)). Solution set = {4}. FLAG: the printed key lists only {(-7 +- sqrt(17))/8}, which are the extraneous values from the a + b branch, and omits the genuine solution x = 4.

Exercise 4.4 (book p. 155)

1. Find the three cube roots of: 8, -8, 27, -27, 64.

The cube roots of a number a are a^(1/3), a^(1/3) x w and a^(1/3) x w^2, where w = (-1 + sqrt(3)i)/2 is a complex cube root of unity. Hence: cube roots of 8: {2, 2w, 2w^2}; of -8: {-2, -2w, -2w^2}; of 27: {3, 3w, 3w^2}; of -27: {-3, -3w, -3w^2}; of 64: {4, 4w, 4w^2}.

2. Evaluate: i) (1 + w - w^2)^8 ii) w^28 + w^29 + 1 iii) (1 + w - w^2)(1 - w + w^2) iv) ((-1 + sqrt(-3))/2)^9 + ((-1 - sqrt(-3))/2)^7 v) (-1 + sqrt(-3))^5 + (-1 - sqrt(-3))^5

Here w = (-1 + sqrt(3)i)/2, so 1 + w + w^2 = 0 and w^3 = 1. i) 1 + w - w^2 = -w^2 - w^2 = -2w^2, since 1 + w = -w^2. So (1 + w - w^2)^8 = (-2w^2)^8 = 256 w^16 = 256w, since 16 = 3 x 5 + 1. ii) w^28 + w^29 + 1 = w + w^2 + 1 = 0, since 28 = 3 x 9 + 1 and 29 = 3 x 9 + 2. iii) 1 + w - w^2 = -2w^2 and 1 - w + w^2 = -2w (since w^2 = -1 - w), so the product is 4w^3 = 4. iv) (-1 + sqrt(-3))/2 = w and (-1 - sqrt(-3))/2 = w^2, so the sum is w^9 + (w^2)^7 = 1 + w^14 = 1 + w^2 = -w. FLAG: the printed key gives -1 for this part, but recomputation gives -w (i.e. (1 - sqrt(3)i)/2); 1 + w^2 = -w, not -1. v) -1 + sqrt(-3) = 2w and -1 - sqrt(-3) = 2w^2, so the sum is (2w)^5 + (2w^2)^5 = 32w^2 + 32w = 32(w + w^2) = -32.

3. Show that: i) x^3 - y^3 = (x - y)(x - wy)(x - w^2 y) ii) x^3 + y^3 + z^3 - 3xyz =

(x + y + z)(x + wy + w^2 z)(x + w^2 y + wz) iii) (1 + w)(1 + w^2)(1 + w^4)(1 + w^8) ... 2n factors = 1

i) The roots of x^3 - y^3 = 0 in x are x = y, x = wy and x = w^2 y, so x^3 - y^3 = (x - y)(x - wy)(x - w^2 y). ii) Expanding the right side (or using the identity with the cube roots of unity) gives x^3 + y^3 + z^3 - 3xyz: the factors vanish exactly when x^3 + y^3 + z^3 = 3xyz, and comparing the coefficient of x^3 (which is 1 on both sides) gives the identity. iii) 1 + w = -w^2, 1 + w^2 = -w, 1 + w^4 = 1 + w = -w^2 (since w^3 = 1), 1 + w^8 = 1 + w^2 = -w, and so on: the factors alternate -w^2, -w, -w^2, -w, ... Each pair multiplies to w^3 = 1, and 2n factors make n pairs, so the product is 1.

4. If w is a root of x^2 + x + 1 = 0, show that its other root is w^2 and prove that w^3 = 1.

Step 1: The sum of the roots is -1 and the product is 1. Step 2: If one root is w, the other is -1 - w, and since w^2 + w + 1 = 0, we have -1 - w = w^2. Step 3: The product of the roots is w x w^2 = w^3, which must equal 1. Alternatively, multiplying w^2 + w + 1 = 0 by (w - 1) gives w^3 - 1 = 0, so w^3 = 1 (and w != 1, since 1 is not a root of x^2 + x + 1 = 0).

5. Prove that complex cube roots of -1 are (1 + sqrt(3)i)/2 and (1 - sqrt(3)i)/2; and hence prove that ((1 + sqrt(-3))/2)^9 + ((1 - sqrt(-3))/2)^9 = -2.

Step 1: x^3 = -1 gives (x + 1)(x^2 - x + 1) = 0, so x = -1 or x = (1 +- sqrt(-3))/2 = (1 +- sqrt(3)i)/2. Step 2: (1 + sqrt(-3))/2 = -w^2 and (1 - sqrt(-3))/2 = -w, where w = (-1 + sqrt(3)i)/2. Step 3: (-w^2)^9 = -w^18 = -1 and (-w)^9 = -w^9 = -1, so the sum is -2.

6. If w is a cube root of unity, form an equation whose roots are 2w and 2w^2.

Step 1: The sum of the roots is 2w + 2w^2 = 2(w + w^2) = -2. Step 2: The product of the roots is 4w^3 = 4. Step 3: Hence the equation is x^2 - (sum)x + (product) = x^2 + 2x + 4 = 0.

7. Find four fourth roots of 16, 81, 625.

The fourth roots of a^4 are a, -a, ai and -ai. Hence: fourth roots of 16: {2, -2, 2i, -2i}; of 81: {3, -3, 3i, -3i}; of 625: {5, -5, 5i, -5i}.

8. Solve the following equations: i) 2x^4 - 32 = 0 ii) 3y^5 - 243y = 0 iii) x^3 + x^2 + x + 1 = 0 iv) 5x^5 - 5x = 0

i) x^4 = 16, so x = 2, -2, 2i, -2i. Solution set = {+- 2, +- 2i}. ii) y(y^4 - 81) = 0, so y = 0 or y^4 = 81, giving y = 0, 3, -3, 3i, -3i. Solution set = {0, +- 3, +- 3i}. iii) x^2(x + 1) + (x + 1) = (x + 1)(x^2 + 1) = 0, so x = -1, i, -i. Solution set = {-1, +- i}. iv) x(x^4 - 1) = 0, so x = 0 or x^4 = 1, giving x = 0, 1, -1, i, -i. Solution set = {0, +- 1, +- i}.

Exercise 4.5 (book pp. 160-161)

Use the remainder theorem to find the remainder when the first polynomial is divided by the second polynomial: 1. x^2 + 3x + 7, x + 1 2. x^3 - x^2 + 5x + 4, x - 2 3. 3x^4 + 4x^3 + x - 5, x + 1 4. x^3 - 2x^2 + 3x + 3, x - 3

1. Remainder = f(-1) = 1 - 3 + 7 = 5. 2. Remainder = f(2) = 8 - 4 + 10 + 4 = 18. 3. Remainder = f(-1) = 3 - 4 - 1 - 5 = -7. 4. Remainder = f(3) = 27 - 18 + 9 + 3 = 21.

Use the factor theorem to determine if the first polynomial is a factor of the second polynomial: 5. x - 1, x^2 + 4x - 5 6. x - 2, x^3 + x^2 - 7x + 1 7. w + 2, 2w^3 + w^2 - 4w + 7 8. x - a, x^n - a^n where n is a positive integer 9. x + a, x^n + a^n, where n is an odd integer

5. f(1) = 1 + 4 - 5 = 0, so x - 1 is a factor. Answer: Yes. 6. f(2) = 8 + 4 - 14 + 1 = -1, which is not 0, so x - 2 is not a factor. Answer: No. 7. Substituting w = -2: 2(-8) + 4 - 4(-2) + 7 = -16 + 4 + 8 + 7 = 3, which is not 0, so w + 2 is not a factor. Answer: No. 8. f(a) = a^n - a^n = 0, so x - a is a factor. Answer: Yes. 9. f(-a) = (-a)^n + a^n = 0 when n is odd, so x + a is a factor. Answer: Yes.

10. When x^4 + 2x^3 + kx^2 + 3 is divided by x - 2, the remainder is 1. Find the

value of k. 11. When the polynomial x^3 + 2x^2 + kx + 4 is divided by x - 2, the remainder is 14. Find the value of k.

10. By the remainder theorem, f(2) = 16 + 16 + 4k + 3 = 35 + 4k = 1, so 4k = -34 and k = -17/2. 11. By the remainder theorem, f(2) = 8 + 8 + 2k + 4 = 20 + 2k = 14, so 2k = -6 and k = -3.

Use synthetic division to show that x is the solution of the polynomial and use the result to factorize the polynomial completely: 12. x^3 - 7x + 6 = 0, x = 2 13. x^3 - 28x - 48 = 0, x = -4 14. 2x^4 + 7x^3 - 4x^2 - 27x - 18, x = 2, x = -3

12. Synthetic division by (x - 2): coefficients 1, 0, -7, 6 give the depressed equation x^2 + 2x - 3 = (x + 3)(x - 1) = 0. So x^3 - 7x + 6 = (x - 2)(x - 1)(x + 3), and the roots are 2, 1, -3. 13. Synthetic division by (x + 4): coefficients 1, 0, -28, -48 give the depressed equation x^2 - 4x - 12 = (x - 6)(x + 2) = 0. So x^3 - 28x - 48 = (x + 4)(x - 6)(x + 2), and the roots are -4, 6, -2. 14. Synthetic division by (x - 2): coefficients 2, 7, -4, -27, -18 give 2, 11, 18, 9, remainder 0. Dividing again by (x + 3): coefficients 2, 11, 18, 9 give 2, 5, 3, remainder 0. The depressed equation is 2x^2 + 5x + 3 = (2x + 3)(x + 1) = 0. So the polynomial is (x - 2)(x + 3)(2x + 3)(x + 1), and the roots are 2, -3, -3/2, -1.

15. Use synthetic division to find the values of p and q if x + 1 and x - 2 are the factors of the polynomial x^3 + px^2 + qx + 6. 16. Find the values of a and b if -2 and 2 are the roots of the polynomial x^3 - 4x^2 + ax + b.

15. f(-1) = -1 + p - q + 6 = 0, so p - q = -5. f(2) = 8 + 4p + 2q + 6 = 0, so 2p + q = -7. Adding: 3p = -12, so p = -4, and then q = 1. 16. f(2) = 8 - 16 + 2a + b = 0, so 2a + b = 8. f(-2) = -8 - 16 - 2a + b = 0, so -2a + b = 24. Adding: 2b = 32, so b = 16, and then a = -4.

Exercise 4.6 (book p. 164)

1. If u, v are the roots of 3x^2 - 2x + 4 = 0, find the values of: i) 1/u^2 + 1/v^2 ii)

u/v + v/u iii) u^4 + v^4 iv) u^3 + v^3 v) 1/u^3 + 1/v^3 vi) u^2 - v^2

Here u + v = 2/3 and uv = 4/3. Also u^2 + v^2 = (u + v)^2 - 2uv = 4/9 - 8/3 = -20/9. i) 1/u^2 + 1/v^2 = (u^2 + v^2)/(uv)^2 = (-20/9)/(16/9) = -5/4. ii) u/v + v/u = (u^2 + v^2)/uv = (-20/9)/(4/3) = -5/3. iii) u^4 + v^4 = (u^2 + v^2)^2 - 2(uv)^2 = 400/81 - 32/9 = 112/81. iv) u^3 + v^3 = (u + v)^3 - 3uv(u + v) = 8/27 - 3(4/3)(2/3) = 8/27 - 8/3 = -64/27. v) 1/u^3 + 1/v^3 = (u^3 + v^3)/(uv)^3 = (-64/27)/(64/27) = -1. vi) u^2 - v^2 = (u - v)(u + v); (u - v)^2 = (u + v)^2 - 4uv = 4/9 - 16/3 = -44/9, so u - v = 2 sqrt(-11)/3 (taking u as the root (1 + sqrt(-11))/3), giving u^2 - v^2 = (2/3)(2 sqrt(-11)/3) = 4 sqrt(-11)/9.

2. If u, v are the roots of x^2 - px - p - c = 0, prove that (1 + u)(1 + v) = 1 - c.

Step 1: u + v = p and uv = -p - c. Step 2: (1 + u)(1 + v) = 1 + (u + v) + uv = 1 + p - p - c = 1 - c, as required.

3. Find the condition that one root of x^2 + px + q = 0 is: i) double the other ii) square of the other iii) additive inverse of the other iv) multiplicative inverse of the other

Let the roots be u and v, so u + v = -p and uv = q. i) If v = 2u, then 3u = -p and 2u^2 = q, so u = -p/3 and 2p^2/9 = q, i.e. 2p^2 = 9q. ii) If v = u^2, then u + u^2 = -p and u^3 = q. Using (u + v)^3 = u^3 + v^3 + 3uv(u + v): (-p)^3 = q + q^2 + 3q(-p), i.e. -p^3 = q + q^2 - 3pq, i.e. p^3 + q + q^2 = 3pq. iii) If v = -u, then u + v = 0, so p = 0. iv) If v = 1/u, then uv = 1, so q = 1.

4. If the roots of the equation x^2 - px + q = 0 differ by unity, prove that p^2 = 4q + 1.

Step 1: Let the roots be u and v, so u + v = p and uv = q. Step 2: (u - v)^2 = (u + v)^2 - 4uv = p^2 - 4q. Step 3: Since the roots differ by unity, (u - v)^2 = 1, so p^2 - 4q = 1, i.e. p^2 = 4q + 1, as required.

5. Find the condition that a/(x - a) + b/(x - b) = 5 may have roots equal in magnitude but opposite in signs.

Step 1: Clearing denominators: a(x - b) + b(x - a) = 5(x - a)(x - b), i.e. (a + b)x - 2ab = 5x^2 - 5(a + b)x + 5ab, i.e. 5x^2 - 6(a + b)x + 7ab = 0. Step 2: Roots equal in magnitude but opposite in sign means their sum is 0. Step 3: The sum of the roots is 6(a + b)/5, so the condition is a + b = 0.

6. If the roots of px^2 + qx + q = 0 are u and v then prove that sqrt(u/v) + sqrt(v/u) + sqrt(q/p) = 0.

Step 1: u + v = -q/p and uv = q/p. Step 2: sqrt(u/v) + sqrt(v/u) = (u + v)/sqrt(uv) = (-q/p)/sqrt(q/p) = -sqrt(q/p) (taking the consistent branch of the square root). Step 3: Hence sqrt(u/v) + sqrt(v/u) + sqrt(q/p) = 0, as required.

7. If u, v are the roots of the equation ax^2 + bx + c = 0, form the equations whose roots are: i) u^2, v^2 ii) 1/u, 1/v iii) 1/u^2, 1/v^2 iv) u^3, v^3 v) 1/u^3, 1/v^3 vi) u + 1/u, v + 1/v vii) (u - v)^2, (u + v)^2 viii) -1/u^3, -1/v^3

Here u + v = -b/a and uv = c/a. i) Sum = u^2 + v^2 = (u + v)^2 - 2uv = (b^2 - 2ac)/a^2; product = (uv)^2 = c^2/a^2. Equation: a^2 y^2 - (b^2 - 2ac)y + c^2 = 0. ii) Sum = (u + v)/uv = -b/c; product = 1/uv = a/c. Equation: cy^2 + by + a = 0. iii) Sum = (u^2 + v^2)/(uv)^2 = (b^2 - 2ac)/c^2; product = a^2/c^2. Equation: c^2 y^2 - (b^2 - 2ac)y + a^2 = 0. iv) Sum = u^3 + v^3 = (u + v)^3 - 3uv(u + v) = (3abc - b^3)/a^3; product = c^3/a^3. Equation: a^3 y^2 - (3abc - b^3)y + c^3 = 0. v) Sum = (u^3 + v^3)/(uv)^3 = (3abc - b^3)/c^3; product = a^3/c^3. Equation: c^3 y^2 - (3abc - b^3)y + a^3 = 0. vi) Sum = (u + v) + (1/u + 1/v) = -b/a - b/c = -b(a + c)/(ac); product = (u + 1/u)(v + 1/v) = uv + (u^2 + v^2)/uv + 1/(uv) = (a^2 + b^2 + c^2 - 2ac)/(ac). Equation: acy^2 + b(a + c)y + (a^2 + b^2 + c^2 - 2ac) = 0. vii) (u + v)^2 = b^2/a^2 and (u - v)^2 = (b^2 - 4ac)/a^2. Sum = 2(b^2 - 2ac)/a^2; product = b^2(b^2 - 4ac)/a^4. Equation: a^4 y^2 - 2a^2(b^2 - 2ac)y + b^2(b^2 - 4ac) = 0. viii) Sum = -(3abc - b^3)/c^3; product = a^3/c^3. Equation: c^3 y^2 - (b^3 - 3abc)y + a^3 = 0.

8. If u, v are the roots of 5x^2 - x - 2 = 0, form the equation whose roots are 3/u

and 3/v. 9. If u and v are the roots of x^2 - 3x + 5 = 0, form the equation whose roots are (1 - u)/(1 + u) and (1 - v)/(1 + v).

8. Here u + v = 1/5 and uv = -2/5. Sum of the new roots = 3(u + v)/uv = 3(1/5)/(-2/5) = -3/2. Product = 9/uv = -45/2. Equation: y^2 + (3/2)y - 45/2 = 0, i.e. 2y^2 + 3y - 45 = 0. 9. Here u + v = 3 and uv = 5. Sum of the new roots = [(1 - u)(1 + v) + (1 - v)(1 + u)]/((1 + u)(1 + v)) = (2 - 2uv)/(1 + (u + v) + uv) = (2 - 10)/(1 + 3 + 5) = -8/9. Product = (1 - (u + v) + uv)/((1 + u)(1 + v)) = (1 - 3 + 5)/9 = 1/3. Equation: y^2 + (8/9)y + 1/3 = 0, i.e. 9y^2 + 8y + 3 = 0.

Exercise 4.7 (book pp. 167-168)

1. Discuss the nature of the roots of the following equations: i) 4x^2 + 6x + 1 = 0 ii) x^2 - 5x + 6 = 0 iii) 2x^2 - 5x + 1 = 0 iv) 25x^2 - 30x + 9 = 0

i) Discriminant = 36 - 16 = 20 > 0, and 20 is not a perfect square, so the roots are irrational and unequal. ii) Discriminant = 25 - 24 = 1 > 0, a perfect square, so the roots are rational and unequal. iii) Discriminant = 25 - 8 = 17 > 0, not a perfect square, so the roots are irrational and unequal. iv) Discriminant = 900 - 900 = 0, so the roots are real and equal.

2. Show that the roots of the following equations will be real: i) x^2 - 2(m + 1/m)x + 3 = 0; m != 0 ii) (b - c)x^2 + (c - a)x + (a - b) = 0; a, b, c in Q

i) Discriminant = 4(m + 1/m)^2 - 12 = 4[(m + 1/m)^2 - 3]. For real m != 0, |m + 1/m| >= 2, so (m + 1/m)^2 >= 4 and the discriminant is at least 4 > 0. Hence the roots are real (and unequal). ii) The sum of the coefficients is (b - c) + (c - a) + (a - b) = 0, so x = 1 is a root. The other root is (a - b)/(b - c) (by the product of roots), which is real since a, b, c are rational. Hence both roots are real.

3. Show that the roots of the following equations will be rational: i) (p + q)x^2 - px - q = 0 ii) px^2 - (p - q)x - q = 0

i) The sum of the coefficients is (p + q) - p - q = 0, so x = 1 is a root. The other root

is -q/(p + q) (by the product of roots), which is rational when p and q are rational.

Hence both roots are rational. ii) The sum of the coefficients is p - (p - q) - q = 0, so

x = 1 is a root. The other root is -q/p (by the product of roots), which is rational

when p and q are rational. Hence both roots are rational.

4. For what values of m will the roots of the following equations be equal? i) (m + 1)x^2 + 2(m + 3)x + m + 8 = 0 ii) x^2 - 2(1 + 3m)x + 7(3 + 2m) = 0 iii) (1 + m)x^2 - 2(1 + 3m)x + (1 + 8m) = 0

i) Discriminant/4 = (m + 3)^2 - (m + 1)(m + 8) = 1 - 3m. Setting it to 0 gives m = 1/3. ii) Discriminant/4 = (1 + 3m)^2 - 7(3 + 2m) = 9m^2 - 8m - 20 = (9m + 10)(m - 2). Setting it to 0 gives m = 2 or m = -10/9. iii) Discriminant/4 = (1 + 3m)^2 - (1 + m)(1 + 8m) = m^2 - 3m = m(m - 3). Setting it to 0 gives m = 0 or m = 3.

5. Show that the roots of x^2 + (mx + c)^2 = a^2 will be equal, if c^2 = a^2(1 + m^2).

Step 1: Expanding: (1 + m^2)x^2 + 2mcx + (c^2 - a^2) = 0. Step 2: Discriminant/4 = m^2 c^2 - (1 + m^2)(c^2 - a^2) = a^2(1 + m^2) - c^2. Step 3: If c^2 = a^2(1 + m^2), the discriminant is 0, so the roots are equal, as required.

6. Show that the roots of (mx + c)^2 = 4ax will be equal, if c = a/m; m != 0.

Step 1: Expanding: m^2 x^2 + 2(mc - 2a)x + c^2 = 0. Step 2: Discriminant/4 = (mc - 2a)^2 - m^2 c^2 = 4a(a - mc). Step 3: If c = a/m, then mc = a and the discriminant is 0, so the roots are equal, as required.

7. Prove that x^2/a^2 + (mx + c)^2/b^2 = 1 will have equal roots, if c^2 = a^2 m^2 + b^2; a != 0, b != 0.

Step 1: Clearing denominators: (b^2 + a^2 m^2)x^2 + 2a^2 mcx + a^2(c^2 - b^2) = 0. Step 2: Discriminant/4 = a^4 m^2 c^2 - (b^2 + a^2 m^2)a^2(c^2 - b^2) = a^2 b^2(a^2 m^2 + b^2 - c^2). Step 3: If c^2 = a^2 m^2 + b^2, the discriminant is 0, so the roots are equal, as required.

8. Show that the roots of the equation (a^2 - bc)x^2 + 2(b^2 - ca)x + c^2 - ab = 0

will be equal, if either a^3 + b^3 + c^3 = 3abc or b = 0.

Step 1: Discriminant/4 = (b^2 - ca)^2 - (a^2 - bc)(c^2 - ab) = b^4 - 2ab^2 c + c^2 a^2 - a^2 c^2 + a^3 b + bc^3 - ab^2 c = b(a^3 + b^3 + c^3 - 3abc). Step 2: The discriminant is 0 exactly when b = 0 or a^3 + b^3 + c^3 = 3abc. Step 3: Hence the roots are equal under either condition, as required.

Exercise 4.8 (book p. 169)

Solve the following systems of equations: 1. 2x - y = 4; 2x^2 - 4xy - y^2 = 6 2. x + y = 5; x^2 + 2y^2 = 17 3. 3x + 2y = 7; 3x^2 = 25 + 2y^2 4. x + y = 5; 2/x + 3/y = 2, x != 0, y != 0 5. x + y = a + b; a/x + b/y = 2 6. 3x + 4y = 25; 3/x + 4/y = 2

1. From the first equation, y = 2x - 4. Substituting: 2x^2 - 4x(2x - 4) - (2x - 4)^2 = 6, i.e. -10x^2 + 32x - 22 = 0, i.e. 5x^2 - 16x + 11 = (5x - 11)(x - 1) = 0. So x = 11/5, y = 2/5; or x = 1, y = -2. Solution set = {(1, -2), (11/5, 2/5)}. 2. From the first equation, y = 5 - x. Substituting: x^2 + 2(5 - x)^2 = 17, i.e. 3x^2 - 20x + 33 = (3x - 11)(x - 3) = 0. So x = 11/3, y = 4/3; or x = 3, y = 2. Solution set = {(3, 2), (11/3, 4/3)}. 3. From the first equation, y = (7 - 3x)/2. Substituting: 3x^2 = 25 + 2((7 - 3x)/2)^2, i.e. 6x^2 = 50 + 49 - 42x + 9x^2, i.e. x^2 - 14x + 33 = (x - 11)(x - 3) = 0. So x = 11, y = -13; or x = 3, y = -1. Solution set = {(3, -1), (11, -13)}. 4. From the second equation: (2y + 3x)/(xy) = 2, i.e. 2y + 3x = 2xy. With y = 5 - x: 2(5 - x) + 3x = 2x(5 - x), i.e. 2x^2 - 9x + 10 = (2x - 5)(x - 2) = 0. So x = 5/2, y = 5/2; or x = 2, y = 3. Solution set = {(2, 3), (5/2, 5/2)}. 5. From the second equation: (ay + bx)/(xy) = 2, i.e. ay + bx = 2xy. With y = a + b - x: a(a + b - x) + bx = 2x(a + b - x), i.e. 2x^2 - (3a + b)x + a(a + b) = 0. The discriminant is (3a + b)^2 - 8a(a + b) = (a - b)^2, so x = (3a + b +- (a - b))/4, i.e. x = a or x = (a + b)/2. So (x, y) = (a, b) or ((a + b)/2, (a + b)/2). Solution set = {(a, b), ((a + b)/2, (a + b)/2)}. 6. From the second equation: (3y + 4x)/(xy) = 2, i.e. 3y + 4x = 2xy. With y = (25 - 3x)/4: 3(25 - 3x)/4 + 4x = 2x(25 - 3x)/4, i.e. 6x^2 - 43x + 75 = (6x - 25)(x - 3) = 0. So x = 25/6,

y = 25/8; or x = 3, y = 4. Solution set = {(3, 4), (25/6, 25/8)}.

Solve the following systems of equations: 7. (x - 3)^2 + y^2 = 5; 2x = y + 6 8. (x + 3)^2 + (y - 1)^2 = 5; x^2 + y^2 + 2x = 9 9. x^2 + (y + 1)^2 = 18; (x + 2)^2 + y^2 = 21 10. x^2 + y^2 + 6x = 1; x^2 + y^2 + 2(x + y) = 3

7. From the second equation, y = 2x - 6. Substituting: (x - 3)^2 + (2x - 6)^2 = 5, i.e. 5x^2 - 30x + 40 = 0, i.e. x^2 - 6x + 8 = (x - 4)(x - 2) = 0. So x = 4, y = 2; or x = 2, y = -2. Solution set = {(4, 2), (2, -2)}. 8. Subtracting the second equation from the expanded first: (x^2 + 6x + 9) + (y^2 - 2y + 1) - (x^2 + y^2 + 2x) = 5 - 9, i.e. 4x - 2y + 14 = 0, i.e. y = 2x + 7. Substituting in the second equation: x^2 + (2x + 7)^2 + 2x = 9, i.e. x^2 + 6x + 8 = (x + 4)(x + 2) = 0. So x = -4, y = -1; or x = -2, y = 3. Solution set = {(-4, -1), (-2, 3)}. 9. Expanding: x^2 + y^2 + 2y + 1 = 18 and x^2 + y^2 + 4x + 4 = 21. Subtracting: 2y - 4x - 3 = -3, i.e. y = 2x. Substituting: x^2 + (2x + 1)^2 = 18, i.e. 5x^2 + 4x - 17 = 0, so x = (-4 +- 2 sqrt(89))/10 = (-2 +- sqrt(89))/5 and y = (-4 +- 2 sqrt(89))/5. Solution set = {((-2 +- sqrt(89))/5, (-4 +- 2 sqrt(89))/5)}. 10. Subtracting the second equation from the first: 4x - 2y = -2, i.e. y = 2x + 1. Substituting in the second equation: x^2 + (2x + 1)^2 + 2(3x + 1) = 3, i.e. 5x^2 + 10x = 0, i.e. x = 0 or x = -2. So (x, y) = (0, 1) or (-2, -3). Solution set = {(0, 1), (-2, -3)}.

Exercise 4.9 (book p. 172)

Solve the following systems of equations: 1. 2x^2 = 6 + 3y^2; 3x^2 - 5y^2 = 7 2. 8x^2 = y^2; x^2 + 2y^2 = 19 3. 2x^2 - 8 = 5y^2; x^2 - 13 = -2y^2 4. x^2 - 5xy + 6y^2 = 0; x^2 + y^2 = 45 5. 12x^2 - 25xy + 12y^2 = 0; 4x^2 + 7y^2 = 148

1. From the first equation, x^2 = (6 + 3y^2)/2. Substituting: 3(6 + 3y^2)/2 - 5y^2 = 7, i.e. 18 + 9y^2 - 10y^2 = 14, so y^2 = 4 and y = +- 2; then x^2 = 6 and x = +- 3. Solution set = {(3, 2), (3, -2), (-3, 2), (-3, -2)}. 2. Substituting y^2 = 8x^2: x^2 + 16x^2 = 19, so x^2 = 19/17 and x = +- sqrt(19/17); then y^2 = 152/17 and

y = +- 2 sqrt(38/17). Solution set = {(+- sqrt(19/17), +- 2 sqrt(38/17))} (with the signs of x and y independent, since only squares occur). FLAG: the printed key writes the y-value as +- 2 sqrt(38/7), but 8 x 19/17 = 152/17, so the correct value is +- 2 sqrt(38/17); 152/7 would not satisfy x^2 + 2y^2 = 19. 3. From the second equation, y^2 = (13 - x^2)/2. Substituting: 2x^2 - 8 = 5(13 - x^2)/2, i.e. 4x^2 - 16 = 65 - 5x^2, so 9x^2 = 81 and x = +- 3; then y^2 = 2 and y = +- sqrt(2). Solution set = {(+- 3, +- sqrt(2))}. 4. The first equation factors: (x - 2y)(x - 3y) = 0, so x = 2y or x = 3y. For x = 2y: 5y^2 = 45, so y = +- 3 and x = +- 6. For x = 3y: 10y^2 = 45, so y = +- 3/sqrt(2) and x = +- 9/sqrt(2). Solution set = {(6, 3), (-6, -3), (9/sqrt(2), 3/sqrt(2)), (-9/sqrt(2), -3/sqrt(2))}. 5. The first equation factors: (3x - 4y)(4x - 3y) = 0, so y = 3x/4 or y = 4x/3. For y = 3x/4: 4x^2 + 7(9x^2/16) = 148, i.e. 127x^2/16 = 148, so x^2 = 2368/127 = 64 x 37/127, giving x = +- 8 sqrt(37/127) and y = +- 6 sqrt(37/127). For y = 4x/3: 4x^2 + 7(16x^2/9) = 148, i.e. 148x^2/9 = 148, so x = +- 3 and y = +- 4. Solution set = {(3, 4), (-3, -4), (8 sqrt(37/127), 6 sqrt(37/127)), (-8 sqrt(37/127), -6 sqrt(37/127))}.

Solve the following systems of equations: 6. 12x^2 - 11xy + 2y^2 = 0; 2x^2 + 7xy = 60 7. x^2 - y^2 = 16; xy = 15 8. x^2 + xy = 9; x^2 - y^2 = 2 9. y^2 - 7 = 2xy; 2x^2 + 3 = xy 10. x^2 + y^2 = 5; xy = 2

6. The first equation factors: (3x - 2y)(4x - y) = 0, so y = 3x/2 or y = 4x. For y = 3x/2: 2x^2 + 7x(3x/2) = 60, i.e. 25x^2/2 = 60, so x^2 = 24/5, giving x = +- 2 sqrt(30)/5 and y = +- 3 sqrt(30)/5. For y = 4x: 2x^2 + 28x^2 = 60, so x^2 = 2, giving x = +- sqrt(2) and y = +- 4 sqrt(2). Solution set = {(+- 2 sqrt(30)/5, +- 3 sqrt(30)/5), (+- sqrt(2), +- 4 sqrt(2))}. FLAG: the printed key lists only the first branch, (+- 2 sqrt(30)/5, +- 3 sqrt(30)/5); the second branch (+- sqrt(2), +- 4 sqrt(2)) also satisfies both equations and is included here. 7. (x^2 + y^2)^2 = (x^2 - y^2)^2 + 4x^2 y^2 = 256 + 900 = 1156, so x^2 + y^2 = 34 or x^2 + y^2 = -34. For x^2 + y^2 = 34: x^2 = 25 and y^2 = 9, giving (5, 3) and (-5, -3) (xy =

15 > 0 forces equal signs). For x^2 + y^2 = -34: x^2 = -9 and y^2 = -25, giving (3i, -5i) and (-3i, 5i) (again xy = 15 forces the signs shown). Solution set = {(5, 3), (-5, -3), (3i, -5i), (-3i, 5i)}. 8. From the first equation, y = (9 - x^2)/x (x != 0). Substituting in the second: x^2 - (9 - x^2)^2/x^2 = 2, i.e. x^4 - 2x^2 - (81 - 18x^2 + x^4) = 0, i.e. 16x^2 = 81, so x = +- 9/4 and y = +- 7/4 (same sign as x). Solution set = {(9/4, 7/4), (-9/4, -7/4)}. 9. From the second equation, y = (2x^2 + 3)/x (x != 0). Substituting in the first: ((2x^2 + 3)/x)^2 - 7 = 2(2x^2 + 3), i.e. 4x^2 + 12 + 9/x^2 - 7 = 4x^2 + 6, i.e. 9/x^2 = 1, so x = +- 3. Then y = 7 for x = 3 and y = -7 for x = -3. Solution set = {(3, 7), (-3, -7)}. 10. (x + y)^2 = x^2 + y^2 + 2xy = 9, so x + y = +- 3; (x - y)^2 = x^2 + y^2 - 2xy = 1, so x - y = +- 1. Combining: (x, y) = (2, 1), (1, 2), (-2, -1) or (-1, -2). Solution set = {(2, 1), (1, 2), (-2, -1), (-1, -2)}.

Exercise 4.10 (book pp. 176-177)

1. The product of one less than a certain positive number and two less than three times the number is 14. Find the number. 2. The sum of a positive number and its square is 380. Find the number. 3. Divide 40 into two parts such that the sum of their squares is greater than 2 times their product by 100. 4. The sum of a positive number and its reciprocal is 26/5. Find the number. 5. A number exceeds its square root by 56. Find the number.

1. Let the number be x: (x - 1)(3x - 2) = 14, i.e. 3x^2 - 5x - 12 = (3x + 4)(x - 3) = 0. Since x is positive, x = 3. 2. Let the number be x: x + x^2 = 380, i.e. x^2 + x - 380 = (x + 20)(x - 19) = 0. Since x is positive, x = 19. 3. Let the parts be x and 40 - x: x^2 + (40 - x)^2 = 2x(40 - x) + 100, i.e. 4x^2 - 160x + 1500 = 0, i.e. x^2 - 40x + 375 = (x - 15)(x - 25) = 0. So the parts are 15 and 25. 4. Let the number be x: x + 1/x = 26/5, i.e. 5x^2 - 26x + 5 = (5x - 1)(x - 5) = 0, so x = 5 or x = 1/5 (the two values are reciprocals of each other; both satisfy the condition). 5. Let the number

be x: x - sqrt(x) = 56. Put t = sqrt(x): t^2 - t - 56 = (t - 8)(t + 7) = 0. Since t >= 0, t = 8 and x = 64.

6. Find two consecutive numbers, whose product is 132. (Hint: Suppose the numbers are x and x + 1). 7. The difference between the cubes of two consecutive even numbers is 296. Find them. (Hint: Let two consecutive even numbers be x and x + 2). 8. A farmer bought some sheep for Rs. 9000. If he had paid Rs. 100 less for each, he would have got 3 sheep more for the same money. How many sheep did he buy, when the rate in each case is uniform? 9. A man sold his stock of eggs for Rs. 240. If he had 2 dozen more, he would have got the same money by selling the whole for Rs. 0.50 per dozen cheaper. How many dozen eggs did he sell? 10. A cyclist travelled 48 km at a uniform speed. Had he travelled 2 km/hour slower, he would have taken 2 hours more to perform the journey. How long did he take to cover 48 km?

6. x(x + 1) = 132, i.e. x^2 + x - 132 = (x + 12)(x - 11) = 0. Taking the positive value, x = 11, so the numbers are 11 and 12. 7. (x + 2)^3 - x^3 = 296, i.e. 6x^2 + 12x + 8 = 296, i.e. x^2 + 2x - 48 = (x + 8)(x - 6) = 0. Taking the positive even value, x = 6, so the numbers are 6 and 8. 8. Let n be the number of sheep: the price per sheep is 9000/n. Then (n + 3)(9000/n - 100) = 9000, i.e. 27000/n - 100n - 300 = 0, i.e. n^2 + 3n - 270 = (n + 18)(n - 15) = 0. So n = 15 sheep. 9. Let d be the number of dozen eggs: the price per dozen is 240/d. Then (d + 2)(240/d - 0.50) = 240, i.e. 480/d - 0.5d - 1 = 0, i.e. d^2 + 2d - 960 = (d + 32)(d - 30) = 0. So d = 30 dozen. 10. Let the speed be v km/hour: the time taken is 48/v hours. Then 48/(v - 2) = 48/v + 2, i.e. 24/(v - 2) - 24/v = 1, i.e. 48 = v(v - 2), i.e. v^2 - 2v - 48 = (v - 8)(v + 6) = 0. Taking v = 8 km/hour, the time taken is 48/8 = 6 hours.

11. The area of a rectangular field is 297 square meters. Had it been 3 meters longer and one meter shorter, the area would have been 3 square meters more. Find its length and breadth. 12. The length of a rectangular piece of paper exceeds

its breadth by 5 cm. If a strip 0.5 cm wide be cut all around the piece of paper, the area of the remaining part would be 500 square cms. Find its original dimensions. 13. A number consists of two digits whose product is 18. If the digits are interchanged, the new number becomes 27 less than the original number. Find the number. 14. A number consists of two digits whose product is 14. If the digits are interchanged, the resulting number will exceed the original number by 45. Find the number. 15. The area of a right triangle is 210 square meters. If its hypotenuse is 37 meters long, find the length of the base and the altitude.

11. Let the length be l and the breadth be b: lb = 297 and (l + 3)(b - 1) = 300. Expanding the second: lb - l + 3b - 3 = 300, i.e. -l + 3b = 6, so l = 3b - 6. Substituting: (3b - 6)b = 297, i.e. b^2 - 2b - 99 = (b - 11)(b + 9) = 0. So b = 11 m and l = 27 m. 12. Let the breadth be b cm: the length is b + 5 cm. After cutting a 0.5 cm strip all around, the remaining piece is (b + 4) cm by (b - 1) cm, so (b + 4)(b - 1) = 500, i.e. b^2 + 3b - 504 = (b + 24)(b - 21) = 0. So the breadth is 21 cm and the length is 26 cm. 13. Let the tens digit be x and the units digit be y: xy = 18 and (10y + x) = (10x + y) - 27, i.e. x - y = 3. So x = y + 3 and (y + 3)y = 18, i.e. y^2 + 3y - 18 = (y + 6)(y - 3) = 0. So y = 3 and x = 6, giving the number 63. 14. Let the tens digit be x and the units digit be y: xy = 14 and (10y + x) = (10x + y) + 45, i.e. y - x = 5. So y = x + 5 and x(x + 5) = 14, i.e. x^2 + 5x - 14 = (x + 7)(x - 2) = 0. So x = 2 and y = 7, giving the number 27. 15. Let the base be a and the altitude be b: (1/2)ab = 210, so ab = 420; and a^2 + b^2 = 37^2 = 1369. Then (a + b)^2 = 1369 + 840 = 2209, so a + b = 47. Hence a and b are the roots of t^2 - 47t + 420 = (t - 35)(t - 12) = 0, so the base is 35 m and the altitude is 12 m (or vice versa).

16. The area of a rectangle is 1680 square meters. If its diagonal is 58 meters long, find the length and the breadth of the rectangle. 17. To do a piece of work, A takes 10 days more than B. Together they finish the work in 12 days. How long would B take to finish it alone? (Hint: If some one takes x days to finish a work, the one day's

work will be 1/x). 18. To complete a job, A and B take 4 days working together. A alone takes twice as long as B alone to finish the same job. How long would each one alone take to do the job? 19. An open box is to be made from a square piece of tin by cutting a piece 2 dm square from each corner and then folding the sides of the remaining piece. If the capacity of the box is to be 128 c.dm, find the length of the side of the piece. 20. A man invests Rs. 100,000 in two companies. His total profit is Rs. 3080. If he receives Rs. 1980 from one company and at the rate 1% more from the other, find the amount of each investment.

16. Let the length be l and the breadth be b: lb = 1680 and l^2 + b^2 = 58^2 = 3364. Then (l + b)^2 = 3364 + 3360 = 6724, so l + b = 82. Hence l and b are the roots of t^2 - 82t + 1680 = (t - 42)(t - 40) = 0, so the length is 42 m and the breadth is 40 m. 17. Let B take x days: A takes x + 10 days. Together: 1/x + 1/(x + 10) = 1/12, i.e. 12(2x + 10) = x(x + 10), i.e. x^2 - 14x - 120 = (x - 20)(x + 6) = 0. So B alone takes 20 days. 18. Let B take x days: A takes 2x days. Together: 1/x + 1/(2x) = 1/4, i.e. 3/(2x) = 1/4, so x = 6. Hence B alone takes 6 days and A alone takes 12 days. 19. Let the side of the square piece be s dm: after cutting 2 dm squares from the corners and folding, the box is (s - 4) dm by (s - 4) dm by 2 dm, so its capacity is 2(s - 4)^2 = 128, i.e. (s - 4)^2 = 64, so s - 4 = 8 (taking the positive value) and s = 12 dm. 20. Let the two investments be Rs. x and Rs. (100000 - x), at rates r% and (r + 1)% respectively, with profits Rs. 1980 and Rs. 3080: x r/100 = 1980 and (100000 - x)(r + 1)/100 = 3080. From the first, r = 198000/x. Substituting: (100000 - x)(198000 + x)/(100x) = 3080, i.e. (100000 - x)(198000 + x) = 308000x, i.e. x^2 + 406000x - 19800000000 = 0, i.e. (x - 44000)(x + 450000) = 0. So x = 44000, and the investments are Rs. 44,000 (at 4.5%, giving profit Rs. 1980) and Rs. 56,000 (at 5.5%, giving profit Rs. 3080), as in the printed key. FLAG: the question as printed says 'total profit is Rs. 3080', but under that literal reading (profits 1980 and 1100) the investments would be about Rs. 70,974 and Rs. 29,026 (or about Rs.

56,297 and Rs. 43,703 if the 1980 profit is at the higher rate), not the key's Rs. 56,000 and Rs. 44,000. The solution above follows the reading that reproduces the printed key exactly: the two profits are Rs. 1980 and Rs. 3080.

Chapter 5: Partial Fractions

Exercise 5.1 (book p. 183)

1. Resolve 1/(x^2 - 1) into partial fractions

Step 1: x^2 - 1 = (x - 1)(x + 1). Write 1/((x - 1)(x + 1)) = A/(x - 1) + B/(x + 1). Step 2: Identity: 1 = A(x + 1) + B(x - 1). Step 3: Put x = 1: 1 = 2A, so A = 1/2. Put x = -1: 1 = -2B, so B = -1/2. Answer: 1/(2(x - 1)) - 1/(2(x + 1)). This agrees with the book's Answers (p. 416).

2. Resolve (x^2 + 1)/((x + 1)(x - 1)) into partial fractions

Step 1: The fraction is improper (numerator and denominator both degree 2). Divide: (x^2 + 1)/(x^2 - 1) = 1 + 2/(x^2 - 1). Step 2: Resolve 2/((x - 1)(x + 1)) = A/(x - 1) + B/(x + 1): identity 2 = A(x + 1) + B(x - 1); x = 1 gives A = 1; x = -1 gives B = -1. Answer: 1 + 1/(x - 1) - 1/(x + 1). This agrees with the book's Answers (p. 416).

3. Resolve (2x + 1)/((x - 1)(x + 2)(x + 3)) into partial fractions

Step 1: Write (2x + 1)/((x - 1)(x + 2)(x + 3)) = A/(x - 1) + B/(x + 2) + C/(x + 3). Step 2: Identity: 2x + 1 = A(x + 2)(x + 3) + B(x - 1)(x + 3) + C(x - 1)(x + 2). Step 3: Put x = 1: 3 = A(3)(4) = 12A, so A = 1/4. Put x = -2: -3 = B(-3)(1) = -3B, so B = 1. Put x = -3: -5 = C(-4)(-1) = 4C, so C = -5/4. Answer: 1/(4(x - 1)) + 1/(x + 2) - 5/(4(x + 3)). This agrees with the book's Answers (p. 416).

4. Resolve (3x^2 - 4x - 5)/((x - 2)(x^2 + 7x + 10)) into partial fractions

Step 1: x^2 + 7x + 10 = (x + 2)(x + 5), so the fraction is (3x^2 - 4x - 5)/((x - 2)(x +

2)(x + 5)) = A/(x - 2) + B/(x + 2) + C/(x + 5). Step 2: Identity: 3x^2 - 4x - 5 = A(x + 2)(x + 5) + B(x - 2)(x + 5) + C(x - 2)(x + 2). Step 3: Put x = 2: 12 - 8 - 5 = -1 = A(4)(7) = 28A, so A = -1/28. Put x = -2: 12 + 8 - 5 = 15 = B(-4)(3) = -12B, so B = -5/4. Put x = -5: 75 + 20 - 5 = 90 = C(-7)(-3) = 21C, so C = 30/7. Answer: -1/(28(x - 2)) - 5/(4(x + 2)) + 30/(7(x + 5)). This agrees with the book's Answers (p. 416).

5. Resolve 1/((x - 1)(2x - 1)(3x - 1)) into partial fractions

Step 1: Write 1/((x - 1)(2x - 1)(3x - 1)) = A/(x - 1) + B/(2x - 1) + C/(3x - 1). Step 2: Identity: 1 = A(2x - 1)(3x - 1) + B(x - 1)(3x - 1) + C(x - 1)(2x - 1). Step 3: Put x = 1: 1 = A(1)(2) = 2A, so A = 1/2. Put x = 1/2: 1 = B(-1/2)(1/2) = -B/4, so B = -4. Put x = 1/3: 1 = C(-2/3)(-1/3) = 2C/9, so C = 9/2. Answer: 1/(2(x - 1)) - 4/(2x - 1) + 9/(2(3x - 1)). This agrees with the book's Answers (p. 416).

6. Resolve x/((x - a)(x - b)(x - c)) into partial fractions

Step 1: Write x/((x - a)(x - b)(x - c)) = A/(x - a) + B/(x - b) + C/(x - c). Step 2: Identity: x = A(x - b)(x - c) + B(x - a)(x - c) + C(x - a)(x - b). Step 3: Put x = a: a = A(a - b)(a - c), so A = a/((a - b)(a - c)). Put x = b: b = B(b - a)(b - c), so B = b/((b - a)(b - c)). Put x = c: c = C(c - a)(c - b), so C = c/((c - a)(c - b)). Answer: a/((a - b)(a - c)(x - a)) + b/((b - a)(b - c)(x - b)) + c/((c - a)(c - b)(x - c)). This agrees with the book's Answers (p. 416).

7. Resolve (6x^3 + 5x^2 - 7)/(2x^2 - x - 1) into partial fractions

Step 1: The fraction is improper. Divide: (3x + 4)(2x^2 - x - 1) = 6x^3 + 5x^2 - 7x - 4, so (6x^3 + 5x^2 - 7) = (3x + 4)(2x^2 - x - 1) + (7x - 3). Step 2: Resolve (7x - 3)/((2x + 1)(x - 1)) = A/(x - 1) + B/(2x + 1): identity 7x - 3 = A(2x + 1) + B(x - 1); x = 1 gives 4 = 3A, so A = 4/3; x = -1/2 gives -13/2 = B(-3/2), so B = 13/3. Step 3: Answer: 3x + 4 + 4/(3(x - 1)) + 13/(3(2x + 1)). This agrees with the book's Answers (p. 416).

8. Resolve (2x^3 + x^2 - 5x + 3)/(2x^3 + x^2 - 3x) into partial fractions

Step 1: The denominator factors as x(2x + 3)(x - 1). The numerator minus the

denominator is -2x + 3, so the fraction equals 1 + (-2x + 3)/(x(2x + 3)(x - 1)). Step 2: Resolve (-2x + 3)/(x(2x + 3)(x - 1)) = A/x + B/(2x + 3) + C/(x - 1): identity -2x + 3 = A(2x + 3)(x - 1) + Bx(x - 1) + Cx(2x + 3). Put x = 0: 3 = A(3)(-1) = -3A, so A = -1. Put x = 1: 1 = C(1)(5) = 5C, so C = 1/5. Put x = -3/2: 6 = B(-3/2)(-5/2) = 15B/4, so B = 8/5. Step 3: Answer: 1 - 1/x + 8/(5(2x + 3)) + 1/(5(x - 1)). This agrees with the book's Answers (p. 416).

9. Resolve (x - 1)(x - 3)(x - 5)/((x - 2)(x - 4)(x - 6)) into partial fractions

Step 1: Both numerator and denominator are monic cubics, so the fraction equals 1 plus a proper fraction. Numerator minus denominator: (x^3 - 9x^2 + 23x - 15) - (x^3 - 12x^2 + 44x - 48) = 3x^2 - 21x + 33 = 3(x^2 - 7x + 11). Step 2: Resolve 3(x^2 - 7x + 11)/((x - 2)(x - 4)(x - 6)) = A/(x - 2) + B/(x - 4) + C/(x - 6). Put x = 2: 3(4 - 14 + 11) = 3 = A(-2)(-4) = 8A, so A = 3/8. Put x = 4: 3(16 - 28 + 11) = -3 = B(2)(-2) = -4B, so B = 3/4. Put x = 6: 3(36 - 42 + 11) = 15 = C(4)(2) = 8C, so C = 15/8. Answer: 1 + 3/(8(x - 2)) + 3/(4(x - 4)) + 15/(8(x - 6)). This agrees with the book's Answers (p. 416).

10. Resolve 1/((1 - ax)(1 - bx)(1 - cx)) into partial fractions

Step 1: Write 1/((1 - ax)(1 - bx)(1 - cx)) = A/(1 - ax) + B/(1 - bx) + C/(1 - cx). Step 2: Identity: 1 = A(1 - bx)(1 - cx) + B(1 - ax)(1 - cx) + C(1 - ax)(1 - bx). Step 3: Put x = 1/a: 1 = A(1 - b/a)(1 - c/a) = A(a - b)(a - c)/a^2, so A = a^2/((a - b)(a - c)). Similarly B = b^2/((b - a)(b - c)) and C = c^2/((c - a)(c - b)). Answer: a^2/((a - b)(a - c)(1 - ax)) + b^2/((b - a)(b - c)(1 - bx)) + c^2/((c - a)(c - b)(1 - cx)). This agrees with the book's Answers (p. 416).

11. Resolve (x^2 + a^2)/((x^2 + b^2)(x^2 + c^2)(x^2 + d^2)) into partial fractions. [Hint: Put x^2 = y to make factors of the denominator linear]

Step 1: Put y = x^2. Then the fraction is (y + a^2)/((y + b^2)(y + c^2)(y + d^2)) = A/(y + b^2) + B/(y + c^2) + C/(y + d^2). Step 2: Identity: y + a^2 = A(y + c^2)(y + d^2) + B(y + b^2)(y + d^2) + C(y + b^2)(y + c^2). Step 3: Put y =

-b^2: a^2 - b^2 = A(c^2 - b^2)(d^2 - b^2), so A = (a^2 - b^2)/((c^2 - b^2)(d^2 - b^2)). Similarly B = (a^2 - c^2)/((b^2 - c^2)(d^2 - c^2)) and C = (a^2 - d^2)/((b^2 - d^2)(c^2 - d^2)). Step 4: Replacing y by x^2 gives the answer: (a^2 - b^2)/((c^2 - b^2)(d^2 - b^2)(x^2 + b^2)) + (a^2 - c^2)/((b^2 - c^2)(d^2 - c^2)(x^2 + c^2)) + (a^2 - d^2)/((b^2 - d^2)(c^2 - d^2)(x^2 + d^2)). This agrees with the book's Answers (p. 416) (the key writes each denominator with the factors in the opposite order, which is the same value since two sign changes cancel).

Exercise 5.2 (book p. 185)

1. Resolve (2x^2 - 3x + 4)/(x - 1)^3 into partial fractions

Step 1: Write (2x^2 - 3x + 4)/(x - 1)^3 = A/(x - 1) + B/(x - 1)^2 + C/(x - 1)^3. Step 2: Identity: 2x^2 - 3x + 4 = A(x - 1)^2 + B(x - 1) + C. Step 3: Put x = 1: 2 - 3 + 4 = 3 = C. Equating x^2 coefficients: A = 2. Equating x coefficients: -2A + B = -3, so B = 1. Answer: 2/(x - 1) + 1/(x - 1)^2 + 3/(x - 1)^3. This agrees with the book's Answers (p. 416).

2. Resolve (5x^2 - 2x + 3)/(x + 2)^3 into partial fractions

Step 1: Write (5x^2 - 2x + 3)/(x + 2)^3 = A/(x + 2) + B/(x + 2)^2 + C/(x + 2)^3. Step 2: Identity: 5x^2 - 2x + 3 = A(x + 2)^2 + B(x + 2) + C. Step 3: Put x = -2: 20 + 4 + 3 = 27 = C. Equating x^2 coefficients: A = 5. Equating x coefficients: 4A + B = -2, so B = -22. Answer: 5/(x + 2) - 22/(x + 2)^2 + 27/(x + 2)^3. This agrees with the book's Answers (p. 416).

3. Resolve 4x/((x + 1)^2 (x - 1)) into partial fractions

Step 1: Write 4x/((x + 1)^2 (x - 1)) = A/(x + 1) + B/(x + 1)^2 + C/(x - 1). Step 2: Identity: 4x = A(x + 1)(x - 1) + B(x - 1) + C(x + 1)^2. Step 3: Put x = -1: -4 = B(-2), so B = 2. Put x = 1: 4 = C(4), so C = 1. Equating x^2 coefficients: 0 = A + C, so A = -1. Answer: -1/(x + 1) + 2/(x + 1)^2 + 1/(x - 1). This agrees with the book's Answers (p. 416).

4. Resolve 9/((x + 2)^2 (x - 1)) into partial fractions

Step 1: Write 9/((x + 2)^2 (x - 1)) = A/(x + 2) + B/(x + 2)^2 + C/(x - 1). Step 2: Identity: 9 = A(x + 2)(x - 1) + B(x - 1) + C(x + 2)^2. Step 3: Put x = -2: 9 = B(-3), so B = -3. Put x = 1: 9 = C(9), so C = 1. Equating x^2 coefficients: 0 = A + C, so A = -1. Answer: -1/(x + 2) - 3/(x + 2)^2 + 1/(x - 1). This agrees with the book's Answers (p. 416).

5. Resolve 1/((x - 3)^2 (x + 1)) into partial fractions

Step 1: Write 1/((x - 3)^2 (x + 1)) = A/(x - 3) + B/(x - 3)^2 + C/(x + 1). Step 2: Identity: 1 = A(x - 3)(x + 1) + B(x + 1) + C(x - 3)^2. Step 3: Put x = 3: 1 = B(4), so B = 1/4. Put x = -1: 1 = C(16), so C = 1/16. Equating x^2 coefficients: 0 = A + C, so A = -1/16. Answer: -1/(16(x - 3)) + 1/(4(x - 3)^2) + 1/(16(x + 1)). This agrees with the book's Answers (p. 417).

6. Resolve x^2/((x - 2)(x - 1)^2) into partial fractions

Step 1: Write x^2/((x - 2)(x - 1)^2) = A/(x - 2) + B/(x - 1) + C/(x - 1)^2. Step 2: Identity: x^2 = A(x - 1)^2 + B(x - 2)(x - 1) + C(x - 2). Step 3: Put x = 2: 4 = A(1), so A = 4. Put x = 1: 1 = C(-1), so C = -1. Equating x^2 coefficients: 1 = A + B, so B = -3. Answer: 4/(x - 2) - 3/(x - 1) - 1/(x - 1)^2. This agrees with the book's Answers (p. 417).

7. Resolve 1/((x - 1)^2 (x + 1)) into partial fractions

Step 1: Write 1/((x - 1)^2 (x + 1)) = A/(x - 1) + B/(x - 1)^2 + C/(x + 1). Step 2: Identity: 1 = A(x - 1)(x + 1) + B(x + 1) + C(x - 1)^2. Step 3: Put x = 1: 1 = B(2), so B = 1/2. Put x = -1: 1 = C(4), so C = 1/4. Equating x^2 coefficients: 0 = A + C, so A = -1/4. Answer: -1/(4(x - 1)) + 1/(2(x - 1)^2) + 1/(4(x + 1)). This agrees with the book's Answers (p. 417).

8. Resolve x^2/((x - 1)^3 (x + 1)) into partial fractions

Step 1: Write x^2/((x - 1)^3 (x + 1)) = A/(x - 1) + B/(x - 1)^2 + C/(x - 1)^3 + D/(x + 1). Step 2: Identity: x^2 = A(x - 1)^2 (x + 1) + B(x - 1)(x + 1) + C(x + 1) + D(x -

1)^3. Step 3: Put x = 1: 1 = C(2), so C = 1/2. Put x = -1: 1 = D(-8), so D = -1/8. Equating x^3 coefficients: 0 = A + D, so A = 1/8. Equating x^2 coefficients: 1 = A + B - 3D? Expanding fully: A(x^3 - x^2 - x + 1) + B(x^2 - 1) + C(x + 1) + D(x^3 - 3x^2 + 3x - 1); the x^2 coefficient is -A + B - 3D = -1/8 + B + 3/8 = B + 1/4 = 1, so B = 3/4. Answer: 1/(8(x - 1)) + 3/(4(x - 1)^2) + 1/(2(x - 1)^3) - 1/(8(x + 1)). FLAG: the printed key gives the last term as -1/(2(x - 1)^3); the cover-up value is x^2/(x + 1) at x = 1, which is +1/2, and recombining the key's form differs from the fraction by -1/(x - 1)^3, so the key's sign is a misprint and the correct term is +1/(2(x - 1)^3).

9. Resolve (x - 1)/((x - 2)(x + 1)^3) into partial fractions

Step 1: Write (x - 1)/((x - 2)(x + 1)^3) = A/(x - 2) + B/(x + 1) + C/(x + 1)^2 + D/(x + 1)^3. Step 2: Identity: x - 1 = A(x + 1)^3 + B(x - 2)(x + 1)^2 + C(x - 2)(x + 1) + D(x - 2). Step 3: Put x = 2: 1 = A(27), so A = 1/27. Put x = -1: -2 = D(-3), so D = 2/3. Equating x^3 coefficients: 0 = A + B, so B = -1/27. Expanding: A(x^3 + 3x^2 + 3x + 1) + B(x^3 - 3x - 2) + C(x^2 - x - 2) + D(x - 2); the x^2 coefficient is 3A + C = 1/9 + C = 0, so C = -1/9. Answer: 1/(27(x - 2)) - 1/(27(x + 1)) - 1/(9(x + 1)^2) + 2/(3(x + 1)^3). This agrees with the book's Answers (p. 417).

10. Resolve 4x^3/((x^2 - 1)(x + 1)^2) into partial fractions

Step 1: (x^2 - 1)(x + 1)^2 = (x - 1)(x + 1)^3. Write 4x^3/((x - 1)(x + 1)^3) = A/(x - 1) + B/(x + 1) + C/(x + 1)^2 + D/(x + 1)^3. Step 2: Identity: 4x^3 = A(x + 1)^3 + B(x - 1)(x + 1)^2 + C(x - 1)(x + 1) + D(x - 1). Step 3: Put x = 1: 4 = A(8), so A = 1/2. Put x = -1: -4 = D(-2), so D = 2. Equating x^3 coefficients: 4 = A + B, so B = 7/2. Equating x^2 coefficients: 0 = 3A - B + C? Expanding: A(x^3 + 3x^2 + 3x + 1) + B(x^3 + x^2 - x - 1) + C(x^2 - 1) + D(x - 1); the x^2 coefficient is 3A + B + C = 3/2 + 7/2 + C = 5 + C = 0, so C = -5. Answer: 1/(2(x - 1)) + 7/(2(x + 1)) - 5/(x + 1)^2 + 2/(x + 1)^3. This agrees with the book's Answers (p. 417).

11. Resolve (2x + 1)/((x + 3)(x - 1)(x + 2)^2) into partial fractions

Step 1: Write (2x + 1)/((x + 3)(x - 1)(x + 2)^2) = A/(x + 3) + B/(x - 1) + C/(x + 2) + D/(x + 2)^2. Step 2: Identity: 2x + 1 = A(x - 1)(x + 2)^2 + B(x + 3)(x + 2)^2 + C(x + 3)(x - 1)(x + 2) + D(x + 3)(x - 1). Step 3: Put x = -3: -5 = A(-4)(1) = -4A, so A = 5/4. Put x = 1: 3 = B(4)(9) = 36B, so B = 1/12. Put x = -2: -3 = D(1)(-3) = -3D, so D = 1. Equating x^3 coefficients: 0 = A + B + C, so C = -5/4 - 1/12 = -4/3. Answer: 5/(4(x + 3)) + 1/(12(x - 1)) - 4/(3(x + 2)) + 1/(x + 2)^2. This agrees with the book's Answers (p. 417).

12. Resolve 2x^4/((x - 3)(x + 2)^2) into partial fractions

Step 1: (x - 3)(x + 2)^2 = x^3 + x^2 - 8x - 12. Dividing: (2x - 2)(x^3 + x^2 - 8x - 12) = 2x^4 - 18x^2 - 8x + 24, so 2x^4 = (2x - 2)(x - 3)(x + 2)^2 + (18x^2 + 8x - 24). Step 2: Resolve (18x^2 + 8x - 24)/((x - 3)(x + 2)^2) = A/(x - 3) + B/(x + 2) + C/(x + 2)^2. Put x = 3: 162 + 24 - 24 = 162 = A(25), so A = 162/25. Put x = -2: 72 - 16 - 24 = 32 = C(-5), so C = -32/5. Equating x^2 coefficients: 18 = A + B, so B = 288/25. Answer: 2x - 2 + 162/(25(x - 3)) + 288/(25(x + 2)) - 32/(5(x + 2)^2). This agrees with the book's Answers (p. 417).

Exercise 5.3 (book p. 187)

1. Resolve (9x - 7)/((x^2 + 1)(x + 3)) into partial fractions

Step 1: Write (9x - 7)/((x^2 + 1)(x + 3)) = (Ax + B)/(x^2 + 1) + C/(x + 3). Step 2: Identity: 9x - 7 = (Ax + B)(x + 3) + C(x^2 + 1). Step 3: Put x = -3: -34 = C(10), so C = -17/5. Equating x^2 coefficients: 0 = A + C, so A = 17/5. Equating constants: -7 = 3B + C, so 3B = -7 + 17/5 = -18/5, B = -6/5. Answer: (17x - 6)/(5(x^2 + 1)) - 17/(5(x + 3)). This agrees with the book's Answers (p. 417).

2. Resolve 1/((x^2 + 1)(x + 1)) into partial fractions

Step 1: Write 1/((x^2 + 1)(x + 1)) = (Ax + B)/(x^2 + 1) + C/(x + 1). Step 2: Identity: 1 = (Ax + B)(x + 1) + C(x^2 + 1). Step 3: Put x = -1: 1 = C(2), so C = 1/2. Equating x^2 coefficients: 0 = A + C, so A = -1/2. Equating constants: 1 = B + C, so

B = 1/2. Answer: (1 - x)/(2(x^2 + 1)) + 1/(2(x + 1)). This agrees with the book's Answers (p. 417) (the key writes (1 - x)/(2(1 + x^2))).

3. Resolve (3x + 7)/((x^2 + 4)(x + 3)) into partial fractions

Step 1: Write (3x + 7)/((x^2 + 4)(x + 3)) = (Ax + B)/(x^2 + 4) + C/(x + 3). Step 2: Identity: 3x + 7 = (Ax + B)(x + 3) + C(x^2 + 4). Step 3: Put x = -3: -2 = C(13), so C = -2/13. Equating x^2 coefficients: 0 = A + C, so A = 2/13. Equating constants: 7 = 3B + 4C = 3B - 8/13, so 3B = 99/13, B = 33/13. Answer: (2x + 33)/(13(x^2 + 4)) - 2/(13(x + 3)). This agrees with the book's Answers (p. 417).

4. Resolve (x^2 + 15)/((x^2 + 2x + 5)(x - 1)) into partial fractions

Step 1: Write (x^2 + 15)/((x^2 + 2x + 5)(x - 1)) = (Ax + B)/(x^2 + 2x + 5) + C/(x - 1). Step 2: Identity: x^2 + 15 = (Ax + B)(x - 1) + C(x^2 + 2x + 5). Step 3: Put x = 1: 16 = C(8), so C = 2. Equating x^2 coefficients: 1 = A + C, so A = -1. Equating constants: 15 = -B + 5C = -B + 10, so B = -5. Answer: -(x + 5)/(x^2 + 2x + 5) + 2/(x - 1). This agrees with the book's Answers (p. 417).

5. Resolve x^2/((x^2 + 4)(x + 2)) into partial fractions

Step 1: Write x^2/((x^2 + 4)(x + 2)) = (Ax + B)/(x^2 + 4) + C/(x + 2). Step 2: Identity: x^2 = (Ax + B)(x + 2) + C(x^2 + 4). Step 3: Put x = -2: 4 = C(8), so C = 1/2. Equating x^2 coefficients: 1 = A + C, so A = 1/2. Equating constants: 0 = 2B + 4C = 2B + 2, so B = -1. Answer: (x - 2)/(2(x^2 + 4)) + 1/(2(x + 2)). This agrees with the book's Answers (p. 417).

6. Resolve (x^2 + 1)/(x^3 + 1) into partial fractions

Step 1: x^3 + 1 = (x + 1)(x^2 - x + 1). Write (x^2 + 1)/((x + 1)(x^2 - x + 1)) = (Ax + B)/(x^2 - x + 1) + C/(x + 1). Step 2: Identity: x^2 + 1 = (Ax + B)(x + 1) + C(x^2 - x + 1). Step 3: Put x = -1: 2 = C(3), so C = 2/3. Equating x^2 coefficients: 1 = A + C, so A = 1/3. Equating constants: 1 = B + C, so B = 1/3. Answer: (x + 1)/(3(x^2 - x + 1)) + 2/(3(x + 1)). This agrees with the book's Answers (p. 417).

7. Resolve (x^2 + 2x + 2)/((x^2 + 3)(x + 1)(x - 1)) into partial fractions

Step 1: (x + 1)(x - 1) = x^2 - 1. Write (x^2 + 2x + 2)/((x^2 + 3)(x^2 - 1)) = (Ax + B)/(x^2 + 3) + C/(x + 1) + D/(x - 1). Step 2: Identity: x^2 + 2x + 2 = (Ax + B)(x^2 - 1) + C(x^2 + 3)(x - 1) + D(x^2 + 3)(x + 1). Step 3: Put x = -1: 1 = C(4)(-2) = -8C, so C = -1/8. Put x = 1: 5 = D(4)(2) = 8D, so D = 5/8. Equating x^3 coefficients: 0 = A + C + D = A + 1/2, so A = -1/2. Equating constants: 2 = -B - 3C + 3D = -B + 3/8 + 15/8 = -B + 9/4, so B = 1/4. Answer: -(2x - 1)/(4(x^2 + 3)) - 1/(8(x + 1)) + 5/(8(x - 1)). FLAG: the printed key's form for this question does not recombine to the printed fraction (it differs by a non-zero rational function), so the key is misprinted here; the answer above recombines exactly to the fraction in the question.

8. Resolve 1/((x - 1)^2 (x^2 + 2)) into partial fractions

Step 1: Write 1/((x - 1)^2 (x^2 + 2)) = A/(x - 1) + B/(x - 1)^2 + (Cx + D)/(x^2 + 2). Step 2: Identity: 1 = A(x - 1)(x^2 + 2) + B(x^2 + 2) + (Cx + D)(x - 1)^2. Step 3: Put x = 1: 1 = B(3), so B = 1/3. Equating x^3 coefficients: 0 = A + C. Equating x coefficients: 0 = 2A + C - 2D. Equating constants: 1 = -2A + 2B + D = -2A + 2/3 + D. Solving with C = -A: from the x coefficient, 2A - A - 2D = 0, so D = A/2; from the constant, -2A + 2/3 + A/2 = 1, so -3A/2 = 1/3, A = -2/9, C = 2/9, D = -1/9. Answer: -2/(9(x - 1)) + 1/(3(x - 1)^2) + (2x - 1)/(9(x^2 + 2)). FLAG: the printed key's form does not recombine to 1/((x - 1)^2 (x^2 + 2)) (it differs by x/(2(x - 1)^2 (x^2 + 2))), so the key is misprinted here; the answer above recombines exactly.

9. Resolve x^4/(1 - x^4) into partial fractions

Step 1: x^4/(1 - x^4) = -1 + 1/(1 - x^4), since x^4 = -(1 - x^4) + 1. Step 2: 1 - x^4 = (1 - x)(1 + x)(1 + x^2). Write 1/((1 - x)(1 + x)(1 + x^2)) = A/(1 - x) + B/(1 + x) + (Cx + D)/(1 + x^2). Step 3: Identity: 1 = A(1 + x)(1 + x^2) + B(1 - x)(1 + x^2) + (Cx + D)(1 - x^2). Put x = 1: 1 = A(2)(2) = 4A, so A = 1/4. Put x = -1: 1 = B(2)(2) = 4B, so B = 1/4. Equating x^3 coefficients: 0 = A - B - C = -C, so C = 0. Equating constants: 1 = A + B + D = 1/2 + D, so D = 1/2. Answer: -1 + 1/(4(1 - x)) + 1/(4(1 + x)) + 1/(2(1 + x^2)). This agrees with the book's Answers (p. 417).

10. Resolve (x^2 - 2x + 3)/(x^4 + x^2 + 1) into partial fractions

Step 1: x^4 + x^2 + 1 = (x^2 + x + 1)(x^2 - x + 1). Write (x^2 - 2x + 3)/((x^2 + x + 1)(x^2 - x + 1)) = (Ax + B)/(x^2 + x + 1) + (Cx + D)/(x^2 - x + 1). Step 2: Identity: x^2 - 2x + 3 = (Ax + B)(x^2 - x + 1) + (Cx + D)(x^2 + x + 1). Step 3: Equating x^3 coefficients: 0 = A + C, so C = -A. Equating x coefficients: -2 = A - B + C + D = -B + D, so D = B - 2. Equating constants: 3 = B + D = 2B - 2, so B = 5/2, D = 1/2. Equating x^2 coefficients: 1 = -A + B - C + D = -2A + 3, so A = 1, C = -1. Answer: (2x + 5)/(2(x^2 + x + 1)) + (-2x + 1)/(2(x^2 - x + 1)). FLAG: the printed key's form recombines to (x^2 - 2x)/(x^4 + x^2 + 1), i.e. the numerator without the +3, so the key is misprinted here; the answer above recombines exactly to the printed fraction.

Exercise 5.4 (book p. 188)

1. Resolve (x^3 + 2x + 2)/(x^2 + x + 1)^2 into partial fractions

Step 1: Write (x^3 + 2x + 2)/(x^2 + x + 1)^2 = (Ax + B)/(x^2 + x + 1) + (Cx + D)/(x^2 + x + 1)^2. Step 2: Identity: x^3 + 2x + 2 = (Ax + B)(x^2 + x + 1) + Cx + D. Step 3: Equating x^3 coefficients: A = 1. Equating x^2 coefficients: A + B = 0, so B = -1. Equating x coefficients: A + B + C = 2, so C = 2. Equating constants: B + D = 2, so D = 3. Answer: (x - 1)/(x^2 + x + 1) + (2x + 3)/(x^2 + x + 1)^2. This agrees with the book's Answers (p. 417).

2. Resolve x^2/((x^2 + 1)^2 (x - 1)) into partial fractions

Step 1: Write x^2/((x^2 + 1)^2 (x - 1)) = (Ax + B)/(x^2 + 1) + (Cx + D)/(x^2 + 1)^2 + E/(x - 1). Step 2: Identity: x^2 = (Ax + B)(x^2 + 1)(x - 1) + (Cx + D)(x - 1) + E(x^2 + 1)^2. Step 3: Put x = 1: 1 = E(4), so E = 1/4. Equating x^4 coefficients: 0 = A + E, so A = -1/4. Equating x^3 coefficients: 0 = -A + B, so B = -1/4. Expanding (Ax + B)(x^3 - x^2 + x - 1) + (Cx + D)(x - 1) + E(x^2 + 1)^2: the x^2 coefficient is A - B + C + 2E = C + 1/2 = 1, so C = 1/2; the x coefficient is -A + B - C

+ D = D - 1/2 = 0, so D = 1/2. Answer: -(x + 1)/(4(x^2 + 1)) + (x + 1)/(2(x^2 + 1)^2) + 1/(4(x - 1)). This agrees with the book's Answers (p. 417).

3. Resolve (2x - 5)/((x^2 + 2)^2 (x - 2)) into partial fractions

Step 1: Write (2x - 5)/((x^2 + 2)^2 (x - 2)) = (Ax + B)/(x^2 + 2) + (Cx + D)/(x^2 + 2)^2 + E/(x - 2). Step 2: Identity: 2x - 5 = (Ax + B)(x^2 + 2)(x - 2) + (Cx + D)(x - 2) + E(x^2 + 2)^2. Step 3: Put x = 2: -1 = E(36), so E = -1/36. Equating x^4 coefficients: 0 = A + E, so A = 1/36. Equating x^3 coefficients: 0 = -2A + B, so B = 2/36 = 1/18. Equating x^2 coefficients: 0 = 2A - 2B + C + 4E = 1/18 - 1/9 + C - 1/9 = C - 1/6, so C = 1/6. Equating constants: -5 = -4B - 2D + 4E = -2/9 - 2D - 1/9 = -1/3 - 2D, so D = 7/3. Answer: (x + 2)/(36(x^2 + 2)) + (x + 14)/(6(x^2 + 2)^2) - 1/(36(x - 2)). This agrees with the book's Answers (p. 417).

4. Resolve 8x^2/((x^2 + 1)^2 (1 - x)) into partial fractions

Step 1: Write 8x^2/((x^2 + 1)^2 (1 - x)) = (Ax + B)/(x^2 + 1) + (Cx + D)/(x^2 + 1)^2 + E/(1 - x). Step 2: Identity: 8x^2 = (Ax + B)(x^2 + 1)(1 - x) + (Cx + D)(1 - x) + E(x^2 + 1)^2. Step 3: Put x = 1: 8 = E(4), so E = 2. Equating x^4 coefficients: 0 = -A + E, so A = 2. Equating x^3 coefficients: 0 = A - B, so B = 2. Expanding (Ax + B)(x^2 + 1)(1 - x) + (Cx + D)(1 - x) + E(x^2 + 1)^2: the x^2 coefficient is -A + B - C + 2E = -C + 4 = 8, so C = -4; the x coefficient is A - B - C + D? The full expansion gives D = -4. Answer: 2(x + 1)/(x^2 + 1) - 4(x + 1)/(x^2 + 1)^2 + 2/(1 - x). FLAG: the printed key's form, 1/(1 - x) + 1/(1 + x) + 2/(x^2 + 1) - 4/(x^2 + 1)^2, does not recombine to the printed fraction (it differs by 8x^3/((x^2 + 1)^2 (1 - x^2))), so the key is misprinted here; the answer above recombines exactly.

5. Resolve (4x^4 + 3x^3 + 6x^2 + 5x)/((x - 1)(x^2 + x + 1)^2) into partial fractions

Step 1: Write the fraction as (Ax + B)/(x^2 + x + 1) + (Cx + D)/(x^2 + x + 1)^2 + E/(x - 1). Step 2: Identity: 4x^4 + 3x^3 + 6x^2 + 5x = (Ax + B)(x^2 + x + 1)(x - 1) + (Cx + D)(x - 1) + E(x^2 + x + 1)^2. Step 3: Put x = 1: 4 + 3 + 6 + 5 = 18 = E(9),

so E = 2. Equating x^4 coefficients: 4 = A + E, so A = 2. Expanding (Ax + B)(x^3 - 1) + (Cx + D)(x - 1) + E(x^2 + x + 1)^2: the x^3 coefficient is B + 2E = 3, so B = -1; the x^2 coefficient is C + 3E = 6, so C = 0; the x coefficient is -A + D - C + 2E = D + 2 = 5, so D = 3. Answer: (2x - 1)/(x^2 + x + 1) + 3/(x^2 + x + 1)^2 + 2/(x - 1). This agrees with the book's Answers (p. 417).

6. Resolve (2x^4 - 3x^3 - 4x)/((x^2 + 2)^2 (x + 1)^2) into partial fractions

Step 1: Write the fraction as (Ax + B)/(x^2 + 2) + (Cx + D)/(x^2 + 2)^2 + E/(x + 1) + F/(x + 1)^2. Step 2: Identity: 2x^4 - 3x^3 - 4x = (Ax + B)(x^2 + 2)(x + 1)^2 + (Cx + D)(x + 1)^2 + E(x^2 + 2)^2 (x + 1) + F(x^2 + 2)^2. Step 3: Put x = -1: 2 + 3 + 4 = 9 = F(9), so F = 1. Equating x^5 coefficients: 0 = A + E. Equating x^4 coefficients: 2 = 2A + B + 4E? Solving the full system gives A = 1, B = 0, C = -2, D = 0, E = -1. Answer: x/(x^2 + 2) - 2x/(x^2 + 2)^2 - 1/(x + 1) + 1/(x + 1)^2. This agrees with the book's Answers (p. 417) (the key writes -1/(x + 1) + 1/(x + 1)^2 + x/(x^2 + 2) - 2x/(x^2 + 2)^2).

Chapter 6: Sequences and Series

Exercise 6.1 (book p. 191)

1. Write the first four terms of the following sequences, if i) a_n = 2n - 3 ii) a_n = (-1)^n n^2 iii) a_n = (-1)^n (2n - 3) iv) a_n = 3n - 5 v) a_n = n/(2n + 1) vi) a_n = 1/2^n vii) a_n - a_{n-1} = n + 2, a_1 = 2 viii) a_n = n a_{n-1}, a_1 = 1 ix) a_n = (n + 1) a_{n-1}, a_1 = 1 x) a_n = 1/(a + (n - 1)d)

i) Putting n = 1, 2, 3, 4: -1, 1, 3, 5. ii) Putting n = 1, 2, 3, 4: -1, 4, -9, 16. iii) Putting n = 1, 2, 3, 4: 1, 1, -3, 5. iv) Putting n = 1, 2, 3, 4: -2, 1, 4, 7. v) Putting n = 1, 2, 3, 4: 1/3, 2/5, 3/7, 4/9. vi) Putting n = 1, 2, 3, 4: 1/2, 1/4, 1/8, 1/16. vii) a_1 = 2; a_2 = a_1 + 4 = 6; a_3 = a_2 + 5 = 11; a_4 = a_3 + 6 = 17. So the first four terms are 2,

6, 11, 17. viii) a_1 = 1; a_2 = 2 x 1 = 2; a_3 = 3 x 2 = 6; a_4 = 4 x 6 = 24. So the first four terms are 1, 2, 6, 24. ix) a_1 = 1; a_2 = 3 x 1 = 3; a_3 = 4 x 3 = 12; a_4 = 5 x 12 = 60. So the first four terms are 1, 3, 12, 60. x) Putting n = 1, 2, 3, 4: 1/a, 1/(a + d), 1/(a + 2d), 1/(a + 3d).

2. Find the indicated terms of the following sequences: i) 2, 6, 11, 17, ... a_7 ii) 1, 3, 12, 60, ... a_6 iii) 1, 3/2, 5/4, 7/8, ... a_7 iv) 1, 1, -3, 5, -7, 9, ... a_8 v) 1, -3, 5, -7, 9, -11, ... a_8

i) The differences are 4, 5, 6, ..., so a_n = (n^2 + 5n - 2)/2 (check: n = 1 gives 2, n = 2 gives 6, n = 3 gives 11, n = 4 gives 17). Hence a_7 = (49 + 35 - 2)/2 = 41. ii) Each term is obtained by multiplying by 3, 4, 5, ..., so a_5 = 60 x 6 = 360 and a_6 = 360 x 7 = 2520. iii) The numerators are 1, 3, 5, 7, ... and the denominators are 1, 2, 4, 8, ..., so a_n = (2n - 1)/2^(n-1). Hence a_7 = 13/64. iv) From the second term onwards the terms are (-1)^n (2n - 3) (check: n = 2 gives 1, n = 3 gives -3, n = 4 gives 5, n = 5 gives -7, n = 6 gives 9), and a_1 = 1. Hence a_8 = (2 x 8 - 3) = 13. v) Here a_n = (-1)^(n+1) (2n - 1), so a_8 = -(2 x 8 - 1) = -15.

3. Find the next two terms of the following sequences: i) 7, 9, 12, 16, ... ii) 1, 3, 7, 15, 31, ... iii) -1, 2, 12, 40, ... iv) 1, -3, 5, -7, 9, -11, ...

i) The differences are 2, 3, 4, ..., so the next differences are 5 and 6, giving 16 + 5 = 21 and 21 + 6 = 27. ii) The differences are 2, 4, 8, 16, ..., so the next differences are 32 and 64, giving 31 + 32 = 63 and 63 + 64 = 127. iii) The terms satisfy a_{n+1} = 2a_n + 2^(n+1) (check: 2 x (-1) + 4 = 2, 2 x 2 + 8 = 12, 2 x 12 + 16 = 40), so the next terms are 2 x 40 + 32 = 112 and 2 x 112 + 64 = 288. iv) Here a_n = (-1)^(n+1) (2n - 1), so the next terms are a_7 = -13 and a_8 = 15. FLAG: the printed key writes these as 13, -15, with the signs reversed; the sequence 1, -3, 5, -7, 9, -11 continues with -13, 15.

Exercise 6.2 (book p. 194)

1. Write the first four terms of the following arithmetic sequences, if i) $a_1 = 5$ and other three consecutive terms are 23, 26, 29 ii) $a_5 = 17$ and $a_9 = 37$ iii) $3a_7 = 7a_4$ and $a_{10} = 33$

i) The three consecutive terms 23, 26, 29 give the common difference $d = 3$. With $a_1 = 5$, the first four terms are 5, 8, 11, 14. ii) $a_9 - a_5 = 4d = 37 - 17 = 20$, so $d = 5$. Then $a_1 = a_5 - 4d = 17 - 20 = -3$, so the first four terms are -3, 2, 7, 12. iii) $3(a_1 + 6d) = 7(a_1 + 3d)$ gives $-4a_1 = 3d$, i.e. $a_1 = -3d/4$. Also $a_{10} = a_1 + 9d = 33$, so $-3d/4 + 9d = 33$, i.e. $33d/4 = 33$, so $d = 4$ and $a_1 = -3$. Hence the first four terms are -3, 1, 5, 9.

2. If $a_{n-3} = 2n - 5$, find the nth term of the sequence. 3. If the 5th term of an A.P. is 16 and the 20th term is 46, what is its 12th term? 4. Find the 13th term of the sequence x, 1, 2 - x, 3 - 2x, ... 5. Find the 18th term of the A.P. if its 6th term is 19 and the 9th term is 31.

2. Put $m = n - 3$, so $n = m + 3$ and $a_m = 2(m + 3) - 5 = 2m + 1$. Hence $a_n = 2n + 1$. 3. $a_{20} - a_5 = 15d = 46 - 16 = 30$, so $d = 2$. Then $a_{12} = a_5 + 7d = 16 + 14 = 30$. 4. The common difference is $1 - x$, so $a_{13} = x + 12(1 - x) = 12 - 11x$. 5. $a_9 - a_6 = 3d = 31 - 19 = 12$, so $d = 4$. Then $a_{18} = a_6 + 12d = 19 + 48 = 67$.

6. Which term of the A.P. 5, 2, -1, ... is -85? 7. Which term of the A.P. -2, 4, 10, ... is 148? 8. How many terms are there in the A.P. in which $a_1 = 11$, $a_n = 68$, $d = 3$? 9. If the nth term of the A.P. is $3n - 1$, find the A.P. 10. Determine whether i) -19, ii) 2 are the terms of the A.P. 17, 13, 9, ... or not.

6. Here $d = -3$. Setting $5 + (n - 1)(-3) = -85$ gives $(n - 1)(-3) = -90$, so $n - 1 = 30$ and $n = 31$. So -85 is the 31st term. 7. Here $d = 6$. Setting $-2 + (n - 1)6 = 148$ gives $(n - 1)6 = 150$, so $n - 1 = 25$ and $n = 26$. So 148 is the 26th term. 8. Setting $11 + (n - 1)3 = 68$ gives $(n - 1)3 = 57$, so $n - 1 = 19$ and $n = 20$. So there are 20 terms. 9. Putting $n = 1, 2, 3, \ldots$: 2, 5, 8, ... So the A.P. is 2, 5, 8, 11, ... 10. Here $d = -4$. i) Setting $17 + (n - 1)(-4) = -19$ gives $(n - 1)(-4) = -36$, so $n = 10$: -19 is the 10th term.

ii) Setting 17 + (n - 1)(-4) = 2 gives (n - 1)(-4) = -15, so n - 1 = 15/4, which is not an integer: 2 is not a term of the A.P.

11. If l, m, n are the pth, qth and rth terms of an A.P., show that i) l(q - r) + m(r - p) + n(p - q) = 0 ii) p(m - n) + q(n - l) + r(l - m) = 0 12. Find the nth term of the sequence, (4/3)^2, (7/3)^2, (10/3)^2, ... 13. If 1/a, 1/b and 1/c are in A.P., show that b = 2ac/(a + c) 14. If 1/a, 1/b and 1/c are in A.P., show that the common difference is (a - c)/(2ac)

11. Write l = a + (p - 1)d, m = a + (q - 1)d, n = a + (r - 1)d. i) l(q - r) + m(r - p) + n(p - q) = a[(q - r) + (r - p) + (p - q)] + d[(p - 1)(q - r) + (q - 1)(r - p) + (r - 1)(p - q)] = 0 + d[0] = 0, since the bracket expands to pq - pr - q + r + qr - pq - r + p + rp - rq - p + q = 0. ii) m - n = (q - r)d, n - l = (r - p)d, l - m = (p - q)d, so p(m - n) + q(n - l) + r(l - m) = d[p(q - r) + q(r - p) + r(p - q)] = d[0] = 0. 12. The bases 4/3, 7/3, 10/3, ... are in A.P. with first term 4/3 and common difference 1, so the nth base is (3n + 1)/3. Hence the nth term is ((3n + 1)/3)^2. 13. Since 1/a, 1/b, 1/c are in A.P., 2/b = 1/a + 1/c = (a + c)/(ac), so b = 2ac/(a + c). 14. The common difference is 1/b - 1/a = (a + c)/(2ac) - 1/a = (a - c)/(2ac).

Exercise 6.3 (book p. 196)

1. Find A.M. between i) 3 sqrt(5) and 5 sqrt(5) ii) x - 3 and x + 5 iii) 1 - x + x^2 and 1 + x + x^2 2. If 5, 8 are two A.Ms between a and b, find a and b. 3. Find 6 A.Ms. between 2 and 5. 4. Find four A.Ms. between sqrt(2) and 12/sqrt(2). 5. Insert 7 A.Ms. between 4 and 8. 6. Find three A.Ms between 3 and 11.

1. i) (3 sqrt(5) + 5 sqrt(5))/2 = 4 sqrt(5). ii) ((x - 3) + (x + 5))/2 = x + 1. iii) ((1 - x + x^2) + (1 + x + x^2))/2 = 1 + x^2. 2. Since a, 5, 8, b are in A.P., the common difference is 8 - 5 = 3, so a = 5 - 3 = 2 and b = 8 + 3 = 11. 3. The common difference is (5 - 2)/7 = 3/7, so the six A.Ms are 17/7, 20/7, 23/7, 26/7, 29/7, 32/7. 4. Since 12/sqrt(2) = 6 sqrt(2), the common difference is (6 sqrt(2) - sqrt(2))/5 =

sqrt(2), so the four A.Ms are 2 sqrt(2), 3 sqrt(2), 4 sqrt(2), 5 sqrt(2). 5. The common difference is (8 - 4)/8 = 1/2, so the seven A.Ms are 9/2, 5, 11/2, 6, 13/2, 7, 15/2. 6. The common difference is (11 - 3)/4 = 2, so the three A.Ms are 5, 7, 9.

7. Find n so that (a^n + b^n)/(a^(n-1) + b^(n-1)) may be the A.M. between a and b. 8. Show that the sum of n A.Ms. between a and b is equal to n times their A.M.

7. Setting (a^n + b^n)/(a^(n-1) + b^(n-1)) = (a + b)/2 gives 2(a^n + b^n) = (a + b)(a^(n-1) + b^(n-1)), i.e. 2a^n + 2b^n = a^n + ab^(n-1) + a^(n-1)b + b^n, i.e. a^n + b^n - ab^(n-1) - a^(n-1)b = 0, i.e. (a - b)(a^(n-1) - b^(n-1)) = 0. For a != b this gives a^(n-1) = b^(n-1), so n - 1 = 0 and n = 1. 8. The n A.Ms are a + d, a + 2d, ..., a + nd, where d = (b - a)/(n + 1). Their sum is na + d(1 + 2 + ... + n) = na + (b - a)n/2 = n(a + b)/2 = n times the A.M. (a + b)/2, as required.

Exercise 6.4 (book pp. 199-200)

1. Find the sum of all the integral multiples of 3 between 4 and 97. 2. Sum the series i) -3 + (-1) + 1 + 3 + 5 + ... + a_16 ii) 3/sqrt(2) + 2 sqrt(2) + 5/sqrt(2) + ... + a_13 iii) 1.11 + 1.41 + 1.71 + ... + a_10 iv) -8 - 3 1/2 + 1 + ... + a_11 v) (x - a) + (x + a) + (x + 3a) + ... to n terms vi) 1/(1 - sqrt(x)) + 1/(1 - x) + 1/(1 + sqrt(x)) + ... to n terms vii) 1/(1 + sqrt(x)) + 1/(1 - x) + 1/(1 - sqrt(x)) + ... to n terms

1. The multiples are 6, 9, ..., 96: an A.P. with a = 6, d = 3 and 96 = 6 + 30 x 3, so n = 31. Sum = (31/2)(6 + 96) = 31 x 51 = 1581. 2. i) Here a = -3, d = 2, a_16 = -3 + 15 x 2 = 27. Sum = (16/2)(-3 + 27) = 192. ii) The terms are 3/sqrt(2), 4/sqrt(2), 5/sqrt(2), ..., so d = 1/sqrt(2) and a_13 = 15/sqrt(2). Sum = (13/2)(3/sqrt(2) + 15/sqrt(2)) = 117/sqrt(2). iii) Here a = 1.11, d = 0.30, a_10 = 1.11 + 9 x 0.30 = 3.81. Sum = 5(1.11 + 3.81) = 24.6. iv) Here a = -8, d = 4.5, a_11 = -8 + 10 x 4.5 = 37. Sum = (11/2)(-8 + 37) = 319/2 = 159.5. v) Here a = x - a, d = 2a. Sum = (n/2)[2(x - a) + (n - 1)2a] = n[x + (n - 2)a]. vi) The common difference is -sqrt(x)/(1 - x) (check: 1/(1 - x) - 1/(1 - sqrt(x)) = -sqrt(x)/(1 - x), and the same for the next pair).

Sum = (n/2)[2/(1 - sqrt(x)) + (n - 1)(-sqrt(x)/(1 - x))] = (n/2)[(2 + (3 - n)sqrt(x))/(1 - x)]. vii) The common difference is +sqrt(x)/(1 - x), so similarly the sum is (n/2)[(2 + (n - 3)sqrt(x))/(1 - x)].

3. How many terms of the series i) -7 + (-5) + (-3) + ... amount to 65? ii) -7 + (-4) + (-1) + ... amount to 114? 4. Sum the series i) 3 + 5 - 7 + 9 + 11 - 13 + ... to 3n terms ii) 1 + 4 - 7 + 10 + 13 - 16 + ... to 3n terms 5. Find the sum of 20 terms of the series whose rth term is 3r + 1. 6. If S_n = n(2n - 1), then find the series. 7. The ratio of the sums of n terms of two series in A.P. is 3n + 2 : n + 1. Find the ratio of their 8th terms. 8. If S_2, S_3, S_5 are the sums of 2n, 3n, 5n terms of an A.P., show that S_5 = 5(S_3 - S_2).

3. i) Here a = -7, d = 2: S_n = (n/2)[-14 + 2(n - 1)] = n(n - 8) = 65, i.e. n^2 - 8n - 65 = (n - 13)(n + 5) = 0, so n = 13 terms. ii) Here a = -7, d = 3: S_n = (n/2)[-14 + 3(n - 1)] = n(3n - 17)/2 = 114, i.e. 3n^2 - 17n - 228 = 0, i.e. (3n + 19)(n - 12) = 0, so n = 12 terms. 4. i) Grouping in threes: (3 + 5 - 7) = 1, (9 + 11 - 13) = 7, (15 + 17 - 19) = 13, ...: an A.P. with first term 1 and common difference 6. Sum of n groups = (n/2)[2 + 6(n - 1)] = n(3n - 2). ii) Grouping in threes: (1 + 4 - 7) = -2, (10 + 13 - 16) = 7, (19 + 22 - 25) = 16, ...: an A.P. with first term -2 and common difference 9. Sum of n groups = (n/2)[-4 + 9(n - 1)] = (n/2)(9n - 13). 5. S_20 = sum_{r=1}^{20} (3r + 1) = 3(20 x 21/2) + 20 = 630 + 20 = 650. 6. a_n = S_n - S_{n-1} = n(2n - 1) - (n - 1)(2n - 3) = 4n - 3. So the series is 1, 5, 9, 13, ... 7. For an A.P., the nth term is S_{2n-1}/(2n - 1) (since S_{2n-1} = (2n - 1)a_n). Hence the ratio of the 8th terms is S_15/S'_15 = (3 x 15 + 2)/(15 + 1) = 47/16. 8. S_2 = n[2a + (2n - 1)d], S_3 = (3n/2)[2a + (3n - 1)d], S_5 = (5n/2)[2a + (5n - 1)d]. Then S_3 - S_2 = 3na + (9n^2 - 3n)d/2 - 2na - (2n^2 - n)d = na + (5n^2 - n)d/2 = (n/2)[2a + (5n - 1)d] = S_5/5, so S_5 = 5(S_3 - S_2), as required.

9. Obtain the sum of all integers in the first 1000 integers which are neither divisible by 5 nor by 2. 10. S_8 and S_9 are the sums of the first eight and nine terms of an

A.P., find S_9 if 50S_9 = 63S_8 and a_1 = 2. (Hint: S_9 = S_8 + a_9) 11. The sum of 9 terms of an A.P. is 171 and its eighth term is 31. Find the series. 12. The sum of S_9 and S_7 is 203 and S_9 - S_7 = 49, S_7 and S_9 being the sums of the first 7 and 9 terms of an A.P. respectively. Determine the series. 13. S_7 and S_9 are the sums of the first 7 and 9 terms of an A.P. respectively. If S_9/S_7 = 18/11 and a_7 = 20, find the series.

9. Sum of 1 to 1000 = 500500. Sum of even numbers = 2 + 4 + ... + 1000 = 250500. Sum of multiples of 5 = 5 + 10 + ... + 1000 = 100500. Sum of multiples of 10 = 50500. By inclusion-exclusion, the required sum = 500500 - 250500 - 100500 + 50500 = 200000. 10. S_8 = 4(4 + 7d) = 16 + 28d and S_9 = (9/2)(4 + 8d) = 18 + 36d. Then 50(18 + 36d) = 63(16 + 28d), i.e. 900 + 1800d = 1008 + 1764d, so 36d = 108 and d = 3. Hence S_9 = 18 + 108 = 126. 11. S_9 = (9/2)(2a + 8d) = 171 gives a + 4d = 19; and a_8 = a + 7d = 31. Subtracting gives 3d = 12, so d = 4 and a = 3. Hence the series is 3, 7, 11, ... 12. Solving S_9 + S_7 = 203 and S_9 - S_7 = 49 gives S_9 = 126 and S_7 = 77. From S_7 = (7/2)(2a + 6d) = 77: a + 3d = 11. From S_9 = (9/2)(2a + 8d) = 126: a + 4d = 14. Hence d = 3 and a = 2, so the series is 2, 5, 8, 11, ... 13. From a_7 = a + 6d = 20 and S_9/S_7 = [9(a + 4d)]/[7(a + 3d)] = 18/11, we get 11(a + 4d) = 14(a + 3d), i.e. 2d = 3a. With a + 6d = 20 this gives d = 3 and a = 2, so the series is 2, 5, 8, 11, ...

14. The sum of three numbers in an A.P. is 24 and their product is 440. Find the numbers. 15. Find four numbers in A.P. whose sum is 32 and the sum of whose squares is 276. 16. Find the five numbers in A.P. whose sum is 25 and the sum of whose squares is 135. 17. The sum of the 6th and 8th terms of an A.P. is 40 and the product of 4th and 7th terms is 220. Find the A.P. 18. If a^2, b^2 and c^2 are in A.P., show that 1/(b + c), 1/(c + a), 1/(a + b) are in A.P.

14. Let the numbers be a - d, a, a + d: their sum is 3a = 24, so a = 8. Their product is 8(64 - d^2) = 440, so 64 - d^2 = 55 and d^2 = 9, i.e. d = +- 3. Hence the

numbers are 5, 8, 11 or 11, 8, 5. 15. Let the numbers be a - 3d, a - d, a + d, a + 3d: their sum is 4a = 32, so a = 8. The sum of their squares is 4a^2 + 20d^2 = 256 + 20d^2 = 276, so d^2 = 1 and d = +- 1. Hence the numbers are 5, 7, 9, 11 or 11, 9, 7, 5. 16. Let the numbers be a - 2d, a - d, a, a + d, a + 2d: their sum is 5a = 25, so a = 5. The sum of their squares is 5a^2 + 10d^2 = 125 + 10d^2 = 135, so d^2 = 1 and d = +- 1. Hence the numbers are 3, 4, 5, 6, 7 or 7, 6, 5, 4, 3. 17. a_6 + a_8 = 2a + 12d = 40, so a + 6d = 20. Also a_7 = a + 6d = 20, and a_4 x a_7 = 220 gives a_4 = 11, i.e. a + 3d = 11. Hence 3d = 9, so d = 3 and a = 2. So the A.P. is 2, 5, 8, 11, ... 18. We need 2/(c + a) = 1/(b + c) + 1/(a + b) = (a + 2b + c)/((a + b)(b + c)), i.e. 2(a + b)(b + c) = (c + a)(a + c + 2b). Expanding: the left side is 2ab + 2b^2 + 2ac + 2bc and the right side is a^2 + 2ac + c^2 + 2ab + 2bc. The two sides are equal exactly when a^2 + c^2 = 2b^2, which is the given A.P. condition. Hence 1/(b + c), 1/(c + a), 1/(a + b) are in A.P., as required.

Exercise 6.5 (book pp. 202-203)

1. A man deposits in a bank Rs. 10 in the first month; Rs. 15 in the second month; Rs. 20 in the third month and so on. Find how much he will have deposited in the bank by the 9th month. 2. 378 trees are planted in rows in the shape of an isosceles triangle, the numbers in successive rows decreasing by one from the base to the top. How many trees are there in the row which forms the base of the triangle? 3. A man borrows Rs. 1100 and agree to repay with a total interest of Rs. 230 in 14 installments, each installment being less than the preceding by Rs. 10. What should be his first installment? 4. A clock strikes once when its hour hand is at one, twice when it is at two and so on. How many times does the clock strike in twelve hours? 5. A student saves Rs. 12 at the end of the first week and goes on increasing his saving Rs. 4 weekly. After how many weeks will he be able to save Rs. 2100?

1. The deposits form an A.P. with a = 10 and d = 5. The total by the 9th month is

S_9 = (9/2)[20 + 8 x 5] = (9/2)(60) = Rs. 270. 2. If the base row has n trees, the rows have n, n - 1, ..., 1 trees, so n(n + 1)/2 = 378, i.e. n^2 + n - 756 = (n + 28)(n - 27) = 0, so n = 27 trees. 3. The total repaid is 1100 + 230 = Rs. 1330. With d = -10 and n = 14: S_14 = 7[2a - 130] = 1330, so 2a - 130 = 190 and a = Rs. 160. So the first installment is Rs. 160. 4. The number of strikes is 1 + 2 + ... + 12 = 12 x 13/2 = 78. 5. The weekly savings form an A.P. with a = 12 and d = 4. Setting S_n = (n/2)[24 + 4(n - 1)] = 2n(n + 5) = 2100 gives n(n + 5) = 1050, i.e. n^2 + 5n - 1050 = (n + 35)(n - 30) = 0, so n = 30 weeks.

6. An object falling from rest, falls 9 meters during the first second, 27 meters during the next second, 45 meters during the third second and so on. i) How far will it fall during the fifth second? ii) How far will it fall up to the fifth second? 7. An investor earned Rs. 6000 for year 1980 and Rs. 12000 for year 1990 on the same investment. If his earning have increased by the same amount each year, how much income he has received from the investment over the past eleven years? 8. The sum of interior angles of polygons having sides 3, 4, 5, ... etc. form an A.P. Find the sum of the interior angles for a 16 sided polygon. 9. The prize money Rs. 60,000 will be distributed among the eight teams according to their positions determined in the match-series. The award increases by the same amount for each higher position. If the last place team is given Rs. 4000, how much will be awarded to the first place team? 10. An equilateral triangular base is filled by placing eight balls in the first row, 7 balls in the second row and so on with one ball in the last row. After this base layer, second layer is formed by placing 7 balls in its first row, 6 balls in its second row and so on with one ball in its last row. Continuing this process, a pyramid of balls is formed with one ball on top. How many balls are there in the pyramid?

6. The distances form an A.P. with a = 9 and d = 18. i) During the fifth second: a_5 = 9 + 4 x 18 = 81 meters. ii) Up to the fifth second: S_5 = (5/2)[18 + 4 x 18] =

(5/2)(90) = 225 meters. 7. The yearly earnings form an A.P. with first term 6000 (1980) and 11th term 12000 (1990). The total over the eleven years is S_11 = (11/2)(6000 + 12000) = Rs. 99000. 8. The sums of interior angles are 180, 360, 540, ... degrees: an A.P. with a = 180 and d = 180. For a 16-sided polygon (the 14th term, since 3 sides give the 1st term): a_14 = 180 + 13 x 180 = 2520 degrees = 14 pi radians. 9. The awards form an A.P. with last term 4000 and S_8 = 60000. Writing the awards from last place to first place: a = 4000, so 4[8000 + 7d] = 60000, i.e. 8000 + 7d = 15000 and d = 1000. Hence the first place award is 4000 + 7 x 1000 = Rs. 11000. 10. The bottom layer has 8 + 7 + ... + 1 = 36 balls; the next layer has 7 + 6 + ... + 1 = 28 balls; then 21, 15, 10, 6, 3, 1 balls. Total = 36 + 28 + 21 + 15 + 10 + 6 + 3 + 1 = 120 balls.

Exercise 6.6 (book pp. 205-206)

1. Find the 5th term of the G.P.: 3, 6, 12, ... 2. Find the 11th term of the sequence, 1 + i, 2, 4/(1 + i) 3. Find the 12th term of 1 + i, 2i, -2 + 2i, ... 4. Find the 11th term of the sequence, 1 + i, 2, 2(1 - i) 5. If an automobile depreciates in values 5% every year, at the end of 4 years what is the value of the automobile purchased for Rs. 12,000? 6. Which term of the sequence: x^2 - y^2, x + y, (x + y)/(x - y), ... is (x + y)/(x - y)^9?

1. Here a = 3 and r = 2, so a_5 = 3 x 2^4 = 48. 2. Here r = 2/(1 + i) = 1 - i, so a_11 = (1 + i)(1 - i)^10. Since (1 - i)^2 = -2i, (1 - i)^10 = (-2i)^5 = -32i, so a_11 = (1 + i)(-32i) = 32(1 - i). 3. Here r = 2i/(1 + i) = 1 + i, so a_12 = (1 + i)(1 + i)^11 = (1 + i)^12 = ((1 + i)^2)^6 = (2i)^6 = -64. 4. Here r = 2/(1 + i) = 1 - i, so a_11 = (1 + i)(1 - i)^10 = 32(1 - i), as in Q2. 5. The value after 4 years is 12000 x (0.95)^4 = 12000 x 0.81450625 = 9774.075, i.e. about Rs. 9774. 6. Here a_1 = (x - y)(x + y) and r = 1/(x - y), so a_n = (x + y)/(x - y)^(n-2). Setting this equal to (x + y)/(x - y)^9 gives n - 2 = 9, so n = 11. Hence it is the 11th term.

7. If a, b, c, d are in G.P., prove that i) a - b, b - c, c - d are in G.P. ii) a^2 - b^2, b^2 - c^2, c^2 - d^2 are in G.P. iii) a^2 + b^2, b^2 + c^2, c^2 + d^2 are in G.P. 8. Show that the reciprocals of the terms of the geometric sequence a_1, a_1 r^2, a_1 r^4, ... form another geometric sequence.

7. Write b = ar, c = ar^2, d = ar^3. i) a - b = a(1 - r), b - c = ar(1 - r), c - d = ar^2(1 - r): a G.P. with common ratio r. ii) a^2 - b^2 = a^2(1 - r^2), b^2 - c^2 = a^2 r^2(1 - r^2), c^2 - d^2 = a^2 r^4(1 - r^2): a G.P. with common ratio r^2. iii) a^2 + b^2 = a^2(1 + r^2), b^2 + c^2 = a^2 r^2(1 + r^2), c^2 + d^2 = a^2 r^4(1 + r^2): a G.P. with common ratio r^2. 8. The reciprocals are 1/a_1, 1/(a_1 r^2), 1/(a_1 r^4), ...: each term is 1/r^2 times the preceding term, so they form a G.P. with common ratio 1/r^2.

9. Find the nth term of the geometric sequence if: a_5/a_3 = 4/9 and a_2 = 4/9 10. Find three consecutive numbers in G.P. whose sum is 26 and their product is 216. 11. If the sum of the four consecutive terms of a G.P. is 80 and A.M. of the second and the fourth of them is 30. Find the terms. 12. If 1/a, 1/b and 1/c are in G.P., show that the common ratio is +- sqrt(a/c)

9. Since a_5/a_3 = r^2 = 4/9, r = 2/3 or r = -2/3. Also a_2 = a_1 r = 4/9. For r = 2/3: a_1 = 2/3 and a_n = (2/3)^n. For r = -2/3: a_1 = -2/3 and a_n = (-2/3)^n = (-1)^n (2/3)^n. (The printed key gives the second form.) 10. Let the numbers be a/r, a, ar: their product is a^3 = 216, so a = 6. Their sum is 6(1/r + 1 + r) = 26, i.e. 1/r + r = 10/3, i.e. r^2 - (10/3)r + 1 = 0, so r = 3 or r = 1/3. Hence the numbers are 2, 6, 18 or 18, 6, 2. 11. Let the terms be a, ar, ar^2, ar^3: a(1 + r + r^2 + r^3) = 80 and (ar + ar^3)/2 = 30, i.e. ar(1 + r^2) = 60. Dividing: (1 + r)/r = 80/60 = 4/3, so 3 + 3r = 4r and r = 3. Then a(1 + 3)(1 + 9) = 80, so a = 2. Hence the terms are 2, 6, 18, 54. 12. Since 1/a, 1/b, 1/c are in G.P., (1/b)^2 = (1/a)(1/c), so b^2 = ac and b = +- sqrt(ac). The common ratio is (1/b)/(1/a) = a/b = a/(+- sqrt(ac)) = +- sqrt(a/c).

13. If the numbers 1, 4 and 3 are subtracted from three consecutive terms of an

A.P., the resulting numbers are in G.P. Find the numbers if their sum is 21. 14. If three consecutive numbers in A.P. are increased by 1, 4, 15 respectively, the resulting numbers are in G.P. Find the original numbers if their sum is 6.

13. Let the A.P. terms be a - d, a, a + d: their sum is 3a = 21, so a = 7. After subtracting: 6 - d, 3, 4 + d are in G.P., so 3^2 = (6 - d)(4 + d), i.e. 9 = 24 + 2d - d^2, i.e. d^2 - 2d - 15 = (d - 5)(d + 3) = 0, so d = 5 or d = -3. Hence the numbers are 2, 7, 12 or 10, 7, 4. 14. Let the A.P. terms be a - d, a, a + d: their sum is 3a = 6, so a = 2. After increasing: 3 - d, 6, 17 + d are in G.P., so 6^2 = (3 - d)(17 + d), i.e. 36 = 51 - 14d - d^2, i.e. d^2 + 14d - 15 = (d + 15)(d - 1) = 0, so d = 1 or d = -15. Hence the original numbers are 1, 2, 3 or 17, 2, -13.

Exercise 6.7 (book p. 209)

1. Find G.M. between i) -2 and 8 ii) -2i and 8i 2. Insert two G.Ms. between i) 1 and 8 ii) 2 and 16 3. Insert three G.Ms. between i) 1 and 16 ii) 2 and 32 4. Insert four real geometric means between 3 and 96.

1. i) The G.M. is +- sqrt((-2)(8)) = +- sqrt(-16) = +- 4i. ii) The G.M. is +- sqrt((-2i)(8i)) = +- sqrt(16) = +- 4. 2. i) With two means, r^3 = 8/1 = 8, so r = 2 and the means are 2, 4. ii) With two means, r^3 = 16/2 = 8, so r = 2 and the means are 4, 8. 3. i) With three means, r^4 = 16, so r = 2, -2, 2i or -2i, giving the sets (2, 4, 8), (-2, 4, -8), (2i, -4, -8i) and (-2i, -4, 8i). ii) With three means, r^4 = 32/2 = 16, so r = 2, -2, 2i or -2i, giving the sets (4, 8, 16), (-4, 8, -16), (4i, -8, -16i) and (-4i, -8, 16i). 4. With four real means, r^5 = 96/3 = 32, so r = 2 and the means are 6, 12, 24, 48. FLAG: the printed key appears to read 8, 12, 24, 48 for this part, but the first mean must be 3 x 2 = 6 (the full sequence is 3, 6, 12, 24, 48, 96, with common ratio 2 throughout); the key's first value is treated as a misprint.

5. If both x and y are positive distinct real numbers, show that the geometric mean between x and y is less than their arithmetic mean. 6. For what value of n, (a^n +

b^n)/(a^(n-1) + b^(n-1)) is the positive geometric mean between a and b? 7. The A.M. of two positive integral numbers exceeds their (positive) G.M. by 2 and their sum is 20, find the numbers. 8. The A.M. between two numbers is 5 and their (positive) G.M. is 4. Find the numbers.

5. Since x != y, (sqrt(x) - sqrt(y))^2 > 0, i.e. x + y > 2 sqrt(xy), i.e. (x + y)/2 > sqrt(xy). So the geometric mean sqrt(xy) is less than the arithmetic mean (x + y)/2, as required. 6. Setting (a^n + b^n)/(a^(n-1) + b^(n-1)) = sqrt(ab) gives a^n + b^n = sqrt(ab)(a^(n-1) + b^(n-1)), i.e. a^n - a^(n-1/2) b^(1/2) - a^(1/2) b^(n-1/2) + b^n = 0, i.e. (a^(n-1/2) - b^(n-1/2))(sqrt(a) - sqrt(b)) = 0. For a != b this gives a^(n-1/2) = b^(n-1/2), so n - 1/2 = 0 and n = 1/2. 7. Let the numbers be x and y: (x + y)/2 - sqrt(xy) = 2 and x + y = 20, so sqrt(xy) = 8 and xy = 64. Hence x and y are the roots of t^2 - 20t + 64 = (t - 16)(t - 4) = 0, so the numbers are 16 and 4. 8. Let the numbers be x and y: x + y = 10 and xy = 16. Hence x and y are the roots of t^2 - 10t + 16 = (t - 8)(t - 2) = 0, so the numbers are 8 and 2.

Exercise 6.8 (book pp. 215-216)

1. Find the sum of first 15 terms of the geometric sequence 1, 1/3, 1/9, ... 2. Sum to n terms the series i) .2 + .22 + .222 + ... ii) 3 + 33 + 333 + ... 3. Sum to n terms the series i) 1 + (a + b) + (a^2 + ab + b^2) + (a^3 + a^2 b + ab^2 + b^3) + ... ii) r + (1 + k) r^2 + (1 + k + k^2) r^3 + ...

1. Here a = 1 and r = 1/3, so S_15 = (1 - (1/3)^15)/(1 - 1/3) = (3/2)(1 - 1/14348907) = 7174453/4782969. 2. i) Each term is 2 x (1 - 10^(-k))/9, so the sum is (2/9)[n - (1/9)(1 - 1/10^n)]. FLAG: the printed key's factor here appears to read 1/n, but the correct factor is 1/9 (from the geometric sum (1/10 + 1/100 + ... + 1/10^n) = (1/9)(1 - 1/10^n)). ii) Each term is 3 x (10^k - 1)/9, so the sum is (1/3)[(10/9)(10^n - 1) - n]. 3. i) The kth term (k = 0, 1, ...) is (a^(k+1) - b^(k+1))/(a - b), so the sum to n terms is [a(a^n - 1)/(a - 1) - b(b^n - 1)/(b - 1)]/(a - b) = [a(b - 1)(a^n - 1) - (a -

1)b(b^n - 1)]/[(a - b)(a - 1)(b - 1)]. ii) The kth term is (1 - k^k)/(1 - k) x r^k, so the sum is (1/(1 - k))[r(1 - r^n)/(1 - r) - kr(1 - (kr)^n)/(1 - kr)] = (r/(1 - k))[(1 - r^n)/(1 - r) - k(1 - k^n r^n)/(1 - kr)].

4. Sum the series 2 + (1 - i) + (1/i) + ... to 8 terms. 5. Find the sums of the following infinite geometric series: i) 1/5 + 1/25 + 1/125 + ... ii) 1/2 + 1/4 + 1/8 + ... iii) 9/4 + 3/2 + 1 + 2/3 + ... iv) 2 + 1 + 0.5 + ... v) 4 + 2 sqrt(2) + 2 + sqrt(2) + 1 + ... vi) 0.1 + 0.05 + 0.025 + ... 6. Find vulgar fractions equivalent to the following recurring decimals: i) 1.3434... ii) 0.777... iii) 0.259259... iv) 1.5353... v) 0.159159... vi) 1.1474747...

4. Here r = (1 - i)/2 (check: (1/i)/(1 - i) = -i/(1 - i) = (1 - i)/2). Since r^2 = -i/2, r^4 = -1/4 and r^8 = 1/16, S_8 = 2(1 - r^8)/(1 - r) = 2(15/16)/((1 + i)/2) = 15(1 - i)/8. 5. i) S = (1/5)/(1 - 1/5) = 1/4. ii) S = (1/2)/(1 - 1/2) = 1. iii) Here a = 9/4 and r = 2/3, so S = (9/4)/(1/3) = 27/4. iv) S = 2/(1 - 1/2) = 4. v) Here a = 4 and r = sqrt(2)/2, so S = 4/(1 - sqrt(2)/2) = 8/(2 - sqrt(2)) = 4(2 + sqrt(2)). vi) S = 0.1/(1 - 1/2) = 0.2. 6. i) 1.3434... = 1 + 34/99 = 133/99. ii) 0.777... = 7/9. iii) 0.259259... = 259/999. iv) 1.5353... = 1 + 53/99 = 152/99. v) 0.159159... = 159/999. vi) 1.1474747... = 1 + (147 - 1)/990 = 1136/990.

7. Find the sum to infinity of the series: r + (1 + k) r^2 + (1 + k + k^2) r^3 + ...; r and k being proper fractions. 8. If y = x/2 + (1/4) x^2 + (1/8) x^3 + ... and if 0 < x < 2, then prove that x = 2y/(1 + y) 9. If y = (2/3) x + (4/9) x^2 + (8/27) x^3 + ... and if 0 < x < 3/2, then show that x = 3y/(2(1 + y))

7. The kth term is (1 - k^k)/(1 - k) x r^k, so the sum to infinity is (1/(1 - k))[r/(1 - r) - kr/(1 - kr)] = r/((1 - r)(1 - kr)), since the bracket simplifies to (1 - k)/((1 - r)(1 - kr)). 8. The series is a G.P. with a = x/2 and r = x/2, so y = (x/2)/(1 - x/2) = x/(2 - x). Then y(2 - x) = x, i.e. 2y = x(1 + y), so x = 2y/(1 + y), as required. 9. The series is a G.P. with a = (2/3)x and r = (2/3)x, so y = ((2/3)x)/(1 - (2/3)x) = 2x/(3 - 2x). Then y(3 - 2x) = 2x, i.e. 3y = 2x(1 + y), so x = 3y/(2(1 + y)), as required.

10. A ball is dropped from a height of 27 meters and it rebounds two-third of the distance it falls. If it continues to fall in the same way what distance will it travel before coming to rest? 11. What distance will a ball travel before coming to rest if it is dropped from a height of 75 meters and after each fall it rebounds 2/5 of the distance it fell? 12. If y = 1 + 2x + 4x^2 + 8x^3 + ... i) Show that x = (y - 1)/(2y) ii) Find the interval in which the series is convergent. 13. If y = 1 + x/2 + x^2/4 + ... i) Show that x = 2((y - 1)/y) ii) Find the interval in which the series is convergent. 14. The sum of an infinite geometric series is 9 and the sum of the squares of its terms is 81/5. Find the series.

10. The ball falls 27 m, then rebounds 18 m and falls 18 m, then rebounds 12 m and falls 12 m, and so on. Total distance = 27 + 2(18 + 12 + 8 + ...) = 27 + 2 x 18/(1 - 2/3) = 27 + 108 = 135 meters. 11. Total distance = 75 + 2(30 + 12 + ...) = 75 + 2 x 30/(1 - 2/5) = 75 + 100 = 175 meters. 12. i) The series is a G.P. with a = 1 and r = 2x, so y = 1/(1 - 2x). Then 1 - 2x = 1/y, so x = (y - 1)/(2y). ii) The series converges when |2x| < 1, i.e. -1/2 < x < 1/2. 13. i) The series is a G.P. with a = 1 and r = x/2, so y = 1/(1 - x/2) = 2/(2 - x). Then 2 - x = 2/y, so x = 2(y - 1)/y. ii) The series converges when |x/2| < 1, i.e. -2 < x < 2. 14. Let the series be a, ar, ar^2, ...: a/(1 - r) = 9 and a^2/(1 - r^2) = 81/5. Dividing: a/(1 + r) = 9/5, so a = (9/5)(1 + r). With a = 9(1 - r) this gives 9(1 - r) = (9/5)(1 + r), i.e. 5 - 5r = 1 + r, so r = 2/3 and a = 3. Hence the series is 3 + 2 + 4/3 + 8/9 + ...

Exercise 6.9 (book p. 218)

1. A man deposits in a bank Rs. 8 in the first year, Rs. 24 in the second year, Rs. 72 in the third year and so on. Find the amount he will have deposited in the bank by the fifth year. 2. A man borrows Rs. 32760 without interest and agrees to repay the loan in installments, each installment being twice the preceding one. Find the amount of the last installment, if the amount of the first installment is Rs. 8. 3. The

population of a certain village is 62500. What will be its population after 3 years if it increases geometrically at the rate of 4% annually? 4. The enrollment of a famous school doubled after every eight years from 1970 to 1994. If the enrollment was 6000 in 1994, what was its enrollment in 1970? 5. A singular cholera bacteria produces two complete bacteria in 1/2 hours. If we start with a colony of A bacteria, how many bacteria will we have in n hours? 6. Joining the mid points of the sides of an equilateral triangle, an equilateral triangle having half the perimeter of the original triangle is obtained. We form a sequence of nested equilateral triangles in the manner described above with the original triangle having perimeter 3/2. What will be the total perimeter of all the triangles formed in this way?

1. The deposits form a G.P. with a = 8 and r = 3. The total by the fifth year is S_5 = 8(3^5 - 1)/(3 - 1) = 4 x 242 = Rs. 968. 2. The installments form a G.P. with a = 8 and r = 2. If there are n installments, 8(2^n - 1) = 32760, so 2^n - 1 = 4095 and 2^n = 4096, giving n = 12. The last installment is 8 x 2^11 = Rs. 16384. 3. The population after 3 years is 62500 x (1.04)^3 = 62500 x 1.124864 = 70304. 4. From 1970 to 1994 is 24 years, i.e. three doublings. So the 1970 enrollment was 6000/2^3 = 750. 5. The bacteria double every half hour, so in n hours there are 2n doublings. Hence the number of bacteria is A x 2^(2n) = A x 4^n. 6. The perimeters form a G.P. with a = 3/2 and r = 1/2. The total perimeter is (3/2)/(1 - 1/2) = 3.

Exercise 6.10 (book pp. 224-225)

1. Find the 9th term of the harmonic sequence i) 1/3, 1/5, 1/7, ... ii) -1/5, -1/3, -1, ... 2. Find the 12th terms of the following harmonic sequences i) 1/2, 1/5, 1/8, ... ii) 1/3, 2/9, 1/6, ... 3. Insert five harmonic means between the following given numbers, i) -2/5 and 2/13 ii) 1/4 and 1/24 4. Insert four harmonic means between the following given numbers. i) 1/3 and 1/23 ii) 7/3 and 7/11 iii) 4 and 20 5. If the 7th and 10th terms of an H.P. are 1/3 and 5/21 respectively, find its 14th term.

1. i) The reciprocals 3, 5, 7, ... form an A.P. with d = 2, so the 9th reciprocal is 3 + 16 = 19 and the 9th term is 1/19. ii) The reciprocals -5, -3, -1, ... form an A.P. with d = 2, so the 9th reciprocal is -5 + 16 = 11 and the 9th term is 1/11. 2. i) The reciprocals 2, 5, 8, ... form an A.P. with d = 3, so the 12th reciprocal is 2 + 33 = 35 and the 12th term is 1/35. ii) The reciprocals 3, 9/2, 6, ... form an A.P. with d = 3/2, so the 12th reciprocal is 3 + 33/2 = 39/2 and the 12th term is 2/39. 3. i) The reciprocals are -5/2 and 13/2; with five means the A.P. has d = (13/2 + 5/2)/6 = 3/2, giving reciprocals -5/2, -1, 1/2, 2, 7/2, 5, 13/2. Hence the harmonic means are -1, 2, 1/2, 2/7, 1/5. ii) The reciprocals are 4 and 24; with five means d = 20/6 = 10/3, giving reciprocals 4, 22/3, 32/3, 14, 52/3, 62/3, 24. Hence the harmonic means are 3/22, 3/32, 1/14, 3/52, 3/62. 4. i) The reciprocals are 3 and 23; with four means d = 20/5 = 4, giving reciprocals 7, 11, 15, 19. Hence the harmonic means are 1/7, 1/11, 1/15, 1/19. ii) The reciprocals are 3/7 and 11/7; with four means d = (8/7)/5 = 8/35, giving reciprocals 23/35, 31/35, 39/35, 47/35. Hence the harmonic means are 35/23, 35/31, 35/39, 35/47. iii) The reciprocals are 1/4 and 1/20; with four means d = (-1/5)/5 = -1/25, giving reciprocals 21/100, 17/100, 13/100, 9/100. Hence the harmonic means are 100/21, 100/17, 100/13, 100/9. 5. The reciprocals of the 7th and 10th terms are 3 and 21/5. In the reciprocal A.P., a + 6d = 3 and a + 9d = 21/5, so 3d = 6/5, i.e. d = 2/5, and a = 3 - 12/5 = 3/5. The 14th reciprocal is 3/5 + 13 x 2/5 = 29/5, so the 14th term is 5/29.

6. The first term of an H.P. is -1/3 and the fifth term is 1/5. Find its 9th term. 7. If 5 is the harmonic mean between 2 and b, find b. 8. If the numbers 1/k, 1/(2k + 1) and 1/(4k - 1) are in harmonic sequence, find k. 9. Find n so that (a^(n+1) + b^(n+1))/(a^n + b^n) may be H.M. between a and b. 10. If a^2, b^2 and c^2 are in A.P., show that a + b, c + a and b + c are in H.P. 11. The sum of the first and fifth terms of the harmonic sequence is 4/7, if the first term is 1/2, find the sequence.

6. The reciprocals form an A.P. with first term -3 and fifth term 5, so -3 + 4d = 5 and

d = 2. The 9th reciprocal is -3 + 16 = 13, so the 9th term is 1/13. 7. The harmonic mean is 2 x 2b/(2 + b) = 4b/(2 + b) = 5, so 4b = 10 + 5b and b = -10. 8. The reciprocals k, 2k + 1, 4k - 1 are in A.P., so 2(2k + 1) = k + (4k - 1), i.e. 4k + 2 = 5k - 1, so k = 3. FLAG: the printed key reads 'k = 67' for this part, but k = 67 does not satisfy the A.P. condition (for k = 67 the reciprocals are 67, 135, 267, and 2 x 135 = 270 is not 67 + 267 = 334); the value forced by the condition is k = 3. 9. Setting (a^(n+1) + b^(n+1))/(a^n + b^n) = 2ab/(a + b) gives (a + b)(a^(n+1) + b^(n+1)) = 2ab(a^n + b^n), i.e. a^(n+2) - a^(n+1)b - ab^(n+1) + b^(n+2) = 0, i.e. (a - b)(a^(n+1) - b^(n+1)) = 0. For a != b this gives a^(n+1) = b^(n+1), so n + 1 = 0 and n = -1. 10. We need the reciprocals 1/(a + b), 1/(c + a), 1/(b + c) to be in A.P., i.e. 2/(c + a) = 1/(a + b) + 1/(b + c) = (a + 2b + c)/((a + b)(b + c)), i.e. 2(a + b)(b + c) = (c + a)(a + c + 2b). Expanding: the left side is 2ab + 2b^2 + 2ac + 2bc and the right side is a^2 + 2ac + c^2 + 2ab + 2bc. The two sides are equal exactly when a^2 + c^2 = 2b^2, which is the given A.P. condition. Hence a + b, c + a, b + c are in H.P., as required. 11. The reciprocal A.P. has first term 2; the fifth reciprocal is 2 + 4d, so the fifth term is 1/(2 + 4d). Then 1/2 + 1/(2 + 4d) = 4/7, so 1/(2 + 4d) = 1/14, i.e. 2 + 4d = 14 and d = 3. Hence the sequence is 1/2, 1/5, 1/8, 1/11, ...

12. If A, G and H are the arithmetic, geometric and harmonic means between a and b respectively, show that G^2 = A.H. 13. Find A, G, H and show that G^2 = A.H. if i) a = -2, b = -6 ii) a = 2i, b = 4i iii) a = 9, b = 4 14. Find A, G, H and verify that A > G > H (G > 0), if i) a = 2, b = 8 ii) a = 2/5, b = 8/5 15. Find A, G, H and verify that A < G < H (G < 0), if i) a = -2, b = -8 ii) a = -2/5, b = -8/5

12. Here A = (a + b)/2, G^2 = ab and H = 2ab/(a + b), so A.H = ab = G^2, as required. 13. i) A = (-2 - 6)/2 = -4; G^2 = (-2)(-6) = 12, so G = +- sqrt(12); H = 2(12)/(-8) = -3. Then A.H = (-4)(-3) = 12 = G^2. ii) A = (2i + 4i)/2 = 3i; G^2 = (2i)(4i) = -8; H = 2(-8)/(6i) = 8i/3. Then A.H = 3i x 8i/3 = -8 = G^2. FLAG: the

printed key writes 'G = -8' for this part, but -8 is G^2, not G (G = 2 sqrt(2) i, taking the principal square root). iii) A = (9 + 4)/2 = 13/2; G^2 = 36, so G = +- 6; H = 2(36)/13 = 72/13. Then A.H = (13/2)(72/13) = 36 = G^2. 14. i) A = 5, G = 4, H = 16/5 = 3.2, so 5 > 4 > 3.2, i.e. A > G > H. ii) A = 1, G = 4/5, H = 16/25 = 0.64, so 1 > 0.8 > 0.64, i.e. A > G > H. 15. i) A = -5, G = -4 (the negative square root, since G < 0), H = -16/5 = -3.2, so -5 < -4 < -3.2, i.e. A < G < H. ii) A = -1, G = -4/5, H = -16/25 = -0.64, so -1 < -0.8 < -0.64, i.e. A < G < H.

16. If the H.M. and A.M. between two numbers are 4 and 9/2 respectively, find the numbers. 17. If the (positive) G.M. and H.M. between two numbers are 4 and 16/5, find the numbers. 18. If the numbers 1/2, 4/21 and 1/36 are subtracted from the three consecutive terms of a G.P., the resulting numbers are in H.P. Find the numbers if their product is 1/27.

16. Let the numbers be x and y: x + y = 9 and 2xy/(x + y) = 4, so xy = 18. Hence x and y are the roots of t^2 - 9t + 18 = (t - 6)(t - 3) = 0, so the numbers are 6 and 3. 17. Let the numbers be x and y: xy = 16 and 2xy/(x + y) = 16/5, so x + y = 10. Hence x and y are the roots of t^2 - 10t + 16 = (t - 8)(t - 2) = 0, so the numbers are 8 and 2. 18. Let the G.P. terms be a/r, a, ar: their product is a^3 = 1/27, so a = 1/3. After subtracting: 1/(3r) - 1/2, 1/3 - 4/21 = 1/7, r/3 - 1/36 are in H.P., so their reciprocals are in A.P.: 2 x 7 = 6r/(2 - 3r) + 36/(12r - 1). Solving: 14(2 - 3r)(12r - 1) = 6r(12r - 1) + 36(2 - 3r), i.e. 144r^2 - 123r + 25 = 0, i.e. (3r - 1)(48r - 25) = 0, so r = 1/3 or r = 25/48. For r = 1/3 the numbers are 1, 1/3, 1/9; for r = 25/48 the numbers are 16/25, 1/3, 25/144.

Exercise 6.11 (book p. 229)

Sum the following series upto n terms. 1. 1 x 1 + 2 x 4 + 3 x 7 + ... 2. 1 x 3 + 3 x 6 + 5 x 9 + ... 3. 1 x 4 + 2 x 7 + 3 x 10 + ... 4. 3 x 5 + 5 x 9 + 7 x 13 + ... 5. 1^2 + 3^2 + 5^2 + ... 6. 2^2 + 5^2 + 8^2 + ... 7. 2 x 1^2 + 4 x 2^2 + 6 x 3^2 + ... 8.

3 x 2^2 + 5 x 3^2 + 7 x 4^2 + ...

1. The nth term is n(3n - 2) = 3n^2 - 2n, so S_n = 3 x n(n + 1)(2n + 1)/6 - 2 x n(n + 1)/2 = n(n + 1)(2n - 1)/2 = n(2n^2 + n - 1)/2. 2. The nth term is (2n - 1)(3n) = 6n^2 - 3n, so S_n = 6 x n(n + 1)(2n + 1)/6 - 3 x n(n + 1)/2 = n(n + 1)(4n - 1)/2. 3. The nth term is n(3n + 1) = 3n^2 + n, so S_n = 3 x n(n + 1)(2n + 1)/6 + n(n + 1)/2 = n(n + 1)^2. 4. The nth term is (2n + 1)(4n + 1) = 8n^2 + 6n + 1, so S_n = 8 x n(n + 1)(2n + 1)/6 + 6 x n(n + 1)/2 + n = n(8n^2 + 21n + 16)/3. 5. The nth term is (2n - 1)^2 = 4n^2 - 4n + 1, so S_n = 4 x n(n + 1)(2n + 1)/6 - 4 x n(n + 1)/2 + n = n(4n^2 - 1)/3. 6. The nth term is (3n - 1)^2 = 9n^2 - 6n + 1, so S_n = 9 x n(n + 1)(2n + 1)/6 - 6 x n(n + 1)/2 + n = n(6n^2 + 3n - 1)/2. 7. The nth term is 2n x n^2 = 2n^3, so S_n = 2 x [n(n + 1)/2]^2 = n^2(n + 1)^2/2. 8. The nth term is (2n + 1)(n + 1)^2 = 2n^3 + 5n^2 + 4n + 1, so S_n = 2 x [n(n + 1)/2]^2 + 5 x n(n + 1)(2n + 1)/6 + 4 x n(n + 1)/2 + n = n(3n^3 + 16n^2 + 30n + 23)/6.

Sum the following series upto n terms. 9. 2 x 4 x 7 + 3 x 6 x 10 + 4 x 8 x 13 + ... 10. 1 x 4 x 6 + 4 x 7 x 10 + 7 x 10 x 14 + ... 11. 1 + (1 + 2) + (1 + 2 + 3) + ... 12. 1^2 + (1^2 + 2^2) + (1^2 + 2^2 + 3^2) + ... 13. 2 + (2 + 5) + (2 + 5 + 8) + ... 14. Sum the series. i) 1^2 - 2^2 + 3^2 - 4^2 + ... + (2n - 1)^2 - (2n)^2 ii) 1^2 - 3^2 + 5^2 - 7^2 + ... + (4n - 3)^2 - (4n - 1)^2 iii) 1^2/1 + (1^2 + 2^2)/2 + (1^2 + 2^2 + 3^2)/3 + ... to n terms

9. The nth term is (n + 1)(2n + 2)(3n + 4) = 6n^3 + 20n^2 + 22n + 8, so S_n = 6 x [n(n + 1)/2]^2 + 20 x n(n + 1)(2n + 1)/6 + 22 x n(n + 1)/2 + 8n = n(9n^3 + 58n^2 + 135n + 134)/6. 10. The nth term is (3n - 2)(3n + 1)(4n + 2) = 36n^3 + 6n^2 - 14n - 4, so S_n = 36 x [n(n + 1)/2]^2 + 6 x n(n + 1)(2n + 1)/6 - 14 x n(n + 1)/2 - 4n = n(9n^3 + 20n^2 + 5n - 10). 11. The nth term is n(n + 1)/2, so S_n = (1/2)[n(n + 1)(2n + 1)/6 + n(n + 1)/2] = n(n + 1)(n + 2)/6. 12. The nth term is n(n + 1)(2n + 1)/6, so S_n = (1/6)[2 x [n(n + 1)/2]^2 + 3 x n(n + 1)(2n + 1)/6 + n(n + 1)/2] = n(n + 1)^2(n + 2)/12. 13. The nth term is the sum of the first n terms of 2,

5, 8, ...: 3n(n + 1)/2 - n = n(3n + 1)/2, so S_n = (1/2)[3 x n(n + 1)(2n + 1)/6 + n(n + 1)/2] = n(n + 1)^2/2. 14. i) Pairing consecutive terms: (2k - 1)^2 - (2k)^2 = -(4k - 1), so S = -sum_{k=1}^n (4k - 1) = -n(2n + 1). ii) Pairing consecutive terms: (4k - 3)^2 - (4k - 1)^2 = -16k + 8, so S = -16 x n(n + 1)/2 + 8n = -8n^2. iii) The kth term is k(k + 1)(2k + 1)/(6k) = (2k^2 + 3k + 1)/6, so S_n = (1/6)[2 x n(n + 1)(2n + 1)/6 + 3 x n(n + 1)/2 + n] = n(4n^2 + 15n + 17)/36.

15. Find the sum to n terms of the series whose nth terms are given. i) 3n^2 + n + 1 ii) n^2 + 4n + 1 16. Given nth terms of the series, find the sum to 2n terms. i) 3n^2 + 2n + 1 ii) n^3 + 2n + 3

15. i) S_n = 3 x n(n + 1)(2n + 1)/6 + n(n + 1)/2 + n = n(n + 1)^2 + n = n(n^2 + 2n + 2). ii) S_n = n(n + 1)(2n + 1)/6 + 4 x n(n + 1)/2 + n = n(2n^2 + 15n + 19)/6.

16. i) Summing 3k^2 + 2k + 1 for k = 1 to 2n: 3 x (2n)(2n + 1)(4n + 1)/6 + 2 x (2n)(2n + 1)/2 + 2n = n(2n + 1)(4n + 1) + 2n(2n + 1) + 2n = n(8n^2 + 10n + 5). ii) Summing k^3 + 2k + 3 for k = 1 to 2n: [(2n)(2n + 1)/2]^2 + 2 x (2n)(2n + 1)/2 + 6n = n^2(2n + 1)^2 + 2n(2n + 1) + 6n = n(4n^3 + 4n^2 + 5n + 8).

Chapter 7: Permutation, Combination and Probability

Exercise 7.1 (book p. 231)

1. Evaluate each of the following: i) 4! ii) 6! iii) 8!/7! iv) 10!/7! v) 11!/(4! 7!) vi) 6!/(3! 3!) vii) 8!/(4! 2!) viii) 11!/(2! 4! 5!) ix) 9!/(2! (9-2)!) x) 15!/(15! (15-15)!) xi) 3!/0! xii) 4! 0! 1!

i) 4! = 4 x 3 x 2 x 1 = 24. ii) 6! = 720. iii) 8!/7! = 8. iv) 10!/7! = 10 x 9 x 8 = 720. v) 11!/(4! 7!) = (11 x 10 x 9 x 8)/(4 x 3 x 2 x 1) = 330. vi) 6!/(3! 3!) = 720/36 = 20. vii) 8!/(4! 2!) = 40320/48 = 840. viii) 11!/(2! 4! 5!) = 39916800/(2 x 24 x 120) = 39916800/5760 = 6930. FLAG: the printed key reads 770 for this part, but 11!/(2! 4!

5!) = 6930, since 2! x 4! x 5! = 5760 and 39916800/5760 = 6930. ix) 9!/(2! 7!) = (9 x 8)/2 = 36. x) 15!/(15! 0!) = 1, since 0! = 1. xi) 3!/0! = 6/1 = 6. xii) 4! x 0! x 1! = 24 x 1 x 1 = 24.

2. Write each of the following in the factorial form: i) 6.5.4. ii) 12.11.10 iii) 20.19.18.17 iv) (10.9)/(2.1) v) (8.7.6)/(3.2.1) vi) (52.51.50.49)/(4.3.2.1) vii) n(n-1)(n-2) viii) (n+2)(n+1)(n) ix) ((n+1)(n)(n-1))/(3.2.1) x) n(n-1)(n-2)...(n-r+1)

i) 6 x 5 x 4 = 6!/3!. ii) 12 x 11 x 10 = 12!/9!. iii) 20 x 19 x 18 x 17 = 20!/16!. iv) (10 x 9)/(2 x 1) = 10!/(2! 8!). v) (8 x 7 x 6)/(3 x 2 x 1) = 8!/(3! 5!). vi) (52 x 51 x 50 x 49)/(4 x 3 x 2 x 1) = 52!/(4! 48!). vii) n(n-1)(n-2) = n!/(n-3)!. viii) (n+2)(n+1)n = (n+2)!/(n-1)!. ix) (n+1)n(n-1)/(3 x 2 x 1) = (n+1)!/(3! (n-2)!). x) n(n-1)(n-2)...(n-r+1) = n!/(n-r)!.

Exercise 7.2 (book pp. 235-236)

1. Evaluate the following: i) 20P3 ii) 16P4 iii) 12P5 iv) 10P7 v) 9P8 2. Find the value of n when: i) nP2 = 30 ii) 11Pn = 11.10.9 iii) nP4 : (n-1)P3 = 9 : 1 3. Prove from the first principle that: i) nPr = n . (n-1)P(r-1) ii) nPr = (n-1)Pr + r . (n-1)P(r-1)

1. i) 20P3 = 20 x 19 x 18 = 6840. ii) 16P4 = 16 x 15 x 14 x 13 = 43680. iii) 12P5 = 12 x 11 x 10 x 9 x 8 = 95040. iv) 10P7 = 10 x 9 x 8 x 7 x 6 x 5 x 4 = 604800. v) 9P8 = 9! = 362880. 2. i) nP2 = n(n-1) = 30, so n = 6 (since 6 x 5 = 30). ii) 11Pn = 11 x 10 x 9, so three factors are used and n = 3. iii) nP4/(n-1)P3 = n(n-1)(n-2)(n-3)/((n-1)(n-2)(n-3)) = n = 9/1, so n = 9. 3. i) By definition nPr = n!/(n-r)! = n x (n-1)!/(n-r)! = n x (n-1)P(r-1), since (n-1)P(r-1) = (n-1)!/((n-1)-(r-1))! = (n-1)!/(n-r)!. ii) (n-1)Pr + r x (n-1)P(r-1) = (n-1)!/(n-1-r)! + r(n-1)!/(n-r)! = (n-1)!/(n-r)! x ((n-r) + r) = n(n-1)!/(n-r)! = n!/(n-r)! = nPr, as required.

4. How many signals can be given by 5 flags of different colours, using 3 flags at a time? 5. How many signals can be given by 6 flags of different colours when any number of flags can be used at a time? 6. How many words can be formed from the

letters of the following words using all letters when no letter is to be repeated: i) PLANE ii) OBJECT iii) FASTING? 7. How many 3-digit numbers can be formed by using each one of the digits 2, 3, 5, 7, 9 only once?

4. The number of signals is 5P3 = 5 x 4 x 3 = 60. 5. Using 1, 2, 3, 4, 5 or 6 flags at a time: 6P1 + 6P2 + 6P3 + 6P4 + 6P5 + 6P6 = 6 + 30 + 120 + 360 + 720 + 720 = 1956 signals. 6. i) PLANE has 5 different letters: 5! = 120 words. ii) OBJECT has 6 different letters: 6! = 720 words. iii) FASTING has 7 different letters: 7! = 5040 words. 7. The number of 3-digit numbers is 5P3 = 5 x 4 x 3 = 60.

8. Find the numbers greater than 23000 that can be formed from the digits 1, 2, 3, 5, 6, without repeating any digit. HINT: The first two digits on L.H.S. will be 23 etc. 9. Find the number of 5-digit numbers that can be formed from the digits 1, 2, 4, 6, 8 (when no digit is repeated), but i) the digits 2 and 8 are next to each other; ii) the digits 2 and 8 are not next to each other. 10. How many 6-digit numbers can be formed, without repeating any digit from the digits 0, 1, 2, 3, 4, 5? In how many of them will 0 be at the tens place? 11. How many 5-digit multiples of 5 can be formed from the digits 2, 3, 5, 7, 9, when no digit is repeated.

8. Only 5-digit numbers can exceed 23000. First digit 3, 5 or 6: 3 x 4! = 72 numbers. First digit 2: the second digit must be 3, 5 or 6 (a second digit 1 gives a number below 23000), giving 3 x 3! = 18 numbers. Total = 72 + 18 = 90. 9. All 5-digit numbers: 5! = 120. i) Treating 2 and 8 as one block: 4! x 2 = 48 numbers. ii) Not next to each other: 120 - 48 = 72 numbers. 10. The first digit cannot be 0: 5 x 5! = 600 numbers. With 0 fixed at the tens place, the first digit has 5 choices and the remaining four places can be filled in 4! ways: 5 x 4! = 120 numbers. 11. A multiple of 5 must end in 5, so the units place is fixed and the remaining four places can be filled in 4! = 24 ways.

12. In how many ways can 8 books including 2 on English be arranged on a shelf in such a way that the English books are never together? 13. Find the number of

arrangements of 3 books on English and 5 books on Urdu for placing them on a shelf such that the books on the same subject are together. 14. In how many ways can 5 boys and 4 girls be seated on a bench so that the girls and the boys occupy alternate seats?

12. All arrangements: 8! = 40320. Arrangements with the two English books together: treat them as one block, giving 7! x 2! = 10080. Hence the English books are never together in 40320 - 10080 = 30240 ways. 13. The 3 English books can be arranged in 3! ways and the 5 Urdu books in 5! ways, and the two subject-blocks can be ordered in 2! ways: 2! x 3! x 5! = 2 x 6 x 120 = 1440 arrangements. 14. With 9 seats alternating, the pattern must start and end with a boy: B G B G B G B G B. The 5 boys can be seated in 5! ways and the 4 girls in 4! ways, giving 5! x 4! = 120 x 24 = 2880 ways.

Exercise 7.3 (book pp. 238-239)

1. How many arrangements of the letters of the following words, taken all together, can be made: i) PAKPATTAN ii) PAKISTAN iii) MATHEMATICS iv) ASSASSINATION? 2. How many permutation of the letters of the word PANAMA can be made, if P is to be the first letter in each arrangement? 3. How many arrangements of the letters of the word ATTACKED can be made, if each arrangement begins with C and ends with K? 4. How many numbers greater than 1000,000 can be formed from the digits 0, 2, 2, 2, 3, 4, 4?

1. i) PAKPATTAN has 9 letters with A 3 times, P 2 times and T 2 times: 9!/(3! 2! 2!) = 362880/24 = 15120. ii) PAKISTAN has 8 letters with A 2 times: 8!/2! = 20160. iii) MATHEMATICS has 11 letters with M, A and T each 2 times: 11!/(2! 2! 2!) = 39916800/8 = 4989600. iv) ASSASSINATION has 13 letters with A 3 times, S 4 times, I 2 times and N 2 times: 13!/(3! 4! 2! 2!) = 6227020800/576 = 10810800. 2. With P fixed first, the remaining letters A, N, A, M, A have A 3 times: 5!/3! = 20

arrangements. 3. With C first and K last, the remaining letters A, T, T, A, E, D have A 2 times and T 2 times: 6!/(2! 2!) = 180 arrangements. 4. All 7-digit numbers formed from 0, 2, 2, 2, 3, 4, 4 exceed 1000,000. Total arrangements: 7!/(3! 2!) = 420. Arrangements with first digit 0: 6!/(3! 2!) = 60. Hence 420 - 60 = 360 numbers.

5. How many 6-digit numbers can be formed from the digits 2, 2, 3, 3, 4, 4? How many of them will lie between 400,000 and 430,000? 6. 11 members of a club form 4 committees of 3, 4, 2, 2 members so that no member is a member of more than one committee. Find the number of committees. 7. The D.C.Os of 11 districts meet to discuss the law and order situation in their districts. In how many ways can they be seated at a round table, when two particular D.C.Os insist on sitting together? 8. The Governor of the Punjab calls a meeting of 12 officers. In how many ways can they be seated at a round table?

5. Total 6-digit numbers: 6!/(2! 2! 2!) = 90. Between 400,000 and 430,000 the first digit must be 4 and the second digit must be 2 (a second digit 3 gives a number above 430,000): the remaining digits 2, 3, 3, 4 can be arranged in 4!/2! = 12 ways. Hence 12 numbers lie between 400,000 and 430,000. 6. Choosing 3, then 4, then 2, then 2 members: 11!/(3! 4! 2! 2!) = 39916800/576 = 69300 ways. NOTE: the key gives 69300, i.e. the two committees of 2 are counted as distinct committees; dividing by 2! for indistinguishable equal-size committees would give 34650. 7. Treating the two particular D.C.Os as one block gives 10 objects at a round table: (10-1)! = 9! = 362880, and the pair can sit in 2 orders, giving 2 x 362880 = 725760 ways. 8. The printed key gives 19958400 = 11!/2 ways, i.e. the key counts a seating and its mirror image as the same arrangement. NOTE: counting rotations only, the usual round-table count is (12-1)! = 11! = 39916800 ways.

9. Fatima invites 14 people to a dinner. There are 9 males and 5 females who are seated at two different tables so that guests of one sex sit at one round table and the guests of the other sex at the second table. Find the number of ways in which all

guests are seated. 10. Find the number of ways in which 5 men and 5 women can be seated at a round table in such a way that no two persons of the same sex sit together. 11. In how many ways can 4 keys be arranged on a circular key ring? 12. How many necklaces can be made from 6 beads of different colours?

9. The 9 males at one round table can be seated in (9-1)! = 8! = 40320 ways and the 5 females at the other in (5-1)! = 4! = 24 ways, giving 40320 x 24 = 967680 ways. 10. Fix one man to remove rotation: the remaining 4 men can be seated in 4! ways in the alternate seats and the 5 women in 5! ways, giving 4! x 5! = 24 x 120 = 2880 ways. 11. For a key ring, flipping gives the same arrangement, so the number is (4-1)!/2 = 3 ways. 12. For a necklace, flipping gives the same arrangement, so the number is (6-1)!/2 = 60 necklaces.

Exercise 7.4 (book p. 242)

1. Evaluate the following: i) 12C3 ii) 20C17 iii) nC4 2. Find the value of n, when i) nC5 = nC4 ii) nC10 = (12 x 11)/(2!) iii) nC12 = nC6 3. Find the values of n and r, when i) nCr = 35 and nPr = 210 ii) (n-1)C(r-1) : nCr : (n+1)C(r+1) = 3 : 6 : 11

1. i) 12C3 = (12 x 11 x 10)/(3 x 2 x 1) = 220. ii) 20C17 = 20C3 = (20 x 19 x 18)/6 = 1140. iii) nC4 = n(n-1)(n-2)(n-3)/4!. 2. i) nC5 = nC4 gives 5 + 4 = n, so n = 9. ii) nC10 = 66 = 12C10, so n = 12. iii) nC12 = nC6 gives 12 + 6 = n, so n = 18. 3. i) nPr/nCr = r! = 210/35 = 6, so r = 3. Then nC3 = 35 gives n(n-1)(n-2) = 210, so n = 7 (since 7 x 6 x 5 = 210). ii) nCr/(n-1)C(r-1) = n/r = 6/3 = 2, so n = 2r. Also (n+1)C(r+1)/nCr = (n+1)/(r+1) = 11/6, so 6n + 6 = 11r + 11; with n = 2r this gives 12r + 6 = 11r + 11, so r = 5 and n = 10.

4. How many (a) diagonals and (b) triangles can be formed by joining the vertices of the polygon having: i) 5 sides ii) 8 sides iii) 12 sides? 5. The members of a club are 12 boys and 8 girls. In how many ways can a committee of 3 boys and 2 girls be formed? 6. How many committees of 5 members can be chosen from a group of 8

persons when each committee must include 2 particular persons? 7. In how many ways can a hockey team of 11 players be selected out of 15 players? How many of them will include a particular player?

4. The number of diagonals of an n-sided polygon is nC2 - n and the number of triangles is nC3. i) 5 sides: diagonals 10 - 5 = 5, triangles 10. ii) 8 sides: diagonals 28 - 8 = 20, triangles 56. iii) 12 sides: diagonals 66 - 12 = 54, triangles 220. NOTE: the printed key lists one number per part (10, 20, 54); these are the pentagon triangles, the octagon diagonals and the 12-gon diagonals respectively, all of which appear above. 5. 12C3 x 8C2 = 220 x 28 = 6160 ways. 6. With 2 particular persons fixed, the remaining 3 members are chosen from the other 6 persons: 6C3 = 20 committees. 7. 15C11 = 1365 teams. Teams including a particular player: choose the remaining 10 from the other 14 players: 14C10 = 1001 teams.

8. Show that: 16C11 + 16C10 = 17C11 9. There are 8 men and 10 women members of a club. How many committees of 7 can be formed, having: i) 4 women ii) at the most 4 women iii) atleast 4 women? 10. Prove that nCr + nC(r-1) = (n+1)Cr.

8. By Pascal's rule, 16C11 + 16C10 = 17C11, since nCr + nC(r-1) = (n+1)Cr with n = 16 and r = 11. 9. A committee of 7 with k women and 7-k men can be formed in 10Ck x 8C(7-k) ways. i) Exactly 4 women: 10C4 x 8C3 = 210 x 56 = 11760. ii) At most 4 women: k = 0 to 4 gives 1 x 8 + 10 x 28 + 45 x 56 + 120 x 70 + 210 x 56 = 8 + 280 + 2520 + 8400 + 11760 = 22968. iii) At least 4 women: k = 4 to 7 gives 11760 + 252 x 28 + 210 x 8 + 120 x 1 = 11760 + 7056 + 1680 + 120 = 20616. NOTE: the committee size 7 is the value forced by the printed key (all three key values 11760, 22968, 20616 are reproduced exactly by committees of 7). 10. nCr + nC(r-1) = n!/(r!(n-r)!) + n!/((r-1)!(n-r+1)!) = n!/((r-1)!(n-r)!) x (1/r + 1/(n-r+1)) = n!/((r-1)!(n-r)!) x (n+1)/(r(n-r+1)) = (n+1)!/(r!(n+1-r)!) = (n+1)Cr, as required.

Exercise 7.5 (book pp. 245-247)

For the following experiments, find the probability in each case: 1. Experiment: From a box containing orange-flavoured sweets, Bilal takes out one sweet without looking. Events Happening: i) the sweet is orange-flavoured ii) the sweet is lemon-flavoured. 2. Experiment: Pakistan and India play a cricket match. The result is: Events Happening: i) Pakistan wins ii) India does not lose. 3. Experiment: There are 5 green and 3 red balls in a box, one ball is taken out. Events Happening: i) the ball is green ii) the ball is red. 4. Experiment: A fair coin is tossed three times. It shows Events Happening: i) One tail ii) atleast one head.

1. The box contains only orange-flavoured sweets. i) P(orange) = 1. ii) P(lemon) = 0. 2. The three equally likely results are: Pakistan wins, India wins, draw. i) P(Pakistan wins) = 1/3. ii) India does not lose means India wins or the match is drawn: P = 2/3. 3. There are 8 balls in all. i) P(green) = 5/8. ii) P(red) = 3/8. 4. The sample space has 2^3 = 8 equally likely outcomes. i) Exactly one tail occurs in HTT, THT, TTH: P = 3/8. ii) At least one head occurs in all outcomes except TTT: P = 7/8.

5. Experiment: A die is rolled. The top shows Events Happening: i) 3 or 4 dots ii) dots less than 5. 6. Experiment: From a box containing slips numbered 1, 2, 3, ..., 5 one slip is picked up. Events Happening: i) the number on the slip is a prime number ii) the number on the slip is a multiple of 3. 7. Experiment: Two dice, one red and the other blue, are rolled simultaneously. The numbers of dots on the tops are added. The total of the two scores is: Events Happening: i) 5 ii) 7 iii) 11

5. The sample space has 6 outcomes. i) P(3 or 4) = 2/6 = 1/3. ii) Dots less than 5 are 1, 2, 3, 4: P = 4/6 = 2/3. 6. The sample space has 5 slips. i) Prime numbers among 1, 2, 3, 4, 5 are 2, 3, 5: P = 3/5. ii) The only multiple of 3 is 3: P = 1/5. 7. The sample space has 36 outcomes. i) Sum 5 occurs as (1,4), (2,3), (3,2), (4,1): P = 4/36 = 1/9. ii) Sum 7 occurs as (1,6), (2,5), (3,4), (4,3), (5,2), (6,1): P = 6/36 = 1/6. iii) Sum 11 occurs as (5,6), (6,5): P = 2/36 = 1/18.

8. Experiment: A bag contains 40 balls out of which 5 are green, 15 are black and

the remaining are yellow. A ball is taken out of the bag. Events Happening: i) The ball is black ii) The ball is green iii) The ball is not green. 9. Experiment: One chit out of 30 containing the names of 30 students of a class of 18 boys and 12 girls is taken out at random, for nomination as the monitor of the class. Events Happening: i) the monitor is a boy ii) the monitor is a girl. 10. Experiment: A coin is tossed four times. The tops show Events Happening: i) all heads ii) 2 heads and 2 tails.

8. The remaining balls are 40 - 5 - 15 = 20 yellow balls. i) P(black) = 15/40 = 3/8. ii) P(green) = 5/40 = 1/8. iii) P(not green) = 35/40 = 7/8. 9. There are 30 chits. i) P(boy) = 18/30 = 3/5. ii) P(girl) = 12/30 = 2/5. 10. The sample space has 2^4 = 16 outcomes. i) All heads occurs only as HHHH: P = 1/16. ii) 2 heads and 2 tails occurs in 4C2 = 6 outcomes: P = 6/16 = 3/8. NOTE: the printed Answers block for Exercise 7.5 lists values for Q1-Q7 only; the values for Q8-Q10 above are computed directly from the experiments.

Exercise 7.6 (book pp. 248-249)

1. A fair coin is tossed 30 times, the result of which is tabulated below. Study the table and answer the questions given below the table: Event: Head, Tally Marks: 14; Event: Tail, Tally Marks: 16. i) How many times does 'head' appear? ii) How many times does 'tail' appear? iii) Estimate the probability of the appearance of head? iv) Estimate the probability of the appearance of tail? 2. A die is tossed 100 times. The result is tabulated below. Study the table and answer the questions given below the table: Frequencies: 1: 14, 2: 17, 3: 20, 4: 18, 5: 15, 6: 16. i) How many times do 3 dots appear? ii) How many times do 5 dots appear? iii) How many times does an even number of dots appear? iv) How many times does a prime number of dots appear? v) Find the probability of each one of the above cases.

1. i) Head appears 14 times. ii) Tail appears 16 times. iii) The estimated probability of head is 14/30 = 7/15. iv) The estimated probability of tail is 16/30 = 8/15. 2. i) 3

dots appear 20 times. ii) 5 dots appear 15 times. iii) An even number of dots (2, 4, 6) appears 17 + 18 + 16 = 51 times. iv) A prime number of dots (2, 3, 5) appears 17 + 20 + 15 = 52 times. v) The estimated probabilities are: 20/100 = 1/5 for 3 dots; 15/100 = 3/20 for 5 dots; 51/100 for an even number; 52/100 = 13/25 for a prime number.

3. The eggs supplied by a poultry farm during a week broke during transit as follows: 1 %, 2 %, 1 1/2 %, 1/2 %, 1 %, 2 %, 1 %. Find the probability of the eggs that broke in a day. Calculate the number of eggs that will be broken in transiting the following number of eggs: i) 7,000 ii) 8,400 iii) 10,500

3. The total breakage over the 7 days is 1 + 2 + 1.5 + 0.5 + 1 + 2 + 1 = 9 %. Hence the probability that an egg breaks in a day is 9/(7 x 100) = 9/700. i) Out of 7,000 eggs: 7000 x 9/700 = 90 eggs. ii) Out of 8,400 eggs: 8400 x 9/700 = 108 eggs. iii) Out of 10,500 eggs: 10500 x 9/700 = 135 eggs.

Exercise 7.7 (book p. 251)

1. If sample space S = {1, 2, 3, ..., 9}, Event A = {2, 4, 6, 8} and Event B = {1, 3, 5}, find P(A union B). 2. A box contains 10 red, 30 white and 20 black marbles. A marble is drawn at random. Find the probability that it is either red or white. 3. A natural number is chosen out of the first fifty natural numbers. What is the probability that the chosen number is a multiple of 3 or of 5? 4. A card is drawn from a deck of 52 playing cards. What is the probability that it is a diamond card or an ace?

1. A and B are disjoint, so P(A union B) = P(A) + P(B) = 4/9 + 3/9 = 7/9. 2. The box has 60 marbles; red or white marbles number 10 + 30 = 40, so P = 40/60 = 2/3. 3. Multiples of 3 among the first 50 numbers: 16; multiples of 5: 10; multiples of both (i.e. of 15): 3. Hence multiples of 3 or 5 number 16 + 10 - 3 = 23, so P = 23/50. 4. P(diamond) = 13/52, P(ace) = 4/52, P(diamond and ace) = 1/52, so P = 13/52 +

4/52 - 1/52 = 16/52 = 4/13.

5. A die is thrown twice. What is the probability that the sum of the number of dots shown is 3 or 11? 6. Two dice are thrown. What is the probability that the sum of the number of dots appearing on them is 4 or 6? 7. Two dice are thrown simultaneously. If the event A is that the sum of the number of dots shown is an odd number and the event B is that the number of dots shown on at least one die is 3. Find P(A union B). 8. There are 10 girls and 20 boys in a class. Half of the boys and half of the girls have blue eyes. Find the probability that one student chosen as monitor is either a girl or has blue eyes.

5. Sum 3 occurs in 2 ways and sum 11 in 2 ways, out of 36 outcomes: P = 4/36 = 1/9. 6. Sum 4 occurs in 3 ways and sum 6 in 5 ways: P = 8/36 = 2/9. 7. P(A) = 18/36 (half of all outcomes have odd sum). P(B) = 11/36 (outcomes with at least one 3). A and B together: outcomes with a 3 and odd sum are (3,2), (3,4), (3,6), (2,3), (4,3), (6,3): 6 outcomes, so P(A and B) = 6/36. Hence P(A union B) = 18/36 + 11/36 - 6/36 = 23/36. 8. The class has 30 students. Girls: 10; students with blue eyes: half of 20 boys (10) plus half of 10 girls (5) = 15; girls with blue eyes: 5. Hence P(girl or blue eyes) = 10/30 + 15/30 - 5/30 = 20/30 = 2/3.

Exercise 7.8 (book pp. 254-255)

1. The probability that a person A will be alive 15 year hence is 5/7 and the probability that another person B will be alive 15 years hence is 7/9. Find the probability that both will be alive 15 year hence. 2. A die is rolled twice: Event E1 is the appearance of even number of dots and event E2 is the appearance of more than 4 dots. Prove that: P(E1 and E2) = P(E1).P(E2). 3. Determine the probability of getting 2 heads in two successive tosses of a balanced coin. 4. Two coins are tossed twice each. Find the probability that the head appears on the first toss and the same faces appear in the two tosses. 5. Two cards are drawn from a deck of 52

playing cards. If one card is drawn and replaced before drawing the second card, find the probability that both the cards are aces.

1. The two lives are independent, so P(both alive) = (5/7) x (7/9) = 5/9. 2. For one roll, P(E1) = 3/6 = 1/2 (even dots 2, 4, 6) and P(E2) = 2/6 = 1/3 (dots more than 4 are 5, 6). The event E1 and E2 together is the dot 6: P = 1/6 = (1/2) x (1/3) = P(E1) x P(E2), as required. 3. P(2 heads) = (1/2) x (1/2) = 1/4. 4. P(head on the first toss) = 1/2. P(the same face in the two tosses) = P(HH or TT) = 1/2. The tosses are independent, so P = (1/2) x (1/2) = 1/4. 5. With replacement, P(both aces) = (4/52) x (4/52) = 1/169.

6. Two cards from a deck of 52 playing cards are drawn in such a way that the card is replaced after the first draw. Find the probabilities in the following cases: i) first card is king and the second is a queen. ii) both the cards are faced cards i.e. king, queen, jack. 7. Two dice are thrown twice. What is probability that sum of the dots shown in the first throw is 7 and that of the second throw is 11? 8. Find the probability that the sum of dots appearing in two successive throws of two dice is every time 7. 9. A fair die is thrown twice. Find the probability that a prime number of dots appear in the first throw and the number of dots in the second throw is less than 5. 10. A bag contains 8 red, 5 white and 7 black balls. 3 balls are drawn from the bag. What is the probability that the first ball is red, the second ball is white and the third ball is black, when every time the ball is replaced? HINT: (8/20)(5/20)(7/20) is the probability.

6. i) P(king then queen) = (4/52) x (4/52) = 1/169. ii) There are 12 face cards, so P(both face cards) = (12/52) x (12/52) = 9/169. FLAG: the printed key reads 3/169 for this part, but (12/52) x (12/52) = 144/2704 = 9/169. 7. P(sum 7 in the first throw) = 6/36 = 1/6 and P(sum 11 in the second throw) = 2/36 = 1/18; the throws are independent, so P = (1/6) x (1/18) = 1/108. 8. P(sum 7 in one throw) = 6/36 = 1/6, so P(sum 7 in both throws) = (1/6) x (1/6) = 1/36. 9. P(prime in the first throw)

= 3/6 = 1/2 (primes 2, 3, 5) and P(less than 5 in the second throw) = 4/6 = 2/3, so P = (1/2) x (2/3) = 1/3. 10. With replacement, P = (8/20) x (5/20) x (7/20) = 280/8000 = 7/200.

Chapter 8: Mathematical Induction and Binomial Theorem

Exercise 8.1 (book pp. 263-265)

Use mathematical induction to prove for every positive integer n: 1. 1 + 5 + 9 + ... + (4n - 3) = n(2n - 1)

Step 1 (base case): For n = 1: LHS = 1, RHS = 1 x (2 - 1) = 1. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume 1 + 5 + ... + (4k - 3) = k(2k - 1). Adding the next term (4(k + 1) - 3) = 4k + 1 to both sides gives k(2k - 1) + 4k + 1 = 2k^2 + 3k + 1 = (k + 1)(2k + 1) = (k + 1)(2(k + 1) - 1), which is the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

2. 1 + 3 + 5 + ... + (2n - 1) = n^2

Step 1 (base case): For n = 1: LHS = 1, RHS = 1. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume 1 + 3 + ... + (2k - 1) = k^2. Adding the next term (2(k + 1) - 1) = 2k + 1 gives k^2 + 2k + 1 = (k + 1)^2, the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

3. 1 + 4 + 7 + ... + (3n - 2) = n(3n - 1)/2

Step 1 (base case): For n = 1: LHS = 1, RHS = 1 x 2/2 = 1. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume 1 + 4 + ... + (3k - 2) = k(3k - 1)/2. Adding the next term (3(k + 1) - 2) = 3k + 1 gives k(3k - 1)/2 + 3k + 1 = (3k^2 - k + 6k + 2)/2 = (3k^2 + 5k + 2)/2 = (k + 1)(3k + 2)/2 = (k + 1)(3(k + 1) - 1)/2, the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

4. 1 + 2 + 4 + ... + 2^(n-1) = 2^n - 1

Step 1 (base case): For n = 1: LHS = 1, RHS = 2 - 1 = 1. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume 1 + 2 + ... + 2^(k-1) = 2^k - 1. Adding the next term 2^k gives 2^k - 1 + 2^k = 2^(k+1) - 1, the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

5. 1 + 1/2 + 1/4 + ... + 1/2^(n-1) = 2[1 - 1/2^n]

Step 1 (base case): For n = 1: LHS = 1, RHS = 2(1 - 1/2) = 1. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume 1 + 1/2 + ... + 1/2^(k-1) = 2(1 - 1/2^k). Adding the next term 1/2^k gives 2 - 2/2^k + 1/2^k = 2 - 1/2^k = 2(1 - 1/2^(k+1)), the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

6. 2 + 4 + 6 + ... + 2n = n(n + 1)

Step 1 (base case): For n = 1: LHS = 2, RHS = 1 x 2 = 2. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume 2 + 4 + ... + 2k = k(k + 1). Adding the next term 2(k + 1) gives k(k + 1) + 2(k + 1) = (k + 1)(k + 2), the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

7. 2 + 6 + 18 + ... + 2 x 3^(n-1) = 3^n - 1

Step 1 (base case): For n = 1: LHS = 2, RHS = 3 - 1 = 2. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume 2 + 6 + ... + 2 x 3^(k-1) = 3^k - 1. Adding the next term 2 x 3^k gives 3^k - 1 + 2 x 3^k = 3^(k+1) - 1, the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

8. 1x3 + 2x5 + 3x7 + ... + n x (2n + 1) = n(n + 1)(4n + 5)/6

Step 1 (base case): For n = 1: LHS = 3, RHS = 1 x 2 x 9/6 = 3. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume the sum up to k is k(k + 1)(4k + 5)/6. Adding the next term (k + 1)(2k + 3) gives (k + 1)[k(4k + 5)/6 + 2k + 3] = (k + 1)(4k^2 + 5k + 12k + 18)/6 = (k + 1)(4k^2 + 17k + 18)/6 = (k + 1)(k + 2)(4k + 9)/6 = (k + 1)(k + 2)(4(k + 1) + 5)/6, the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

9. 1x2 + 2x3 + 3x4 + ... + n x (n + 1) = n(n + 1)(n + 2)/3

Step 1 (base case): For n = 1: LHS = 2, RHS = 1 x 2 x 3/3 = 2. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume the sum up to k is k(k + 1)(k + 2)/3. Adding the next term (k + 1)(k + 2) gives (k + 1)(k + 2)(k/3 + 1) = (k + 1)(k + 2)(k + 3)/3, the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

10. 1x2 + 3x4 + 5x6 + ... + (2n - 1) x 2n = n(n + 1)(4n - 1)/3

Step 1 (base case): For n = 1: LHS = 2, RHS = 1 x 2 x 3/3 = 2. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume the sum up to k is k(k + 1)(4k - 1)/3. Adding the next term (2k + 1)(2k + 2) = 2(k +

1)(2k + 1) gives (k + 1)[k(4k - 1)/3 + 2(2k + 1)] = (k + 1)(4k^2 - k + 12k + 6)/3 = (k + 1)(4k^2 + 11k + 6)/3 = (k + 1)(k + 2)(4k + 3)/3 = (k + 1)(k + 2)(4(k + 1) - 1)/3, the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

11. 1/(1x2) + 1/(2x3) + 1/(3x4) + ... + 1/(n(n + 1)) = 1 - 1/(n + 1)

Step 1 (base case): For n = 1: LHS = 1/2, RHS = 1 - 1/2 = 1/2. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume the sum up to k is 1 - 1/(k + 1) = k/(k + 1). Adding the next term 1/((k + 1)(k + 2)) gives k/(k + 1) + 1/((k + 1)(k + 2)) = (k(k + 2) + 1)/((k + 1)(k + 2)) = (k + 1)^2/((k + 1)(k + 2)) = (k + 1)/(k + 2) = 1 - 1/(k + 2), the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

12. 1/(1x3) + 1/(3x5) + 1/(5x7) + ... + 1/((2n - 1)(2n + 1)) = n/(2n + 1)

Step 1 (base case): For n = 1: LHS = 1/3, RHS = 1/3. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume the sum up to k is k/(2k + 1). Adding the next term gives k/(2k + 1) + 1/((2k + 1)(2k + 3)) = (k(2k + 3) + 1)/((2k + 1)(2k + 3)) = (2k^2 + 3k + 1)/((2k + 1)(2k + 3)) = (k + 1)(2k + 1)/((2k + 1)(2k + 3)) = (k + 1)/(2k + 3) = (k + 1)/(2(k + 1) + 1), the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

13. 1/(2x5) + 1/(5x8) + 1/(8x11) + ... + 1/((3n - 1)(3n + 2)) = n/(2(3n + 2))

Step 1 (base case): For n = 1: LHS = 1/10, RHS = 1/(2 x 5) = 1/10. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume the sum up to k is k/(2(3k + 2)). Note 1/((3k + 2)(3k + 5)) = (1/3)(1/(3k + 2) - 1/(3k + 5)). Adding the next term: k/(2(3k + 2)) + 1/((3k + 2)(3k + 5)) = (k(3k + 5) + 2)/(2(3k + 2)(3k + 5)) = (3k^2 + 5k + 2)/(2(3k + 2)(3k + 5)) = (k + 1)(3k + 2)/(2(3k + 2)(3k + 5)) = (k + 1)/(2(3k + 5)) = (k + 1)/(2(3(k + 1) + 2)), the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

14. r + r^2 + r^3 + ... + r^n = r(1 - r^n)/(1 - r), (r != 1)

Step 1 (base case): For n = 1: LHS = r, RHS = r(1 - r)/(1 - r) = r. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume r + ... + r^k = r(1 - r^k)/(1 - r). Adding the next term r^(k+1) gives r(1 - r^k)/(1 - r) + r^(k+1) = r[(1 - r^k) + r^k(1 - r)]/(1 - r) = r(1 - r^(k+1))/(1 - r), the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

15. a + (a + d) + (a + 2d) + ... + [a + (n - 1)d] = n/2 [2a + (n - 1)d]

Step 1 (base case): For n = 1: LHS = a, RHS = (1/2)(2a) = a. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume the sum up to k is k/2 [2a + (k - 1)d]. Adding the next term a + kd gives k/2 [2a + (k - 1)d] + a + kd = (2ak + k(k - 1)d + 2a + 2kd)/2 = ((k + 1)(2a) + k(k + 1)d)/2 = (k + 1)/2 [2a + kd], the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

16. 1 x |1 + 2 x |2 + 3 x |3 + ... + n x |n = |n+1 - 1 (here |n means n!)

Step 1 (base case): For n = 1: LHS = 1 x 1 = 1, RHS = 2! - 1 = 1. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume 1 x 1! + ... + k x k! = (k + 1)! - 1. Adding the next term (k + 1) x (k + 1)! gives (k + 1)! - 1 + (k + 1)(k + 1)! = (k + 2)(k + 1)! - 1 = (k + 2)! - 1, the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

17. a_n = a_1 + (n - 1)d when a_1, a_1 + d, a_1 + 2d, ... form an A.P.

Step 1 (base case): For n = 1: a_1 = a_1 + 0 x d. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume a_k = a_1 + (k - 1)d. The next term of the A.P. is a_(k+1) = a_k + d = a_1 + (k - 1)d + d = a_1 + kd, the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

18. a_n = a_1 r^(n-1) when a_1, a_1 r, a_1 r^2, ... form a G.P.

Step 1 (base case): For n = 1: a_1 = a_1 r^0. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume a_k = a_1 r^(k-1). The next term of the G.P. is a_(k+1) = a_k x r = a_1 r^(k-1) x r = a_1 r^k, the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

19. 1^2 + 3^2 + 5^2 + ... + (2n - 1)^2 = n(4n^2 - 1)/3

Step 1 (base case): For n = 1: LHS = 1, RHS = 1 x 3/3 = 1. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume the sum up to k is k(4k^2 - 1)/3. Adding the next term (2k + 1)^2 gives k(4k^2 - 1)/3 + (2k + 1)^2 = (4k^3 - k + 3(4k^2 + 4k + 1))/3 = (4k^3 + 12k^2 + 11k + 3)/3 = (k + 1)(4k^2 + 8k + 3)/3 = (k + 1)(4(k + 1)^2 - 1)/3, the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

20. C(3,3) + C(4,3) + C(5,3) + ... + C(n + 2,3) = C(n + 3,4) (hockey-stick identity)

Step 1 (base case): For n = 1: LHS = C(3,3) = 1, RHS = C(4,4) = 1. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume C(3,3) + ... + C(k + 2,3) = C(k + 3,4). Adding the next term C(k + 3,3) and using Pascal's identity C(m,4) + C(m,3) = C(m + 1,4) with m = k + 3 gives C(k + 3,4) + C(k + 3,3) = C(k + 4,4), the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

21. Prove by mathematical induction that for all positive integral values of n: i) n^2 + n is divisible by 2. ii) 5^n - 2^n is divisible by 3. iii) 5^n - 1 is divisible by 4. iv) 8 x 10^n - 2 is divisible by 6. v) n^3 - n is divisible by 6.

(i) Step 1: For n = 1: 1 + 1 = 2, divisible by 2. Step 2: Assume k^2 + k = 2m. Then (k + 1)^2 + (k + 1) = k^2 + 3k + 2 = (k^2 + k) + 2(k + 1) = 2m + 2(k + 1), divisible by 2. (ii) Step 1: For n = 1: 5 - 2 = 3, divisible by 3. Step 2: Assume 5^k - 2^k = 3m. Then 5^(k+1) - 2^(k+1) = 5(5^k - 2^k) + 3 x 2^k = 15m + 3 x 2^k, divisible by 3. (iii) Step 1: For n = 1: 5 - 1 = 4, divisible by 4. Step 2: Assume 5^k - 1 = 4m. Then 5^(k+1) - 1 = 5(5^k - 1) + 4 = 20m + 4, divisible by 4. (iv) Step 1: For n = 1: 80 - 2 = 78 = 6 x 13, divisible by 6. Step 2: Assume 8 x 10^k - 2 = 6m. Then 8 x 10^(k+1) - 2 = 10(8 x 10^k - 2) + 18 = 60m + 18, divisible by 6. (v) Step 1: For n = 1: 1 - 1 = 0, divisible by 6. Step 2: Assume k^3 - k = 6m. Then (k + 1)^3 - (k + 1) = k^3 + 3k^2 + 2k = (k^3 - k) + 3k(k + 1) = 6m + 3k(k + 1); k(k + 1) is even, so 3k(k + 1) is divisible by 6, and the sum is divisible by 6.

22. 1/3 + 1/3^2 + ... + 1/3^n = (1/2)[1 - 1/3^n]

Step 1 (base case): For n = 1: LHS = 1/3, RHS = (1/2)(1 - 1/3) = 1/3. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume the

sum up to k is (1/2)(1 - 1/3^k). Adding the next term 1/3^(k+1) gives (1/2)(1 - 1/3^k) + 1/3^(k+1) = 1/2 - 1/(2 x 3^k) + 1/3^(k+1) = 1/2 - 3/(2 x 3^(k+1)) + 2/(2 x 3^(k+1)) = (1/2)(1 - 1/3^(k+1)), the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

23. 1^2 - 2^2 + 3^2 - 4^2 + ... + (-1)^(n-1) x n^2 = (-1)^(n-1) x n(n + 1)/2

Step 1 (base case): For n = 1: LHS = 1, RHS = 1 x 2/2 = 1. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume the sum up to k is (-1)^(k-1) k(k + 1)/2. Adding the next term (-1)^k (k + 1)^2 gives (-1)^k (k + 1)[-(k/2) + (k + 1)] = (-1)^k (k + 1)(k + 2)/2, the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

24. 1^3 + 3^3 + 5^3 + ... + (2n - 1)^3 = n^2[2n^2 - 1]

Step 1 (base case): For n = 1: LHS = 1, RHS = 1 x 1 = 1. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume the sum up to k is k^2(2k^2 - 1). Adding the next term (2k + 1)^3 = 8k^3 + 12k^2 + 6k + 1 gives 2k^4 - k^2 + 8k^3 + 12k^2 + 6k + 1 = 2k^4 + 8k^3 + 11k^2 + 6k + 1 = (k + 1)^2(2(k + 1)^2 - 1), since (k + 1)^2(2k^2 + 4k + 1) = 2k^4 + 8k^3 + 11k^2 + 6k + 1, the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

25. x + 1 is a factor of x^(2n) - 1; (x != -1)

Step 1 (base case): For n = 1: x^2 - 1 = (x - 1)(x + 1), so x + 1 is a factor. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume x + 1 is a factor of x^(2k) - 1. Then x^(2(k+1)) - 1 = x^2 x^(2k) - 1 = x^2(x^(2k) - 1) + (x^2 - 1); both terms are divisible by x + 1, so x + 1 is a factor of x^(2k+2) - 1, the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

26. x - y is a factor of x^n - y^n; (x != y)

Step 1 (base case): For n = 1: x - y is a factor of x - y. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume x - y is a factor of x^k - y^k. Then x^(k+1) - y^(k+1) = x(x^k - y^k) + y^k(x - y); both terms are divisible by x - y, so x - y is a factor of x^(k+1) - y^(k+1), the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

27. x + y is a factor of x^(2n-1) + y^(2n-1); (x != -y)

Step 1 (base case): For n = 1: x + y is a factor of x + y. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume x + y is a factor of x^(2k-1) + y^(2k-1). Then x^(2k+1) + y^(2k+1) = x^2(x^(2k-1) + y^(2k-1)) + y^(2k-1)(y^2 - x^2) = x^2(x^(2k-1) + y^(2k-1)) - y^(2k-1)(x - y)(x + y); both terms are divisible by x + y, so x + y is a factor of x^(2(k+1)-1) + y^(2(k+1)-1), the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

28. Use mathematical induction to show that 1 + 2 + 2^2 + ... + 2^n = 2^(n+1) - 1 for all non-negative integers n.

Step 1 (base case): For n = 0: LHS = 1, RHS = 2 - 1 = 1. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume 1 + 2 + ... + 2^k = 2^(k+1) - 1. Adding the next term 2^(k+1) gives 2^(k+1) - 1 + 2^(k+1) = 2^(k+2) - 1, the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

29. If A and B are square matrices and AB = BA, then show by mathematical induction that AB^n = B^n A for any positive integer n.

Step 1 (base case): For n = 1: AB = BA by hypothesis. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume AB^k = B^k A. Then AB^(k+1) = (AB^k)B = (B^k A)B = B^k(AB) = B^k(BA) = B^(k+1)A, the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

30. Prove by the Principle of mathematical induction that n^2 - 1 is divisible by 8 when n is an odd positive integer.

Write n = 2m + 1? Induction on odd integers: for the first odd positive integer n = 1: 1 - 1 = 0, divisible by 8. Step: assume k^2 - 1 is divisible by 8 for odd k; the next odd integer is k + 2, and (k + 2)^2 - 1 = k^2 + 4k + 3 = (k^2 - 1) + 4(k + 1); since k is odd, k + 1 is even, so 4(k + 1) is divisible by 8; hence (k + 2)^2 - 1 is divisible by 8. By induction on odd integers, n^2 - 1 is divisible by 8 for every odd positive integer n.

31. Use the principle of mathematical induction to prove that ln x^n = n ln x for any integer n >= 0 if x is a positive number.

Step 1 (base case): For n = 0: ln x^0 = ln 1 = 0 = 0 x ln x. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume ln x^k = k ln x. Then ln x^(k+1) = ln(x^k x x) = ln x^k + ln x = k ln x + ln x = (k + 1) ln x, the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

Use the principle of extended mathematical induction to prove that: 32. n! > 2^n - 1 for integral values of n >= 4.

Step 1 (base case): For n = 4: 4! = 24 > 15 = 2^4 - 1. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume k! > 2^k - 1 for some k >= 4. Then (k + 1)! = (k + 1) k! > (k + 1)(2^k - 1) = (k + 1)2^k - (k + 1). Since (k + 1)2^k - (k + 1) >= 2^(k+1) - 1 exactly when (k - 1)2^k >= k, which holds for k >= 4 (as 3 x 16 = 48 >= 4), we get (k + 1)! > 2^(k+1) - 1, the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

33. n^2 > n + 3 for integral values of n >= 3.

Step 1 (base case): For n = 3: 9 > 6. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume k^2 > k + 3 for some k >= 3. Then (k + 1)^2 = k^2 + 2k + 1 > k + 3 + 2k + 1 = 3k + 4 > (k + 1) + 3 = k + 4, since 3k + 4 > k + 4 for k > 0, the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

34. 4^n > 3^n + 2^(n-1) for integral values of n >= 2.

Step 1 (base case): For n = 2: 16 > 9 + 2 = 11. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume 4^k > 3^k + 2^(k-1) for some k >= 2. Then 4^(k+1) = 4 x 4^k > 4(3^k + 2^(k-1)) = 4 x 3^k + 2^(k+1) = 3^(k+1) + 3^k + 2^(k+1) > 3^(k+1) + 2^k, since 3^k + 2^(k+1) > 2^k, the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

35. 3^n < n! for integral values of n > 6.

Step 1 (base case): For n = 7: 3^7 = 2187 < 5040 = 7!. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume 3^k < k! for some k >= 7. Then 3^(k+1) = 3 x 3^k < 3 x k! < (k + 1) k! = (k + 1)!, since k + 1 > 3, the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

36. n! > n^2 for integral values of n >= 4.

Step 1 (base case): For n = 4: 24 > 16. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume k! > k^2 for some k >= 4. Then (k + 1)! = (k + 1) k! > (k + 1) k^2 = k^3 + k^2 > (k + 1)^2 = k^2 + 2k + 1, since k^3 > 2k + 1 for k >= 4 (64 > 9), the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

37. 3 + 5 + 7 + ... + (2n + 5) = (n + 2)(n + 4) for integral values of n >= -1.

Step 1 (base case): For n = -1: the sum runs from 3 to 2(-1) + 5 = 3, so LHS = 3; RHS = (1)(3) = 3. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume the sum up to k (k >= -1) is (k + 2)(k + 4). Adding the next term (2(k + 1) + 5) = 2k + 7 gives (k + 2)(k + 4) + 2k + 7 = k^2 + 6k + 8 + 2k + 7 = k^2 + 8k + 15 = (k + 3)(k + 5) = ((k + 1) + 2)((k + 1) + 4), the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

38. 1 + nx <= (1 + x)^n for n >= 2 and x > -1 (Bernoulli's inequality).

Step 1 (base case): For n = 2: (1 + x)^2 = 1 + 2x + x^2 >= 1 + 2x. True.

Step 2 (induction hypothesis): Assume the statement true for n = k. Assume 1 + kx <= (1 + x)^k for some k >= 2. Since x > -1, we have 1 + x > 0, so (1 + x)^(k+1) = (1 + x)^k (1 + x) >= (1 + kx)(1 + x) = 1 + (k + 1)x + kx^2 >= 1 + (k + 1)x, the statement for n = k + 1.

Step 3: By the principle of mathematical induction, the statement is true for every positive integer n.

Exercise 8.2 (book pp. 273-274)

1. Using binomial theorem, expand the following: i) (a + 2b)^5 ii) (x/2 - 2/x^2)^6 iii) (3a - x/(3a))^4 iv) (2a - x/a)^7 v) (x/(2y) - 2y/x)^8 vi) (sqrt(a/x) - sqrt(x/a))^6

i) a^5 + 5a^4(2b) + 10a^3(2b)^2 + 10a^2(2b)^3 + 5a(2b)^4 + (2b)^5 = a^5 + 10a^4 b + 40a^3 b^2 + 80a^2 b^3 + 80a b^4 + 32b^5. ii) (x/2)^6 - 6(x/2)^5(2/x^2) + 15(x/2)^4(2/x^2)^2 - 20(x/2)^3(2/x^2)^3 + 15(x/2)^2(2/x^2)^4 - 6(x/2)(2/x^2)^5 + (2/x^2)^6 = x^6/64 - 3x^3/8 + 15/4 -

20/x^3 + 60/x^6 - 96/x^9 + 64/x^12. iii) (3a)^4 - 4(3a)^3(x/(3a)) + 6(3a)^2(x/(3a))^2 - 4(3a)(x/(3a))^3 + (x/(3a))^4 = 81a^4 - 36a^2 x + 6x^2 - 4x^3/(9a^2) + x^4/(81a^4). iv) (2a)^7 - 7(2a)^6(x/a) + 21(2a)^5(x/a)^2 - 35(2a)^4(x/a)^3 + 35(2a)^3(x/a)^4 - 21(2a)^2(x/a)^5 + 7(2a)(x/a)^6 - (x/a)^7 = 128a^7 - 448a^5 x + 672a^3 x^2 - 560a x^3 + 280x^4/a - 84x^5/a^3 + 14x^6/a^5 - x^7/a^7. v) (x/(2y))^8 - 8(x/(2y))^7(2y/x) + 28(x/(2y))^6(2y/x)^2 - 56(x/(2y))^5(2y/x)^3 + 70(x/(2y))^4(2y/x)^4 - 56(x/(2y))^3(2y/x)^5 + 28(x/(2y))^2(2y/x)^6 - 8(x/(2y))(2y/x)^7 + (2y/x)^8 = x^8/(256y^8) - x^6/(8y^6) + 7x^4/(4y^4) - 14x^2/y^2 + 70 - 224y^2/x^2 + 448y^4/x^4 - 512y^6/x^6 + 256y^8/x^8. vi) (sqrt(a/x))^6 - 6(sqrt(a/x))^5(sqrt(x/a)) + 15(sqrt(a/x))^4(sqrt(x/a))^2 - 20(sqrt(a/x))^3(sqrt(x/a))^3 + 15(sqrt(a/x))^2(sqrt(x/a))^4 - 6(sqrt(a/x))(sqrt(x/a))^5 + (sqrt(x/a))^6 = a^3/x^3 - 6a^2/x^2 + 15a/x - 20 + 15x/a - 6x^2/a^2 + x^3/a^3. All expansions were checked by exact symbolic expansion.

2. Calculate the following by means of binomial theorem: i) (0.97)^3 ii) (2.02)^4 iii) (9.98)^4 iv) (21)^5

i) (1 - 0.03)^3 = 1 - 3(0.03) + 3(0.03)^2 - (0.03)^3 = 1 - 0.09 + 0.0027 - 0.000027 = 0.912673. ii) (2 + 0.02)^4 = 16 + 4(8)(0.02) + 6(4)(0.02)^2 + 4(2)(0.02)^3 + (0.02)^4 = 16 + 0.64 + 0.0096 + 0.000064 + 0.00000016 = 16.64966416. iii) (10 - 0.02)^4 = 10000 - 4(1000)(0.02) + 6(100)(0.02)^2 - 4(10)(0.02)^3 + (0.02)^4 = 10000 - 80 + 0.24 - 0.00032 + 0.00000016 = 9920.23968016. iv) (20 + 1)^5 = 3200000 + 5(160000) + 10(8000) + 10(400) + 5(20) + 1 = 3200000 + 800000 + 80000 + 4000 + 100 + 1 = 4084101.

3. Expand and simplify the following: i) (a + sqrt(2) x)^4 + (a - sqrt(2) x)^4 ii) (2 + sqrt(3))^5 + (2 - sqrt(3))^5 iii) (2 + i)^5 - (2 - i)^5 iv) (x + sqrt(x^2 - 1))^3 + (x - sqrt(x^2 - 1))^3

i) The odd powers cancel: 2[a^4 + 6a^2(2x^2) + (2x^2)^2] = 2a^4 + 24a^2 x^2

+ 8x^4. ii) The odd powers cancel: 2[2^5 + 10(2^3)(3) + 5(2)(3^2)] = 2[32 + 240 + 90] = 724. iii) (2 + i)^2 = 3 + 4i; (2 + i)^4 = (3 + 4i)^2 = -7 + 24i; (2 + i)^5 = (-7 + 24i)(2 + i) = -38 + 41i. The difference with its conjugate is 2 x 41i = 82i. iv) The odd powers of sqrt(x^2 - 1) cancel: 2[x^3 + 3x(x^2 - 1)] = 2[4x^3 - 3x] = 8x^3 - 6x.

4. Expand the following in ascending powers of x: i) (2 + x - x^2)^4 ii) (1 - x + x^2)^4 iii) (1 - x - x^2)^4

i) Treating (2 + x - x^2) as a trinomial and expanding (or expanding by the binomial theorem twice) gives: 16 + 32x - 8x^2 - 40x^3 + x^4 + 20x^5 - 2x^6 - 4x^7 + x^8. In ascending powers: 16 + 32x - 8x^2 - 40x^3 + x^4 + 20x^5 - 2x^6 - 4x^7 + x^8. ii) 1 - 4x + 10x^2 - 16x^3 + 19x^4 - 16x^5 + 10x^6 - 4x^7 + x^8. iii) 1 - 4x + 2x^2 + 8x^3 - 5x^4 - 8x^5 + 2x^6 + 4x^7 + x^8. All three were checked by exact symbolic expansion.

5. Expand the following in descending powers of x: i) (x^2 + x - 1)^3 ii) (x - 1 - 1/x)^3

i) (x^2 + x - 1)^2 = x^4 + 2x^3 - x^2 - 2x + 1. Multiplying by (x^2 + x - 1) gives x^6 + 3x^5 + 0x^4 - 5x^3 + 0x^2 + 3x - 1, i.e. x^6 + 3x^5 - 5x^3 + 3x - 1. ii) Write it as (x + (-1 - 1/x))^3 = x^3 + 3x^2(-1 - 1/x) + 3x(-1 - 1/x)^2 + (-1 - 1/x)^3 = x^3 - 3x^2 - 3x + 3x + 6 + 3/x - 1 - 3/x - 3/x^2 - 1/x^3 = x^3 - 3x^2 + 5 - 3/x^2 - 1/x^3.

6. Find the term involving: i) x^4 in the expansion of (3 - 2x)^7 ii) x^(-2) in the expansion of (x - 2/x^2)^13 iii) a^4 in the expansion of (2/x - a)^9 iv) y^3 in the expansion of (x - sqrt(y))^11

i) General term: C(7, r) 3^(7-r) (-2x)^r. For x^4, r = 4: C(7, 4) x 27 x 16 x^4 = 35 x 432 x^4 = 15120x^4. ii) General term: C(13, r) x^(13-r) (-2/x^2)^r = C(13, r)(-2)^r x^(13-3r). For x^(-2): 13 - 3r = -2, so r = 5: C(13, 5)(-32) x^(-2) = 1287 x (-32) x^(-2) = -41184x^(-2). iii) General term: C(9, r)(2/x)^(9-r)(-a)^r. For a^4, r =

4: C(9, 4)(2/x)^5 a^4 = 126 x 32 a^4/x^5 = 4032a^4/x^5. iv) General term: C(11, r) x^(11-r)(-sqrt(y))^r. For y^3, r = 6: C(11, 6) x^5 y^3 = 462x^5 y^3 (the sign is positive since (-1)^6 = 1).

7. Find the coefficient of: i) x^5 in the expansion of (x^2 - 3/(2x))^10 ii) x^n in the expansion of (x^2 - 1/x)^21

i) General term: C(10, r) x^(2(10-r)) (-3/(2x))^r = C(10, r)(-3/2)^r x^(20-3r). For x^5: 20 - 3r = 5, so r = 5: C(10, 5)(-3/2)^5 = 252 x (-243/32) = -15309/8. ii) General term: C(21, r) x^(2(21-r)) (-1/x)^r = C(21, r)(-1)^r x^(42-3r). For x^n: 42 - 3r = n, so r = (42 - n)/3. The coefficient is (-1)^r C(21, r) with r = (42 - n)/3, valid when 42 - n is divisible by 3 and 0 <= r <= 21 (i.e. n is a multiple of 3 between -21 and 42); otherwise x^n does not occur.

8. Find 6th term in the expansion of (x^2 - 3/(2x))^10

Step 1: The 6th term has r = 5: C(10, 5) x^(2 x 5) (-3/(2x))^5. Step 2: C(10, 5) = 252 and (-3/2)^5 = -243/32. Step 3: The term is 252 x (-243/32) x^5 = -15309/8 x^5. (This is the same term as in Q7(i), as expected.)

9. Find the term independent of x in the following expansions: i) (x - 2/x)^10 ii) (sqrt(x) + 1/(2x^2))^10 iii) (1 + x^2)^3 (1 + 1/x^2)^4

i) General term: C(10, r) x^(10-r)(-2/x)^r = C(10, r)(-2)^r x^(10-2r). For the constant term: 10 - 2r = 0, so r = 5: C(10, 5)(-32) = 252 x (-32) = -8064. ii) General term: C(10, r)(sqrt(x))^(10-r)(1/(2x^2))^r = C(10, r)(1/2)^r x^(5 - r/2 - 2r) = C(10, r)(1/2)^r x^(5 - 5r/2). For the constant term: 5 - 5r/2 = 0, so r = 2: C(10, 2)(1/4) = 45/4. iii) (1 + x^2)^3 (1 + 1/x^2)^4 = (1 + x^2)^3 (x^2 + 1)^4/x^8 = (1 + x^2)^7/x^8. The constant term comes from the x^8 term of (1 + x^2)^7, which is C(7, 4) = 35. So the term independent of x is 35.

10. Determine the middle term in the following expansions: i) (1/x - x^2/2)^12 ii) ((3/2)x - 1/(3x))^11 iii) (2x - 1/(2x))^(2m+1)

i) There are 13 terms, so the middle term is the 7th (r = 6): C(12,

6)(1/x)^6(-x^2/2)^6 = 924 x^6/64 = 231x^6/16. ii) There are 12 terms, so there are two middle terms, the 6th (r = 5) and 7th (r = 6): r = 5: C(11, 5)((3/2)x)^6(-1/(3x))^5 = 462 x (729/64) x (-1/243) = -693x/32. r = 6: C(11, 6)((3/2)x)^5(-1/(3x))^6 = 462 x (243/32) x (1/729) x (1/x) = 77/(16x). iii) There are 2m + 2 terms, so the middle terms are the (m+1)th (r = m) and (m+2)th (r = m+1): r = m: C(2m+1, m)(2x)^(m+1)(-1/(2x))^m = (-1)^m 2x C(2m+1, m). r = m+1: C(2m+1, m+1)(2x)^m(-1/(2x))^(m+1) = (-1)^(m+1) C(2m+1, m)/(2x), since C(2m+1, m+1) = C(2m+1, m).

11. Find (2n + 1)th term from the end in the expansion of (x - 1/(2x))^(3n)

Step 1: The expansion has 3n + 1 terms. Step 2: The (2n + 1)th term from the end is the (3n + 1) - (2n + 1) + 1 = (n + 1)th term from the beginning, i.e. r = n. Step 3: General term: C(3n, r) x^(3n-r)(-1/(2x))^r = C(3n, r)(-1)^r 2^(-r) x^(3n-2r). Step 4: For r = n: C(3n, n)(-1)^n 2^(-n) x^n. So the required term is (-1)^n C(3n, n) x^n/2^n.

12. Show that the middle term of (1 + x)^(2n) is (1.3.5....(2n - 1))/n! x 2^n x^n

Step 1: (1 + x)^(2n) has 2n + 1 terms, so the middle term is the (n + 1)th term, r = n: C(2n, n) x^n = (2n)!/(n! n!) x^n. Step 2: (2n)! = (2 x 4 x ... x 2n) x (1 x 3 x ... x (2n - 1)) = 2^n n! x (1 x 3 x ... x (2n - 1)). Step 3: Dividing by n! n! gives (1 x 3 x ... x (2n - 1)) x 2^n/n!, so the middle term is (1.3.5....(2n - 1))/n! x 2^n x^n, as required.

13. Show that: C(n,1) + C(n,3) + C(n,5) + ... + C(n,n-1) = 2^(n-1)

Step 1: (1 + 1)^n = C(n,0) + C(n,1) + C(n,2) + ... + C(n,n) = 2^n. Step 2: (1 - 1)^n = C(n,0) - C(n,1) + C(n,2) - ... = 0 (for n >= 1). Step 3: Subtracting the second from the first gives 2[C(n,1) + C(n,3) + C(n,5) + ...] = 2^n. Step 4: Dividing by 2 gives C(n,1) + C(n,3) + C(n,5) + ... = 2^(n-1). Note: the printed last term C(n,n-1) is the last odd-index term when n is even; when n is odd the series ends at C(n,n), and the

identity still holds.

14. Show that: C(n,0) + (1/2)C(n,1) + (1/3)C(n,2) + (1/4)C(n,3) + ... + (1/(n+1))C(n,n) = (2^(n+1) - 1)/(n + 1)

Step 1: C(n,r)/(r + 1) = C(n+1,r+1)/(n + 1), since C(n+1,r+1) = (n + 1)!/((r + 1)!(n - r)!) = (n + 1)/(r + 1) x C(n,r). Step 2: So the sum equals (1/(n + 1)) x [C(n+1,1) + C(n+1,2) + ... + C(n+1,n+1)]. Step 3: The bracket is (1 + 1)^(n+1) - C(n+1,0) = 2^(n+1) - 1. Step 4: Hence the sum is (2^(n+1) - 1)/(n + 1), as required.

Exercise 8.3 (book pp. 283-284)

1. Expand the following upto 4 terms, taking the values of x such that the expansion in each case is valid: i) (1 - x)^(1/2) ii) (1 + 2x)^(-1) iii) (1 + x)^(-1/3) iv) (4 - 3x)^(1/2) v) (8 - 2x)^(-1) vi) (2 - 3x)^(-2) vii) (1 - x)/(1 + x)^2 viii) sqrt(1 + 2x)/(1 - x) ix) (4 + 2x)^(1/2)/(2 - x) x) (1 + x - 2x^2)^(1/2) xi) (1 - 2x + 3x^2)^(1/2)

i) 1 - x/2 - x^2/8 - x^3/16, valid for |x| < 1. ii) 1 - 2x + 4x^2 - 8x^3, valid for |x| < 1/2. iii) 1 - x/3 + 2x^2/9 - 14x^3/81, valid for |x| < 1. iv) 2(1 - 3x/4)^(1/2) = 2 - 3x/4 - 9x^2/64 - 27x^3/512, valid for |x| < 4/3. v) (1/8)(1 - x/4)^(-1) = 1/8 + x/32 + x^2/128 + x^3/512, valid for |x| < 4. vi) (1/4)(1 - 3x/2)^(-2) = 1/4 + 3x/4 + 27x^2/16 + 27x^3/8, valid for |x| < 2/3. vii) (1 - x)(1 - 2x + 3x^2 - 4x^3 + ...) = 1 - 3x + 5x^2 - 7x^3, valid for |x| < 1. viii) (1 + x - x^2/2 + ...)(1 + x + x^2 + x^3 + ...) = 1 + 2x + (3/2)x^2 + 2x^3, valid for |x| < 1/2. ix) (4 + 2x)^(1/2) = 2(1 + x/2)^(1/2) = 2 + x/2 - x^2/16 + x^3/64 - ...; 1/(2 - x) = (1/2)(1 + x/2 + x^2/4 + x^3/8 + ...). Product up to x^3: 1 + 3x/4 + 11x^2/32 + 23x^3/128, valid for |x| < 2. x) (1 + x - 2x^2)^(1/2) = 1 + x/2 - 9x^2/8 + 9x^3/16, valid for |x| < 1/2 (since 1 + x - 2x^2 = (1 + 2x)(1 - x)). xi) (1 - 2x + 3x^2)^(1/2) = 1 - x + x^2 + x^3, valid when |-2x + 3x^2| < 1. All expansions were checked by exact series computation.

2. Using Binomial theorem find the value of the following to three places of decimals: i) sqrt(99) ii) (0.98)^(1/2) iii) (1.03)^(1/3) iv) (65)^(1/3) v) (17)^(1/4) vi) (31)^(1/5) vii) 1/(998)^(1/3) viii) 1/(252)^(1/5) ix) sqrt(7)/sqrt(8) x) (0.998)^(-1/3) xi) 1/(486)^(1/6) xii) (1280)^(1/4)

i) sqrt(99) = 10(1 - 0.01)^(1/2) ≈ 10(1 - 0.005 - 0.0000125) = 9.950 (exact 9.9499). ii) (0.98)^(1/2) = (1 - 0.02)^(1/2) ≈ 1 - 0.01 - 0.00005 = 0.990 (exact 0.9899). iii) (1.03)^(1/3) = (1 + 0.03)^(1/3) ≈ 1 + 0.01 - 0.0001 = 1.010 (exact 1.0099). iv) (65)^(1/3) = 4(1 + 1/64)^(1/3) ≈ 4(1 + 1/192) = 4.021 (exact 4.0207). v) (17)^(1/4) = 2(1 + 1/16)^(1/4) ≈ 2(1 + 1/64) = 2.031 (exact 2.0305). vi) (31)^(1/5) = 2(1 - 1/32)^(1/5) ≈ 2(1 - 1/160 - 1/12800) = 1.987 (exact 1.9873). vii) (998)^(1/3) = 10(1 - 0.002)^(1/3) ≈ 10(1 - 0.002/3) = 9.9933, so 1/(998)^(1/3) ≈ 0.100 (exact 0.1001). viii) 1/(252)^(1/5): (252)^(1/5) ≈ 3.022, so the reciprocal is 0.331 (exact 0.3309). ix) sqrt(7)/sqrt(8) = sqrt(7/8) = (1 - 1/8)^(1/2) ≈ 1 - 1/16 - 1/512 = 0.935 (exact 0.9354). x) (0.998)^(-1/3) = (1 - 0.002)^(-1/3) ≈ 1 + 0.002/3 = 1.001 (exact 1.0007). xi) 1/(486)^(1/6): (486)^(1/6) ≈ 2.804, so the reciprocal is 0.357 (exact 0.3566). xii) 6^4 = 1296, so (1280)^(1/4) = 6(1280/1296)^(1/4) = 6(1 - 16/1296)^(1/4) ≈ 6(1 - 1/324) = 5.981 (exact 5.9814).

3. Find the coefficient of x^n in the expansion of: i) (1 + x^2)/(1 + x)^2 ii) (1 + x)^2/(1 - x)^2 iii) (1 + x)^3/(1 - x)^2 iv) (1 + x)^2/(1 - x)^3 v) (1 - x + x^2 - x^3 + ...)^2

i) (1 + x)^(-2) = 1 - 2x + 3x^2 - 4x^3 + ... + (-1)^r (r + 1)x^r + ... Multiplying by (1 + x^2): the coefficient of x^n is (-1)^n (n + 1) + (-1)^(n-2)(n - 1) = (-1)^n 2n for n >= 1; for n = 0 it is 1. ii) (1 - x)^(-2) = 1 + 2x + 3x^2 + ... + (r + 1)x^r + ... Multiplying by (1 + 2x + x^2): the coefficient of x^n is (n + 1) + 2n + (n - 1) = 4n for n >= 1; for n = 0 it is 1. iii) Multiplying (1 + 3x + 3x^2 + x^3) by (1 - x)^(-2) = sum (r + 1)x^r: the coefficient of x^n is (n + 1) + 3n + 3(n - 1) + (n - 2) = 8n - 4 for n >= 2; for n = 0 it is 1 and for n = 1 it is 5. iv) (1 - x)^(-3) = sum ((r + 1)(r + 2)/2)

x^r. Multiplying by (1 + 2x + x^2): the coefficient of x^n is [(n + 1)(n + 2) + 2n(n + 1) + n(n - 1)]/2 = 2n^2 + 2n + 1 (this formula also gives 1 for n = 0). v) 1 - x + x^2 - x^3 + ... = 1/(1 + x) for |x| < 1, so its square is (1 + x)^(-2) = sum (-1)^n (n + 1) x^n; the coefficient of x^n is (-1)^n (n + 1).

4. If x is so small that its square and higher powers can be neglected, then show that: i) (1 - x)/sqrt(1 + x) ≈ 1 - (3/2)x ii) sqrt(1 + 2x)/sqrt(1 - x) ≈ 1 + (3/2)x iii) ((9 + 7x)^(1/2) - (16 + 3x)^(1/4))/(4 + 5x) ≈ 1/4 - (17/384)x iv) sqrt(4 + x)/(1 - x)^3 ≈ 2 + (25/4)x v) (1 + x)^(1/2)(4 - 3x)^(3/2)/(8 + 5x)^(1/3) ≈ 4(1 - 5x/6) vi) (1 - x)^(1/2)(9 - 4x)^(1/2)/(8 + 3x)^(1/3) ≈ 3/2 - (61/48)x vii) (sqrt(4 - x) + (8 - x)^(1/3))/(8 - x)^(1/3) ≈ 2 - x/12

i) (1 - x)(1 + x)^(-1/2) ≈ (1 - x)(1 - x/2) ≈ 1 - 3x/2. ii) (1 + 2x)^(1/2)(1 - x)^(-1/2) ≈ (1 + x)(1 + x/2) ≈ 1 + 3x/2. iii) Numerator: 3(1 + 7x/9)^(1/2) - 2(1 + 3x/16)^(1/4) ≈ 3(1 + 7x/18) - 2(1 + 3x/64) = 1 + 103x/96. Dividing by 4 + 5x = 4(1 + 5x/4): (1/4)(1 + 103x/96)(1 - 5x/4) ≈ (1/4)(1 - 17x/96) = 1/4 - 17x/384. iv) sqrt(4 + x) = 2(1 + x/4)^(1/2) ≈ 2 + x/4; (1 - x)^(-3) ≈ 1 + 3x. Product: (2 + x/4)(1 + 3x) ≈ 2 + 25x/4. v) (1 + x)^(1/2) ≈ 1 + x/2; (4 - 3x)^(3/2) = 8(1 - 3x/4)^(3/2) ≈ 8(1 - 9x/8); (8 + 5x)^(1/3) = 2(1 + 5x/8)^(1/3) ≈ 2(1 + 5x/24). So the fraction is approximately 4(1 + x/2)(1 - 9x/8)(1 - 5x/24) ≈ 4(1 - 5x/6). vi) (1 - x)^(1/2) ≈ 1 - x/2; (9 - 4x)^(1/2) = 3(1 - 4x/9)^(1/2) ≈ 3(1 - 2x/9); (8 + 3x)^(1/3) = 2(1 + 3x/8)^(1/3) ≈ 2(1 + x/8). So the fraction is approximately (3/2)(1 - x/2 - 2x/9)(1 - x/8) ≈ (3/2)(1 - 61x/72) = 3/2 - 61x/48. vii) sqrt(4 - x) = 2(1 - x/4)^(1/2) ≈ 2 - x/4; (8 - x)^(1/3) = 2(1 - x/8)^(1/3) ≈ 2 - x/12. So the numerator is approximately 4 - x/3, and dividing by (8 - x)^(1/3) ≈ 2 - x/12 gives (4 - x/3)/(2 - x/12) = 2(1 - x/12)/(1 - x/24) ≈ 2(1 - x/24) = 2 - x/12.

5. If x is so small that its cube and higher powers can be neglected, then show that: i) sqrt(1 - x - 2x^2) ≈ 1 - (1/2)x - (9/8)x^2 ii) sqrt((1 + x)/(1 - x)) ≈ 1 + x + (1/2)x^2

i) Put u = -x - 2x^2. Then (1 + u)^(1/2) ≈ 1 + u/2 - u^2/8. To order x^2: u/2 = -x/2

- x^2 and u^2/8 = x^2/8 (since u^2 = x^2 + higher terms). So sqrt(1 - x - 2x^2) ≈ 1 - x/2 - x^2 - x^2/8 = 1 - x/2 - 9x^2/8. ii) (1 + x)^(1/2) ≈ 1 + x/2 - x^2/8; (1 - x)^(-1/2) ≈ 1 + x/2 + 3x^2/8. Product to order x^2: 1 + x + (3/8 - 1/8 + 1/4)x^2 = 1 + x + x^2/2.

6. If x is very nearly equal 1, then prove that p x^p - q x^q ≈ (p - q) x^(p+q)

Step 1: Put x = 1 + h with h small. Step 2: (1 + h)^p ≈ 1 + ph and (1 + h)^q ≈ 1 + qh. Step 3: p x^p - q x^q ≈ p(1 + ph) - q(1 + qh) = (p - q) + (p^2 - q^2)h. Step 4: (p - q) x^(p+q) ≈ (p - q)(1 + (p + q)h) = (p - q) + (p - q)(p + q)h = (p - q) + (p^2 - q^2)h. Step 5: Both sides agree to first order in h, so p x^p - q x^q ≈ (p - q) x^(p+q) when x is nearly 1.

7. If p - q is small when compared with p or q, show that ((2n + 1)p + (2n - 1)q)/((2n - 1)p + (2n + 1)q) ≈ ((p + q)/(2q))^(1/n)

Step 1: Put r = p/q = 1 + d, where d is small. Step 2: The left side becomes ((2n + 1)r + (2n - 1))/((2n - 1)r + (2n + 1)) = (4n + (2n + 1)d)/(4n + (2n - 1)d) ≈ (1 + (2n + 1)d/(4n))(1 - (2n - 1)d/(4n)) ≈ 1 + d/(2n). Step 3: The right side is ((r + 1)/2)^(1/n) = (1 + d/2)^(1/n) ≈ 1 + d/(2n). Step 4: Both sides agree to first order in d, which proves the approximation.

8. Show that [n/(2(n + N))]^(1/2) ≈ 8n/(9n - N) - (n + N)/(4n), where n and N are nearly equal

Step 1: Put N = n + d with d small compared with n. Step 2: Left side: [n/(2(2n + d))]^(1/2) = (1/2)(1 + d/(2n))^(-1/2) ≈ (1/2)(1 - d/(4n)) = 1/2 - d/(8n). Step 3: Right side: 8n/(9n - N) - (n + N)/(4n) = 8n/(8n - d) - (2n + d)/(4n) ≈ (1 + d/(8n)) - 1/2 - d/(4n) = 1/2 - d/(8n). Step 4: Both sides agree to first order in d, which proves the approximation.

9. Identify the following series as binomial expansion and find the sum in each case: i) 1 - (1/2)(1/4) + (1.3)/(2.4)(1/4)^2 - (1.3.5)/(2.4.6)(1/4)^3 + ... ii) 1 - (1/2)(1/2) + (1.3)/(2.4)(1/2)^2 - (1.3.5)/(2.4.6)(1/2)^3 + ... iii) 1 + 3/4 + (3.5)/(4.8) +

(3.5.7)/(4.8.12) + ... iv) 1 - (1/2)(1/3) + (1.3)/(2.4)(1/3)^2 - (1.3.5)/(2.4.6)(1/3)^3 + ...

i) This is (1 + x)^(-1/2) = 1 - x/2 + (1.3)/(2.4) x^2 - (1.3.5)/(2.4.6) x^3 + ... with x = 1/4. Sum = (1 + 1/4)^(-1/2) = (5/4)^(-1/2) = 2/sqrt(5). ii) The same series with x = 1/2. Sum = (1 + 1/2)^(-1/2) = (3/2)^(-1/2) = sqrt(2/3). iii) This is (1 - x)^(-3/2) = 1 + (3/2)x + (3.5)/(2.4) x^2 + (3.5.7)/(2.4.6) x^3 + ... with x = 1/2 (check: (3/2)(1/2) = 3/4; (3.5)/(2.4)(1/4) = 15/32 = (3.5)/(4.8)). Sum = (1 - 1/2)^(-3/2) = 2^(3/2) = 2 sqrt(2). iv) This is (1 + x)^(-1/2) with x = 1/3. Sum = (1 + 1/3)^(-1/2) = (4/3)^(-1/2) = sqrt(3)/2.

10. Use binomial theorem to show that 1 + 1/4 + (1.3)/(4.8) + (1.3.5)/(4.8.12) + ... = sqrt(2)

Step 1: (1 - x)^(-1/2) = 1 + (1/2)x + (1.3)/(2.4) x^2 + (1.3.5)/(2.4.6) x^3 + ... Step 2: Put x = 1/2: the terms become 1 + 1/4 + (1.3)/(2.4)(1/4) + (1.3.5)/(2.4.6)(1/8) + ... = 1 + 1/4 + (1.3)/(4.8) + (1.3.5)/(4.8.12) + ..., exactly the given series. Step 3: The sum is (1 - 1/2)^(-1/2) = (1/2)^(-1/2) = sqrt(2).

11. If y = 1/3 + (1.3)/(2!)(1/3)^2 + (1.3.5)/(3!)(1/3)^3 + ..., then prove that y^2 + 2y - 2 = 0

Step 1: (1.3....(2r - 1))/r! = C(2r, r)/2^r, since C(2r, r) = (2r)!/(r! r!) = (1.3....(2r - 1)) x 2^r/r!. Step 2: So y = sum_{r>=1} C(2r, r)(1/6)^r = (1 - 4/6)^(-1/2) - 1 = (1/3)^(-1/2) - 1 = sqrt(3) - 1. Step 3: Then y + 1 = sqrt(3), so (y + 1)^2 = 3, i.e. y^2 + 2y + 1 = 3, i.e. y^2 + 2y - 2 = 0.

12. If 2y = 1/2^2 + (1.3)/(2!) x 1/2^4 + (1.3.5)/(3!) x 1/2^6 + ..., then prove that 4y^2 + 4y - 1 = 0

Step 1: The r-th term is (1.3....(2r - 1))/r! x (1/4)^r = C(2r, r)(1/8)^r. Step 2: So 2y = sum_{r>=1} C(2r, r)(1/8)^r = (1 - 4/8)^(-1/2) - 1 = (1/2)^(-1/2) - 1 = sqrt(2) - 1. Step 3: Then 2y + 1 = sqrt(2), so (2y + 1)^2 = 2, i.e. 4y^2 + 4y + 1 = 2, i.e. 4y^2 + 4y - 1 = 0.

13. If y = 2/5 + (1.3)/(2!)(2/5)^2 + (1.3.5)/(3!)(2/5)^3 + ..., then prove that y^2 + 2y - 4 = 0

Step 1: The r-th term is (1.3....(2r - 1))/r! x (2/5)^r = C(2r, r)(1/5)^r. Step 2: So y = sum_{r>=1} C(2r, r)(1/5)^r = (1 - 4/5)^(-1/2) - 1 = (1/5)^(-1/2) - 1 = sqrt(5) - 1. Step 3: Then y + 1 = sqrt(5), so (y + 1)^2 = 5, i.e. y^2 + 2y + 1 = 5, i.e. y^2 + 2y - 4 = 0.

Chapter 9: Fundamentals of Trigonometry

Exercise 9.1 (book pp. 292-293)

1. Express the following sexagesimal measures of angles in radians: i) 30 deg ii) 45 deg iii) 60 deg iv) 75 deg v) 90 deg vi) 105 deg vii) 120 deg viii) 135 deg ix) 150 deg x) 10 deg 15' xi) 35 deg 20' xii) 75 deg 6' 30'' xiii) 120' 40'' xiv) 154 deg 20'' xv) 0 deg xvi) 3''

Multiply by pi/180. i) 30 pi/180 = pi/6. ii) 45 pi/180 = pi/4. iii) 60 pi/180 = pi/3. iv) 75 pi/180 = 5pi/12. v) 90 pi/180 = pi/2. vi) 105 pi/180 = 7pi/12. vii) 120 pi/180 = 2pi/3. viii) 135 pi/180 = 3pi/4. ix) 150 pi/180 = 5pi/6. x) 10 deg 15' = 10.25 deg = (41/4) pi/180 = 41pi/720. xi) 35 deg 20' = (106/3) deg, giving (106/3) pi/180 = 53pi/270. xii) 75 deg 6' 30'' = (9013/120) deg, giving 9013pi/21600. xiii) 120' 40'' = (181/90) deg, giving 181pi/16200. FLAG: the printed key appears to read 181pi/15200 for this part, but (181/90) x (pi/180) = 181pi/16200, so 15200 is arithmetically impossible and is treated as a misprint (or a scan misread of 16200). xiv) 154 deg 20'' = (138605/900) deg, giving 27721pi/32400. xv) 0 deg = 0. xvi) 3'' = (1/1200) deg, giving pi/216000.

2. Convert the following radian measures of angles into the measures of sexagesimal system: i) pi/8 ii) pi/6 iii) pi/4 iv) pi/3 v) pi/2 vi) 2pi/3 vii) 3pi/4 viii) 5pi/6

ix) 7pi/12 x) 9pi/5 xi) 11pi/27 xii) 13pi/16 xiii) 17pi/24 xiv) 25pi/36 xv) 19pi/32

Multiply by 180/pi. i) (pi/8)(180/pi) = 22.5 deg = 22 deg 30'. ii) 30 deg. iii) 45 deg. iv) 60 deg. v) 90 deg. vi) 120 deg. vii) 135 deg. viii) 150 deg. ix) 105 deg. x) 324 deg. xi) (11pi/27)(180/pi) = 220/3 deg = 73 deg 20'. xii) (13pi/16)(180/pi) = 585/4 deg = 146 deg 15'. xiii) (17pi/24)(180/pi) = 127.5 deg = 127 deg 30'. FLAG: the printed key reads 127 deg 37' 59'' for this part, but 17 x 180/24 = 127.5 exactly, so 127 deg 30' is the correct value and the key's figure is flagged. xiv) 125 deg. xv) (19pi/32)(180/pi) = 106.875 deg = 106 deg 52' 30''.

3. What is the circular measure of the angle between the hands of a watch at 4 O'clock?

At 4 O'clock the minute hand points at 12 and the hour hand at 4, i.e. 4 hour-divisions apart. Each hour-division is 30 deg, so the angle is 4 x 30 deg = 120 deg = 120 pi/180 = 2pi/3 radians. Answer: 2pi/3.

4. Find theta, when: i) l = 1.5 cm, r = 2.5 cm ii) l = 3.2 m, r = 2 m

Use theta = l/r. i) theta = 1.5/2.5 = 0.6 rad. ii) theta = 3.2/2 = 1.6 rad. Answers: i) 0.6 rad ii) 1.6 rad.

5. Find l, when: i) theta = pi radians, r = 6 cm ii) theta = 65 deg 20', r = 18 mm

Use l = r theta. i) l = 6 x pi = 6pi = 18.85 cm (the printed key reads 18.86 cm, a 0.01 rounding difference; 6 x 3.1416 = 18.8496). ii) 65 deg 20' = (196/3) deg = 49pi/135 = 1.1404 rad, so l = 18 x 1.1404 = 20.5 mm. Answers: i) 18.85 cm ii) 20.5 mm.

6. Find r, when: i) l = 5 cm, theta = 1/2 radian ii) l = 56 cm, theta = 45 deg

Use r = l/theta. i) r = 5/(1/2) = 10 cm. ii) 45 deg = pi/4 rad, so r = 56/(pi/4) = 224/pi = 71.3 cm. Answers: i) 10 cm ii) 71.3 cm.

7. What is the length of the arc intercepted on a circle of radius 14 cms by the arms of a central angle of 45 deg?

45 deg = pi/4 rad, so l = r theta = 14 x pi/4 = 3.5pi = 10.99 cm. Answer: 10.99 cm.

8. Find the radius of the circle, in which the arms of a central angle of measure 1 radian cut off an arc of length 35 cm.

r = l/theta = 35/1 = 35 cm. Answer: 35 cm.

9. A railway train is running on a circular track of radius 500 meters at the rate of 30 km per hour. Through what angle will it turn in 10 sec.?

Speed = 30 km/h = 30000/3600 = 8.333 m/s. Distance covered in 10 s = 8.333 x 10 = 83.33 m. Hence theta = l/r = 83.33/500 = 1/6 rad. Answer: 1/6 rad.

10. A horse is tethered to a peg by a rope of 9 meters length and it can move in a circle with the peg as centre. If the horse moves along the circumference of the circle, keeping the rope tight, how far will it have gone when the rope has turned through an angle of 70 deg?

70 deg = 7pi/18 rad, so the distance = r theta = 9 x 7pi/18 = 3.5pi = 10.9966 m. Answer: 10.9966 m.

11. The pendulum of a clock is 20 cm long and it swings through an angle of 20 deg each second. How far does the tip of the pendulum move in 1 second?

20 deg = pi/9 rad, so the tip moves l = r theta = 20 x pi/9 = 6.983 cm. Answer: 6.983 cm.

12. Assuming the average distance of the earth from the sun to be 148 x 10^6 km and the angle subtended by the sun at the eye of a person on the earth of measure 9.3 x 10^(-3) radian. Find the diameter of the sun.

Diameter = distance x angle = 148 x 10^6 x 9.3 x 10^(-3) = 148 x 9.3 x 10^3 = 1376400 km. Answer: 1376400 km. FLAG: the printed key reads 1390000 km, which would require an angle of about 9.4 x 10^(-3) rad (or a distance of about 149.5 x 10^6 km); from the printed givens the product is 1376400 km, and the key's figure is flagged.

13. A circular wire of radius 6 cm is cut straightened and then bent so as to lie along the circumference of a hoop of radius 24 cm. Find the measure of the angle which it

subtends at the centre of the hoop.

The wire's length is its circumference = 2pi x 6 = 12pi cm. On the hoop of radius 24 cm it subtends theta = l/r = 12pi/24 = pi/2 rad. Answer: pi/2 rad.

14. Show that the area of a sector of a circular region of radius r is (1/2) r^2 theta, where theta is the circular measure of the central angle of the sector.

The area of the whole circle is pi r^2, corresponding to the central angle 2pi. The sector's area is the same fraction of the whole area as its angle is of 2pi: area = (theta/(2pi)) x pi r^2 = (1/2) r^2 theta, as required.

15. Two cities A and B lie on the equator such that their longitudes are 45 deg E and 25 deg W respectively. Find the distance between the two cities, taking radius of the earth as 6400 kms.

The angular separation is 45 deg + 25 deg = 70 deg = 7pi/18 rad. Hence the distance along the equator = r theta = 6400 x 7pi/18 = 6400 x 1.2217 = 7819 km. Answer: 7819 km.

16. The moon subtends an angle of 0.5 deg at the eye of an observer on earth. The distance of the moon from the earth is 3.844 x 10^5 km approx. What is the length of the diameter of the moon?

0.5 deg = pi/360 = 0.008727 rad. Hence the diameter = distance x angle = 3.844 x 10^5 x 0.008727 = 3354.53 km. Answer: 3354.53 km.

17. The angle subtended by the earth at the eye of a spaceman, landed on the moon, is 1 deg 54'. The radius of the earth is 6400 km. Find the approximate distance between the moon and the earth.

The angle subtended by the earth's diameter is 1 deg 54' = 1.9 deg = 0.03316 rad. Hence the distance = diameter/angle = 12800/0.03316 = 386007 km approx. (Equivalently, using the radius and the half-angle: 6400/tan 0 deg 57' = 386007 km.) Answer: 386007 km approx.

Exercise 9.2 (book pp. 301-302)

1. Find the signs of the following: i) sin 160 deg ii) cos 190 deg iii) tan 115 deg iv) sec 245 deg v) cot 80 deg vi) cosec 297 deg

i) 160 deg lies in quadrant II, where sine is positive: +ve. ii) 190 deg lies in quadrant III, where cosine is negative: -ve. iii) 115 deg lies in quadrant II, where tangent is negative: -ve. iv) 245 deg lies in quadrant III, where secant (1/cos) is negative: -ve. v) 80 deg lies in quadrant I, where all ratios are positive: +ve. vi) 297 deg lies in quadrant IV, where cosecant (1/sin) is negative: -ve. Answers: i) +ve ii) -ve iii) -ve iv) -ve v) +ve vi) -ve.

2. Fill in the blanks: i) sin(-310 deg) = ... sin 310 deg ii) cos(-75 deg) = ... cos 75 deg iii) tan(-182 deg) = ... tan 182 deg iv) cot(-173 deg) = ... cot 173 deg v) sec(-216 deg) = ... sec 216 deg vi) cosec(-15 deg) = ... cosec 15 deg

Sine, tangent, cotangent and cosecant are odd functions (f(-x) = -f(x)); cosine and secant are even (f(-x) = f(x)). Hence: i) sin(-310 deg) = - sin 310 deg. ii) cos(-75 deg) = + cos 75 deg. iii) tan(-182 deg) = - tan 182 deg. iv) cot(-173 deg) = - cot 173 deg. v) sec(-216 deg) = + sec 216 deg. vi) cosec(-15 deg) = - cosec 15 deg. Answers: i) - ii) + iii) - iv) - v) + vi) -.

3. In which quadrant are the terminal arms of the angle lie when i) sin theta < 0 and cos theta > 0 ii) cot theta > 0 and cosec theta > 0 iii) tan theta < 0 and cos theta > 0 iv) sec theta < 0 and sin theta < 0 v) cot theta > 0 and sin theta < 0 vi) cos theta < 0 and tan theta < 0?

i) sin < 0 in III, IV; cos > 0 in I, IV: common quadrant IV. ii) cot > 0 in I, III; cosec > 0 in I, II: common quadrant I. iii) tan < 0 in II, IV; cos > 0 in I, IV: common quadrant IV. iv) sec < 0 in II, III; sin < 0 in III, IV: common quadrant III. v) cot > 0 in I, III; sin < 0 in III, IV: common quadrant III. vi) cos < 0 in II, III; tan < 0 in II, IV: common quadrant II. Answers: i) IV ii) I iii) IV iv) III v) III vi) II.

4. Find the values of the remaining trigonometric functions: i) sin theta = 12/13 and

the terminal arm of the angle is in quad. I ii) cos theta = 9/41 and the terminal arm of the angle is in quad. IV iii) cos theta = -sqrt(3)/2 and the terminal arm of the angle is in quad. III iv) tan theta = -1/3 and the terminal arm of the angle is in quad. II v) sin theta = -1/sqrt(2) and the terminal arm of the angle is not in quad. III

i) In quad. I all ratios are positive. cos theta = sqrt(1 - (12/13)^2) = 5/13. tan theta = (12/13)/(5/13) = 12/5. cosec theta = 13/12. sec theta = 13/5. cot theta = 5/12. ii) In quad. IV cosine is positive and sine negative. sin theta = -sqrt(1 - (9/41)^2) = -40/41. tan theta = (-40/41)/(9/41) = -40/9. cosec theta = -41/40. sec theta = 41/9. cot theta = -9/40. iii) In quad. III sine and cosine are negative, tangent positive. sin theta = -sqrt(1 - 3/4) = -1/2. tan theta = (-1/2)/(-sqrt(3)/2) = 1/sqrt(3). cosec theta = -2. sec theta = -2/sqrt(3). cot theta = sqrt(3). iv) In quad. II sine is positive and cosine negative. From tan theta = -1/3, take opposite 1, adjacent -3, hypotenuse sqrt(10): sin theta = 1/sqrt(10), cos theta = -3/sqrt(10). cosec theta = sqrt(10). sec theta = -sqrt(10)/3. cot theta = -3. v) sin theta negative means quad. III or IV; not III, so quad. IV. cos theta = sqrt(1 - 1/2) = 1/sqrt(2). tan theta = (-1/sqrt(2))/(1/sqrt(2)) = -1. cosec theta = -sqrt(2). sec theta = sqrt(2). cot theta = -1.

5. If cot theta = 15/8 and the terminal arm of the angle is not in quad. I, find the values of cos theta and cosec theta.

Cotangent is positive in quads. I and III; not I, so the terminal arm is in quad. III, where sine and cosine are negative. From cot theta = 15/8, take adjacent 15, opposite 8, hypotenuse 17: cos theta = -15/17 and sin theta = -8/17, so cosec theta = -17/8. Answers: cos theta = -15/17, cosec theta = -17/8.

6. If cosec theta = (m^2 + 1)/(2m) and m > 0 (0 < theta < pi/2), find the values of the remaining trigonometric ratios.

sin theta = 1/cosec theta = 2m/(m^2 + 1). Then cos theta = sqrt(1 - sin^2 theta) = sqrt(((m^2 + 1)^2 - 4m^2)/(m^2 + 1)^2) = (m^2 - 1)/(m^2 + 1), taking the

positive root since theta is in quad. I. Hence tan theta = sin theta/cos theta = 2m/(m^2 - 1) and cot theta = (m^2 - 1)/(2m). Answers: sin theta = 2m/(1 + m^2), cos theta = (m^2 - 1)/(m^2 + 1), tan theta = 2m/(m^2 - 1), cot theta = (m^2 - 1)/(2m). Note: the cosine and tangent forms as printed in the key assume m > 1, so that (m^2 - 1) is positive as required in quad. I; for 0 < m < 1 the quad. I cosine would be (1 - m^2)/(m^2 + 1).

7. If tan theta = 1/sqrt(7) and the terminal arm of the angle is not in the III quad., find the values of (cosec^2 theta - sec^2 theta)/(cosec^2 theta + sec^2 theta).

Tangent is positive in quads. I and III; not III, so the terminal arm is in quad. I. From tan theta = 1/sqrt(7), take opposite 1, adjacent sqrt(7), hypotenuse sqrt(8): sin theta = 1/sqrt(8), cos theta = sqrt(7)/sqrt(8). Hence cosec^2 theta = 8 and sec^2 theta = 8/7. The required value = (8 - 8/7)/(8 + 8/7) = (48/7)/(64/7) = 48/64 = 3/4. Answer: 3/4.

8. If cot theta = 5/2 and the terminal arm of the angle is in the I quad., find the value of (3 sin theta + 4 cos theta)/(cos theta - sin theta).

In quad. I, from cot theta = 5/2, take adjacent 5, opposite 2, hypotenuse sqrt(29): sin theta = 2/sqrt(29), cos theta = 5/sqrt(29). Hence (3 sin theta + 4 cos theta)/(cos theta - sin theta) = (6/sqrt(29) + 20/sqrt(29))/(5/sqrt(29) - 2/sqrt(29)) = 26/3. Answer: 26/3. Note: the printed key labels this value '9.'; the exercise as printed on book pp. 301-302 ends at Question 8, so the key's numbering is treated as a misnumbering of Question 8.

Exercise 9.3 (book pp. 308-309)

1. Verify the following: i) sin 60 deg cos 30 deg - cos 60 deg sin 30 deg = sin 30 deg ii) sin^2 pi/6 + sin^2 pi/3 + tan^2 pi/4 = 2 iii) 2 sin 45 deg + (1/2) cosec 45 deg = 3/sqrt(2) iv) sin^2 pi/6 : sin^2 pi/4 : sin^2 pi/3 : sin^2 pi/2 = 1 : 2 : 3 : 4

i) L.H.S. = (sqrt(3)/2)(sqrt(3)/2) - (1/2)(1/2) = 3/4 - 1/4 = 1/2 = sin 30 deg. ii) L.H.S.

= (1/2)^2 + (sqrt(3)/2)^2 + 1^2 = 1/4 + 3/4 + 1 = 2. iii) L.H.S. = 2(1/sqrt(2)) + (1/2)(sqrt(2)) = sqrt(2) + sqrt(2)/2 = 3sqrt(2)/2 = 3/sqrt(2). iv) The values are 1/4, 1/2, 3/4, 1; multiplying by 4 gives 1 : 2 : 3 : 4, as required.

2. Evaluate the following: i) (tan pi/3 - tan pi/6)/(1 + tan pi/3 tan pi/6) ii) (1 - tan^2 pi/3)/(1 + tan^2 pi/3)

i) This is tan(pi/3 - pi/6) = tan pi/6 = 1/sqrt(3). Directly: (sqrt(3) - 1/sqrt(3))/(1 + sqrt(3) x 1/sqrt(3)) = ((3 - 1)/sqrt(3))/2 = 1/sqrt(3). ii) tan pi/3 = sqrt(3), so the value = (1 - 3)/(1 + 3) = -2/4 = -1/2. Answers: i) 1/sqrt(3) ii) -1/2.

3. Verify the following when theta = 30 deg, 45 deg: i) sin 2theta = 2 sin theta cos theta ii) cos 2theta = cos^2 theta - sin^2 theta iii) cos 2theta = 2cos^2 theta - 1 iv) cos 2theta = 1 - 2sin^2 theta v) tan 2theta = 2tan theta/(1 - tan^2 theta)

For theta = 30 deg: i) sin 60 deg = sqrt(3)/2; 2 sin 30 deg cos 30 deg = 2(1/2)(sqrt(3)/2) = sqrt(3)/2. ii) cos 60 deg = 1/2; cos^2 30 deg - sin^2 30 deg = 3/4 - 1/4 = 1/2. iii) 2cos^2 30 deg - 1 = 2(3/4) - 1 = 1/2. iv) 1 - 2sin^2 30 deg = 1 - 2(1/4) = 1/2. v) tan 60 deg = sqrt(3); 2tan 30 deg/(1 - tan^2 30 deg) = 2(1/sqrt(3))/(1 - 1/3) = (2/sqrt(3))/(2/3) = sqrt(3). For theta = 45 deg: i) sin 90 deg = 1; 2(1/sqrt(2))(1/sqrt(2)) = 1. ii) cos 90 deg = 0; 1/2 - 1/2 = 0. iii) 2(1/2) - 1 = 0. iv) 1 - 2(1/2) = 0. v) tan 90 deg is undefined, and 2tan 45 deg/(1 - tan^2 45 deg) = 2/(1 - 1) is likewise undefined, so both sides are undefined together. All five identities are verified for both angles.

4. Find x, if tan^2 45 deg - cos^2 60 deg = x sin 45 deg cos 45 deg tan 60 deg.

L.H.S. = 1^2 - (1/2)^2 = 1 - 1/4 = 3/4. R.H.S. = x (1/sqrt(2))(1/sqrt(2))(sqrt(3)) = x sqrt(3)/2. Hence x = (3/4)/(sqrt(3)/2) = 3/(2sqrt(3)) = sqrt(3)/2. Answer: x = sqrt(3)/2.

5. Find the values of the trigonometric functions of the following quadrantal angles: i) -pi ii) -3pi iii) 5pi/2 iv) -9pi/2 v) -15pi vi) 1530 deg vii) -2430 deg viii) 235pi/2 ix) 407pi/2

Reduce each angle to its coterminal angle in [0, 2pi), then read the functions. i) -pi is coterminal with pi: sin = 0, cos = -1, tan = 0, cosec undefined, sec = -1, cot undefined. ii) -3pi is coterminal with pi: same values as (i). iii) 5pi/2 is coterminal with pi/2: sin = 1, cos = 0, tan undefined, cosec = 1, sec undefined, cot = 0. iv) -9pi/2 is coterminal with 3pi/2: sin = -1, cos = 0, tan undefined, cosec = -1, sec undefined, cot = 0. v) -15pi is coterminal with pi: same values as (i). vi) 1530 deg = 1440 deg + 90 deg, coterminal with 90 deg = pi/2: same values as (iii). vii) -2430 deg = -2520 deg + 90 deg, coterminal with 90 deg: same values as (iii). viii) 235pi/2 = 116pi + 3pi/2, coterminal with 3pi/2: same values as (iv). ix) 407pi/2 = 202pi + 3pi/2, coterminal with 3pi/2: same values as (iv). Reduced angles (as given in the key): i) pi ii) pi iii) pi/2 iv) 3pi/2 v) pi vi) 90 deg vii) 90 deg viii) 3pi/2 ix) 3pi/2.

6. Find the values of the trigonometric functions of the following angles: i) 390 deg ii) -330 deg iii) 765 deg iv) -675 deg v) -17pi/3 vi) 13pi/3 vii) 25pi/6 viii) -71pi/6 ix) -1035 deg

Reduce each angle, then read the functions. i) 390 deg = 360 deg + 30 deg: sin = 1/2, cos = sqrt(3)/2, tan = 1/sqrt(3), cosec = 2, sec = 2/sqrt(3), cot = sqrt(3). ii) -330 deg = -360 deg + 30 deg: same values as (i). iii) 765 deg = 720 deg + 45 deg: sin = 1/sqrt(2), cos = 1/sqrt(2), tan = 1, cosec = sqrt(2), sec = sqrt(2), cot = 1. iv) -675 deg = -720 deg + 45 deg: same values as (iii). v) -17pi/3 = -6pi + pi/3: sin = sqrt(3)/2, cos = 1/2, tan = sqrt(3), cosec = 2/sqrt(3), sec = 2, cot = 1/sqrt(3). vi) 13pi/3 = 4pi + pi/3: same values as (v). vii) 25pi/6 = 4pi + pi/6: sin = 1/2, cos = sqrt(3)/2, tan = 1/sqrt(3), cosec = 2, sec = 2/sqrt(3), cot = sqrt(3). viii) -71pi/6 = -12pi + pi/6: same values as (vii). ix) -1035 deg = -1080 deg + 45 deg: same values as (iii). Reduced angles (as given in the key): i) 30 deg ii) 30 deg iii) 45 deg iv) 45 deg v) pi/3 vi) pi/3 vii) pi/6 viii) pi/6 ix) 45 deg.

Exercise 9.4 (book pp. 311-312)

Prove the following identities, state the domain of theta in each case: 1. tan theta + cot theta = cosec theta sec theta 2. sec theta cosec theta sin theta cos theta = 1 3. cos theta + tan theta sin theta = sec theta 4. cosec theta + tan theta sec theta = cosec theta sec^2 theta 5. sec^2 theta - cosec^2 theta = tan^2 theta - cot^2 theta 6. cot^2 theta - cos^2 theta = cot^2 theta cos^2 theta 7. (sec theta + tan theta)(sec theta - tan theta) = 1 8. 2cos^2 theta - 1 = 1 - 2sin^2 theta 9. cos^2 theta - sin^2 theta = (1 - tan^2 theta)/(1 + tan^2 theta) 10. (cos theta - sin theta)/(cos theta + sin theta) = (cot theta - 1)/(cot theta + 1) 11. sin theta/(1 + cos theta) + cot theta = cosec theta

1. L.H.S. = sin theta/cos theta + cos theta/sin theta = (sin^2 theta + cos^2 theta)/(sin theta cos theta) = 1/(sin theta cos theta) = cosec theta sec theta. Domain: sin theta != 0 and cos theta != 0, i.e. theta is not a multiple of pi/2. 2. L.H.S. = (1/cos theta)(1/sin theta) sin theta cos theta = 1. Domain: theta is not a multiple of pi/2. 3. L.H.S. = cos theta + (sin theta/cos theta) sin theta = (cos^2 theta + sin^2 theta)/cos theta = 1/cos theta = sec theta. Domain: cos theta != 0, i.e. theta != (2n + 1)pi/2. 4. L.H.S. = 1/sin theta + sin theta/cos^2 theta = (cos^2 theta + sin^2 theta)/(sin theta cos^2 theta) = 1/(sin theta cos^2 theta) = cosec theta sec^2 theta. Domain: sin theta != 0 and cos theta != 0. 5. L.H.S. = (1 + tan^2 theta) - (1 + cot^2 theta) = tan^2 theta - cot^2 theta. Domain: theta is not a multiple of pi/2. 6. L.H.S. = cos^2 theta/sin^2 theta - cos^2 theta = cos^2 theta (1 - sin^2 theta)/sin^2 theta = cos^2 theta x cos^2 theta/sin^2 theta = cot^2 theta cos^2 theta. Domain: sin theta != 0, i.e. theta != n pi. 7. L.H.S. = sec^2 theta - tan^2 theta = 1. Domain: cos theta != 0. 8. Both sides equal cos^2 theta - sin^2 theta: 2cos^2 theta - 1 = cos^2 theta - (1 - cos^2 theta) = cos^2 theta - sin^2 theta, and 1 - 2sin^2 theta = (sin^2 theta + cos^2 theta) - 2sin^2 theta = cos^2 theta - sin^2 theta. Domain: all theta. 9. R.H.S. = (1 - sin^2 theta/cos^2 theta)/(1 + sin^2 theta/cos^2 theta) = (cos^2 theta - sin^2 theta)/(cos^2 theta + sin^2

theta) = cos^2 theta - sin^2 theta. Domain: cos theta != 0. 10. Dividing numerator and denominator of the L.H.S. by sin theta gives (cot theta - 1)/(cot theta + 1). Domain: sin theta != 0 and cos theta + sin theta != 0. 11. sin theta/(1 + cos theta) = sin theta (1 - cos theta)/(1 - cos^2 theta) = sin theta (1 - cos theta)/sin^2 theta = (1 - cos theta)/sin theta = cosec theta - cot theta. Adding cot theta gives cosec theta. Domain: sin theta != 0.

12. (cot^2 theta - 1)/(1 + cot^2 theta) = 2cos^2 theta - 1 13. (1 + cos theta)/(1 - cos theta) = (cosec theta + cot theta)^2 14. (sec theta - tan theta)^2 = (1 - sin theta)/(1 + sin theta) 15. 2tan theta/(1 + tan^2 theta) = 2 sin theta cos theta 16. (1 - sin theta)/cos theta = cos theta/(1 + sin theta) 17. (tan theta + cot theta)^2 = sec^2 theta cosec^2 theta 18. (tan theta + sec theta - 1)/(tan theta - sec theta + 1) = tan theta + sec theta 19. 1/(cosec theta - cot theta) - 1/sin theta = 1/sin theta - 1/(cosec theta + cot theta) 20. sin^3 theta - cos^3 theta = (sin theta - cos theta)(1 + sin theta cos theta) 21. sin^6 theta - cos^6 theta = (sin^2 theta - cos^2 theta)(1 - sin^2 theta cos^2 theta) 22. sin^6 theta + cos^6 theta = 1 - 3sin^2 theta cos^2 theta 23. 1/(1 + sin theta) + 1/(1 - sin theta) = 2sec^2 theta 24. (cos theta + sin theta)/(cos theta - sin theta) + (cos theta - sin theta)/(cos theta + sin theta) = 2/(1 - 2sin^2 theta)

12. L.H.S. = (cos^2 theta/sin^2 theta - 1)/(1 + cos^2 theta/sin^2 theta) = (cos^2 theta - sin^2 theta)/(sin^2 theta + cos^2 theta) = cos^2 theta - sin^2 theta = 2cos^2 theta - 1. Domain: sin theta != 0. 13. R.H.S. = (1/sin theta + cos theta/sin theta)^2 = (1 + cos theta)^2/sin^2 theta = (1 + cos theta)^2/((1 - cos theta)(1 + cos theta)) = (1 + cos theta)/(1 - cos theta). Domain: sin theta != 0 and cos theta != 1. 14. sec theta - tan theta = (1 - sin theta)/cos theta, so its square is (1 - sin theta)^2/cos^2 theta = (1 - sin theta)^2/((1 - sin theta)(1 + sin theta)) = (1 - sin theta)/(1 + sin theta). Domain: cos theta != 0 and sin theta != -1. 15. L.H.S. = 2(sin theta/cos theta)/(1 + sin^2 theta/cos^2 theta) = 2 sin theta cos theta/(cos^2 theta

+ sin^2 theta) = 2 sin theta cos theta. Domain: cos theta != 0. 16. Cross-multiplying, (1 - sin theta)(1 + sin theta) = 1 - sin^2 theta = cos^2 theta, which is the numerator of the R.H.S. times cos theta. Domain: cos theta != 0 and sin theta != -1. 17. From Question 1, tan theta + cot theta = 1/(sin theta cos theta) = sec theta cosec theta; squaring gives sec^2 theta cosec^2 theta. Domain: theta is not a multiple of pi/2. 18. Put t = tan theta + sec theta. Then (tan theta + sec theta)(tan theta - sec theta) = tan^2 theta - sec^2 theta = -1, so tan theta - sec theta = -1/t. Hence the denominator tan theta - sec theta + 1 = 1 - 1/t = (t - 1)/t, and the numerator is t - 1, so the fraction equals t = tan theta + sec theta. Domain: cos theta != 0 and tan theta - sec theta + 1 != 0. 19. Note 1/(cosec theta - cot theta) = sin theta/(1 - cos theta) = (1 + cos theta)/sin theta = cosec theta + cot theta. Hence L.H.S. = (cosec theta + cot theta) - cosec theta = cot theta. Also 1/(cosec theta + cot theta) = sin theta/(1 + cos theta) = (1 - cos theta)/sin theta = cosec theta - cot theta, so R.H.S. = cosec theta - (cosec theta - cot theta) = cot theta. So both sides equal cot theta. Domain: sin theta != 0. 20. R.H.S. = (sin theta - cos theta)(sin^2 theta + cos^2 theta + sin theta cos theta) = sin^3 theta - cos^3 theta by the difference-of-cubes formula a^3 - b^3 = (a - b)(a^2 + ab + b^2). Domain: all theta. 21. sin^6 theta - cos^6 theta = (sin^2 theta)^3 - (cos^2 theta)^3 = (sin^2 theta - cos^2 theta)(sin^4 theta + sin^2 theta cos^2 theta + cos^4 theta) = (sin^2 theta - cos^2 theta)((sin^2 theta + cos^2 theta)^2 - sin^2 theta cos^2 theta) = (sin^2 theta - cos^2 theta)(1 - sin^2 theta cos^2 theta). Domain: all theta. 22. sin^6 theta + cos^6 theta = (sin^2 theta)^3 + (cos^2 theta)^3 = (sin^2 theta + cos^2 theta)(sin^4 theta - sin^2 theta cos^2 theta + cos^4 theta) = (sin^2 theta + cos^2 theta)^2 - 3sin^2 theta cos^2 theta = 1 - 3sin^2 theta cos^2 theta. Domain: all theta. 23. L.H.S. = (1 - sin theta + 1 + sin theta)/(1 - sin^2 theta) = 2/cos^2 theta = 2sec^2 theta. Domain: cos theta != 0. 24. L.H.S. = ((cos theta + sin theta)^2 + (cos theta - sin theta)^2)/(cos^2 theta -

sin^2 theta) = (2cos^2 theta + 2sin^2 theta)/(cos^2 theta - sin^2 theta) = 2/(cos^2 theta - sin^2 theta) = 2/(1 - 2sin^2 theta). Domain: cos theta != sin theta and cos theta != -sin theta.

Chapter 10: Trigonometric Identities Sum and Difference of Angles

Exercise 10.1 (book pp. 321-322)

1. Without using the tables, find the values of: i) sin(-780 deg) ii) cot(-855 deg) iii) cosec(2040 deg) iv) sec(-960 deg) v) tan(1110 deg) vi) sin(-300 deg)

Reduce each angle by whole revolutions. i) sin(-780 deg) = -sin 780 deg = -sin(780 deg - 720 deg) = -sin 60 deg = -sqrt(3)/2. ii) cot(-855 deg) = -cot 855 deg = -cot(855 deg - 720 deg) = -cot 135 deg = -(-1) = 1. iii) cosec(2040 deg) = cosec(2040 deg - 1800 deg) = cosec 240 deg = 1/sin 240 deg = 1/(-sqrt(3)/2) = -2/sqrt(3). iv) sec(-960 deg) = sec 960 deg = sec(960 deg - 720 deg) = sec 240 deg = 1/cos 240 deg = 1/(-1/2) = -2. v) tan(1110 deg) = tan(1110 deg - 1080 deg) = tan 30 deg = 1/sqrt(3). vi) sin(-300 deg) = -sin 300 deg = -(-sqrt(3)/2) = sqrt(3)/2. Answers: i) -sqrt(3)/2 ii) 1 iii) -2/sqrt(3) iv) -2 v) 1/sqrt(3) vi) sqrt(3)/2.

2. Express each of the following as a trigonometric function of an angle of positive degree measure of less than 45 deg: i) sin 196 deg ii) cos 147 deg iii) sin 319 deg iv) cos 254 deg v) tan 294 deg vi) cos 728 deg vii) sin(-625 deg) viii) cos(-435 deg) ix) sin 150 deg

i) sin 196 deg = -sin(196 deg - 180 deg) = -sin 16 deg. ii) cos 147 deg = -cos(180 deg - 147 deg) = -cos 33 deg. iii) sin 319 deg = -sin(360 deg - 319 deg) = -sin 41 deg. iv) cos 254 deg = cos(270 deg - 16 deg) = -sin 16 deg. v) tan 294 deg = tan(270 deg + 24 deg) = -cot 24 deg. vi) cos 728 deg = cos(728 deg - 720 deg) =

cos 8 deg. vii) sin(-625 deg) = -sin 625 deg = -sin(625 deg - 360 deg) = -sin 265 deg = -(-cos 5 deg) = cos 5 deg. viii) cos(-435 deg) = cos 435 deg = cos(435 deg - 360 deg) = cos 75 deg = sin 15 deg. ix) sin 150 deg = sin(180 deg - 150 deg) = sin 30 deg. Answers: i) -sin 16 deg ii) -cos 33 deg iii) -sin 41 deg iv) -sin 16 deg v) -cot 24 deg vi) cos 8 deg vii) cos 5 deg viii) sin 15 deg ix) sin 30 deg. Note: the printed key labels the last answer 'x)'; the exercise as printed has nine parts, so the key's numbering is treated as a misnumbering of part (ix).

3. Prove the following: i) sin(180 deg + alpha) sin(90 deg - alpha) = -sin alpha cos alpha ii) sin 780 deg sin 480 deg + cos 120 deg sin 30 deg = 1/2 iii) cos 306 deg + cos 234 deg + cos 162 deg + cos 18 deg = 0 iv) cos 330 deg sin 600 deg + cos 120 deg sin 150 deg = -1

i) sin(180 deg + alpha) = -sin alpha and sin(90 deg - alpha) = cos alpha, so the product is -sin alpha cos alpha. ii) sin 780 deg = sin 60 deg = sqrt(3)/2 and sin 480 deg = sin 120 deg = sqrt(3)/2, so the first product is 3/4; cos 120 deg sin 30 deg = (-1/2)(1/2) = -1/4; the sum is 3/4 - 1/4 = 1/2. iii) cos 306 deg = cos 54 deg, cos 234 deg = -cos 54 deg, cos 162 deg = -cos 18 deg, so the sum is cos 54 deg - cos 54 deg - cos 18 deg + cos 18 deg = 0. iv) cos 330 deg = cos 30 deg = sqrt(3)/2 and sin 600 deg = sin 240 deg = -sqrt(3)/2, so the first product is -3/4; cos 120 deg sin 150 deg = (-1/2)(1/2) = -1/4; the sum is -1.

4. Prove that: i) [sin^2(pi + theta) tan(3pi/2 + theta)] / [cot^2(3pi/2 - theta) cos^2(pi - theta) cosec(2pi - theta)] = cos theta ii) [cos(90 deg + theta) sec(-theta) tan(180 deg - theta)] / [sec(360 deg - theta) sin(180 deg + theta) cot(90 deg - theta)] = -1

i) Numerator: sin^2(pi + theta) = sin^2 theta and tan(3pi/2 + theta) = -cot theta, so the numerator is -sin^2 theta cot theta = -sin theta cos theta. Denominator: cot^2(3pi/2 - theta) = tan^2 theta, cos^2(pi - theta) = cos^2 theta, cosec(2pi - theta) = -cosec theta, so the denominator is tan^2 theta x cos^2 theta x (-cosec

theta) = -sin theta. The ratio is (-sin theta cos theta)/(-sin theta) = cos theta. ii) Numerator: cos(90 deg + theta) = -sin theta, sec(-theta) = sec theta, tan(180 deg - theta) = -tan theta, so the numerator is sin theta sec theta tan theta. Denominator: sec(360 deg - theta) = sec theta, sin(180 deg + theta) = -sin theta, cot(90 deg - theta) = tan theta, so the denominator is -sin theta sec theta tan theta. The ratio is -1.

5. If alpha, beta, gamma are the angles of a triangle ABC, then prove that i) sin(alpha + beta) = sin gamma ii) cos((alpha + beta)/2) = sin(gamma/2) iii) cos(alpha + beta) = -cos gamma iv) tan(alpha + beta) + tan gamma = 0

Since alpha + beta + gamma = 180 deg, we have alpha + beta = 180 deg - gamma. i) sin(alpha + beta) = sin(180 deg - gamma) = sin gamma. ii) (alpha + beta)/2 = 90 deg - gamma/2, so cos((alpha + beta)/2) = cos(90 deg - gamma/2) = sin(gamma/2). iii) cos(alpha + beta) = cos(180 deg - gamma) = -cos gamma. iv) tan(alpha + beta) = tan(180 deg - gamma) = -tan gamma, so tan(alpha + beta) + tan gamma = 0.

Exercise 10.2 (book pp. 326-328)

1. Prove that: i) sin(180 deg + theta) = -sin theta ii) cos(180 deg + theta) = -cos theta iii) tan(270 deg - theta) = cot theta iv) cos(theta - 180 deg) = -cos theta v) cos(270 deg + theta) = sin theta vi) sin(theta + 270 deg) = -cos theta vii) tan(180 deg + theta) = tan theta viii) cos(360 deg - theta) = cos theta

These are the allied-angle formulas. i) sin(180 deg + theta) = -sin theta since the angle is in the third quadrant with reference angle theta. ii) cos(180 deg + theta) = -cos theta by the same reasoning. iii) tan(270 deg - theta) = cot theta (the 270 deg shift changes tangent to cotangent with a positive sign). iv) cos(theta - 180 deg) = cos(180 deg - theta) with sign: cos(theta - 180 deg) = -cos theta. v) cos(270 deg + theta) = sin theta. vi) sin(theta + 270 deg) = -cos theta. vii) tan(180 deg + theta) =

tan theta (period 180 deg). viii) cos(360 deg - theta) = cos theta (cosine is even and periodic).

2. Find the values of the following: i) sin 15 deg ii) cos 15 deg iii) tan 15 deg iv) sin 105 deg v) cos 105 deg vi) tan 105 deg (Hint: 15 deg = (45 deg - 30 deg) and 105 deg = (60 deg + 45 deg))

i) sin 15 deg = sin(45 deg - 30 deg) = sin 45 deg cos 30 deg - cos 45 deg sin 30 deg = (1/sqrt(2))(sqrt(3)/2) - (1/sqrt(2))(1/2) = (sqrt(3) - 1)/(2sqrt(2)). ii) cos 15 deg = cos(45 deg - 30 deg) = cos 45 deg cos 30 deg + sin 45 deg sin 30 deg = (sqrt(3) + 1)/(2sqrt(2)). iii) tan 15 deg = sin 15 deg/cos 15 deg = (sqrt(3) - 1)/(sqrt(3) + 1). iv) sin 105 deg = sin(60 deg + 45 deg) = sin 60 deg cos 45 deg + cos 60 deg sin 45 deg = (sqrt(3) + 1)/(2sqrt(2)). v) cos 105 deg = cos(60 deg + 45 deg) = cos 60 deg cos 45 deg - sin 60 deg sin 45 deg = (1 - sqrt(3))/(2sqrt(2)). vi) tan 105 deg = sin 105 deg/cos 105 deg = (1 + sqrt(3))/(1 - sqrt(3)).

3. Prove that: i) sin(45 deg + alpha) = (1/sqrt(2))(sin alpha + cos alpha) ii) cos(alpha + 45 deg) = (1/sqrt(2))(cos alpha - sin alpha)

i) sin(45 deg + alpha) = sin 45 deg cos alpha + cos 45 deg sin alpha = (1/sqrt(2)) cos alpha + (1/sqrt(2)) sin alpha = (1/sqrt(2))(sin alpha + cos alpha). ii) cos(alpha + 45 deg) = cos alpha cos 45 deg - sin alpha sin 45 deg = (1/sqrt(2))(cos alpha - sin alpha).

4. Prove that: i) tan(45 deg + A) tan(45 deg - A) = 1 ii) tan(pi/4 - theta) + tan(3pi/4 + theta) = 0 iii) sin(theta + pi/6) + cos(theta + pi/3) = cos theta iv) [sin theta - cos theta tan(theta/2)] / [cos theta + sin theta tan(theta/2)] = tan(theta/2) v) (1 - tan theta tan phi)/(1 + tan theta tan phi) = cos(theta + phi)/cos(theta - phi)

i) tan(45 deg + A) = (1 + tan A)/(1 - tan A) and tan(45 deg - A) = (1 - tan A)/(1 + tan A); their product is 1. ii) tan(3pi/4 + theta) = tan(pi - (pi/4 - theta)) = -tan(pi/4 - theta), so the sum is 0. Note: the right-hand side is cut off at the foot of book p. 326 in the scan; 0 is the value forced by the computation. iii) sin(theta + pi/6) = sin

theta cos pi/6 + cos theta sin pi/6 = (sqrt(3)/2) sin theta + (1/2) cos theta. cos(theta + pi/3) = cos theta cos pi/3 - sin theta sin pi/3 = (1/2) cos theta - (sqrt(3)/2) sin theta. Adding gives cos theta. iv) Put t = tan(theta/2), so tan theta = 2t/(1 - t^2). Dividing numerator and denominator by cos theta gives (tan theta - t)/(1 + tan theta x t) = (2t/(1 - t^2) - t)/(1 + 2t^2/(1 - t^2)) = (t(1 + t^2)/(1 - t^2))/((1 + t^2)/(1 - t^2)) = t = tan(theta/2). v) Multiplying numerator and denominator by cos theta cos phi gives (cos theta cos phi - sin theta sin phi)/(cos theta cos phi + sin theta sin phi) = cos(theta + phi)/cos(theta - phi).

5. Show that: cos(alpha + beta) cos(alpha - beta) = cos^2 alpha - sin^2 beta = cos^2 beta - sin^2 alpha

cos(alpha + beta) cos(alpha - beta) = (cos alpha cos beta - sin alpha sin beta)(cos alpha cos beta + sin alpha sin beta) = cos^2 alpha cos^2 beta - sin^2 alpha sin^2 beta. Replacing cos^2 beta by 1 - sin^2 beta and sin^2 alpha by 1 - cos^2 alpha gives cos^2 alpha - sin^2 beta. Replacing cos^2 alpha by 1 - sin^2 alpha and sin^2 beta by 1 - cos^2 beta gives cos^2 beta - sin^2 alpha.

6. Show that: [sin(alpha + beta) + sin(alpha - beta)] / [cos(alpha + beta) + cos(alpha - beta)] = tan alpha

The numerator is 2 sin alpha cos beta and the denominator is 2 cos alpha cos beta; their ratio is tan alpha.

7. Show that: i) cot(alpha + beta) = (cot alpha cot beta - 1)/(cot alpha + cot beta) ii) cot(alpha - beta) = (cot alpha cot beta + 1)/(cot beta - cot alpha) iii) (tan alpha + tan beta)/(tan alpha - tan beta) = sin(alpha + beta)/sin(alpha - beta)

i) cot(alpha + beta) = cos(alpha + beta)/sin(alpha + beta) = (cos alpha cos beta - sin alpha sin beta)/(sin alpha cos beta + cos alpha sin beta); dividing numerator and denominator by sin alpha sin beta gives (cot alpha cot beta - 1)/(cot beta + cot alpha). ii) Similarly, cot(alpha - beta) = (cos alpha cos beta + sin alpha sin beta)/(sin alpha cos beta - cos alpha sin beta) = (cot alpha cot beta + 1)/(cot beta - cot alpha).

iii) Multiplying numerator and denominator by cos alpha cos beta gives (sin alpha cos beta + cos alpha sin beta)/(sin alpha cos beta - cos alpha sin beta) = sin(alpha + beta)/sin(alpha - beta).

8. If sin alpha = 4/5 and cos beta = 40/41, where 0 < alpha < pi/2 and 0 < beta < pi/2, show that sin(alpha - beta) = 133/205.

In the first quadrant, cos alpha = 3/5 and sin beta = 9/41. Hence sin(alpha - beta) = sin alpha cos beta - cos alpha sin beta = (4/5)(40/41) - (3/5)(9/41) = 160/205 - 27/205 = 133/205.

9. If sin alpha = 4/5 and sin beta = 12/13, where pi/2 < alpha < pi and pi/2 < beta < pi, find i) sin(alpha + beta) ii) cos(alpha + beta) iii) tan(alpha + beta) iv) sin(alpha - beta) v) cos(alpha - beta) vi) tan(alpha - beta). In which quadrants do the terminal sides of the angles of measures (alpha + beta) and (alpha - beta) lie?

In the second quadrant, cos alpha = -3/5 and cos beta = -5/13. i) sin(alpha + beta) = (4/5)(-5/13) + (-3/5)(12/13) = -20/65 - 36/65 = -56/65. ii) cos(alpha + beta) = (-3/5)(-5/13) - (4/5)(12/13) = 15/65 - 48/65 = -33/65. iii) tan(alpha + beta) = (-56/65)/(-33/65) = 56/33. iv) sin(alpha - beta) = (4/5)(-5/13) - (-3/5)(12/13) = -20/65 + 36/65 = 16/65. v) cos(alpha - beta) = (-3/5)(-5/13) + (4/5)(12/13) = 15/65 + 48/65 = 63/65. vi) tan(alpha - beta) = 16/63. Since sin(alpha + beta) < 0 and cos(alpha + beta) < 0, the terminal side of alpha + beta lies in quadrant III; since sin(alpha - beta) > 0 and cos(alpha - beta) > 0, the terminal side of alpha - beta lies in quadrant I.

10. Find sin(alpha + beta) and cos(alpha + beta), given that i) tan alpha = 3/4, cos beta = 5/13 and neither the terminal side of the angle of measure alpha nor that of beta is in the I quadrant ii) tan alpha = -15/8 and sin beta = -7/25 and neither the terminal side of the angle of measure alpha nor that of beta is in the IV quadrant

i) tan alpha > 0 means quadrant I or III; not I, so quadrant III: sin alpha = -3/5, cos alpha = -4/5. cos beta > 0 means quadrant I or IV; not I, so quadrant IV: sin beta =

-12/13. Hence sin(alpha + beta) = (-3/5)(5/13) + (-4/5)(-12/13) = -15/65 + 48/65 = 33/65, and cos(alpha + beta) = (-4/5)(5/13) - (-3/5)(-12/13) = -20/65 - 36/65 = -56/65. ii) tan alpha < 0 means quadrant II or IV; not IV, so quadrant II: sin alpha = 15/17, cos alpha = -8/17. sin beta < 0 means quadrant III or IV; not IV, so quadrant III: cos beta = -24/25. Hence sin(alpha + beta) = (15/17)(-24/25) + (-8/17)(-7/25) = -360/425 + 56/425 = -304/425, and cos(alpha + beta) = (-8/17)(-24/25) - (15/17)(-7/25) = 192/425 + 105/425 = 297/425. FLAG: the printed key reads (416/425, 3/5) for part (ii), but those two numbers are internally inconsistent ((416/425)^2 + (3/5)^2 = 1.318, not 1) and do not follow from the printed conditions; the computed values (-304/425, 297/425) are given instead.

11. Prove that: (cos 8 deg - sin 8 deg)/(cos 8 deg + sin 8 deg) = tan 37 deg

Dividing numerator and denominator by cos 8 deg gives (1 - tan 8 deg)/(1 + tan 8 deg) = tan(45 deg - 8 deg) = tan 37 deg.

12. If alpha, beta, gamma are the angles of a triangle ABC, show that cot(alpha/2) + cot(beta/2) + cot(gamma/2) = cot(alpha/2) cot(beta/2) cot(gamma/2)

Since (alpha + beta)/2 = 90 deg - gamma/2, we have cot((alpha + beta)/2) = tan(gamma/2) = 1/cot(gamma/2). But cot((alpha + beta)/2) = (cot(alpha/2) cot(beta/2) - 1)/(cot(alpha/2) + cot(beta/2)). Hence (cot(alpha/2) cot(beta/2) - 1)/(cot(alpha/2) + cot(beta/2)) = 1/cot(gamma/2), which cross-multiplies to cot(alpha/2) cot(beta/2) cot(gamma/2) - cot(gamma/2) = cot(alpha/2) + cot(beta/2), i.e. cot(alpha/2) + cot(beta/2) + cot(gamma/2) = cot(alpha/2) cot(beta/2) cot(gamma/2).

13. If alpha + beta + gamma = 180 deg, show that cot alpha cot beta + cot beta cot gamma + cot gamma cot alpha = 1

Since alpha + beta = 180 deg - gamma, tan(alpha + beta) = -tan gamma. Hence (tan alpha + tan beta)/(1 - tan alpha tan beta) = -tan gamma, so tan alpha + tan beta = -tan gamma + tan alpha tan beta tan gamma. Dividing by tan alpha tan

beta tan gamma gives cot beta cot gamma + cot alpha cot gamma = -cot alpha cot beta + 1, i.e. cot alpha cot beta + cot beta cot gamma + cot gamma cot alpha = 1.

14. Express the following in the form r sin(theta + phi): i) 12 sin theta + 5 cos theta ii) 3 sin theta - 4 cos theta iii) sin theta - cos theta iv) 5 sin theta - 4 cos theta v) sin theta + cos theta vi) 3 sin theta - 5 cos theta

Write a sin theta + b cos theta = r sin(theta + phi) with r = sqrt(a^2 + b^2), cos phi = a/r, sin phi = b/r. i) r = 13, cos phi = 12/13, sin phi = 5/13: 13 sin(theta + phi) with tan phi = 5/12 (phi in quadrant I). ii) r = 5, cos phi = 3/5, sin phi = -4/5: 5 sin(theta + phi) with tan phi = -4/3, phi in quadrant IV; equivalently 5 sin(theta - psi) with tan psi = 4/3, psi in quadrant I. FLAG: the printed key reads tan phi = 4/3 for this part (unsigned), while the signed value in the form r sin(theta + phi) is -4/3; the key gives signed values (-1, -4/5, -5/3) for parts (iii), (iv), (vi). iii) r = sqrt(2), cos phi = 1/sqrt(2), sin phi = -1/sqrt(2): sqrt(2) sin(theta + phi) with tan phi = -1. iv) r = sqrt(41), cos phi = 5/sqrt(41), sin phi = -4/sqrt(41): sqrt(41) sin(theta + phi) with tan phi = -4/5. v) r = sqrt(2), cos phi = 1/sqrt(2), sin phi = 1/sqrt(2): sqrt(2) sin(theta + phi) with tan phi = 1. vi) r = sqrt(34), cos phi = 3/sqrt(34), sin phi = -5/sqrt(34): sqrt(34) sin(theta + phi) with tan phi = -5/3. Note: the question's phrase 'terminal sides of theta and phi in the first quadrant' cannot hold for the minus cases (ii), (iii), (iv), (vi), where phi lies in quadrant IV; the signed-phi convention above is used throughout.

Exercise 10.3 (book p. 332)

1. Find the values of sin 2alpha, cos 2alpha and tan 2alpha, when: i) sin alpha = 12/13 ii) sin alpha = 12/13 iii) cos alpha = 3/5, where 0 < alpha < pi/2

i) Taking alpha acute, cos alpha = 5/13. Then sin 2alpha = 2 sin alpha cos alpha = 2(12/13)(5/13) = 120/169; cos 2alpha = 1 - 2sin^2 alpha = 1 - 288/169 = -119/169; tan 2alpha = -120/119. ii) Part (ii) is printed identically to part (i) (a duplication in

the book, with no distinguishing condition); the notes give the same values as (i).

FLAG: the printed key's second set of values (24/25, -7/25, -24/7) is exactly the

answer to part (iii), not to a distinct part (ii). iii) With cos alpha = 3/5 and 0 < alpha

< pi/2, sin alpha = 4/5. Then sin 2alpha = 2(4/5)(3/5) = 24/25; cos 2alpha =

(3/5)^2 - (4/5)^2 = -7/25; tan 2alpha = -24/7.

2. Prove the following identities: 2. cot alpha - tan alpha = 2 cot 2alpha 3. sin 2alpha/(1 + cos 2alpha) = tan alpha 4. (1 - cos alpha)/sin alpha = tan(alpha/2) 5. (cos alpha - sin alpha)/(cos alpha + sin alpha) = sec 2alpha - tan 2alpha 6. sqrt((1 + sin alpha)/(1 - sin alpha)) = (sin(alpha/2) + cos(alpha/2))/(sin(alpha/2) - cos(alpha/2)) 7. (cosec theta + 2 cosec 2theta)/sec theta = cot(theta/2)

2. cot alpha - tan alpha = cos alpha/sin alpha - sin alpha/cos alpha = (cos^2 alpha - sin^2 alpha)/(sin alpha cos alpha) = cos 2alpha/((1/2) sin 2alpha) = 2 cot 2alpha. 3. sin 2alpha/(1 + cos 2alpha) = 2 sin alpha cos alpha/(2cos^2 alpha) = tan alpha. 4. 1 - cos alpha = 2sin^2(alpha/2) and sin alpha = 2 sin(alpha/2) cos(alpha/2), so the ratio is tan(alpha/2). 5. sec 2alpha - tan 2alpha = (1 - sin 2alpha)/cos 2alpha = (sin alpha - cos alpha)^2/((cos alpha - sin alpha)(cos alpha + sin alpha)) = (cos alpha - sin alpha)/(cos alpha + sin alpha). 6. 1 + sin alpha = (sin(alpha/2) + cos(alpha/2))^2 and 1 - sin alpha = (sin(alpha/2) - cos(alpha/2))^2, so the square root is (sin(alpha/2) + cos(alpha/2))/(sin(alpha/2) - cos(alpha/2)) with the sign choice as printed. 7. cosec theta + 2 cosec 2theta = 1/sin theta + 1/(sin theta cos theta) = (1 + cos theta)/(sin theta cos theta); dividing by sec theta (multiplying by cos theta) gives (1 + cos theta)/sin theta = (2cos^2(theta/2))/(2 sin(theta/2) cos(theta/2)) = cot(theta/2).

8. 1 + tan alpha tan 2alpha = sec 2alpha 9. (2 sin theta sin 2theta)/(cos theta + cos 3theta) = tan 2theta tan theta 10. sin 3theta/sin theta - cos 3theta/cos theta = 2 11. cos 3theta/cos theta + sin 3theta/sin theta = 4 cos 2theta 12. (tan(theta/2) + cot(theta/2))/(cot(theta/2) - tan(theta/2)) = sec theta 13. sin 3theta/cos theta + cos

3theta/sin theta = 2 cot 2theta

8. tan alpha tan 2alpha = tan alpha x 2tan alpha/(1 - tan^2 alpha) = 2tan^2 alpha/(1 - tan^2 alpha), so 1 + tan alpha tan 2alpha = (1 + tan^2 alpha)/(1 - tan^2 alpha) = sec 2alpha. 9. The denominator cos theta + cos 3theta = 2 cos 2theta cos theta; the numerator 2 sin theta sin 2theta = 4sin^2 theta cos theta; the ratio is 2sin^2 theta/cos 2theta = tan 2theta tan theta (since tan 2theta tan theta = 2sin^2 theta/(cos^2 theta - sin^2 theta)). 10. Combining over sin theta cos theta gives (sin 3theta cos theta - cos 3theta sin theta)/(sin theta cos theta) = sin 2theta/(sin theta cos theta) = 2. 11. Combining over sin theta cos theta gives (cos 3theta sin theta + sin 3theta cos theta)/(sin theta cos theta) = sin 4theta/(sin theta cos theta) = 2 sin 4theta/sin 2theta = 4 cos 2theta. 12. tan(theta/2) + cot(theta/2) = 2/sin theta and cot(theta/2) - tan(theta/2) = 2cos theta/sin theta, so the ratio is sec theta. 13. Combining over sin theta cos theta gives (sin 3theta sin theta + cos 3theta cos theta)/(sin theta cos theta) = cos 2theta/(sin theta cos theta) = 2 cot 2theta.

14. Reduce sin^4 theta to an expression involving only function of multiples of theta, raised to the first power.

sin^2 theta = (1 - cos 2theta)/2, so sin^4 theta = (1 - cos 2theta)^2/4 = (1 - 2cos 2theta + cos^2 2theta)/4. Since cos^2 2theta = (1 + cos 4theta)/2, this becomes (1 - 2cos 2theta + (1 + cos 4theta)/2)/4 = (3 - 4cos 2theta + cos 4theta)/8. Answer: sin^4 theta = (3 - 4cos 2theta + cos 4theta)/8.

15. Find the values of sin theta and cos theta without using table or calculator, when theta is i) 18 deg ii) 36 deg iii) 54 deg iv) 72 deg. Hence prove that: cos 36 deg cos 72 deg cos 108 deg cos 144 deg = 1/16

Let theta = 18 deg. Then 2theta = 36 deg and 3theta = 54 deg = 90 deg - 2theta, so sin 2theta = cos 3theta, i.e. 2 sin theta cos theta = 4cos^3 theta - 3 cos theta. Cancelling cos theta gives 2 sin theta = 4(1 - sin^2 theta) - 3, i.e. 4sin^2 theta + 2

sin theta - 1 = 0, so sin theta = (sqrt(5) - 1)/4 (positive root). Hence: i) sin 18 deg = (sqrt(5) - 1)/4, cos 18 deg = sqrt(10 + 2sqrt(5))/4. ii) sin 36 deg = cos 54 deg = sqrt(10 - 2sqrt(5))/4, cos 36 deg = sin 54 deg = (sqrt(5) + 1)/4. iii) sin 54 deg = (sqrt(5) + 1)/4, cos 54 deg = sqrt(10 - 2sqrt(5))/4. iv) sin 72 deg = cos 18 deg = sqrt(10 + 2sqrt(5))/4, cos 72 deg = sin 18 deg = (sqrt(5) - 1)/4. For the product: cos 108 deg = -cos 72 deg and cos 144 deg = -cos 36 deg, so the product equals cos^2 36 deg cos^2 72 deg = ((sqrt(5) + 1)/4)^2 ((sqrt(5) - 1)/4)^2 = ((5 - 1)/16)^2 = (1/4)^2 = 1/16.

Exercise 10.4 (book p. 336)

1. Express the following products as sums or differences: i) 2 sin 3theta cos theta ii) 2 cos 5theta sin 3theta iii) sin 5theta cos 2theta iv) 2 sin 7theta sin 2theta v) cos(x + y) sin(x - y) vi) cos(2x + 30 deg) cos(2x - 30 deg) vii) sin 12 deg sin 46 deg viii) sin(x + 45 deg) sin(x - 45 deg)

Using 2 sin A cos B = sin(A + B) + sin(A - B), 2 cos A sin B = sin(A + B) - sin(A - B), 2 sin A sin B = cos(A - B) - cos(A + B), 2 cos A cos B = cos(A + B) + cos(A - B): i) 2 sin 3theta cos theta = sin 4theta + sin 2theta. ii) 2 cos 5theta sin 3theta = sin 8theta - sin 2theta. iii) sin 5theta cos 2theta = (1/2)(sin 7theta + sin 3theta). iv) 2 sin 7theta sin 2theta = cos 5theta - cos 9theta. v) cos(x + y) sin(x - y) = (1/2)(sin 2x - sin 2y). vi) cos(2x + 30 deg) cos(2x - 30 deg) = (1/2)(cos 4x + cos 60 deg). vii) sin 12 deg sin 46 deg = (1/2)(cos 34 deg - cos 58 deg). viii) sin(x + 45 deg) sin(x - 45 deg) = (1/2)(cos 90 deg - cos 2x).

2. Express the following sums or differences as products: i) sin 5theta + sin 3theta ii) sin 8theta - sin 4theta iii) cos 6theta + cos 3theta iv) cos 7theta - cos theta v) cos 12 deg + cos 48 deg vi) sin(x + 30 deg) + sin(x - 30 deg)

Using sin C + sin D = 2 sin((C + D)/2) cos((C - D)/2), sin C - sin D = 2 cos((C + D)/2) sin((C - D)/2), cos C + cos D = 2 cos((C + D)/2) cos((C - D)/2), cos C - cos D = -2

sin((C + D)/2) sin((C - D)/2): i) sin 5theta + sin 3theta = 2 sin 4theta cos theta. ii) sin 8theta - sin 4theta = 2 cos 6theta sin 2theta. iii) cos 6theta + cos 3theta = 2 cos(9theta/2) cos(3theta/2). iv) cos 7theta - cos theta = -2 sin 4theta sin 3theta. v) cos 12 deg + cos 48 deg = 2 cos 30 deg cos 18 deg. vi) sin(x + 30 deg) + sin(x - 30 deg) = 2 sin x cos 30 deg.

3. Prove the following identities: i) (sin 3x - sin x)/(cos x - cos 3x) = cot 2x ii) (sin 8x + sin 2x)/(cos 8x + cos 2x) = tan 5x iii) (sin alpha - sin beta)/(sin alpha + sin beta) = tan((alpha - beta)/2) cot((alpha + beta)/2)

i) The numerator sin 3x - sin x = 2 cos 2x sin x; the denominator cos x - cos 3x = -2 sin 2x sin(-x) = 2 sin 2x sin x; the ratio is cot 2x. ii) The numerator sin 8x + sin 2x = 2 sin 5x cos 3x; the denominator cos 8x + cos 2x = 2 cos 5x cos 3x; the ratio is tan 5x. iii) The numerator sin alpha - sin beta = 2 cos((alpha + beta)/2) sin((alpha - beta)/2); the denominator sin alpha + sin beta = 2 sin((alpha + beta)/2) cos((alpha - beta)/2); the ratio is tan((alpha - beta)/2) cot((alpha + beta)/2).

4. Prove that: i) cos 20 deg + cos 100 deg + cos 140 deg = 0 ii) sin(pi/4 - theta) sin(pi/4 + theta) = (1/2) cos 2theta iii) (sin theta + sin 3theta + sin 5theta + sin 7theta)/(cos theta + cos 3theta + cos 5theta + cos 7theta) = tan 4theta

i) cos 100 deg + cos 140 deg = 2 cos 120 deg cos 20 deg = 2(-1/2) cos 20 deg = -cos 20 deg, so adding cos 20 deg gives 0. ii) sin(pi/4 - theta) sin(pi/4 + theta) = (1/2)[cos(-2theta) - cos(pi/2)] = (1/2) cos 2theta. iii) The numerator groups as (sin 7theta + sin theta) + (sin 5theta + sin 3theta) = 2 sin 4theta cos 3theta + 2 sin 4theta cos theta = 2 sin 4theta (cos 3theta + cos theta); the denominator groups as 2 cos 4theta (cos 3theta + cos theta); the ratio is tan 4theta.

5. Prove that: i) cos 20 deg cos 40 deg cos 60 deg cos 80 deg = 1/16 ii) sin(pi/9) sin(2pi/9) sin(pi/3) sin(4pi/9) = 3/16 iii) sin 10 deg sin 30 deg sin 50 deg sin 70 deg = 1/16

i) cos 60 deg = 1/2. Also cos 20 deg cos 40 deg cos 80 deg = 1/8: multiply by 8 sin

20 deg and use sin 2x = 2 sin x cos x three times (8 sin 20 deg cos 20 deg cos 40 deg cos 80 deg = sin 160 deg = sin 20 deg). Hence the product is (1/2)(1/8) = 1/16. ii) sin(pi/3) = sqrt(3)/2. Also sin 20 deg sin 40 deg sin 80 deg = sqrt(3)/8 by the identity sin x sin(60 deg - x) sin(60 deg + x) = (1/4) sin 3x with x = 20 deg (giving (1/4) sin 60 deg = sqrt(3)/8). Hence the product is (sqrt(3)/2)(sqrt(3)/8) = 3/16. iii) sin 30 deg = 1/2. Also sin 10 deg sin 50 deg sin 70 deg = 1/8 by the same identity with x = 10 deg (sin 30 deg/4 = (1/2)/4 = 1/8). Hence the product is (1/2)(1/8) = 1/16.

Chapter 11: Trigonometric Functions and their Graphs

Exercise 11.1 (book p. 341)

Find the periods of the following functions: 1. sin 3x 2. cos 2x 3. tan 4x 4. cot(x/2) 5. sin(x/3) 6. cosec(x/4) 7. sin(x/5) 8. cos(x/6) 9. tan(x/7) 10. cot 8x 11. sec 9x 12. cosec 10x 13. 3 sin x 14. 2 cos x 15. 3 cos(x/5)

The period of sin, cos, sec and cosec is 2pi, and of tan and cot is pi; the period of f(kx) is the base period divided by |k|, and a constant factor does not change the period. 1. sin 3x: 2pi/3. 2. cos 2x: 2pi/2 = pi. 3. tan 4x: pi/4. 4. cot(x/2): pi/(1/2) = 2pi. 5. sin(x/3): 2pi/(1/3) = 6pi. 6. cosec(x/4): 2pi/(1/4) = 8pi. 7. sin(x/5): 2pi/(1/5) = 10pi. 8. cos(x/6): 2pi/(1/6) = 12pi. 9. tan(x/7): pi/(1/7) = 7pi. 10. cot 8x: pi/8. 11. sec 9x: 2pi/9. 12. cosec 10x: 2pi/10 = pi/5. 13. 3 sin x: 2pi (the factor 3 changes the amplitude, not the period). 14. 2 cos x: 2pi. 15. 3 cos(x/5): 2pi/(1/5) = 10pi. Note: the running header on book pp. 341 and 351 repeats Chapter 9's title ('Fundamentals of Trigonometry'); the Contents title 'Trigonometric Functions and their Graphs' is used here.

Exercise 11.2 (book p. 351)

1. Draw the graph of each of the following function for the intervals mentioned against each: i) y = -sin x, x in [-2pi, 2pi] ii) y = 2cos x, x in [0, 2pi] iii) y = tan 2x, x in [-pi, pi] iv) y = tan x, x in [-2pi, 2pi] v) y = sin(x/2), x in [0, 2pi] vi) y = cos(x/2), x in [-pi, pi]

Construction instructions (plot the computed points on squared paper, join with a smooth curve, and mark the asymptotes as dashed vertical lines; no fabricated plot is printed here). i) y = -sin x is the sine wave reflected in the x-axis, period 2pi, amplitude 1. Key points: (-2pi, 0), (-3pi/2, -1), (-pi, 0), (-pi/2, 1), (0, 0), (pi/2, -1), (pi, 0), (3pi/2, 1), (2pi, 0). ii) y = 2cos x is the cosine wave stretched to amplitude 2, period 2pi. Key points: (0, 2), (pi/2, 0), (pi, -2), (3pi/2, 0), (2pi, 2). iii) y = tan 2x has period pi/2. Zeros at x = 0, +-pi/2, +-pi; vertical asymptotes at x = +-pi/4, +-3pi/4; the curve rises from -infinity to +infinity between consecutive asymptotes and passes through (pi/8, 1), (-pi/8, -1). iv) y = tan x has period pi. Zeros at x = 0, +-pi, +-2pi; vertical asymptotes at x = +-pi/2, +-3pi/2; the curve rises from -infinity to +infinity between consecutive asymptotes and passes through (pi/4, 1), (-pi/4, -1), (5pi/4, 1). v) y = sin(x/2) has period 4pi, so on [0, 2pi] only half a wave appears. Key points: (0, 0), (pi/2, sin pi/4 = 0.707), (pi, 1), (3pi/2, 0.707), (2pi, 0). vi) y = cos(x/2) has period 4pi. Key points: (-pi, 0), (-pi/2, cos(-pi/4) = 0.707), (0, 1), (pi/2, 0.707), (pi, 0).

2. On the same axes and to the same scale, draw the graphs of the following function for their complete period: i) y = sin x and y = sin 2x ii) y = cos x and y = cos 2x

Construction instructions. i) Draw y = sin x (period 2pi, amplitude 1) and y = sin 2x (period pi, amplitude 1) on the same axes for one complete period of sin x, i.e. x in [0, 2pi]. Both pass through (0, 0); sin 2x completes two full waves while sin x completes one. They cross where sin x = sin 2x, i.e. sin x (1 - 2cos x) = 0: x = 0,

pi/3, pi, 5pi/3, 2pi. Mark these crossings and the peaks: sin x peaks at pi/2 (value 1) and troughs at 3pi/2 (value -1); sin 2x peaks at pi/4 and 5pi/4 (value 1) and troughs at 3pi/4 and 7pi/4 (value -1). ii) Draw y = cos x (period 2pi) and y = cos 2x (period pi) on the same axes for x in [0, 2pi]. Both start at (0, 1); cos 2x completes two full waves while cos x completes one. They cross where cos x = cos 2x, i.e. 2cos^2 x - cos x - 1 = 0, i.e. cos x = 1 or cos x = -1/2: x = 0, 2pi/3, 4pi/3, 2pi. Mark these crossings and the peaks and troughs: cos x peaks at 0 and 2pi (value 1) and troughs at pi (value -1); cos 2x peaks at 0, pi, 2pi (value 1) and troughs at pi/2 and 3pi/2 (value -1).

3. Solve graphically: i) sin x = cos x, x in [0, pi] ii) sin x = x, x in [0, pi]

i) Draw y = sin x and y = cos x on the same axes for x in [0, pi]. The curves cross where sin x = cos x, i.e. tan x = 1, i.e. x = pi/4 (the only solution in [0, pi]). Answer: x = pi/4. ii) Draw y = sin x and the line y = x on the same axes for x in [0, pi]. The line y = x lies above the sine curve for every x > 0 in this interval (since sin x < x for x > 0), so the only intersection is at the origin. Answer: x = 0.

Chapter 12: Application of Trigonometry

Exercise 12.1 (book pp. 353-354)

1. Find the values of: i) sin 53 deg 40' ii) cos 36 deg 20' iii) tan 19 deg 30' iv) cot 33 deg 50' v) cos 42 deg 38' vi) tan 25 deg 34' vii) sin 18 deg 31' viii) cos 52 deg 13' ix) cot 89 deg 9'

Read from the tables of natural sines, cosines and tangents (values to four decimal places, as printed in the book's Answers): i) sin 53 deg 40' = 0.8055. ii) cos 36 deg 20' = 0.8055. iii) tan 19 deg 30' = 0.3541. iv) cot 33 deg 50' = 1.4919. v) cos 42 deg 38' = 0.7357. vi) tan 25 deg 34' = 0.4784. vii) sin 18 deg 31' = 0.3175. viii) cos

52 deg 13' = 0.6127. ix) FLAG: the printed key gives 0.1736 for this part, but 0.1736 is cot 80 deg 9' (since cot 80 deg 9' = tan 9 deg 51' = 0.1736), whereas cot 89 deg 9' as printed in the question = tan 0 deg 51' = 0.0148. The key's value therefore belongs to 80 deg 9', not to the printed 89 deg 9'; both values are recorded here and the discrepancy is the book's, not a silent correction. Note on last digits: a calculator gives sin 53 deg 40' = 0.8056 and sin 18 deg 31' = 0.3176, one unit in the last place above the book's table readings 0.8055 and 0.3175; the table values are given as the answers.

2. Find theta, if: i) sin theta = 0.5791 ii) cos theta = 0.9316 iii) cos theta = 0.5257 iv) tan theta = 1.705 v) tan theta = 21.943 vi) sin theta = 0.5186

Read inversely from the tables (angles to the nearest minute, as printed in the book's Answers): i) sin theta = 0.5791 gives theta = 35 deg 23'. ii) cos theta = 0.9316 gives theta = 21 deg 18' (a calculator gives 21 deg 19'; the one-minute difference is table rounding, and the table value is given). iii) cos theta = 0.5257 gives theta = 58 deg 17'. iv) tan theta = 1.705 gives theta = 59 deg 36'. v) tan theta = 21.943 gives theta = 87 deg 23'. vi) sin theta = 0.5186 gives theta = 31 deg 14'.

Exercise 12.2 (book p. 355)

1. Find the unknown angles and sides of the following triangles (all right-angled): (i) angle 45 deg, vertical side 4 (ii) angle 60 deg, hypotenuse 12 (iii) top side 5, hypotenuse 10 (iv) vertical side 8, angle 40 deg (v) angle 56 deg, hypotenuse 15 (vi) top side 8, vertical side 8

(i) Right angle at the third vertex, so the other acute angle = 90 deg - 45 deg = 45 deg. The side opposite 45 deg is 4, so tan 45 deg = 4/base gives base = 4/1 = 4, and hypotenuse = sqrt(4^2 + 4^2) = 4 sqrt(2). Answer: 4, 4 sqrt(2), 45 deg. (ii) Other acute angle = 90 deg - 60 deg = 30 deg. Side opposite 60 deg = 12 sin 60

deg = 12 x (sqrt(3)/2) = 6 sqrt(3). Side adjacent to 60 deg = 12 cos 60 deg = 12 x (1/2) = 6. Answer: 6, 6 sqrt(3), 30 deg. (iii) The leg 5 with hypotenuse 10 gives sin of the angle opposite the leg 5 = 5/10 = 1/2, so that angle = 30 deg and the third angle = 60 deg. The other leg = sqrt(10^2 - 5^2) = sqrt(75) = 5 sqrt(3). Answer: 5 sqrt(3), 30 deg, 60 deg. (iv) Other acute angle = 90 deg - 40 deg = 50 deg. The side opposite 40 deg is 8, so hypotenuse = 8/sin 40 deg = 8/0.6428 = 12.4, and the adjacent side = 8/tan 40 deg = 8/0.8391 = 9.5. Answer: 9.5, 12.4, 50 deg. FLAG: the printed key reads 10.4 for the hypotenuse here, but 9.5 and 8 give hypotenuse sqrt(9.5^2 + 8^2) = 12.4, and 8/sin 40 deg = 12.4; the key's 10.4 is inconsistent with its own 9.5 and 50 deg, and is treated as a misprint for 12.4. (v) Other acute angle = 90 deg - 56 deg = 34 deg. Side opposite 56 deg = 15 sin 56 deg = 15 x 0.8290 = 12.4. Side adjacent to 56 deg = 15 cos 56 deg = 15 x 0.5592 = 8.4. Answer: 8.4, 12.4, 34 deg. (vi) Legs 8 and 8 give an isosceles right triangle, so both acute angles = 45 deg and hypotenuse = sqrt(8^2 + 8^2) = 8 sqrt(2). Answer: 8 sqrt(2), 45 deg, 45 deg.

2. Solve the right triangle ABC, in which gamma = 90 deg, alpha = 37 deg 20', a = 243

beta = 90 deg - 37 deg 20' = 52 deg 40'. Then b = a/tan alpha = 243/tan 37 deg 20' = 243/0.7630 = 318, and c = a/sin alpha = 243/sin 37 deg 20' = 243/0.6065 = 400. Answer: beta = 52 deg 40', b = 318, c = 400.

3. Solve the right triangle ABC, in which gamma = 90 deg, alpha = 62 deg 40', b = 796

beta = 90 deg - 62 deg 40' = 27 deg 20'. Then a = b tan alpha = 796 tan 62 deg 40' = 796 x 1.9347 = 1540, and c = b/cos alpha = 796/cos 62 deg 40' = 796/0.4592 = 1733. Answer: beta = 27 deg 20', a = 1540, c = 1733.

4. Solve the right triangle ABC, in which gamma = 90 deg, a = 3.28, b = 5.74

c = sqrt(a^2 + b^2) = sqrt(3.28^2 + 5.74^2) = sqrt(10.7584 + 32.9476) =

sqrt(43.706) = 6.61. Then tan alpha = a/b = 3.28/5.74 = 0.5714, so alpha = 29 deg 45', and beta = 90 deg - 29 deg 45' = 60 deg 15'. Answer: c = 6.61, alpha = 29 deg 45', beta = 60 deg 15'.

5. Solve the right triangle ABC, in which gamma = 90 deg, b = 68.4, c = 96.2

a = sqrt(c^2 - b^2) = sqrt(96.2^2 - 68.4^2) = sqrt(9254.44 - 4678.56) = sqrt(4575.88) = 67.7. Then cos alpha = b/c = 68.4/96.2 = 0.7110, so alpha = 44 deg 40', and beta = 90 deg - 44 deg 40' = 45 deg 20'. Answer: a = 67.7, alpha = 44 deg 40', beta = 45 deg 20'.

6. Solve the right triangle ABC, in which gamma = 90 deg, a = 5429, c = 6294

b = sqrt(c^2 - a^2) = sqrt(6294^2 - 5429^2) = sqrt(39614436 - 29474041) = sqrt(10140395) = 3184. Then sin alpha = a/c = 5429/6294 = 0.8626, so alpha = 59 deg 36', and beta = 90 deg - 59 deg 36' = 30 deg 24'. Answer: b = 3184, alpha = 59 deg 36', beta = 30 deg 24'.

7. Solve the right triangle ABC, in which gamma = 90 deg, beta = 50 deg 10', c = 0.832

alpha = 90 deg - 50 deg 10' = 39 deg 50'. Then a = c sin alpha = 0.832 sin 39 deg 50' = 0.832 x 0.6406 = 0.533, and b = c cos alpha = 0.832 cos 39 deg 50' = 0.832 x 0.7679 = 0.639. Answer: alpha = 39 deg 50', a = 0.533, b = 0.639.

Exercise 12.3 (book pp. 359-360)

1. A vertical pole is 8 m high and the length of its shadow is 6 m. What is the angle of elevation of the sun at that moment?

Let theta be the angle of elevation. Then tan theta = height/shadow = 8/6 = 1.3333, so theta = 53 deg 8'. Answer: 53 deg 8'.

2. A man 18 dm tall observes that the angle of elevation of the top of a tree at a distance of 12 m from him is 32 deg. What is the height of the tree?

18 dm = 1.8 m. The part of the tree above the man's eye level = 12 tan 32 deg =

12 x 0.6249 = 7.5 m. Hence the height of the tree = 1.8 + 7.5 = 9.3 m. Answer: 9.3 m.

3. At the top of a cliff 80 m high, the angle of depression of a boat is 12 deg. How far is the boat from the cliff?

The angle of depression equals the angle of elevation from the boat to the top of the cliff. If d is the distance, tan 12 deg = 80/d, so d = 80/tan 12 deg = 80/0.2126 = 376 m. Answer: 376 m.

4. A ladder leaning against a vertical wall makes an angle of 24 deg with the wall. Its foot is 5 m from the wall. Find its length.

FLAG on question versus key: with the angle measured from the wall as printed, the foot's distance 5 m is opposite the 24 deg angle, so the ladder length L satisfies sin 24 deg = 5/L, giving L = 5/sin 24 deg = 5/0.4067 = 12.29 m. The printed key gives 5.47 m, which is 5/cos 24 deg, i.e. the value obtained if 24 deg were the angle with the ground (horizontal). The solution for the question as printed is 12.29 m; the key's 5.47 m follows only from the ground-angle reading, and the two are not reconciled silently.

5. A kite flying at a height of 67.2 m is attached to a fully stretched string inclined at an angle of 55 deg to the horizontal. Find the length of the string.

If L is the string length, sin 55 deg = 67.2/L, so L = 67.2/sin 55 deg = 67.2/0.8192 = 82 m. Answer: 82 m.

6. When the angle between the ground and the sun is 30 deg, a flag pole casts a shadow of 40 m long. Find the height of the top of the flag.

Height = 40 tan 30 deg = 40 x 0.5774 = 23 m. Answer: 23 m.

7. A plane flying directly above a post 6000 m away from an anti-aircraft gun observes the gun at an angle of depression of 27 deg. Find the height of the plane.

Height = 6000 tan 27 deg = 6000 x 0.5095 = 3057 m. Answer: 3057 m.

8. A man on the top of a 100 m high light-house is in line with two ships on the

same side of it, whose angles of depression from the man are 17 deg and 19 deg respectively. Find the distance between the ships.

Distance of the farther ship = 100/tan 17 deg = 100/0.3057 = 327.1 m. Distance of the nearer ship = 100/tan 19 deg = 100/0.3443 = 290.5 m. Hence the distance between the ships = 327.1 - 290.5 = 36.6 m. Answer: 36.6 m.

9. P and Q are two points in line with a tree. If the distance between P and Q be 30 m and the angles of elevation of the top of the tree at P and Q be 12 deg and 15 deg respectively, find the height of the tree.

Let h be the height and let P and Q lie on the same side of the tree (the reading confirmed by the key). Then the distances from the foot of the tree are h/tan 12 deg and h/tan 15 deg, and their difference is 30 m: h(1/tan 12 deg - 1/tan 15 deg) = h(4.7046 - 3.7321) = 30, so h = 30/0.9725 = 30.8 m. Answer: 30.8 m.

10. Two men are on the opposite sides of a 100 m high tower. If the measures of the angles of elevation of the top of the tower are 18 deg and 22 deg respectively, find the distance between them.

Distance of the first man = 100/tan 18 deg = 100/0.3249 = 307.8 m. Distance of the second man = 100/tan 22 deg = 100/0.4040 = 247.5 m. Since they are on opposite sides, the distance between them = 307.8 + 247.5 = 555 m. Answer: 555 m.

11. A man standing 60 m away from a tower notices that the angles of elevation of the top and the bottom of a flag staff on the top of the tower are 64 deg and 62 deg respectively. Find the length of the flag staff.

Height to the top of the staff = 60 tan 64 deg = 60 x 2.0503 = 123.0 m. Height to the bottom of the staff (top of the tower) = 60 tan 62 deg = 60 x 1.8807 = 112.8 m. Hence the length of the flag staff = 123.0 - 112.8 = 10.2 m. Answer: 10.2 m.

12. The angle of elevation of the top of a 60 m high tower from a point A, on the same level as the foot of the tower, is 25 deg. Find the angle of elevation of the top

of the tower from a point B, 20 m nearer to A from the foot of the tower.

Distance of A from the foot = 60/tan 25 deg = 60/0.4663 = 128.7 m. Distance of B from the foot = 128.7 - 20 = 108.7 m. Hence the required angle theta satisfies tan theta = 60/108.7 = 0.5520, so theta = 28 deg 54'. Answer: 28 deg 54'.

13. Two buildings A and B are 100 m apart. The angle of elevation from the top of the building A to the top of the building B is 20 deg. The angle of elevation from the base of the building B to the top of the building A is 50 deg. Find the height of the building B.

Height of A = 100 tan 50 deg = 100 x 1.1918 = 119.2 m. The top of B stands above the top of A by 100 tan 20 deg = 100 x 0.3640 = 36.4 m. Hence the height of B = 119.2 + 36.4 = 155.5 m. Answer: 155.5 m.

14. A window washer is working in a hotel building. An observer at a distance of 20 m from the building finds the angle of elevation of the worker to be of 30 deg. The worker climbs up 12 m and the observer moves 4 m farther away from the building. Find the new angle of elevation of the worker.

At first the worker's height = 20 tan 30 deg = 20 x 0.5774 = 11.5 m. After climbing 12 m his height = 23.5 m, and the observer is now 24 m from the building. Hence the new angle theta satisfies tan theta = 23.5/24 = 0.9792, so theta = 44 deg 27' (a calculator gives 44 deg 26'; the one-minute difference is table rounding and the key value 44 deg 27' is given). Answer: 44 deg 27'.

15. A man standing on the bank of a canal observes that the measure of the angle of elevation of a tree on the other side of the canal is 60 deg. On retreating 40 meters from the bank, he finds the measure of the angle of elevation of the tree as 30 deg. Find the height of the tree and the width of the canal.

Let h be the height and w the width. Then tan 60 deg = h/w and tan 30 deg = h/(w + 40). From the first, h = w sqrt(3); from the second, h = (w + 40)/sqrt(3). Hence w sqrt(3) = (w + 40)/sqrt(3), so 3w = w + 40, giving w = 20 m and h = 20 sqrt(3) =

34.64 m. Answer: height 34.64 m, width 20 m.

Exercise 12.4 (book p. 368)

Solve the triangle ABC, if 1. beta = 60 deg, gamma = 15 deg, b = sqrt(6)

alpha = 180 deg - 60 deg - 15 deg = 105 deg. By the sine law, a = b sin alpha/sin beta = sqrt(6) sin 105 deg/sin 60 deg. Now sin 105 deg = sin(60 deg + 45 deg) = sin 60 deg cos 45 deg + cos 60 deg sin 45 deg = (sqrt(3)/2)(sqrt(2)/2) + (1/2)(sqrt(2)/2) = (sqrt(6) + sqrt(2))/4. Hence a = sqrt(6) x ((sqrt(6) + sqrt(2))/4)/(sqrt(3)/2) = (6 + sqrt(12))/(2 sqrt(3)) = sqrt(3) + 1. Similarly c = b sin gamma/sin beta = sqrt(6) sin 15 deg/sin 60 deg = sqrt(3) - 1, since sin 15 deg = (sqrt(6) - sqrt(2))/4. Answer: alpha = 105 deg, a = sqrt(3) + 1, c = sqrt(3) - 1.

2. beta = 52 deg, gamma = 89 deg 35', a = 89.35

alpha = 180 deg - 52 deg - 89 deg 35' = 38 deg 25'. By the sine law, b = a sin beta/sin alpha = 89.35 sin 52 deg/sin 38 deg 25' = 89.35 x 0.7880/0.6211 = 113.32, and c = a sin gamma/sin alpha = 89.35 sin 89 deg 35'/sin 38 deg 25' = 89.35 x 1.0000/0.6211 = 143.78. Answer: alpha = 38 deg 25', b = 113.32, c = 143.78.

3. b = 125, gamma = 53 deg, alpha = 47 deg

beta = 180 deg - 47 deg - 53 deg = 80 deg. By the sine law, a = b sin alpha/sin beta = 125 sin 47 deg/sin 80 deg = 125 x 0.7314/0.9848 = 93, and c = b sin gamma/sin beta = 125 sin 53 deg/sin 80 deg = 125 x 0.7986/0.9848 = 101. Answer: beta = 80 deg, a = 93, c = 101.

4. c = 16.1, alpha = 42 deg 45', gamma = 74 deg 32'

beta = 180 deg - 42 deg 45' - 74 deg 32' = 62 deg 43'. By the sine law, a = c sin alpha/sin gamma = 16.1 sin 42 deg 45'/sin 74 deg 32' = 16.1 x 0.6788/0.9638 = 11.3, and b = c sin beta/sin gamma = 16.1 sin 62 deg 43'/sin 74 deg 32' = 16.1 x 0.8890/0.9638 = 14.8. Answer: beta = 62 deg 43', a = 11.3, b = 14.8.

5. a = 53, beta = 88 deg 36', gamma = 31 deg 54'

alpha = 180 deg - 88 deg 36' - 31 deg 54' = 59 deg 30'. By the sine law, b = a sin beta/sin alpha = 53 sin 88 deg 36'/sin 59 deg 30' = 53 x 0.9997/0.8616 = 61.5, and c = a sin gamma/sin alpha = 53 sin 31 deg 54'/sin 59 deg 30' = 53 x 0.5285/0.8616 = 32.5. Answer: alpha = 59 deg 30', b = 61.5, c = 32.5.

Exercise 12.5 (book p. 371)

Solve the triangle ABC in which: 1. b = 95, c = 34 and alpha = 52 deg

By the cosine law, a^2 = b^2 + c^2 - 2bc cos alpha = 95^2 + 34^2 - 2(95)(34)cos 52 deg = 9025 + 1156 - 6460 x 0.6157 = 10181 - 3977 = 6204, so a = 78.8. By the sine law, sin beta = b sin alpha/a = 95 sin 52 deg/78.8 = 95 x 0.7880/78.8 = 0.9504. Since b > a, beta is obtuse: beta = 180 deg - 71 deg 53' = 108 deg 7'. Then gamma = 180 deg - 52 deg - 108 deg 7' = 19 deg 53'. Answer: a = 78.8, beta = 108 deg 7', gamma = 19 deg 53'. FLAG: the printed key gives a = 36.01, beta = 80 deg, gamma = 48 deg for this question, but those are exactly the values for b = 45, not b = 95 (with b = 45: a^2 = 45^2 + 34^2 - 2(45)(34)cos 52 deg = 1297, a = 36.01, and the angles follow as 80 deg and 48 deg). The question as printed says b = 95 (confirmed at 300 DPI), so the solution above follows the printed question and the key's values are flagged as belonging to b = 45.

2. b = 12.5, c = 23, and alpha = 38 deg 20'

By the cosine law, a^2 = 12.5^2 + 23^2 - 2(12.5)(23)cos 38 deg 20' = 156.25 + 529 - 575 x 0.7846 = 685.25 - 451.1 = 234.1, so a = 15.3. By the sine law, sin beta = b sin alpha/a = 12.5 sin 38 deg 20'/15.3 = 12.5 x 0.6202/15.3 = 0.5067, so beta = 30 deg 26'. Then gamma = 180 deg - 38 deg 20' - 30 deg 26' = 111 deg 14'. Answer: a = 15.3, beta = 30 deg 26', gamma = 111 deg 14'.

3. a = sqrt(3) - 1, b = sqrt(3) + 1 and gamma = 60 deg

By the cosine law, c^2 = a^2 + b^2 - 2ab cos 60 deg = (4 - 2 sqrt(3)) + (4 + 2

sqrt(3)) - 2(3 - 1)(1/2) = 8 - 2 = 6, so c = sqrt(6). By the sine law, sin alpha = a sin gamma/c = (sqrt(3) - 1)(sqrt(3)/2)/sqrt(6) = 0.2588, so alpha = 15 deg. Then beta = 180 deg - 15 deg - 60 deg = 105 deg. Answer: c = sqrt(6), alpha = 15 deg, beta = 105 deg.

4. a = 3, c = 6 and beta = 36 deg 20'

By the cosine law, b^2 = a^2 + c^2 - 2ac cos beta = 9 + 36 - 2(3)(6)cos 36 deg 20' = 45 - 36 x 0.8058 = 45 - 29.0 = 16.0, so b = 4. By the sine law, sin alpha = a sin beta/b = 3 sin 36 deg 20'/4 = 3 x 0.5920/4 = 0.4440, so alpha = 26 deg 23'. Then gamma = 180 deg - 36 deg 20' - 26 deg 23' = 117 deg 17'. Answer: b = 4, alpha = 26 deg 23', gamma = 117 deg 17'.

5. a = 7, b = 3, and gamma = 38 deg 13'

By the cosine law, c^2 = a^2 + b^2 - 2ab cos gamma = 49 + 9 - 2(7)(3)cos 38 deg 13' = 58 - 42 x 0.7859 = 58 - 33.0 = 25.0, so c = 5. For alpha, cos alpha = (b^2 + c^2 - a^2)/(2bc) = (9 + 25 - 49)/(2 x 3 x 5) = -15/30 = -1/2, so alpha = 120 deg (obtuse, as it must be, since a is the longest side). Then beta = 180 deg - 120 deg - 38 deg 13' = 21 deg 47'. Answer: c = 5, alpha = 120 deg, beta = 21 deg 47'.

Solve the following triangles, using first Law of tangents and then Law of sines: 6. a = 36.21, b = 42.09 and gamma = 44 deg 29'

alpha + beta = 180 deg - 44 deg 29' = 135 deg 31', so (alpha + beta)/2 = 67 deg 45.5'. By the law of tangents, tan((alpha - beta)/2) = ((a - b)/(a + b)) tan((alpha + beta)/2) = ((36.21 - 42.09)/(36.21 + 42.09)) tan 67 deg 45.5' = (-5.88/78.30) x 2.4367 = -0.1830, so (alpha - beta)/2 = -10 deg 23', i.e. alpha - beta = -20 deg 46'. Hence alpha = (135 deg 31' - 20 deg 46')/2 = 57 deg 21' and beta = (135 deg 31' + 20 deg 46')/2 = 78 deg 10'. By the sine law, c = a sin gamma/sin alpha = 36.21 sin 44 deg 29'/sin 57 deg 21' = 36.21 x 0.7004/0.8420 = 30.14. Answer: c = 30.14, alpha = 57 deg 21', beta = 78 deg 10'.

7. a = 93, c = 101 and beta = 80 deg

By the cosine law, b^2 = a^2 + c^2 - 2ac cos beta = 93^2 + 101^2 - 2(93)(101)cos 80 deg = 8649 + 10201 - 18786 x 0.1736 = 18850 - 3262 = 15588, so b = 125. By the sine law, sin alpha = a sin beta/b = 93 sin 80 deg/125 = 93 x 0.9848/125 = 0.7327, so alpha = 47 deg. Then gamma = 180 deg - 80 deg - 47 deg = 53 deg. Answer: b = 125, alpha = 47 deg, gamma = 53 deg.

8. b = 14.8, c = 16.1 and alpha = 42 deg 45'

By the cosine law, a^2 = b^2 + c^2 - 2bc cos alpha = 14.8^2 + 16.1^2 - 2(14.8)(16.1)cos 42 deg 45' = 219.04 + 259.21 - 476.56 x 0.7346 = 478.25 - 350.1 = 128.2, so a = 11.3. By the sine law, sin beta = b sin alpha/a = 14.8 sin 42 deg 45'/11.3 = 14.8 x 0.6788/11.3 = 0.8890, so beta = 62 deg 43'. Then gamma = 180 deg - 42 deg 45' - 62 deg 43' = 74 deg 32'. Answer: a = 11.3, beta = 62 deg 43', gamma = 74 deg 32'. Note: the printed key labels the found side 'b = 11.3', but the side found from b, c and alpha is a, so it is given here as a = 11.3.

9. a = 319, b = 168 and gamma = 110 deg 22'

By the cosine law, c^2 = a^2 + b^2 - 2ab cos gamma = 319^2 + 168^2 - 2(319)(168)cos 110 deg 22' = 101761 + 28224 - 107184 x (-0.3473) = 129985 + 37225 = 167210, so c = 409. By the sine law, sin alpha = a sin gamma/c = 319 sin 110 deg 22'/409 = 319 x 0.9378/409 = 0.7315, so alpha = 46 deg 59'. Then beta = 180 deg - 110 deg 22' - 46 deg 59' = 22 deg 39'. Answer: c = 409, alpha = 46 deg 59', beta = 22 deg 39'.

10. b = 61, a = 32 and alpha = 59 deg 30'

By the sine law, sin beta = b sin alpha/a = 61 sin 59 deg 30'/32 = 61 x 0.8616/32 = 1.6424. Since sin beta cannot exceed 1, NO triangle exists with the measures as printed: for a triangle to exist, b sin alpha must not exceed a, but here b sin alpha = 61 x 0.8616 = 52.6, which exceeds a = 32. FLAG: the printed key's line for this question (alpha = 53, beta = 88 deg 36', gamma = 31 deg 54') does not solve these givens at all; it repeats the Exercise 12.4 Question 5 triangle (a = 53, beta =

88 deg 36', gamma = 31 deg 54', alpha = 59 deg 30'). The correct statement for Question 10 as printed is that no such triangle exists.

11. Measures of two sides of a triangle are in the ratio 3 : 2 and they include an angle of measure 57 deg. Find the remaining two angles.

Let the sides be 3k and 2k with included angle 57 deg. Then the sum of the other two angles is 180 deg - 57 deg = 123 deg. By the law of tangents, tan((alpha - beta)/2) = ((3 - 2)/(3 + 2)) tan((123 deg)/2) = (1/5) tan 61 deg 30' = 0.2 x 1.8418 = 0.3684, so (alpha - beta)/2 = 20 deg 13', i.e. alpha - beta = 40 deg 26'. Hence alpha = (123 deg + 40 deg 26')/2 = 81 deg 43' and beta = (123 deg - 40 deg 26')/2 = 41 deg 17'. Answer: 81 deg 43' and 41 deg 17'.

12. Two forces of 40 N and 30 N are represented by AB and BC which are inclined at an angle of 147 deg 25'. Find AC, the resultant of AB and BC.

In the force triangle with sides AB = 40 N and BC = 30 N and included angle 147 deg 25', the resultant AC is the side opposite that angle. By the cosine law, AC^2 = 40^2 + 30^2 - 2(40)(30)cos 147 deg 25' = 1600 + 900 - 2400 x (-0.8425) = 2500 + 2022 = 4522, so AC = 67.25 N. Answer: 67.25 N (the book's value, obtained by treating 147 deg 25' as the included angle of the force triangle).

Exercise 12.6 (book pp. 372-373)

Solve the following triangles, in which 1. a = 7, b = 7, c = 9

By the cosine law, cos gamma = (a^2 + b^2 - c^2)/(2ab) = (49 + 49 - 81)/(2 x 7 x 7) = 17/98 = 0.1735, so gamma = 80 deg. Since a = b, the triangle is isosceles and alpha = beta = (180 deg - 80 deg)/2 = 50 deg. Answer: alpha = 50 deg, beta = 50 deg, gamma = 80 deg.

2. a = 32, b = 40, c = 66

By the cosine law, cos alpha = (b^2 + c^2 - a^2)/(2bc) = (1600 + 4356 - 1024)/(2 x 40 x 66) = 4932/5280 = 0.9341, so alpha = 20 deg 55'. Then cos beta = (a^2 +

c^2 - b^2)/(2ac) = (1024 + 4356 - 1600)/(2 x 32 x 66) = 3780/4224 = 0.8949, so beta = 26 deg 30'. Then gamma = 180 deg - 20 deg 55' - 26 deg 30' = 132 deg 35'. Answer: alpha = 20 deg 55', beta = 26 deg 30', gamma = 132 deg 35'.

3. a = 28.3, b = 31.7, c = 42.8

By the cosine law, cos alpha = (b^2 + c^2 - a^2)/(2bc) = (31.7^2 + 42.8^2 - 28.3^2)/(2 x 31.7 x 42.8) = (1004.89 + 1831.84 - 800.89)/2713.52 = 2035.84/2713.52 = 0.7503, so alpha = 41 deg 23'. Then cos beta = (a^2 + c^2 - b^2)/(2ac) = (800.89 + 1831.84 - 1004.89)/(2 x 28.3 x 42.8) = 1627.84/2422.48 = 0.6720, so beta = 47 deg 47'. Then gamma = 180 deg - 41 deg 23' - 47 deg 47' = 90 deg 50'. Answer: alpha = 41 deg 23', beta = 47 deg 47', gamma = 90 deg 50'. FLAG: the printed key gives alpha = 38 deg 22', beta = 47 deg 46', gamma = 93 deg 52' for this question, but those angles imply side ratios sin 38 deg 22' : sin 47 deg 46' : sin 93 deg 52' = 0.620 : 0.740 : 0.998, i.e. with c = 42.8 they require a = 26.6, not the printed a = 28.3 (with a = 28.3 the angle opposite c is 90 deg 50', not 93 deg 52'). The solution above follows the printed sides and the key's values are flagged as belonging to a = 26.6.

4. a = 31.9, b = 56.31, c = 40.27

By the cosine law, cos alpha = (b^2 + c^2 - a^2)/(2bc) = (56.31^2 + 40.27^2 - 31.9^2)/(2 x 56.31 x 40.27) = (3170.82 + 1621.67 - 1017.61)/4535.21 = 3774.88/4535.21 = 0.8323, so alpha = 33 deg 40'. Then cos beta = (a^2 + c^2 - b^2)/(2ac) = (1017.61 + 1621.67 - 3170.82)/(2 x 31.9 x 40.27) = -531.54/2569.23 = -0.2069, so beta = 101 deg 58'. Then gamma = 180 deg - 33 deg 40' - 101 deg 58' = 44 deg 22'. Answer: alpha = 33 deg 40', beta = 101 deg 58', gamma = 44 deg 22' (a calculator gives beta = 101 deg 56' and gamma = 44 deg 24'; the two-minute differences are table rounding and the key values are given).

5. a = 4584, b = 5140, c = 3624

By the cosine law, cos alpha = (b^2 + c^2 - a^2)/(2bc) = (5140^2 + 3624^2 -

4584^2)/(2 x 5140 x 3624) = (26419600 + 13133376 - 21013056)/37254720 = 18539920/37254720 = 0.4977, so alpha = 60 deg 10'. Then cos beta = (a^2 + c^2 - b^2)/(2ac) = (21013056 + 13133376 - 26419600)/(2 x 4584 x 3624) = 7726832/33224832 = 0.2326, so beta = 76 deg 32'. Then gamma = 180 deg - 60 deg 10' - 76 deg 32' = 43 deg 18'. Answer: alpha = 60 deg 10', beta = 76 deg 32', gamma = 43 deg 18' (a calculator gives alpha = 60 deg 9' and beta = 76 deg 33'; the one-minute differences are table rounding and the key values are given).

6. Find the smallest angle of the triangle ABC, when a = 37.34, b = 3.24, c = 35.06

The smallest angle is opposite the smallest side, which is b = 3.24, so it is beta. By the cosine law, cos beta = (a^2 + c^2 - b^2)/(2ac) = (37.34^2 + 35.06^2 - 3.24^2)/(2 x 37.34 x 35.06) = (1394.28 + 1229.20 - 10.50)/2618.28 = 2612.98/2618.28 = 0.9980, so beta = 3 deg 39'. Answer: beta = 3 deg 39'.

7. Find the measure of the greatest angle, if sides of the triangle are 16, 20, 33

The greatest angle is opposite the longest side, 33. By the cosine law, cos theta = (16^2 + 20^2 - 33^2)/(2 x 16 x 20) = (256 + 400 - 1089)/640 = -433/640 = -0.6766, so theta = 132 deg 35'. Answer: 132 deg 35'.

8. The sides of a triangle are x^2 + x + 1, 2x + 1 and x^2 - 1. Prove that the greatest angle of the triangle is 120 deg

For x > 1, x^2 + x + 1 exceeds both 2x + 1 (difference x^2 - x = x(x - 1) > 0) and x^2 - 1 (difference x + 2 > 0), so the greatest angle theta is opposite the side x^2 + x + 1. By the cosine law, cos theta = ((2x + 1)^2 + (x^2 - 1)^2 - (x^2 + x + 1)^2)/(2(2x + 1)(x^2 - 1)). The numerator is (4x^2 + 4x + 1) + (x^4 - 2x^2 + 1) - (x^4 + 2x^3 + 3x^2 + 2x + 1) = -2x^3 - x^2 + 2x + 1 = -(2x + 1)(x^2 - 1), since 2x^3 + x^2 - 2x - 1 = (2x + 1)(x^2 - 1). Hence cos theta = -(2x + 1)(x^2 - 1)/(2(2x + 1)(x^2 - 1)) = -1/2, so theta = 120 deg, as required.

9. The measures of side of a triangular plot are 413, 214 and 375 meters. Find the measures of the corner angles of the plot

Let the sides be a = 413, b = 214, c = 375. By the cosine law, cos alpha = (b^2 + c^2 - a^2)/(2bc) = (214^2 + 375^2 - 413^2)/(2 x 214 x 375) = (45796 + 140625 - 170569)/160500 = 15852/160500 = 0.0988, so alpha = 84 deg 20'. Then cos beta = (a^2 + c^2 - b^2)/(2ac) = (170569 + 140625 - 45796)/(2 x 413 x 375) = 265398/309750 = 0.8568, so beta = 31 deg 2'. Then gamma = 180 deg - 84 deg 20' - 31 deg 2' = 64 deg 38'. Answer: 84 deg 20', 31 deg 2', 64 deg 38'. FLAG: the printed key gives 43 deg 17', 64 deg 26', 72 deg 17' for this question, but those angles imply side ratios sin 43 deg 17' : sin 64 deg 26' : sin 72 deg 17' = 0.686 : 0.902 : 0.952, which are inconsistent with the printed sides 413 : 214 : 375 (in particular the angle opposite 214 would have to be 31 deg 2', not 43 deg 17' or 72 deg 17'). The solution above follows the printed sides and the key's values are flagged as not following from them.

10. Three villages A, B and C are connected by straight roads 6 km, 9 km and 13 km. What angles these roads make with each other?

Let the roads be a = 6, b = 9, c = 13 km. By the cosine law, cos alpha = (b^2 + c^2 - a^2)/(2bc) = (81 + 169 - 36)/(2 x 9 x 13) = 214/234 = 0.9145, so alpha = 23 deg 52'. Then cos beta = (a^2 + c^2 - b^2)/(2ac) = (36 + 169 - 81)/(2 x 6 x 13) = 124/156 = 0.7949, so beta = 37 deg 22'. Then gamma = 180 deg - 23 deg 52' - 37 deg 22' = 118 deg 46'. Answer: alpha = 23 deg 52', beta = 37 deg 22', gamma = 118 deg 46' (a calculator gives beta = 37 deg 21' and gamma = 118 deg 47'; the one-minute differences are table rounding and the key values are given).

Exercise 12.7 (book pp. 376-377)

1. Find the area of the triangle ABC, given two sides and their included angle: i) a = 200, b = 120, gamma = 150 deg ii) b = 37, c = 45, alpha = 30 deg 50' iii) a = 4.33, b = 9.25, gamma = 56 deg 44'

Use Delta = (1/2) x (product of the two sides) x sin(included angle). i) Delta =

(1/2)(200)(120) sin 150 deg = 12000 x 0.5 = 6000 Sq. Unit. ii) Delta = (1/2)(37)(45) sin 30 deg 50' = 832.5 x 0.5125 = 426.69 Sq. Unit. iii) Delta = (1/2)(4.33)(9.25) sin 56 deg 44' = 20.026 x 0.8360 = 16.74 Sq. Unit. Answers: i) 6000 Sq. Unit ii) 426.69 Sq. Unit iii) 16.74 Sq. Unit.

2. Find the area of the triangle ABC, given one side and two angles: i) b = 25.4, gamma = 36 deg 41', alpha = 45 deg 17' ii) c = 32, alpha = 47 deg 24', beta = 70 deg 16' iii) a = 4.8, alpha = 83 deg 42', gamma = 37 deg 12'

Use Delta = (1/2) x (given side)^2 x sin(first angle) x sin(second angle)/sin(third angle). i) beta = 180 deg - 36 deg 41' - 45 deg 17' = 98 deg 2'. Delta = (1/2)(25.4)^2 sin 36 deg 41' sin 45 deg 17'/sin 98 deg 2' = 322.58 x 0.5976 x 0.7108/0.9902 = 138.29 Sq. Unit. ii) gamma = 180 deg - 47 deg 24' - 70 deg 16' = 62 deg 20'. Delta = (1/2)(32)^2 sin 47 deg 24' sin 70 deg 16'/sin 62 deg 20' = 512 x 0.7361 x 0.9415/0.8855 = 400.57 Sq. Unit (a calculator gives 400.55; the 0.02 difference is table rounding and the key value is given). iii) beta = 180 deg - 83 deg 42' - 37 deg 12' = 59 deg 6'. Delta = (1/2)(4.8)^2 sin 59 deg 6' sin 37 deg 12'/sin 83 deg 42' = 11.52 x 0.8585 x 0.6048/0.9940 = 6.01 Sq. Unit. Answers: i) 138.29 Sq. Unit ii) 400.57 Sq. Unit iii) 6.01 Sq. Unit.

3. Find the area of the triangle ABC, given three sides: i) a = 18, b = 24, c = 30 ii) a = 524, b = 276, c = 315 iii) a = 32.65, b = 42.81, c = 64.92

Use Heron's formula Delta = sqrt(s(s - a)(s - b)(s - c)) with s = (a + b + c)/2. i) s = (18 + 24 + 30)/2 = 36. Delta = sqrt(36 x 18 x 12 x 6) = sqrt(46656) = 216 Sq. Unit. ii) s = (524 + 276 + 315)/2 = 557.5. Delta = sqrt(557.5 x 33.5 x 281.5 x 242.5) = sqrt(1274913056) = 35705.89 Sq. Unit. iii) s = (32.65 + 42.81 + 64.92)/2 = 70.19. Delta = sqrt(70.19 x 37.54 x 27.38 x 5.27) = sqrt(380201) = 616.60 Sq. Unit. Answers: i) 216 Sq. Unit ii) 35705.89 Sq. Unit iii) 616.60 Sq. Unit.

4. The area of triangle is 2437. If a = 79, and c = 97, then find angle beta

Delta = (1/2) a c sin beta, so sin beta = 2 Delta/(a c) = 2 x 2437/(79 x 97) =

4874/7663 = 0.6360, so beta = 39 deg 30'. Answer: beta = 39 deg 30'. (The sine equation also admits the supplement 140 deg 30'; the book takes the acute value.)

5. The area of triangle is 121.34. If alpha = 32 deg 15', beta = 65 deg 37', then find c and angle gamma

gamma = 180 deg - 32 deg 15' - 65 deg 37' = 82 deg 8'. By the area formula Delta = (1/2) c^2 sin alpha sin beta/sin gamma, so c^2 = 2 Delta sin gamma/(sin alpha sin beta) = 2 x 121.34 x sin 82 deg 8'/(sin 32 deg 15' x sin 65 deg 37') = 242.68 x 0.9907/(0.5340 x 0.9114) = 240.41/0.4866 = 494.0, so c = 22.24. Answer: c = 22.24, gamma = 82 deg 8'.

6. One side of a triangular garden is 30 m. If its two corner angles are 22 deg 1/2 and 112 deg 1/2, find the cost of planting the grass at the rate of Rs. 5 per square meter

The third angle = 180 deg - 22 deg 30' - 112 deg 30' = 45 deg, and the 30 m side lies between the two given angles, so it is opposite the 45 deg angle. By the area formula, Delta = (1/2)(30)^2 sin 22 deg 30' sin 112 deg 30'/sin 45 deg = 450 x 0.3827 x 0.9239/0.7071 = 450 x 0.3536/0.7071 = 225.0 square meters. Hence the cost = 225.0 x 5 = Rs. 1125. Answer: Rs. 1125.

Exercise 12.8 (book pp. 385-386)

1. Show that: i) r = 4R sin(alpha/2) sin(beta/2) sin(gamma/2) ii) s = 4R cos(alpha/2) cos(beta/2) cos(gamma/2)

Write a = 2R sin alpha, b = 2R sin beta, c = 2R sin gamma. Then 2s = a + b + c = 2R(sin alpha + sin beta + sin gamma). Now sin alpha + sin beta = 2 sin((alpha + beta)/2) cos((alpha - beta)/2) = 2 cos(gamma/2) cos((alpha - beta)/2), since (alpha + beta)/2 = 90 deg - gamma/2. Adding sin gamma = 2 sin(gamma/2) cos(gamma/2) gives sin alpha + sin beta + sin gamma = 2 cos(gamma/2)[cos((alpha - beta)/2) + sin(gamma/2)] = 2 cos(gamma/2)[cos((alpha - beta)/2) + cos((alpha + beta)/2)] = 4

cos(alpha/2) cos(beta/2) cos(gamma/2). Hence 2s = 8R cos(alpha/2) cos(beta/2) cos(gamma/2), i.e. (ii) s = 4R cos(alpha/2) cos(beta/2) cos(gamma/2). For (i), Delta = (1/2) b c sin alpha = (1/2)(2R sin beta)(2R sin gamma) sin alpha = 2R^2 sin alpha sin beta sin gamma = 2R^2 x 8 sin(alpha/2) cos(alpha/2) sin(beta/2) cos(beta/2) sin(gamma/2) cos(gamma/2). Since r = Delta/s, dividing by (ii) gives r = 4R sin(alpha/2) sin(beta/2) sin(gamma/2), as required.

2. Show that: r = a sin(beta/2) sin(gamma/2) sec(alpha/2) = b sin(gamma/2) sin(alpha/2) sec(beta/2) = c sin(alpha/2) sin(beta/2) sec(gamma/2)

From Question 1(i), r = 4R sin(alpha/2) sin(beta/2) sin(gamma/2). Now a = 2R sin alpha = 4R sin(alpha/2) cos(alpha/2), so 4R sin(alpha/2) = a/cos(alpha/2) = a sec(alpha/2). Substituting gives r = a sin(beta/2) sin(gamma/2) sec(alpha/2). The other two forms follow cyclically from b = 4R sin(beta/2) cos(beta/2) and c = 4R sin(gamma/2) cos(gamma/2), as required.

3. Show that: i) r1 = 4R sin(alpha/2) cos(beta/2) cos(gamma/2) ii) r2 = 4R cos(alpha/2) sin(beta/2) cos(gamma/2) iii) r3 = 4R cos(alpha/2) cos(beta/2) sin(gamma/2)

Use r1 = Delta/(s - a). From Question 1(ii), s = 4R cos(alpha/2) cos(beta/2) cos(gamma/2), and a = 4R sin(alpha/2) cos(alpha/2). Hence s - a = 4R cos(alpha/2)[cos(beta/2) cos(gamma/2) - sin(alpha/2)]. Since alpha/2 = 90 deg - (beta + gamma)/2, sin(alpha/2) = cos((beta + gamma)/2) = cos(beta/2) cos(gamma/2) - sin(beta/2) sin(gamma/2). Substituting gives s - a = 4R cos(alpha/2) sin(beta/2) sin(gamma/2). With Delta = 2R^2 x 8 sin(alpha/2) cos(alpha/2) sin(beta/2) cos(beta/2) sin(gamma/2) cos(gamma/2) from Question 1, dividing gives r1 = Delta/(s - a) = 4R sin(alpha/2) cos(beta/2) cos(gamma/2), which is (i). The forms (ii) and (iii) follow cyclically, as required.

4. Show that: i) r1 = s tan(alpha/2) ii) r2 = s tan(beta/2) iii) r3 = s tan(gamma/2)

We know Delta = r s and also Delta = r1 (s - a), so r1 = r s/(s - a). But tan(alpha/2)

= r/(s - a) (from the right triangle formed by the in-centre, the touch point and vertex A, where the adjacent side is s - a and the opposite side is r). Hence r1 = s tan(alpha/2), which is (i). The forms (ii) and (iii) follow cyclically, as required.

5. Prove that: i) r1 r2 + r2 r3 + r3 r1 = s^2 ii) r r1 r2 r3 = Delta^2 iii) r1 + r2 + r3 - r = 4R iv) r1 r2 r3 = r s^2

i) From Question 4, r1 r2 = s^2 tan(alpha/2) tan(beta/2), and cyclically. Hence r1 r2 + r2 r3 + r3 r1 = s^2 [tan(alpha/2) tan(beta/2) + tan(beta/2) tan(gamma/2) + tan(gamma/2) tan(alpha/2)]. The bracket equals 1: since (alpha + beta)/2 = 90 deg - gamma/2, tan((alpha + beta)/2) = cot(gamma/2), i.e. (tan(alpha/2) + tan(beta/2))/(1 - tan(alpha/2) tan(beta/2)) = 1/tan(gamma/2), which rearranges to tan(alpha/2) tan(gamma/2) + tan(beta/2) tan(gamma/2) = 1 - tan(alpha/2) tan(beta/2). Hence the sum is s^2. ii) r r1 r2 r3 = (Delta/s)(Delta/(s - a))(Delta/(s - b))(Delta/(s - c)) = Delta^4/[s(s - a)(s - b)(s - c)] = Delta^4/Delta^2 = Delta^2, by Heron's formula. iii) From Questions 1 and 3, r1 + r2 + r3 - r = 4R[sin(alpha/2) cos(beta/2) cos(gamma/2) + cos(alpha/2) sin(beta/2) cos(gamma/2) + cos(alpha/2) cos(beta/2) sin(gamma/2) - sin(alpha/2) sin(beta/2) sin(gamma/2)]. The bracket is sin(alpha/2 + beta/2 + gamma/2) = sin 90 deg = 1, so the sum is 4R. iv) r1 r2 r3 = Delta^3/[(s - a)(s - b)(s - c)] = Delta^3 x s/Delta^2 = Delta x s = r s^2, since Delta = r s, as required.

6. Find R, r, r1, r2 and r3, if measures of the sides of triangle ABC are i) a = 13, b = 14, c = 15 ii) a = 34, b = 20, c = 42

Use s = (a + b + c)/2, Delta = sqrt(s(s - a)(s - b)(s - c)), R = abc/(4 Delta), r = Delta/s, r1 = Delta/(s - a), r2 = Delta/(s - b), r3 = Delta/(s - c). i) s = (13 + 14 + 15)/2 = 21. Delta = sqrt(21 x 8 x 7 x 6) = sqrt(7056) = 84. R = 13 x 14 x 15/(4 x 84) = 2730/336 = 8.125. r = 84/21 = 4. r1 = 84/(21 - 13) = 84/8 = 10.5. r2 = 84/(21 - 14) = 84/7 = 12. r3 = 84/(21 - 15) = 84/6 = 14. Answer: R = 8.125, r = 4, r1 = 10.5, r2 = 12, r3 = 14. FLAG: the printed key reads r3 = 12 for this part, but

Delta/(s - c) = 84/6 = 14, and 12 is the value of r2; the key's r3 duplicates r2 and is treated as a misprint. ii) s = (34 + 20 + 42)/2 = 48. Delta = sqrt(48 x 14 x 28 x 6) = sqrt(112896) = 336. R = 34 x 20 x 42/(4 x 336) = 28560/1344 = 21.25. r = 336/48 = 7. r1 = 336/(48 - 34) = 336/14 = 24. r2 = 336/(48 - 20) = 336/28 = 12. r3 = 336/(48 - 42) = 336/6 = 56. Answer: R = 21.25, r = 7, r1 = 24, r2 = 12, r3 = 56.

7. Prove that in an equilateral triangle, i) r : R : r1 = 1 : 2 : 3 ii) r : R : r1 : r2 : r3 = 1 : 2 : 3 : 3 : 3

In an equilateral triangle alpha = beta = gamma = 60 deg. From Question 1(i), r = 4R sin 30 deg sin 30 deg sin 30 deg = 4R x (1/2)^3 = R/2, so r : R = 1 : 2. From Question 3(i), r1 = 4R sin 30 deg cos 30 deg cos 30 deg = 4R x (1/2) x (sqrt(3)/2) x (sqrt(3)/2) = 3R/2, so r1 : R = 3 : 2, i.e. r : R : r1 = 1 : 2 : 3, which is (i). By symmetry r1 = r2 = r3, so r : R : r1 : r2 : r3 = 1 : 2 : 3 : 3 : 3, which is (ii), as required.

8. Prove that: i) Delta = r^2 cot(alpha/2) cot(beta/2) cot(gamma/2) ii) r = s tan(alpha/2) tan(beta/2) tan(gamma/2) iii) Delta = 4R r cos(alpha/2) cos(beta/2) cos(gamma/2)

i) From tan(alpha/2) = r/(s - a) we get s - a = r cot(alpha/2), and cyclically s - b = r cot(beta/2), s - c = r cot(gamma/2). Heron's formula gives Delta^2 = s(s - a)(s - b)(s - c) = s r^3 cot(alpha/2) cot(beta/2) cot(gamma/2). But Delta = r s, so Delta^2 = r^2 s^2. Equating gives r^2 s^2 = s r^3 cot(alpha/2) cot(beta/2) cot(gamma/2), so s = r cot(alpha/2) cot(beta/2) cot(gamma/2), and Delta = r s = r^2 cot(alpha/2) cot(beta/2) cot(gamma/2), as required. ii) From (i), s = r cot(alpha/2) cot(beta/2) cot(gamma/2), so r = s tan(alpha/2) tan(beta/2) tan(gamma/2), as required. iii) From Question 1(i), r = 4R sin(alpha/2) sin(beta/2) sin(gamma/2), so 4R r cos(alpha/2) cos(beta/2) cos(gamma/2) = 16R^2 sin(alpha/2) cos(alpha/2) sin(beta/2) cos(beta/2) sin(gamma/2) cos(gamma/2) = 2R^2 sin alpha sin beta sin gamma = (1/2)(2R sin beta)(2R sin gamma) sin alpha = (1/2) b c sin alpha = Delta, as

required.

9. Show that: i) 1/(2rR) = 1/(ab) + 1/(bc) + 1/(ca) ii) 1/r = 1/r1 + 1/r2 + 1/r3

i) The right side is (c + a + b)/(abc) = 2s/(abc). Now abc = (2R sin alpha)(2R sin beta)(2R sin gamma) = 8R^3 sin alpha sin beta sin gamma, and Delta = 2R^2 sin alpha sin beta sin gamma, so abc = 4R Delta. Since Delta = r s, abc = 4R r s. Hence 2s/(abc) = 2s/(4R r s) = 1/(2rR), as required. ii) 1/r1 + 1/r2 + 1/r3 = (s - a)/Delta + (s - b)/Delta + (s - c)/Delta = (3s - (a + b + c))/Delta = (3s - 2s)/Delta = s/Delta = 1/r, since Delta = r s, as required.

10. Prove that: r = (a sin(beta/2) sin(gamma/2))/cos(alpha/2) = (b sin(gamma/2) sin(alpha/2))/cos(beta/2) = (c sin(alpha/2) sin(beta/2))/cos(gamma/2)

This is Question 2 written in fractional form: from Question 1(i), r = 4R sin(alpha/2) sin(beta/2) sin(gamma/2), and a = 4R sin(alpha/2) cos(alpha/2), so 4R sin(alpha/2) = a/cos(alpha/2). Substituting gives r = a sin(beta/2) sin(gamma/2)/cos(alpha/2). The other two forms follow cyclically, as required.

11. Prove that: abc (sin alpha + sin beta + sin gamma) = 4 Delta s

From Question 1, abc = 4R Delta, and sin alpha + sin beta + sin gamma = 4 cos(alpha/2) cos(beta/2) cos(gamma/2) = s/R, since s = 4R cos(alpha/2) cos(beta/2) cos(gamma/2). Hence abc (sin alpha + sin beta + sin gamma) = 4R Delta x s/R = 4 Delta s, as required.

12. Prove that: i) (r1 + r2) tan(gamma/2) = c ii) (r3 - r) cot(gamma/2) = c

i) From Question 4, r1 + r2 = s[tan(alpha/2) + tan(beta/2)]. The half-angle identity from Question 5(i) gives (tan(alpha/2) + tan(beta/2)) tan(gamma/2) = 1 - tan(alpha/2) tan(beta/2). Now tan(alpha/2) tan(beta/2) = r^2/((s - a)(s - b)), and r^2 = (s - a)(s - b)(s - c)/s, so 1 - tan(alpha/2) tan(beta/2) = 1 - (s - c)/s = c/s. Hence (r1 + r2) tan(gamma/2) = s x c/s = c, as required. ii) From Question 4, r3 = s tan(gamma/2), and tan(gamma/2) = r/(s - c) gives r = (s - c) tan(gamma/2). Hence r3 - r = (s - (s - c)) tan(gamma/2) = c tan(gamma/2), so (r3 - r) cot(gamma/2) = c,

as required.

Chapter 13: Inverse Trigonometric Functions

Exercise 13.1 (book p. 398)

1. Evaluate without using tables / calculator: i) sin^-1(1) ii) sin^-1(-1) iii) cos^-1(sqrt(3)/2) iv) tan^-1(-1/sqrt(3)) v) cos^-1(1/2) vi) tan^-1(1/sqrt(3)) vii) cot^-1(-1) viii) cosec^-1(-2/sqrt(3)) ix) sin^-1(-1/sqrt(2))

Using the principal-value ranges (sin^-1 in [-pi/2, pi/2], cos^-1 in [0, pi], tan^-1 in (-pi/2, pi/2), cot^-1 in (0, pi), cosec^-1 in [-pi/2, pi/2] except 0): i) sin^-1(1) = pi/2. ii) sin^-1(-1) = -pi/2. iii) cos^-1(sqrt(3)/2) = pi/6. iv) tan^-1(-1/sqrt(3)) = -pi/6. v) cos^-1(1/2) = pi/3. FLAG: the key's figure for this part appears to read pi/6 in the scan, but cos^-1(1/2) = pi/3 exactly. vi) tan^-1(1/sqrt(3)) = pi/6. vii) cot^-1(-1) = 3pi/4. viii) cosec^-1(-2/sqrt(3)) = -pi/3. ix) sin^-1(-1/sqrt(2)) = -pi/4.

2. Without using table/ Calculator show that: i) tan^-1(5/12) = sin^-1(5/13) ii) 2cos^-1(4/5) = sin^-1(24/25) iii) cos^-1(4/5) = cot^-1(4/3)

i) Let theta = tan^-1(5/12), so tan theta = 5/12 with theta acute; then sin theta = 5/13, so theta = sin^-1(5/13). ii) Let phi = cos^-1(4/5), so cos phi = 4/5 and sin phi = 3/5 (phi acute); then sin 2phi = 2 sin phi cos phi = 2(3/5)(4/5) = 24/25, and 2phi is acute (about 73.7 deg), so sin^-1(24/25) = 2phi = 2cos^-1(4/5). iii) Let phi = cos^-1(4/5), so cos phi = 4/5 and sin phi = 3/5 (phi acute); then cot phi = 4/3, so phi = cot^-1(4/3).

3. Find the value of each expression: i) cos(sin^-1(1/sqrt(2))) ii) sec(cos^-1(1/2)) iii) tan(cos^-1(sqrt(3)/2)) iv) csc(tan^-1(-1)) v) sec(sin^-1(-1/2)) vi) tan(tan^-1(-1)) vii) sin(sin^-1(1/2)) viii) tan(sin^-1(-1/2)) ix) sin(tan^-1(-1))

i) sin^-1(1/sqrt(2)) = pi/4, so cos(pi/4) = 1/sqrt(2). ii) cos^-1(1/2) = pi/3, so

sec(pi/3) = 2. iii) cos^-1(sqrt(3)/2) = pi/6, so tan(pi/6) = 1/sqrt(3). iv) tan^-1(-1) = -pi/4, so csc(-pi/4) = -sqrt(2). v) sin^-1(-1/2) = -pi/6, so sec(-pi/6) = 2/sqrt(3). vi) tan(tan^-1(-1)) = -1. vii) sin(sin^-1(1/2)) = 1/2. viii) sin^-1(-1/2) = -pi/6, so tan(-pi/6) = -1/sqrt(3). ix) tan^-1(-1) = -pi/4, so sin(-pi/4) = -1/sqrt(2). Answers: i) 1/sqrt(2) ii) 2 iii) 1/sqrt(3) iv) -sqrt(2) v) 2/sqrt(3) vi) -1 vii) 1/2 viii) -1/sqrt(3) ix) -1/sqrt(2).

Exercise 13.2 (book p. 400)

Prove the following: 1. sin^-1(5/13) + sin^-1(7/25) = cos^-1(253/325) 2. tan^-1(1/4) + tan^-1(1/5) = tan^-1(9/19) 3. 2tan^-1(2/3) = sin^-1(12/13) [Hint: Let tan^-1(2/3) = x, and show sin 2x = 12/13] 4. tan^-1(120/119) = 2cos^-1(12/13) 5. sin^-1(1/sqrt(5)) + cot^-1(3) = pi/4 6. sin^-1(3/5) + sin^-1(8/17) = sin^-1(77/85) 7. sin^-1(77/85) - sin^-1(3/5) = cos^-1(15/17) 8. cos^-1(63/65) + 2tan^-1(1/5) = sin^-1(3/5)

1. Let A = sin^-1(5/13) (cos A = 12/13) and B = sin^-1(7/25) (cos B = 24/25). Then cos(A + B) = cos A cos B - sin A sin B = (12/13)(24/25) - (5/13)(7/25) = 288/325 - 35/325 = 253/325. Since A + B lies in (0, pi/2), it equals cos^-1(253/325). 2. tan(A + B) = (1/4 + 1/5)/(1 - 1/20) = (9/20)/(19/20) = 9/19, and A + B is acute, so A + B = tan^-1(9/19). 3. Let x = tan^-1(2/3); then sin x = 2/sqrt(13) and cos x = 3/sqrt(13), so sin 2x = 2(2/sqrt(13))(3/sqrt(13)) = 12/13. Since 2x is acute (about 67.4 deg), 2x = sin^-1(12/13). 4. Let phi = cos^-1(12/13); then tan phi = 5/12 and tan 2phi = 2(5/12)/(1 - 25/144) = (5/6)/(119/144) = 120/119. Since 2phi is acute (about 45.2 deg), tan^-1(120/119) = 2phi. 5. Let theta = sin^-1(1/sqrt(5)), so tan theta = 1/2, and phi = cot^-1(3), so tan phi = 1/3. Then tan(theta + phi) = (1/2 + 1/3)/(1 - 1/6) = 1, and theta + phi lies in (0, pi/2), so theta + phi = pi/4. 6. sin(A + B) = (3/5)(15/17) + (4/5)(8/17) = 45/85 + 32/85 = 77/85, and A + B is acute (about 64.9 deg), so A + B = sin^-1(77/85). 7. Let A = sin^-1(77/85) (cos A = 36/85) and B =

sin^-1(3/5) (cos B = 4/5). Then cos(A - B) = (36/85)(4/5) + (77/85)(3/5) = 144/425 + 231/425 = 375/425 = 15/17, and A - B lies in (0, pi/2), so A - B = cos^-1(15/17).

8. Let phi = tan^-1(1/5); then tan 2phi = 5/12, so sin 2phi = 5/13 and cos 2phi = 12/13. Let psi = cos^-1(63/65); then sin psi = 16/65. Hence sin(psi + 2phi) = (16/65)(12/13) + (63/65)(5/13) = 192/845 + 315/845 = 507/845 = 3/5, and psi + 2phi is acute (about 36.9 deg), so psi + 2phi = sin^-1(3/5).

9. tan^-1(3/4) + tan^-1(3/5) - tan^-1(8/19) = pi/4 [Hint: First add tan^-1(3/4) + tan^-1(3/5) and then proceed] 10. sin^-1(4/5) + sin^-1(5/13) + sin^-1(16/65) = pi/2 11. tan^-1(1/11) + tan^-1(5/6) = tan^-1(1/3) + tan^-1(1/2) 12. 2tan^-1(1/3) + tan^-1(1/7) = pi/4

9. tan^-1(3/4) + tan^-1(3/5) = tan^-1((3/4 + 3/5)/(1 - 9/20)) = tan^-1(27/11) (the sum is acute, about 67.8 deg). Then tan^-1(27/11) - tan^-1(8/19) = tan^-1((27/11 - 8/19)/(1 + 27 x 8/(11 x 19))) = tan^-1(425/425) = tan^-1(1) = pi/4 (the difference lies in (0, pi/2)). 10. Let A = sin^-1(4/5) (cos A = 3/5) and B = sin^-1(5/13) (cos B = 12/13). Then sin(A + B) = (4/5)(12/13) + (3/5)(5/13) = 63/65 and cos(A + B) = (3/5)(12/13) - (4/5)(5/13) = 16/65, so A + B = cos^-1(16/65) = pi/2 - sin^-1(16/65). Hence A + B + sin^-1(16/65) = pi/2. 11. The left side has tangent (1/11 + 5/6)/(1 - 5/66) = 1 and lies in (0, pi/2) (about 5.2 deg + 39.8 deg = 45 deg), so it equals pi/4. The right side has tangent (1/3 + 1/2)/(1 - 1/6) = 1 and likewise equals pi/4. Hence both sides are equal. 12. 2tan^-1(1/3) = tan^-1(2(1/3)/(1 - 1/9)) = tan^-1(3/4). Then tan^-1(3/4) + tan^-1(1/7) has tangent (3/4 + 1/7)/(1 - 3/28) = 1 and lies in (0, pi/2) (about 36.9 deg + 8.1 deg = 45 deg), so it equals pi/4.

13. Show that cos(sin^-1 x) = sqrt(1 - x^2) 14. Show that sin(2cos^-1 x) = 2x sqrt(1 - x^2) 15. Show that cos(2sin^-1 x) = 1 - 2x^2 16. Show that tan^-1(-x) = -tan^-1 x 17. Show that sin^-1(-x) = -sin^-1 x 18. Show that cos^-1(-x) = pi - cos^-1 x 19. Show that tan(sin^-1 x) = x/sqrt(1 - x^2)

13. Let theta = sin^-1 x, so theta lies in [-pi/2, pi/2] where cosine is non-negative;

cos theta = sqrt(1 - sin^2 theta) = sqrt(1 - x^2). 14. Let theta = cos^-1 x, so cos theta = x and sin theta = sqrt(1 - x^2); then sin 2theta = 2 sin theta cos theta = 2x sqrt(1 - x^2). 15. Let theta = sin^-1 x; then cos 2theta = 1 - 2sin^2 theta = 1 - 2x^2. 16. Tangent is an odd function and its principal range is symmetric, so tan^-1(-x) = -tan^-1 x. 17. Sine is an odd function and its principal range is symmetric, so sin^-1(-x) = -sin^-1 x. 18. Let theta = cos^-1 x; then cos(pi - theta) = -cos theta = -x and pi - theta lies in [0, pi], so cos^-1(-x) = pi - theta = pi - cos^-1 x. 19. Let theta = sin^-1 x; then tan theta = sin theta/cos theta = x/sqrt(1 - x^2).

20. Given that x = sin^-1(1/2), find the values of following trigonometric functions: sin x, cos x, tan x, cot x, sec x and csc x.

x = sin^-1(1/2) = pi/6. Hence sin x = 1/2, cos x = sqrt(3)/2, tan x = 1/sqrt(3), cot x = sqrt(3), sec x = 2/sqrt(3), csc x = 2.

Chapter 14: Solutions of Trigonometric Equations

Exercise 14 (book p. 407)

1. Find the solutions of the following equations which lie in [0, 2pi]: i) sin x = -sqrt(3)/2 ii) cosec theta = 2 iii) sec x = -2 iv) cot theta = 1/sqrt(3)

i) sin x = -sqrt(3)/2 with x in [0, 2pi]: x = 4pi/3, 5pi/3. ii) cosec theta = 2 means sin theta = 1/2: theta = pi/6, 5pi/6. iii) sec x = -2 means cos x = -1/2: x = 2pi/3, 4pi/3. iv) cot theta = 1/sqrt(3) means tan theta = sqrt(3): theta = pi/3, 4pi/3.

2. Solve the following trigonometric equations: i) tan^2 theta = 1/3 ii) cosec^2 theta = 4/3 iii) sec^2 theta = 4/3 iv) cot^2 theta = 1/3

i) tan theta = +-1/sqrt(3), so theta = npi +- pi/6. In [0, 2pi] the values are pi/6, 5pi/6, 7pi/6, 11pi/6. ii) sin^2 theta = 3/4, so sin theta = +-sqrt(3)/2, giving theta = npi +- pi/3. In [0, 2pi] the values are pi/3, 2pi/3, 4pi/3, 5pi/3. iii) cos^2 theta = 3/4,

so cos theta = +-sqrt(3)/2, giving theta = npi +- pi/6. In [0, 2pi] the values are pi/6, 5pi/6, 7pi/6, 11pi/6. iv) tan^2 theta = 3, so tan theta = +-sqrt(3), giving theta = npi +- pi/3. In [0, 2pi] the values are pi/3, 2pi/3, 4pi/3, 5pi/3.

Find the values of theta satisfying the following equations: 3. 3tan^2 theta + 2sqrt(3) tan theta + 1 = 0 4. tan^2 theta - sec theta - 1 = 0 5. 2sin theta + cos^2 theta - 1 = 0 6. 2sin^2 theta - sin theta = 0 7. 3cos^2 theta - 2sqrt(3) sin theta cos theta - 3sin^2 theta = 0 [Hint: Divide by sin^2 theta] 8. 4sin^2 theta - 8cos theta + 1 = 0

3. The equation is (sqrt(3) tan theta + 1)^2 = 0, so tan theta = -1/sqrt(3), giving theta = npi - pi/6. In [0, 2pi] the values are 5pi/6, 11pi/6. 4. Replacing tan^2 theta by sec^2 theta - 1 gives sec^2 theta - sec theta - 2 = 0, i.e. (sec theta - 2)(sec theta + 1) = 0. Hence sec theta = 2, giving theta = 2npi +- pi/3, or sec theta = -1, giving theta = (2n + 1)pi. In [0, 2pi] the values are pi/3, 5pi/3, pi. 5. Replacing cos^2 theta by 1 - sin^2 theta gives 2sin theta - sin^2 theta = 0, i.e. sin theta (2 - sin theta) = 0, so sin theta = 0 (the factor 2 - sin theta is never zero). Hence theta = npi. In [0, 2pi] the values are 0, pi (and 2pi as the endpoint). 6. sin theta (2sin theta - 1) = 0, so sin theta = 0, giving theta = npi, or sin theta = 1/2, giving theta = 2npi + pi/6 or 2npi + 5pi/6. In [0, 2pi) the values are 0, pi, pi/6, 5pi/6 (as in the key). 7. Dividing by sin^2 theta gives 3cot^2 theta - 2sqrt(3) cot theta - 3 = 0, i.e. (3cot theta + sqrt(3))(cot theta - sqrt(3)) = 0. Hence cot theta = -1/sqrt(3), giving theta = 2pi/3, 5pi/3 in [0, 2pi), or cot theta = sqrt(3), giving theta = pi/6, 7pi/6 in [0, 2pi). 8. Replacing sin^2 theta by 1 - cos^2 theta gives 4cos^2 theta + 8cos theta - 5 = 0, i.e. (2cos theta - 1)(2cos theta + 5) = 0. Hence cos theta = 1/2 (the root cos theta = -5/2 is impossible), giving theta = 2npi +- pi/3. In [0, 2pi] the values are pi/3, 5pi/3.

Find the solution sets of the following equations: 9. sqrt(3) tan x - sec x - 1 = 0 10. cos 2x = sin 3x [Hint: sin 3x = 3sin x - 4sin^3 x] 11. sec 3theta = sec theta 12. tan 2theta + cot theta = 0 13. sin 2x + sin x = 0 14. sin 4x - sin 2x = cos 3x

9. Multiplying by cos x gives sqrt(3) sin x - 1 - cos x = 0, i.e. sqrt(3) sin x - cos x = 1, i.e. 2 sin(x - pi/6) = 1, so sin(x - pi/6) = 1/2. Hence x - pi/6 = pi/6 + 2npi or 5pi/6 + 2npi, giving x = pi/3 + 2npi or pi + 2npi. (Neither value makes cos x zero, so no solution is lost.) Solution set: {pi + 2npi} union {pi/3 + 2npi}. 10. Write cos 2x as sin(pi/2 - 2x); then sin 3x = sin(pi/2 - 2x) gives 3x = npi + (-1)^n (pi/2 - 2x). For n even this gives x = pi/10 + 2npi/5 (five residue classes modulo 2pi: pi/10, pi/2, 9pi/10, 13pi/10, 17pi/10); for n odd it gives x = pi/2 + 2npi, already included. Solution set: {pi/10 + 2npi/5} union {pi/2 + 2npi} (the key lists the five residue classes plus pi/2 + 2npi). 11. cos 3theta = cos theta (where both secants are defined) gives 3theta = 2npi +- theta, i.e. theta = npi or theta = npi/2. FLAG: the key lists {npi} union {npi/2}, but the odd multiples of pi/2 make sec theta and sec 3theta undefined and are inadmissible; the admissible solution set is {npi} only. 12. tan 2theta + cot theta = (sin 2theta sin theta + cos 2theta cos theta)/(cos 2theta sin theta) = cos theta/(cos 2theta sin theta). Setting this to zero gives cos theta = 0, i.e. theta = pi/2 + npi (with cos 2theta = -1 and sin theta = +-1, both non-zero, so no solution is lost). FLAG: the key's line for this question reads {pi/6 + npi} union {5pi/6 + npi}, but those values do not satisfy the equation (e.g. at theta = pi/6 the sum is 2sqrt(3), not 0); that set is the solution of tan^2 theta = 1/3 (Question 2(i)). The correct solution set is {pi/2 + npi}. 13. sin 2x + sin x = 2 sin(3x/2) cos(x/2) = 0, so 3x/2 = npi, i.e. x = 2npi/3, or cos(x/2) = 0, i.e. x = pi + 2npi. Solution set: {2npi/3} union {pi + 2npi}. FLAG: the key reads {npi} union {2pi/3 + 2npi} union {2pi/3 + 2npi} (the last set duplicated); it omits the {4pi/3 + 2npi} family of {2npi/3}. 14. sin 4x - sin 2x = 2 cos 3x sin x, so the equation becomes cos 3x (2 sin x - 1) = 0. Hence cos 3x = 0, giving x = pi/6 + npi/3, or sin x = 1/2, giving x = pi/6 + 2npi or 5pi/6 + 2npi. FLAG: the key's first family reads {pi/6 + 2npi/3}, which covers only half of the cos 3x = 0 solutions (it omits pi/2 + 2npi/3, i.e. pi/2, 7pi/6, 11pi/6 modulo 2pi); the complete family is {pi/6 + npi/3}.

15. sin x + cos 3x = cos 5x 16. sin 3x + sin 2x + sin x = 0 17. sin 7x - sin x = sin 3x

18. sin x + sin 3x + sin 5x = 0 19. sin theta + sin 3theta + sin 5theta + sin 7theta

= 0 20. cos theta + cos 3theta + cos 5theta + cos 7theta = 0

15. sin x = cos 5x - cos 3x = -2 sin 4x sin x, so sin x (1 + 2 sin 4x) = 0. Hence sin x = 0, giving x = npi, or sin 4x = -1/2, giving 4x = 7pi/6 + 2npi or 11pi/6 + 2npi, i.e. x = 7pi/24 + npi/2 or 11pi/24 + npi/2. Solution set: {npi} union {7pi/24 + npi/2} union {11pi/24 + npi/2} (the key writes {npi} as {2npi} union {pi + 2npi}). 16. (sin 3x + sin x) + sin 2x = 2 sin 2x cos x + sin 2x = sin 2x (2cos x + 1) = 0. Hence sin 2x = 0, giving x = npi/2, or cos x = -1/2, giving x = 2pi/3 + 2npi or 4pi/3 + 2npi. Solution set: {npi/2} union {2pi/3 + 2npi} union {4pi/3 + 2npi} (the key writes {npi/2} as {npi} union {pi/2 + npi}). 17. sin 7x - sin x = 2 cos 4x sin 3x, so the equation becomes sin 3x (2cos 4x - 1) = 0. Hence sin 3x = 0, giving x = npi/3, or cos 4x = 1/2, giving 4x = +-pi/3 + 2npi, i.e. x = pi/12 + npi/2 or 5pi/12 + npi/2 (since -pi/12 + npi/2 covers the same values as 5pi/12 + npi/2). Solution set: {npi/3} union {pi/12 + npi/2} union {5pi/12 + npi/2} (the key writes {npi/3} as {(2/3)npi} union {pi/3 + (2/3)npi}). 18. (sin 5x + sin x) + sin 3x = 2 sin 3x cos 2x + sin 3x = sin 3x (2cos 2x + 1) = 0. Hence sin 3x = 0, giving x = npi/3, or cos 2x = -1/2, giving x = pi/3 + npi or 2pi/3 + npi. The cosine families are subsets of {npi/3}, so the solution set is {npi/3} (the key writes it in expanded form as {2npi} union {pi + 2npi} union {pi/3 + npi} union {2pi/3 + npi}). 19. (sin 7theta + sin theta) + (sin 5theta + sin 3theta) = 2 sin 4theta cos 3theta + 2 sin 4theta cos theta = 4 sin 4theta cos 2theta cos theta = 0. Hence sin 4theta = 0, giving theta = npi/4, or cos 2theta = 0, giving theta = pi/4 + npi/2, or cos theta = 0, giving theta = pi/2 + npi. Both cosine families are subsets of {npi/4}, so the solution set is {npi/4} (the key writes it in expanded redundant form). 20. (cos 7theta + cos theta) + (cos 5theta + cos 3theta) = 2 cos 4theta cos 3theta + 2 cos 4theta cos theta = 4 cos 4theta cos 2theta cos theta = 0. Hence cos 4theta = 0, giving theta = pi/8 + npi/4, or cos

2theta = 0, giving theta = pi/4 + npi/2, or cos theta = 0, giving theta = pi/2 + npi.

Solution set: {pi/8 + npi/4} union {pi/4 + npi/2} union {pi/2 + npi}. FLAG: the key lists only {pi/8 + npi/2} union {pi/4 + npi} union {pi/2 + 2npi}, i.e. only the even-indexed half of each family; the odd residues (e.g. 3pi/8, 3pi/4, 3pi/2 modulo 2pi) are omitted.